THE COMPLETE RETAINED RESEARCH RECORD

Zenon Taproot Puzzle Master Solve

From the original public puzzle
through 6 November 2026

TminusZ

The authenticated treasure solve. The subsequent investigation. Successful and unsuccessful tests. Corrections, exact inputs, and the evidence needed to continue.

Includes the complete October 13 technical reference and a guide to the independently checked preservation collection. In this print edition, [EYES] represents the eyes emoji; the manuscript and source files retain the original Unicode character.

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TminusZ

Research record through 6 November 2026

Preface

This book brings the Zenon treasure hunt and the subsequent investigation into one readable record. It begins with the public puzzle, explains the Bitcoin artwork and the authenticated first solution, and follows the later work through signatures, historical documents, poems, image construction, geometry, and competing readers. It preserves the experiments that failed as well as those that succeeded. A correction changes the conclusion attached to a result; it does not erase the result or the path that led to it.

The original treasure objective has a reproducible solution. The later investigation has established a substantial body of exact arithmetic, source relationships, geometric mechanisms, and bounded negative results. A complete authenticated secondary message has not been recovered. Those conclusions can coexist: the first solve is secure, and the later research contains real findings whose intended use is still being investigated. (B:Authenticated primary solve; R1815)

The readable account and the preservation collection serve different purposes. The book explains the investigation. The collection retains the exact reconstructed source texts, data, scripts, reports, figures, and package contents needed to inspect it. The source catalogue connects the two. The evidence can be read without continuing this conversation or trusting a model's recollection.

The present result

The strongest results fall into several distinct groups.

The public treasure has been solved. Nineteen Bitcoin transaction payloads form a 19 by 38 character artwork. Four visible Base64 fields provide an AES-GCM nonce, ciphertext, and authentication tag. With the key derived from the exact word `Taproot`, the ciphertext authenticates and yields `wall nation drive smoke`. Together with the preserved eight-word prefix, this gives the public twelve-word quest mnemonic and its native Zenon account. (B:Authenticated primary solve; V01)

The construction has a common signing authority. The retained native verification attributes forty construction-signature events to public key Q. The terminal transaction commits to `ZENON NETWORK`. This proves a relationship between the construction and its signing key. It does not identify the human or organization controlling that key. (B:K09; B:Authenticated primary solve)

The artwork has reproducible internal structure. Its punctuation, q/1 markers, empty regions, repeated tiles, and mirrored boundaries support exact descriptions. A four-character `qzzq` move joins two original blank regions into a fifty-five-cell region. The original lower opening maps into the upper opening under a specified half-turn, leaving eleven upper roof cells unmatched. These calculations can be checked directly against the published characters. (V01; R1674)

The parity readers have a precise information limit. Three pairs of different printed glyphs give the same retained LOW/FULL code: z/comma, 1/semicolon, and q/period. Across the artwork these account for 285 literal choices. Adding the original ASCII bit-1 values resolves those choices when occupancy and the four native fields are retained. This identifies information discarded by the reader; it does not establish that all 285 choices were independently authored message bits. (V01; R1803)

Several later candidate readers are coherent but remain inferred. The mirror-XOR construction producing FOLLOW GOLD retains all five variable choices in the latest thirty-two-pattern calibration family. It therefore depends on the particular glyph choices, rather than following inevitably from the silhouette alone. Its input relocation, operation, and serialization still require an independently grounded selection rule. (R1815)

Historical-authority claims remain a separate question. The 145 historical Bitcoin-message signatures, OpenPGP certificate signatures, and public-key calculations have their own valid mathematical or native relationships. None of those results, by itself, proves that historical Satoshi controlled Q, that the historical-message publisher constructed the artwork, or that a later symbolic SATOSHI label is an external signature of authorship. (B:K24-K27; B:Current proof gaps and supersession rules)

How to use this book

Read the opening account and the historical narrative first for the original problem and its solution. Continue into the October developments and the visual investigation to understand how the research changed. The technical reference preserves the detailed October atlas, including constants and complete branch descriptions. The correction ledger explains which parts of that older atlas require a narrower present reading.

Source labels are deliberately simple. **B** identifies a section of the retained October 13 technical atlas. **S** identifies one of its reconstructed source records. **R** identifies a later recovered file. A package-member identifier adds **M** and a member number. **V01** identifies the independent offline checks performed while preparing this edition. The preservation collection includes the catalogues behind these labels.

An experiment can be completely reproducible without being a complete solution. For example, turning a selected shape may give an exact bit string. That establishes the output of the stated operation. It leaves a different question: what selected that shape, turn, direction, and interpretation before the output was seen? The book keeps both questions visible so a useful calculation is neither lost nor overstated.

Dates identify the evidence being discussed. A later report is not automatically more reliable; its new source, counterexample, or corrected calculation must explain why it changes an earlier conclusion. Several reports contain the words complete, verified, proof, or breakthrough in their historical titles. Those titles are retained for identification. The current conclusions depend on the substance and controls, not the title.

The investigation in chronological order

Period or event	What entered the record	What the reader should retain
Earlier public quests	The preserved eight-word prefix, associated with the project's earlier hunt stages	The prefix is usable public puzzle input; original issuer/backend delivery remains a distinct provenance question.
14 November 2021	Bitcoin block 709632 and the nineteen-row construction	This is the native source for the exact ASCII object and its transaction commitments.
November 2021 clue publications	The key theme, block reference, prize, and later quantum-computer wording	The later source audit dates the quantum clue to November 17; a November 12 compilation heading must not be substituted for that timestamp.
24 November 2021 project genesis setting	The genesis anchor and the project quotation	The configuration/account relationships have native checks; their publication chronology must be established separately.
December 2021 to January 2022	Quest-account action, later Q history, and retained release/configuration records	Account-key, Bitcoin-Q, and configuration-authority claims are separate.
Earlier 2026 investigation	ART arithmetic, symbolic interpreters, historical points, certificates, correspondence, and external-source routes	Many branches reproduce under declared rules; their input selection and execution assumptions must remain attached.
October 13 consolidation	The detailed technical atlas and its 8,024 source identities	The atlas is preserved completely at its declared normalized/literal-text level.
October 14 to 20	GLASS, writable-space hypotheses, residuals, historical panels, PULL/PIPE, and mirror readers	Positive local results and negative controls define a conditional continuation.
October 21 to 29	Source chronology, E1/E2, native-witness comparisons, donor749, physical readers, and historical rendering	Exact witnesses and pointers constrain proposed routes without automatically proving authorial intent.
October 30 to November 2	Direct sightlines, occlusion, paired boundary faces, qzzq movement, and projected eyes	Direct observation and transported-coordinate readings must be distinguished.
November 3 to 4	Literal-choice information, renderer families, broad angular scans, and native proof assembly	A fitted generator and a historically identified generator are different claims.
November 5 to 6	Frame calibration, withheld marker reconstruction, thirty-two-pattern tests, original-source instruction audit, and video frame comparison	The latest results sharpen the geometry and source constraints while leaving the full secondary reading rule open.

This timeline summarizes the retained sources. It does not resolve every missing original post, historical software state, or abandoned experiment. The full catalogue preserves the finer chronology. (B:Authenticated primary solve; B:Reconciled later findings; R1812; R1815; R1823)

The object everyone must begin with

The source image is a rectangle of characters, not an approximate visual logo. There are nineteen rows, each containing thirty-eight ASCII bytes. Spaces, punctuation, lower-case q, the digit 1, and repeated z characters all matter. The eighty characters belonging to the four visible Base64 fields must be distinguished from the structural glyphs outside those fields.

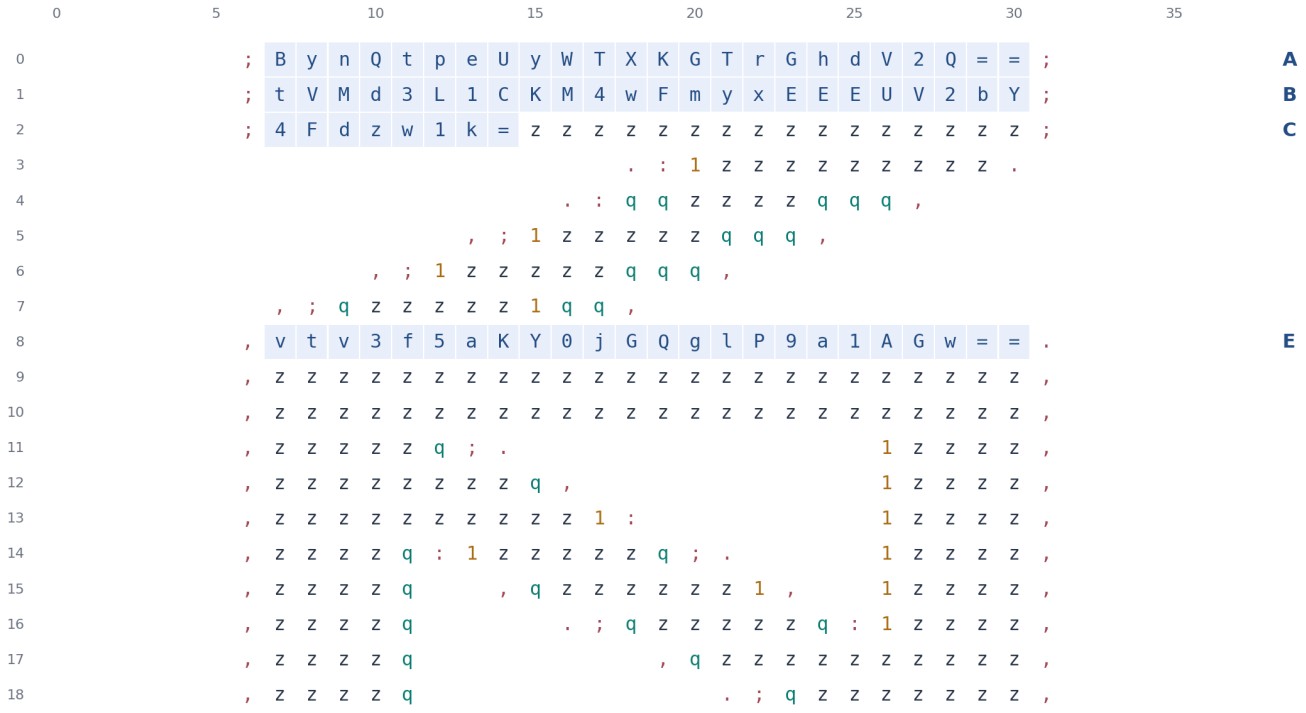


Figure 1. The exact nineteen-row source, with editorial colours identifying the four native fields and structural glyph classes. Colour is explanatory and is not part of the Bitcoin payload. Row and column numbers are zero-based. (V01)

The separator-free byte sequence has SHA-256:

```
869020c3e598955bfc71244923c1fb557a5e0d4cadd89443012c8232bfa618d7
```

The version with a line feed after every row has 741 bytes and a different SHA-256:

```
96534d9faf3b6260987939fdb6994cd5b97eed18a175b0620c4ea5256c0c5607
```

Both are useful serializations. They must not be compared as though they were the same byte string. A screenshot, proportional font, trimmed text field, or changed newline convention can preserve a visual resemblance while changing the object being tested. (V01)

The preservation collection contains the row strings, the 722-byte flat form, and the line-delimited display form. The main narrative uses zero-based coordinates unless a source explicitly requires a different convention. A half-open interval such as columns [7:31) includes column 7 and stops before column 31.

A short glossary before the technical chapters

Term	Meaning in this investigation
ASCII byte	The numerical encoding of one printed character or space in the original artwork.
Base64	A printable encoding used by the four visible fields. Decoding gives bytes; it does not decrypt them.
AES-GCM	The authenticated-encryption scheme used in the primary solve. A successful tag check validates the exact key, nonce, ciphertext, tag, and associated-data combination.
Nonce	An input to an encryption or signature operation. A nonce is not automatically a password or an instruction.
Authentication tag	The GCM value checked when decrypting. It is stronger evidence for that decryption than an attractive-looking plaintext alone.
BIP39	The word-list and checksum system used by the public quest mnemonic and by many later experimental numerical readers.
U11	An eleven-bit unsigned value from 0 to 2047. It can index the BIP39 English list; a pair of such indexes is not automatically a valid mnemonic.
ART	The BIP39 label at index 102, used as a retained arithmetic constant in several branches.
Q	The common public key recovered and verified for the Bitcoin construction. It is a mathematical authority label, not a person's identity.
r and s	The two scalar components of an ECDSA signature. Their representation and ordering matter in experimental readers.
DER	The structured encoding used for the construction's transaction ECDSA signatures. The 145 historical Bitcoin Signed Message signatures use a separate 65-byte compact encoding: header, r, and s. Byte offsets depend on the exact encoded object.
Witness	Transaction data carrying signatures and related spending information. A different valid signature can still be a different committed witness.
OP_RETURN	A Bitcoin script operation used here to publish data that cannot be spent through that output.
Hash	A digest of exact bytes. Equal text appearance does not guarantee equal byte hashes.
Commitment	A relationship binding an object to data, subject to the actual hash/signature/consensus rules that establish it.
GLOBAL40	A retained forty-item numerical table derived under a declared field organization. Every lookup still requires an explicit index and representation.
CONTROL98	The ninety-eight two-bit structural controls, giving a 192-bit body and four-bit trailing check under the retained reader.
LOW and FULL	Parity readers applied to the low nibble and full ASCII byte in the retained model. They are two views of a character's encoding, not two photographs.

Term	Meaning in this investigation
Occupancy	Whether a cell is a space or contains a printed character. This is different from the ink shape within the cell.
Source owner	The identity of the original character that supplied a later displayed cell. Identical glyphs can have different owners.
Aperture or cavity	A connected region of blank cells through or from which a geometric reader is defined.
First hit	The first opaque cell or glyph encountered along a declared ray. Later characters can be occluded.
T30	The specific half-turn T(r,c) = (30-r,37-c), used in the larger frame comparisons.
U	The separate local tile half-turn U(r,c) = (27-r,30-c). It must not be substituted for T30.
Projected or virtual cell	A position produced by an extension, reflection, or display construction. It is not an extra published Bitcoin character.
Renderer	A procedure turning inputs into a character image. Reproducing the final image does not alone identify its historical production process.
Holdout	Information withheld from a fitting procedure. A later mask over a previously inspected target is a useful isolation test, but not a historically blind discovery.
Counterexample	An explicit alternative satisfying a proposed rule while defeating a claimed conclusion. Its value is bounded by that rule's stated scope.
Consumer	The operation or object that is supposed to use a previously calculated result. A meaningful word without a selected consumer can remain an endpoint.
Supersession	A documented correction replacing an earlier interpretation while preserving the earlier evidence and calculation.

How the evidence labels should be read

Label	What is established	What is not automatically established
Authenticated	A specified native check succeeds in its own domain.	Every surrounding interpretation, a human identity, or a second protocol's authority.
Direct observation	A feature is present in an identified source.	Its intended meaning.
Mechanical	A specified computation gives the stated result.	Why that computation should have been selected.
Supported interpretation	Evidence favours a reading with explicit alternatives.	Certainty or cryptographic authentication.
Conditional	The result follows if named assumptions are admitted.	Those assumptions themselves.
Rejected within scope	A defined test or family fails.	The failure of every possible encoding or route.
Open	A required source, operation, selection, or verification relationship is missing.	That the missing relationship cannot exist.

The later chapters use these distinctions where they affect a conclusion. They are not a reason to stop documenting useful results. They make it possible to continue from the actual frontier instead of repeatedly rediscovering an old success or reviving a rejected assumption. (B:Evidence language)

What was rechecked for this edition

The edition's offline verification reconstructs the canonical artwork and checks the AES-GCM solution, the structural counts, CONTROL98's four-bit check, the equal-code glyph-pair counts, the additional literal-choice reconstruction, the original blank components, the qzzq move, the eleven unmatched roof cells under T30, and the local rotated A/C tile equality. The complete inputs, script, and results are retained as V01.

The historical archive was independently decoded and all 8,024 reconstructed source texts matched their declared normalized or literal-text hashes. This verifies the preservation process. It does not mean every historical experiment was rerun, every original binary file was recovered, or every interpretation in those records is correct. Later chapters identify results supported by prior receipts rather than silently calling them new execution results.

The archive's original-file hashes remain useful provenance. Where the old master preserved an anonymized or normalized text representation of a DOCX, log, or source file, the reconstruction verifies that representation. The collection does not mislabel it as the original binary container. This distinction lets the research remain both readable and honest about what has survived.

Part I — The original puzzle and the historical investigation

This part explains how the original treasure was solved, why investigation continued, and what the historical and cryptographic branches actually established. It consolidates the results retained in the October 13 baseline. The bracketed references identify the original baseline sections; they also distinguish preserved experimental receipts from experiments performed for a later edition. Later chapters govern developments after this cutoff. An earlier open question is not silently treated as current when a later test has answered it.

The central result is substantial: the published Bitcoin construction contains an authenticated decryption that completes a public twelve-word mnemonic, and that mnemonic derives the account receiving the documented Zenon prize allocation. Further numerical, visual, literary and signature relationships also reproduce. Their status depends on whether the rule selecting them is established, not merely whether their arithmetic works. Keeping those distinctions visible preserves the useful work without turning provisional links into historical facts. (B:Authenticated primary solve; B:Evidence language; B:Current proof gaps and supersession rules)

1. The original object is a transaction construction

The starting artifact is a set of linked Bitcoin transactions. A funding transaction creates nineteen staging outputs. Nineteen subsequent transactions each publish one line of artwork and leave a spendable output. A terminal transaction spends those nineteen outputs in the order that defines the picture. This gives a native, recoverable ordering: the image is not dependent on the order in which an explorer happens to display transactions. Each published row is exactly 38 ASCII bytes, giving nineteen rows and 722 artwork cells. (B:Authenticated primary solve; B:K34; B:K35)

Construction component	Retained native result	Why it matters
Funding	Block 709594; nineteen staging outputs of 75,000 satoshis	Establishes the common construction and the inputs available to the row transactions
Nineteen rows	Block 709632; each output 0 contains 38 ASCII bytes	Provides the exact artwork, including spaces and punctuation
Row outputs and fees	Nineteen outputs of 69,240 satoshis; 5,760 satoshis fee per row transaction	Preserves the complete amount pattern without presuming it is another code
Terminal	Block 709632; spends the row outputs in row order	Fixes the assembly and publishes the literal project name
Other fees	Funding fee 25,000; terminal fee 1,008,000 satoshis	Separates ordinary transaction accounting from later numerical hypotheses

Construction component	Retained native result	Why it matters
Core signatures	Forty P2WPKH/SIGHASH_ALL events verifying under Q	Authenticates the Bitcoin construction in its actual signing domain
Later sweep	Two further Q signature events	Creates a separate 42-event population; it must not be confused with the original forty

These are SegWit-v0 transactions using BIP143 signature hashing. Taproot supplies the activation setting for the puzzle; it does not change these particular spends into Taproot spends. Uniform transaction version 2, funding locktime 709593, and non-final sequences have ordinary protocol functions. A proposed hidden channel in these fields has to demonstrate actual varying, author-controlled information. Their presence alone is insufficient. (B:Authenticated primary solve; B:What the steganography book audit actually adds)

The terminal output script is `6a0d5a454e4f4e204e4554574f524b`. Its payload is the thirteen ASCII bytes `ZENON NETWORK`. All nineteen terminal signatures commit to this output. The Bitcoin-to-Zenon connection therefore begins with a literal signed construction, before any literary interpretation or dictionary arithmetic. The funding transaction and terminal transaction are the preserved public references. (B:Authenticated primary solve)

The exact row strings are retained in the constants and raw evidence volumes. Two serialization lengths must remain distinct: the bare artwork is 722 bytes; adding an LF after every row produces 741 bytes. The latter has SHA256 `96534d9faf3b6260987939fdb6994cd5b97eed18a175b0620c4ea5256c0c5607`. Trailing spaces belong to the source. A screenshot, a rewrapped code block, or a copied image is a presentation of that source rather than a substitute for its bytes. (B:K35)

1.1 Four fields complete the seed

Four Base64 substrings are embedded in the artwork. Coordinates below use zero-based rows and half-open column intervals: `[7:31)` includes columns 7 through 30. Decoding is performed independently for each field. (B:Authenticated primary solve; B:K35)

Field	Location	Base64 text	Decoded role and length
A	Row 0, columns `[7:31)`	`BynQtpeUyWTXKGTrGhdV2Q==`	Nonce, 16 bytes
B	Row 1, columns `[7:31)`	`tVMd3L1CKM4wFmyxEEEUV2bY`	First ciphertext segment, 18 bytes
C	Row 2, columns `[7:15)`	`4Fdzw1k=`	Second ciphertext segment, 5 bytes
E	Row 8, columns `[7:31)`	`vtv3f5aKY0jGQglP9a1AGw==`	Authentication tag, 16 bytes

The decryption uses AES-256-GCM, with a key equal to SHA256 of the exact ASCII text `Taproot`, nonce A, ciphertext B followed by C, tag E, and no additional authenticated data. It produces:

```
wall nation drive smoke
```

GCM matters because it checks more than whether a candidate output looks like English. Its authentication tag checks the ciphertext under the specified key and nonce. The successful decryption is the firmest decoding step in the solve. It authenticates this completion in its cryptographic domain; it does not authenticate every later interpretation of the same bytes. (B:Authenticated primary solve)

The public eight-word prefix and the recovered four-word completion form:

```
oblige dilemma hurry disorder happy spoil shiver key wall nation drive smoke
```

This is public puzzle data. Its full twelve-word form passes the BIP39 checksum. The eight-word prefix, normalized to single spaces with no terminal newline, has SHA256 `e0468f21f2640a0a01a30a864d115bda60ce686eedc225b228625cb4cac463dd`. Keeping the prefix and completion separate preserves their different provenance: one was publicly disclosed; the other was recovered through authenticated decryption. (B:Authenticated primary solve)

1.2 The mnemonic reaches the documented prize

The accepted derivation uses an empty mnemonic passphrase and the native hardened Ed25519 path `m/44'/73404'/0'`. Zenon's SHA3-256 and Bech32 encoding then produces:

```
z1qrn3jeapt848zxg3akf2ewhrxxwsa945sj798s
```

In the pinned official genesis configuration, this is a unique typed match at `GenesisBlocks.Blocks[9190].Address`, among 14,090 account records. Its allocation is 474,513,400,000 raw QSR units, equivalent to 4,745.134 QSR. The retained audit reproduces the native account-block hashes and verifies the height-2 Ed25519 signature under the derived quest key. The pinned SDK, pinned genesis configuration, and quest preservation copy anchor this result. (B:Authenticated primary solve)

There are several different authorities here. Bitcoin Q authenticates the original transaction construction. The derived quest key authenticates its Zenon account actions. An unsigned genesis record and a node's reported inclusion are not additional Q signatures, and reproducing selected native objects is narrower than independently validating an entire consensus chain. Those qualifications leave the original treasure derivation intact. (B:Authenticated primary solve)

The retained burn records close a further part of the account history. The send `ac3fb28796c0aec1c512396e54f23fe5882246c44372cc23897d455cfe88f7eb` was confirmed on 23 December 2021 at 14:55:10 UTC; its receive `c65cfdc5db8818729f19acea9289a965c0c5e95b1e15289700cce7b484f35272` followed at 14:56:10 UTC. It burns the 4,745.134 QSR allocation and contains no scripted continuation. A signature under a now-public puzzle mnemonic does not uniquely identify which person used it. (B:Parked infrastructure and settlement findings retain their narrower scope)

1.3 The block anchor and the prefix disclosure have separate histories

The configuration's ExtraData is exactly:

```
00000000000000000004dd040595540d43ce8ff5946eeaa403fb13d0e582d8f#We are all Satoshi#Don't trust.
 Verify
```

The first component names Bitcoin block 710968. Its retained 80-byte header reproduces the double-SHA256 block hash and encoded proof-of-work target. The two phrases are genuine project-source data. The pinned release commit is dated 22 January 2022; the file's contents alone do not establish public availability of that final configuration in November 2021. (B:Authenticated primary solve)

Later source recovery corrected the origin of the eight-word screenshot. Native metadata identifies community post 1460002976907595779, dated 14 November 2021 at 21:53:29 UTC, containing the exact eight words. It has no native attached image. Screenshot line wrapping therefore cannot be treated as an issuer-designed spatial alphabet. The original issuer-controlled first/middle-four disclosure, and the original links from a deleted first post, were still missing in the baseline. The source correction narrows what the screenshot can prove; it does not undo the successful mnemonic derivation. (B:The preserved eight-word image has a community Twitter origin)

2. Why investigation continued after the treasure

The next clue was numerical. BIP39 assigns each English word an index from 0 to 2047, so a word can represent eleven bits. The accepted mnemonic's final entropy byte is 102, whose word is ART. Subtracting 102 from the four recovered word indices, modulo 2048, gives:

```
WALL NATION DRIVE SMOKE
          minus ART
TRY  MAKE   CUSTOM SCALE
```

The arithmetic is exact and uses the authenticated completion. Reading the result as an invitation to investigate another scale is reasonable. The unresolved question is which scale, which input, and which next operation the source establishes. (B:K04)

2.1 The older word machine is a conditional model

The older proposal subtracts ART across all twelve mnemonic positions and treats the resulting ordered vector as a program. Its first label is MERGE. The later audit separated three steps that earlier narratives had run together: producing a vector, showing that it is executable code, and choosing its initial instruction. MERGE is an English label until a rule makes it an operation. A `4|4|4` arrangement is an ordered organization of data until a consumer uses it as code. (B:K04)

The final four positions reproduce TRY MAKE CUSTOM SCALE because those four positions supplied the original transformation. They are training examples. Reproducing them is a useful consistency check, but not an independent forty-four-bit prediction. This matters because later branches can be perfectly reproducible while remaining conditional on this initial program interpretation. (B:K04)

Within the installed model, operations include addition, both subtraction directions, XOR, AND and OR. Addition and subtraction wrap modulo 2048; the bitwise operations act on eleven-bit values. A state can carry an operation, and that operation can affect the subsequent path. Thus `BEGIN[ADD]` and `BEGIN[AND]` are different occurrences, despite sharing the same word. A checksum repair called CHECK changes the checksum part of a candidate representation; it does not by itself authorize a new program, source or identity claim. (B:K07; B:K10; B:K34)

The graph became large: one retained archive has 8,120 definitions and 113,990 anchors. Those counts measure an installed research model, not thousands of independently authenticated puzzle instructions. A reentry study makes the distinction concrete: of 2,445 reentries, 300 retained the operation, 2,145 changed it, and 2,115 changed the complete path. No provenance transfers or restored parent histories were recorded. Reaching the same label again does not reconstruct a completed earlier join. (B:K10)

2.2 The forty-word table is exact source data

Concatenating the decoded bytes in artwork order—A, then B, then C, then E—gives 55 bytes, or 440 bits. That is exactly forty eleven-bit values, with no remainder. This source concatenation is different from the argument order used by AES-GCM. Displayed as five rows of eight, the values are: (B:K05; B:K35)

Positions	Eight words in source order
0–7	ALREADY · EXCESS · COLOR · CONNECT · ERODE · RAVEN · FREQUENT · ARRIVE
8–15	TWELVE · SPAWN · STEP · RECALL · FESTIVAL · MOMENT · INDICATE · PORTION
16–23	CHURN · IMMENSE · AIRPORT · SUNNY · BABY · CALM · EYEBROW · UNDER
24–31	SUGGEST · ACTOR · TABLE · MAMMAL · ORBIT · HOSPITAL · TARGET · VERSION
32–39	MEDIA · BOTTOM · COVER · LOTTERY · NEUTRAL · WIDTH · RELAX · ADDRESS

The conspicuous final column contains UNDER, VERSION and ADDRESS, and the last row contains several plausible instructions. The table is therefore worth preserving exactly. Its eight-column display is a proposed organization of forty values, however; it is not the artwork's physical width of 38 characters. (B:K05)

Decoding each field separately gives a different boundary structure. A provides eleven words and seven remaining bits `1011001`; B provides thirteen words and one remaining bit `0`; C provides three words and seven remaining bits `1011001`; E provides eleven words and seven remaining bits `0011011`. The twenty-two residual bits concatenate to RAZOR/NATURE. These are residuals of the eleven-bit reading, not unused AES data. The field-local and concatenated views reuse the same bytes and cannot be counted as independent witnesses. (B:K05; B:K35)

2.3 The source-local punctuation proposal remains distinct

A later proposal derives an ordered four-symbol alphabet from punctuation fields and their own attached suffixes. It addresses CUSTOM SCALE without needing the whole older program interpretation. The baseline retains it as a source-local avenue with its own tests and limitations. The physical chapters give its complete construction and subsequent developments. It does not retroactively validate every descendant of the older interpreter, and a later stronger physical result must not be used to rewrite the provenance of an older historical branch. (B:K04; B:K12; B:Current proof gaps and supersession rules)

3. Q authenticates the construction; historical points require separate selection

The construction public key is serialized directly in the witnesses:

```
Q = 0398cf668fe678cc6d8e2b20b5f574c9f398c7f031a1e91e9b85029e4cde597ad7
```

It matches the P2WPKH commitments and verifies the native signatures. This is stronger than discovering a suggestive dictionary label in a coordinate: Q's relationship to the construction is a protocol fact. (B:K09; B:K35)

An ECDSA signature can also be used to enumerate candidate public keys once the correct message digest is known. In the retained terminal study, each of nineteen signatures has two recovery candidates. The proposed pair 11/18 intersects at Q, but so do all 171 distinct terminal pairs. Across the forty core signatures, all 780 distinct pairs likewise isolate Q. Recovery demonstrates the verification relationship; choosing a special-looking pair does not make Q an exceptional hidden discovery. (B:K09; B:K14)

The proposed 11/18 selector uses two representations of a public recovery-gap point: raw DESIGN478 and normalized CLIMB344. Division by forty gives quotient 11 and remainder 38; division by nineteen gives quotient 18 and remainder 2. Those computations reproduce, but the choice of representation and divisor remains part of the selector's contract. English usefulness after division cannot choose the rule retroactively. (B:K09)

3.1 P2095 and P3928 are not interchangeable

Two historical Bitcoin public points recur in the record. P2095 is the older historical point. P3928 is the corrected point obtained from input 145 of the retained 205-input consolidation, through a route associated with the block 3928 coinbase. Each creates a different public difference point:

```
R2095 = Q − P2095
R3928 = Q − P3928
```

Public elliptic-curve subtraction is legitimate arithmetic. It does not demonstrate knowledge of either corresponding private scalar. Substituting one P or R for the other changes the object, its reductions, and its ancestry. Both complete SEC encodings remain in the constants volume. (B:K07; B:K08; B:K35)

The historical consolidation is transaction `29591d211054478ed22c93571955cb92ef4ec2b40d914a43ba3a91d52a171c61`. Its native list has 205 ordered, zero-based inputs. The retained raw transaction is 23,446 bytes, SHA256 `24f2ff30597e2821e83f7c42a7f9c5d0b144f5f509364ba21e49139f22f4abe6`. In the older route, the block 2905 coinbase outpoint identifies input 134; nonce-coordinate reductions give ACQUIRE/ADD; GLOBAL40's RECALL at position 11 supplies the proposed offset to input 145. The spent P2PK output there supplies P3928. (B:K08; B:Historical205-input pointers and the Entry39 alternatives)

The native historical transaction and public points are genuine. The original punctuation packing used to select block 2905 was later superseded. Consequently this remains a conditional route to genuine public objects, rather than the established continuation of the corrected punctuation scale. Keeping

that status avoids the false choice between discarding real ledger data and treating every route to it as proven. (B:K08)

3.2 The complete P3928/Q/R3928 reduction census

A coordinate census begins with an exact 32-byte coordinate, produces its canonical BIP39 representation, and tests eighteen fixed reductions: three pairings times six operations. Every checksum-valid result is retained with its endpoint and checksum. The six coordinates of P3928, Q and R3928 produced these nine valid results; P3928.x produced none. (B:K08; B:K34)

Coordinate	Pairing and operation	Endpoint	Checksum
P3928.y	Adjacent, right minus left	ASSAULT108	14
P3928.y	Mirror, ADD	AFFAIR34	0
Q.x	Halves, OR	BUYER251	9
Q.y	Mirror, OR	ART102	9
Q.y	Halves, ADD	ABOVE4	7
R3928.x	Adjacent, ADD	ASPECT107	12
R3928.x	Adjacent, OR	ALTER59	6
R3928.y	Adjacent, ADD	AGENT39	13
R3928.y	Halves, left minus right	ATTEND116	1

This is a complete result within a declared finite family. It demonstrates the arithmetic and checksum outcomes. It does not show that the dictionary meanings select a CALL definition, or that a four-bit checksum is a historical signature. Q's ART/ABOVE and BUYER relationships are retained without upgrading them into an identity assertion. (B:K08; B:K09)

4. What the P2095 compiler branch accomplished

The P2095 branch deserves a connected account because it contains real, reproducible work that later strict audits qualified. The August 28 reports began from the numbered-poem route, an ACTOR poem-line operand, P2095, and R2095. They assumed the whole-state six-operation CUSTOM SCALE and the CHECK normalizer. Later auditing did not validate that boot as an unconditional instruction from the root. Every result below therefore retains this historical compiler as an explicit dependency. (B:K07)

4.1 Dispatch and the ENVELOPE connection

The scale outputs were ADD→AISLE45, left-minus-right→ADVANCE31, right-minus-left→ABLE2, XOR→ABUSE9, AND→ADMIT29, and OR→ACTUAL23. Using the compressed R object as twenty-four complete eleven-bit cells gave the following retained dispatch. (B:K07)

Selector	Quotient on division by 24	Selected position	Label there	Carried result
AISLE45	1	21	DENY	DOCTOR
ADVANCE31	1	7	FIRE	PROUD
ABLE2	0	2	MERIT	MERCY
ABUSE9	0	9	STING	STATE
ADMIT29	1	5	PAVE	ACCOUNT
ACTUAL23	0	23	ENERGY	ENOUGH

Packing the six quotients gives binary `110010`, decimal 50, ALIEN. The report uses ENERGY with ALIEN to produce EXPECT and then FINAL, with contextual FILE/FINISH checks. Division is mechanical; packing the quotients and promoting them to a state are further rules of this compiler. They are not a general license to reinterpret any residual from a different branch. (B:K07)

For P2095.x, the eighteen pair/operator reductions have two CHECK-valid rows: adjacent/AND ending ANCHOR67 and mirror/ADD ending BONUS203. Their retained concatenation and normalization gives endpoint ANOTHER. The proposed connector installs BONUS into the final value of the mirrored R2095.x ADD result, appends the EXPECT checkpoint, and CHECK-normalizes the result. The checked foot becomes ENVELOPE and the entropy endpoint becomes BEGIN. These are separate outputs with different subsequent consumers. (B:K07)

The accompanying controls are useful. Arbitrary final-slot replacements across eighteen reductions and all 2,048 anchors give 167 ENVELOPE hits among 36,864 candidates. Requiring the same P.x pairing, operation and endpoint to be carried to R.x gives one hit among eighteen: mirror/ADD/BONUS. These numbers describe exactly what was compared. They do not supply a probability that the puzzle author intended the compiler, because the installation and appended checkpoint remain assumed steps. (B:K07)

The checked ENVELOPE foot was routed using several proposed divisors, including ACCOUNT12, 26 and 22. One retained route gives FESTIVAL and then `ENVELOPE − FESTIVAL = WALL`, returning to an authenticated root value. The BEGIN endpoint, carrying ADD, follows a longer path through AGAIN, EDGE, ANGER, BLOOD, LITTLE, CUSHION, DIARY, FADE, BODY, AUTHOR and DEBRIS to AREA. Returning to a known authenticated word is a recognizable loop; it does not authenticate every edge used to get there. (B:K07)

4.2 Input 6 and public verifier points

A separate consumer uses selectors as native byte offsets in 32-byte coordinates. AISLE45 is out of range and is excluded without a modulo repair. The remaining native byte order is `[2,9,23,29,31]`, whereas scale-generation order is `[31,2,9,29,23]`. The source retained both orders rather than using the decoded words to choose one. (B:K07)

At terminal input 6, the two relevant signature-coordinate byte positions yield BULLET/CABBAGE and AROUND/ATTEND. Cross-applying the operations associated with selectors 2 and 9 gives ACCOUNT at position 2 under both tested operations, ACT at position 9 under right-minus-left, and ACTOR at position 9 under XOR. The resulting arithmetic fingerprint is unique among the nineteen terminal witnesses. Cross-applying two operators as a two-by-two fanout remains a conditional instruction, even though its result is selective. (B:K07)

The public point construction uses the native signature values `r`, `s` and digest `z`, with `u1=z/s` and `u2=r/s` modulo the curve order. It then forms `A=u1G`, `C=u2P2095`, `D=u2R2095`, `K=A+u2Q`, and `V=A+D=K−C`. These are all publicly computable verification objects. This A is a local point name and must not be confused with the artwork's nonce field A. No private scalar is recovered. (B:K07)

Under the retained uniform-AND interpretation, V.x/V.y at the five selected positions return ABANDON, ACCOUNT, ABOVE, BEGIN and ABILITY in native order. A census of nineteen witnesses, five points and seven procedures gives 665 candidates; one matches the entire target and the same candidate matches its first/last pair. The competing per-selector operator carry instead returns VISA, BLUR, ARROW, BEGIN and VOCAL. These are different contracts, not two executions of the same instruction. (B:K07)

All child outcomes were kept, including cycles. ABANDON/AND remains ABANDON; ACCOUNT/AND reaches ABLE; ABOVE/AND cycles through ACTRESS; BEGIN/AND cycles through ABSURD and ACOUSTIC; ABILITY/AND enters the ABOVE/ACTRESS cycle. BEGIN/AND cannot be spliced onto the BEGIN/ADD route. Nor do nonadjacent ABANDON and ABILITY automatically form a call-zero instruction. (B:K07)

A later report joins three historical parents at ENVELOPE/ADD and follows ENVELOPE→KISS→BETRAY. Its installed BETRAY/AND handler emits BOOK/ABANDON; the old call-zero handler maps zero through cosine to ABILITY. Two complete thirty-seven-state rounds reproduce the same states, operations, external requests, output and acknowledgement, without new requests. This closes an earlier question inside that model. It does not establish its CALL-zero handler or restore the rejected first/last-word shortcut. (B:K07)

5. Native software execution taught an important boundary

A decoded label can name a real source concept without selecting a callable method. The go-zenon audit illustrates this. The retained commit `1baa7c4e057da4f2708a970b4fedf70a8de77fbe` contains 240 Go files and a fifty-five-term census. Its date is 23 November 2021, nine days after the artwork. Its merge and join functions operate on typed protocol or software objects. They do not accept seven BIP39 cells merely because the source contains MERGE, NETWORK or ZENON. The sole seven-argument ABI method, RegisterLegacy, lacks both a selected destination/method and valid argument types at the proposed root. (B:K10)

The selected call 1024 has what that root proposal lacks: an actual destination, ABI selector `47f12c81`, two typed RetrieveAssets fields, and native links. Its complete trace is the user's send, Swap receipt and Mint child, Token receipt and transfer, and the original caller's signed receive. The transfer is 469,136,612 raw ZNN units. Six hashes, two Ed25519 signatures and legacy compact recovery verify in the retained audit, with thirty-two controls. Final data and descendants are empty. The native settlement is complete without returning another puzzle opcode. (B:K10)

Under the older TABLE/JUDGE hypothesis, the four projections of public ordinal 1024 are CALL/ARTWORK/ARCTIC/ADULT, a unique satisfying fingerprint. The old continuation promoted the first field to an opcode and the second to call 105. Source genealogy found this promotion earlier than the WALK/LENGTH justification. It was therefore not a prospectively established adapter. A useful fingerprint and a successfully settled native call can both be real while that continuation fails. (B:K10)

Two later deliveries reinforce the distinction. The selected MOUNTAIN operation was followed to a signed receive with empty data and zero descendants. The ARTIST-related CALL104 authorizes an ordinary 2.35985 ZNN swap, matches genesis entry 2470, and reaches a newly authenticated height-2 receive at `z1qztc5y325kf5nawlzfy63yd939cn6j2u9cm7qp`. Its exact amount, destination and paired links pass; altered digest/preimage controls fail. Its final data is also empty. These receipts close their selected settlement routes; none is a new Q relationship or historical attribution statement. (B:The145-address board, GAS closure and object-bound evidence test; B:Entry 123, collective-key controls and native ARTIST delivery)

Terminal input 18 is another example of preserving the object before interpreting its labels. The raw-s object is the exact 32-byte, unnormalized DER s scalar, read as twenty-three eleven-bit cells plus three residual bits. P08 is BROOM, P05 QUANTUM and P12 INSTALL. QUANTUM and INSTALL coincide with already named source-local phrase positions in the conditional research. BROOM has no carried operation or member selector. It cannot authorize sweeping an arbitrary database simply because that is a natural English reading. (B:K24)

6. External works were investigated as possible instructions and proof analogies

The external branches are not disposable decoration. They preserve the reasoning that connected the puzzle to original/copy distinctions, signatures, visible demonstrations and the difference between a creator and a character. Their evidentiary role must be stated individually. A work can be correctly identified while its proposed return to the puzzle remains unearned. (B:K06; B:K28)

Source family	What the retained work established	Boundary that remains
Many Facez	The 1997 release, seventeen-track edition and contemporary five-persona description support the retained media interpretation	Choosing editions, LOTTERY divisors, operators and parent return needs the relevant local rule
Rush promotional pair	Earthshine and One Little Victory fit the retained EARTH/GLARE and VICTORY/AMOUNT interpretation	GLARE→SHINE is semantic, disc order differs, and targets/operands were already known
Numbered poem and letter 439	The selected J1336/F1371 poem, manuscript/catalogue context and “little Book” editorial note are recoverable	Edition-specific counts and an editorial conjecture do not become source-native commands
Sevenoaks	The selected text contains identity recognition, original letters, signature comparison, demonstration and record	Narrative correspondence does not identify a controller of a historical private key
Yellow Woman	Author/character names match selected whitepaper references conditionally	Name matches do not establish a reference-number dispatcher or transfer identities
The Endless	The retained synopsis/interview set supplies the sixteen tested film words and predecessor-title context	Proper names remain proper names; character and historical person are not interchangeable
Nehemiah	Walls, rebuilding, books, census and missing genealogy support a proof-themed reading	Numeric or narrative parallels do not prove identity
Hashcash first-cut source	Package and detached signature can authenticate that source's author	That authority is not automatically Satoshi or Q

For Many Facez, the five-by-six table-operator census found a unique bottom-row AND result passing the installed checksum. It was tempting to read NEUTRAL as a license for AND, but that argument followed observation. The main TABLE/JUDGE acceptance also preceded the sibling census. Later corrections rejected the claim that depth-first audit order itself established a return barrier. The album remains a genuine external object; the stronger control-flow interpretation is conditional. Preserved references include the edition listing and contemporary review. (B:K06)

The Rush arithmetic `RIGHT1486 − DEGREE462 = LENGTH1024` is exact. Reversing it modulo 2048 gives the same result because 1024 is half the modulus; that is not a second independent confirmation. The retained promotional-disc reference, Earthshine and One Little Victory can acknowledge an already pending LENGTH interpretation. They do not establish a generic media-return rule. (B:K06)

The poem route has an exact numerical attraction: `710968 − 709632 = 1336`, corresponding to the retained numbered poem. Its poem, letter, editorial note, and manuscript catalogue remain separate sources. The retained transcription lengths 24/21/30/20 sum to 95, ARMOR. Nearby poems also yield dense puzzle vocabulary, so the successful count is not strong enough by itself to select an edition and character-count convention. (B:K28)

The chosen Sevenoaks text has SHA256 `94750ec4ab19b081c4eab43a286bd666fb3a05b5049bc34f79cf801a835baf5f`. Its thirty chapters provide a coherent proof-themed narrative. Retained numerical reductions of 10,000,000 and 500 give SOURCE1664 and DIRECT500, whose conditional sum returns ATTEND116. These are preserved correspondences, not a cryptographic signature by the person the research ultimately sought to identify. (B:K28)

The four-video audit inspected containers, timelines, contact sheets, audio spectra and public metadata. It found ordinary H264/AAC presentation and no obvious subtitle, attachment or data-stream payload yielding a signature, key or transaction hash. Later recovery added genuine promotional, rollout and wallet-transfer footage. The December 23 burn video's 600,064 decoded audio samples were all zero, with independent controls. None of this is a universal proof that no steganography exists; it records the tested observable channels and the absence of a new literal instruction in them. (B:K06; B:Other long-horizon source recoveries and finite closures)

7. The exact message and its 145 signatures

The 2020 publication contributes a different authenticated object: an exact 362-byte message and 145 compact Bitcoin message signatures. The message uses four paragraphs separated by LF LF, no enclosing quotation marks, and no final newline. Its SHA256 is `bead321c006c7bfc62ab1cd9f585cb1138cbfe7cc78d419411ecc9949cfc033d`; its native Bitcoin Signed Message digest is `c4259881ceecf395c853020810dd6d5a6334ace42703f86464d3df58bac6c34e`. (B:K26)

All 145 compact signatures verify those exact bytes. Frozen quoted/newline and other variants do not. This verification uses Bitcoin's message framing, compact recovery, public-key serialization and address comparison. Hashing the text with raw SHA256 and calling that the same test would be an error. The signatures prove control of the respective signing authorities for this message. They do not prove 145 separate people, a particular civil identity, or Q's control of any of those keys. (B:K26)

The address publication also provides an ordered carrier for the word-board experiments. Removing each Base58 version byte leaves twenty HASH160 bytes and four checksum bytes. A fixed inverse-LSB reading down the 145 records partitions each column into thirteen eleven-bit words and a two-bit trailer, producing the retained thirteen-by-twenty-four board. The signatures authenticate the message, not the order of the list. The order has separate publication provenance. The earlier claim of an independently authenticated artwork entrance into that external publication was withdrawn. (B:The145-address board, GAS closure and object-bound evidence test)

7.1 Entry 123 and the tagged attribution proposal

One packet, displayed as Entry 123, produces a selective collection of equations, including TEAM AND SEEK=SEARCH, PATROL−TEAM=SECRET, VALID−PINK=ENVELOPE, and SEEK+RECORD=ITEM. Later reading preserves terminal fields `11|30` and `0|123`, corresponding to RECORD/AUTHOR. Replacing the first payload while retaining tag 11 yields `11|123`, SATOSHI1531. The operands have a source-local explanation; assigning that replacement and choosing a consumer remain conditional. (B:Tagged attribution and the Entry123 certificate; B:Entry 123, collective-key controls and native ARTIST delivery)

The complete fixed population has 145 packets and 1,595 adjacent pairs. Exactly one packet has terminal tags 11/0: Entry 123, raw-checksum valid and self-addressing under displayed one-based numbering. Five pairs qualify anywhere; the other four fail raw checksum. Zero-based self-reference has no match. These counts quantify the observed specificity. They are not a blind historical attribution proof, and displayed Entry 123 must not be silently converted into zero-based input 123 of another transaction. (B:Entry 123, collective-key controls and native ARTIST delivery)

The message's final phrase `We are all Satoshi` occupies bytes `[344:362)`, with its sole Satoshi token at `[355:362)`. A signature authenticates that assertion under the selected key. It does not bind that key to Q or certify the statement as a civil-identity claim. The later 145-key collective test is negative: adding all signer public keys does not produce Q, and neither does the standard empty-tree Taproot tweak of that sum. All 145 leave-one-out sums also miss Q. These fixed relationships are excluded; ordinary point addition is not a secure multisignature protocol. (B:Entry 123, collective-key controls and native ARTIST delivery)

7.2 Entry 39, nonce coordinates and recovery alternatives

Entry 39 uses address `12CTHhyJtr49LgoUShbWgebLBviLAFj6nj`. Its compact header fixes a public effective nonce point K satisfying the native ECDSA equation. Among its eighteen nonce-x reductions, one is checksum-valid: HALVES/right-minus-left. It yields POND SHORT DISH BARGAIN IMPOSE INPUT LUNAR TOGETHER COMIC OFTEN LOCAL PATROL, endpoint ANXIETY80 and checksum 9. Across all 145 nonce coordinates, 166 of 2,610 reductions validate, and fifty-seven records have exactly one valid lane. Entry 39 alone has the frozen combined signer/nonce tuple. (B:Historical205-input pointers and the Entry39 alternatives)

A million random coordinate-pair controls gave sixty-eight signer hits, seventeen nonce hits, two sole-valid nonce hits, and zero joint hits. The marginal expected joint count was only 0.001156, so zero joint hits adds little evidence. These random coordinate pairs were not valid ECDSA records and do not provide an authorship probability. The distinction prevents a dramatic-looking zero from being used beyond its test. (B:Historical205-input pointers and the Entry39 alternatives)

The alternative-recovery study forms GAP as accepted public key minus the opposite recovery candidate. Its public relation `GAP=(2s/r)K` verifies for all 145 records. Entry 39 has a sole raw33 HALVES/OR reduction ending ANTENNA78, checksum 15 and foot OVER. Only that entry has endpoint 78 in the frozen census. The appealing “command followed by the next two cells” reading was proposed after observing the result. DENY SOURCE rejects unreliable attribution while retaining the exact signed witness; it does not select the opposite recovery candidate or import P3928's procedure. (B:Historical205-input pointers and the Entry39 alternatives)

7.3 Direct historical-key joins failed

The conditional word selector found input 145 alone satisfying its predeclared BONUS/MESSAGE conjunction across the 205-input historical transaction; four alternative digest populations had no such conjunction. Native identity verification gave a different answer: the 205 distinct historical authorities and 145 message authorities have zero key intersection. Five frozen digest variants cover 148,625 comparison cells, all zero; a 10,000-digest null family also has no joins. These close the tested equality/recovery predicates. Empty recovery intersections are not positive evidence. (B:K26)

A separate pointer walk illustrates a clean stop. Historical input 87's fixed MIRROR/right-minus-left reducer gives endpoint APOLOGY83, which the existing same-population map consumes as input 83. Input 83 retains endpoint BUILD238 after CHECK repairs only its four checksum bits. Since 238 lies outside the valid input range 0–204, the native pointer walk stops. The later BUILD→CALL→ANOTHER→TEST/XOR→SATOSHI route requires a new reducer-to-CALL admission. Its readability cannot turn 238 into a valid historical input. (B:Historical205-input pointers and the Entry39 alternatives)

8. Natural signature channels were tested without finding a payload

The October 4 independent audit verified all 145 signatures with two implementations. All r values are unique and below 2^{255}; all s values are low-S. These features are compatible with ordinary low-R/low-S signing. A pre-May-2020 Electrum low-R procedure offers a routine explanation, but does not prove which software signed this particular publication. Header parity and recovered effective nonce-point y parity are the same channel, not two independent bits of evidence. (B:Independent145-signature carrier and steganography audit)

The constrained channel search used the recovery header, the lowest one through eight bits of r and s, declared interleavings, forward/reverse record order, polarities and alignments. It did not permit arbitrary sorts, subsets, permutations, guessed decryption or an unrestricted XOR-key search. Its 4,160 labeled variants reduce to 3,880 distinct streams containing 504,664 bytes. None matches the 583 declared targets or validates as a file, whole DER object or whole printable UTF8 string. The longest ASCII run is twelve bytes. (B:Independent145-signature carrier and steganography audit)

Framing test	Retained population	Authenticated hits
Distinct 32-byte scalar windows against signer x coordinates	387,024 windows	0
Raw 64-byte ECDSA windows	280,888 windows; 561,776 recovery attempts	0
Compact 65-byte windows	277,840	0
Embedded compressed-key/ECDSA frames	3,280	0
Embedded x-only-key/BIP340 frames	387,742	0

Framing test	Retained population	Authenticated hits
Binary address/WIF Base58Check attempts	4,874	0

Valid scalar parsing and generic ECDSA recovery can occur in random data; they are not extracted signatures without a committed verifier and message. BIP340 against existing signer keys without an embedded key was not tested in that run, and external Q/native witness digests were not supplied. Those scope limits remain part of the result. (B:Independent145-signature carrier and steganography audit)

There was a parity imbalance worth recording: `header XOR r_LSB XOR s_LSB` has 91 zeros and 54 ones. A reproducible null using 2,000 sets of 145 accepted low-R/low-S signatures gives a plus-one probability of about 0.45% for that exact triple and about 1.60% for an equally extreme imbalance somewhere among seven tested combinations. The latter raw family frequency has a Wilson 95% interval of 1.09–2.19%. These values do not account for the broader research search, and are not the probability of steganography. No payload, key or witness was extracted. (B:Independent145-signature carrier and steganography audit)

9. The historical OpenPGP certificate is real; the proposed new claim is missing

The preserved historical public certificate is 1,714 bytes of ASCII armor, decoding to 1,159 packet bytes. Its primary fingerprint is:

```
DE4EFCA3E1AB9E41CE96CECB18C09E865EC948A1
```

It contains a DSA signing primary, user ID, positive certification, ElGamal encryption subkey, and binding signature. Both existing signatures validate the certificate's internal relationships. There are no secret-key packets. The packet's asserted creation time is 30 October 2008 at 18:19:19 UTC. ASCII armor is a transport encoding; its CRC24 detects accidental corruption, and an armor Version header is outside the signed packet stream. An encryption subkey cannot substitute for the signing primary. (B:K25)

The three retained copies are byte-identical. References include archived bitcoin.org, the Nakamoto Institute copy, the Bitcoin Forum copy, and the historical preservation post. The binary SHA256 is `4dc1d9f55a9be2b279d725bc83cd4fd7049fbfb32513872be6446dea8419cff4`. (B:K25; B:K35)

The native UID certification signs a 505-byte preimage; the subkey binding signs 970 bytes. Their preimages share the 421-byte framed primary and then diverge into their respective subject types. Crossed signatures fail. These are signature types `0x13` and `0x18`; neither is a `0x00` binary-document signature over Q or the exact 362-byte message. The UID packet's last byte labels AUTHOR123; the binding endpoint labels BUNKER243. Those endpoint correspondences are retained, while their native signature roles remain unchanged. (B:Historical PRIMARY: packet anatomy, missing payload and finite DSA tests)

The old proposal requested a new binary-document signature by this primary over exact362. Such a signature would be meaningful evidence of control of that credential. The strict audit did not locate

an occurrence that co-addressed this key and this message as a new-signature demand. The 145 Bitcoin signatures already satisfy a literal reading of signing that message. Thus the historical-primary challenge remains a proposed conditional verification contract, rather than the proved single missing ending. (B:K25)

Finite candidate testing found no response. The DSA framing matrix totals 7,554 checks, combining 6,528 earlier checks and a 1,026-check extension. The extension used 171 valid pairs and six preimages; two implementations agreed on all 768 candidate records and the extension checks. Native certificate signatures supplied positive controls and trailer mutations supplied negatives. Binary signing retains 362 bytes; text signing converts six LF characters to CRLF, giving 368 bytes without a final newline. High160, low160, modulo-q and exact-raw32 hashing mappings remain distinct; rejected out-of-range values were not repaired after failure. The missing older runner is disclosed, so user-described omissions are recovery guidance rather than independently verified extra coverage. (B:Historical PRIMARY: packet anatomy, missing payload and finite DSA tests)

10. Email 260, source chronology and retrospective recognition

Email 260 is central to the historical branch because its preserved encrypted reply has striking field correspondences. Two streams must remain separate: Recipient A is 692 bytes; Recipient B is a 434-byte RSA control. Recipient A's first packet is an ElGamal public-key-encrypted session-key packet and its second is an integrity-protected encrypted-data packet. Its exact SHA256 is `a1efc74bcc5f052e2f230b12ee1bd26ae4de3a3eba02a1581b93dd6bb4d3cc78`. (B:Email260: exact native fields, chronology and conditional selection)

Recipient A native field	Exact result	Dictionary label
Second MPI endpoint / first packet body endpoint, offset 528	138	BACON
Encrypted-data body endpoint, offset 691	43	AIR
Complete stream length, including headers	692	FINAL
First MPI endpoint, offset 270	109	ASSET

These fields are real parser results. Recipient A alone matches all four within a nineteen-container comparison population. That population contains fifteen certificates, two detached signatures and only two encrypted messages; it cannot honestly be described as nineteen equivalent encrypted-message competitors. Wider counts of four AIR endpoints among 644 packets and five AIR occurrences among 679 merged objects also retain their own namespaces. (B:Email260: exact native fields, chronology and conditional selection)

Chronology changes the evidentiary weight. Complete armor was exposed on 27 August 2026 at 20:58:57 UTC. The first retained parse of endpoint 138 is at 22:08:01, with BACON recognition six seconds later. The first recovered exact ordered expression `BUNKER243 – ARTWORK105` appears on 7 October. No earlier original selector receipt was recovered. FINAL and the FINAL[XOR] path toward ASSET predated the cipher inspection, but CHOOSE AIR's trailer reading followed AIR's earlier

recognition. The combined four-field result is therefore a mixture of earlier context and retrospective recognition, not a wholly anticipated independent prediction. That chronology neither proves deliberate fitting nor permits the stronger precommitment claim. (B:Email260: exact native fields, chronology and conditional selection)

The retained Q/ElGamal exponent comparison gives zero hits in 210 cases. Recipient A and the historical certificate use different public subkeys: their p and y values differ even though g agrees. A full 256-byte MPI does not become an admitted 32-byte entropy scalar simply because some metadata endpoints fit the dictionary. No full-MPI consumer was recovered. The earliest independently anchored exact public ciphertext copy is dated 23 February 2024; bounded pre-2021/pre-2022 searches found none, which is an availability finding rather than universal absence or exclusive possession. (B:Email260: exact native fields, chronology and conditional selection)

The late October 7 audit therefore demoted AIR/ACTUAL to a conditional research route. The first missing entrance is the selection of BUNKER and ARTWORK, their order, subtraction and comparison domain before matching the ciphertext. Even granting that step, AIR43 alone does not provide CHOOSE322 or an ADD register. Native mailbox recovery would improve source custody, but does not automatically repair either missing instruction. (B:Cross-object negatives and the controlling current boundary)

The existing downstream model is still preserved. A fresh ADD closure has 1,864 templates, 7,395 role edges, 5,592 side/join path checks and 1,864 installed join-arithmetic checks. It has no native SATOSHI or OUTPUT state, output or join seed. ACTUAL and NURSE share the template closure. NURSE's QUOTE and BODY→AUTHOR roles are compatible with the interpretation, but no occurrence-local RECALL replacement, source-object binding and acceptance consumer were recovered. This is a missing assembly step, not a calculation that can be repaired merely by substituting a same-spelled state from elsewhere. (B:The AIR/ACTUAL descendants and later admission failures; B:Tagged attribution and the Entry123 certificate)

Email 244 is a separate source gap: the described encrypted administrator-secret attachment was not recovered natively. A frozen prospective comparison would use its exact bytes and strict canonical OpenPGP binary form, with narrowly specified SHA256 halves and recipient identity. Rendered correspondence cannot reconstruct unavailable ciphertext, and the candidate is not declared a mandatory puzzle output. The 2015 “Not this again” email is another distinct case: preserved transport inconsistencies and contemporary requests for a signature make it useful evidence about the weakness of sender labels. It does not establish a causal 2015→2020→2021 protocol. The preserved mail and contemporary discussion retain that context. (B:K24; B:K25; B:Historical PRIMARY: packet anatomy, missing payload and finite DSA tests)

11. The mixed Q packet and why its selection history matters

The later sweep's two Q signatures support a reproducible twelve-projection packet containing AUTHOR, ENVELOPE, SOURCE, PROOF, SCALE, VERIFY, FINAL, DISCOVER, WALL and ARTIST. The preserved formulas use native signature scalars/digests and publicly derived points. Some cells are eleven-bit values, two are byte endpoints, and one includes checksum material. Several depend on imported P2095. This is a heterogeneous projection recipe, not twelve uniform extractions from a single native packet. (B:The Q packet: exact reproduction and controlling provenance correction)

The twelve-order arrangement supplies a twenty-four-bit trailer, splitting into CHOOSE322, AIR43 and a comma. But swapping the two ENVELOPE-producing projections does not change the words, and an omitted ordinary V.x XOR cell would create a thirteen-entry, twenty-six-bit trailer. No independent boundary justifies that omission. The earliest retained selection history already contained desired features and order; later copies are descendants of that history, not independent discoveries. (B:The Q packet: exact reproduction and controlling provenance correction)

Within a fixed round-four census of 861 unordered signature pairs and 5,166 pair/operator cases, the frozen predicates select the final sweep pair. Each predicate already selects that same pair individually, so combining them does not add independent filtering. The reported 1/861 is descriptive within that family, not a discovery p-value. The AND interpretation remains separately unselected. Those qualifications preserve a selective exact pattern without using it to manufacture the entrance required by Email 260. (B:The Q packet: exact reproduction and controlling provenance correction)

Conditional AUTHOR[SUB] and ARTIST[SUB] closures each contain 3,626 definitions and share a selective STUMBLE[SUB] structure. Their immediate return sets meet at BUNKER. Across the full model, ENVELOPE occurs as a join terminal 3,061 times, with forty-five paths containing the exact GRAB token. Such structural counts are useful for finding and comparing model behavior. Same-spelled BUNKER, GRAB or PRIMARY occurrences do not inherit certificate, email or physical-frame provenance without a recorded transfer. A later check confirms Q.x's final byte is BRAIN215, but actual GRAB[SUB] outputs CARPET/PULSE/CATALOG, not PRIMARY. (B:Conditional Q-SUB closure and the frame seam; B:Parked infrastructure and settlement findings retain their narrower scope)

12. Source recovery produced authentic infrastructure, with narrower conclusions

Several lengthy investigations were worthwhile because they replaced vague historical claims with exact objects and explicit source limits. Their success should not be confused with an undiscovered continuation or attribution proof.

12.1 Genesis quotation and native commitments

The final eighteen bytes of exact362 are byte-for-byte equal to the middle component of genesis ExtraData: `We are all Satoshi`. The retained 3,110,907-byte genesis configuration, SHA256 `be1ea67ecc33f1effe3bac86c7b23f94916bdc654cbf7db0a80e3244e6038206`, supplies a 103-byte Data object and the native content needed to reproduce momentum hash `9e204601d1b7b1427fe12bc82622e610d8a6ad43c40abf020eb66e538bb8eeb0`. Its content includes the prize-account genesis header; the Burn references that header as previousHash. (B:Native proof representations and the genesis quotation reconstruction)

An audit-created quotation capsule packages these relationships. All 145 source signatures pass; altering the quote or a source byte rejects the source signature. Rotating intact address/signature pairs leaves native message verification unchanged but changes the publication-order hash. These controls explain exactly which properties are authenticated. They do not show that the signers endorsed Zenon genesis, that genesis copied specifically from this publication, or that the constructor intended the capsule as its serializer. (B:Native proof representations and the genesis quotation reconstruction)

A separate configuration test changes SporkAddress while preserving the same genesis hash, with both configurations passing native checks. Therefore that particular data commitment does not bind the configuration's upgrade authority. Ordinary account, momentum, history and enforcement receipts remain valid, but must not be promoted into a historical identity binding. (B:Parked infrastructure and settlement findings retain their narrower scope)

12.2 Rootstock and the Bitcoin anchor

The Rootstock investigation identified an exact uncle rather than the canonical block at the selected height. Uncle `0x2f896267d08cad258a82b8c81d48b97e1457264017c56458db6b5b4449511fc7`, height 3,870,562, verifies its native RLP hash, mining prefix, the exact Bitcoin 710968 header, a ten-hash coinbase branch and stated proof of work. Changed proof/prefix controls fail. Its time is 23 November 2021 at 11:00:17 UTC; the IRIS build text is ordinary client identity. (B:Parked infrastructure and settlement findings retain their narrower scope)

The selected contracts describe signed BTCUSD publication and aggregation, including five historical signers in a nine-member round and zero altered-message members. Some executable bytes match while metadata differs; one contract build remains unreproduced, and historical RPC slots/rounds lack independent state proofs. An earlier Bitcoin block already has an RSK tag from another pool. These findings establish real infrastructure. They do not select a new puzzle field or make unavailable sidechain bytes an active mandatory blocker. (B:Parked infrastructure and settlement findings retain their narrower scope)

12.3 Release packages and old software

The recovered February 2011 release announcement and initial Linux package contain real commitment information. The 9,833,353-byte Linux archive matches the original announced SHA1 `7dfbc05b36112f59886a29f044cfd21c6c253169`. A signed-tag ancestry capsule verifies twenty-two raw Git trees, three commits and two parent links. Of 113 package source members, 106 match committed bytes, four differ and three are absent. That authenticates matching source bytes relative to the signed anchor, not the entire unsigned archive or its binaries. (B:The historical release trail adds real commitments, not Q authority)

Only one of sixteen cross-comparisons between two four-file commitment lists matches both filename and full SHA1: the original Mac archive. Its bytes remain missing. Later recovered certification and release signatures implement an ordinary three-of-five software release policy. They provide an ordinary reason for the public key's packaging; they do not sign Email 260, Q or the artwork. (B:The historical release trail adds real commitments, not Q authority; B:A new certification and an ordinary software-verification role)

The Zenon v0.0.1 bundle investigation likewise distinguishes two release assets with an unexplained fourteen-byte difference. Ordinary repackaging is plausible, but no changed input was causally tied to that difference. The exact tag target/history, package assembly scripts, first controller tree, missing local SDK contents and some release-linked hashes prevent deterministic historical reconstruction. A later v0.0.3 checksum cannot authenticate v0.0.1. No qualifying historical-primary statement binding Q, the artwork or terminal was found in the native literal scans. (B:Version0.0.1 bundle recovery and the unexplained fourteen-byte change)

The historical received VERSION log supplies an exact little-endian version 100 and a source-conformant reconstruction of a 46-byte payload. Only twenty-five raw bytes are printed; the stock source supplies fifteen intervening candidate bytes. The endpoint 192.168.0.1:1531 is a receiver-side accepted-socket observation, not proof of the remote server's own address. An ephemeral port is plausible, while the private-address topology remains unresolved. A missing packet capture, modified build or runtime state cannot be recovered by interpreting local variable names as identity evidence. (B:K28)

13. Broad tests, rejected routes and the work that remains preserved

The baseline's negative results are valuable because they limit repeated searches and identify exactly what a future proposal must improve. The table below groups the principal families. A zero belongs to its declared population and predicate; it is never a worldwide absence claim. (B:K29; B:K30)

Family	Retained result	Correct scope
Backward Keccak fit	One reported match among 69,120 state/stage candidates and 51,200 walk formulas	Known target, analyst-selected layout/stage/walk; rejected as a forward solve

Family	Retained result	Correct scope
ART/APES cultural equations	Two matched tweet controls give target rates 25.10% and 24.22%; uniform sampling gives 25.11%	Exact arithmetic is compatible with ordinary word availability; slogan completion remains interpretive
Literal q/1 carrier	Fixed direct/order/polarity and clue-XOR families yield no supported instruction	Six printable XOR outputs among 640 are near the simple null expectation; printability is not a message
Calendar interpretation	Seven of 256 constrained alternatives reproduce the selected calendar properties	Observed pair is nonunique under that geometry model
Signature DER lengths	Tested group/order/polarity readers fail	DER length aliases component high bits, not a separate free channel
Historical/message/Q joins	No same-role or P/K cross-role match in 44,425 pairs across 392 records	Rejects those equality predicates across Q42 + historical205 + message145
Native authorizations	Zero matches in the registered RetrieveAssets and legacy-allocation/pillar comparisons	Large finite comparison populations do not exclude unrelated author-controlled keys
Eight final mechanisms	E001–E008 complete without a new instruction, point or historical claim	Prior exposure remains disclosed; no blind-prediction claim
Storage formats	Dedicated Berkeley DB/wallet.dat/addr.dat/blkindex.dat receipts not fully recovered	A named coverage gap, not an invented exhaustive negative

The eight final mechanisms cover first/last-letter selection by q/1; parity or one of eleven source bits; unsalted SHA256 watermarking of fixed native inputs; an exact Q-derived empty-tree Taproot address; every affine Boolean and integer-threshold rule within four fixed systems; punctuation/q/1 prefix selection; Hamming(15,11) packets; and the complete mixed 156-bit stream. E002 gives zero exact matches among ninety-six candidates; E003 zero among eight; E006 zero meaningful complete readings among sixteen; E007 zero complete valid packets among sixty-four. The results and full unhelpful outputs are preserved. (B:Eight final mechanism tests add bounded negatives)

E005 is particularly strong within its scope: all 4,096 affine rules and 4,098 threshold rules were checked per fixed system. Each affine system has rank 12 and augmented rank 13, proving inconsistency. A four-cell source XOR relation gives zero while the corresponding glyph bits XOR to one; no affine reader can satisfy both. This rules out that entire fixed family, not arbitrary nonlinear codes or a rendering process. (B:Eight final mechanism tests add bounded negatives)

E004's exact standard Q-derived Taproot address has empty confirmed and mempool history in two indexes at the recorded 11 October 2026 snapshot. Seven official BIP341 vectors and independent public-tweak checks pass. The existing Q-pubkeyhash history remains twenty-five confirmed transactions, forty-two funded and forty-two spent outputs, with the latest known sweep on 26 January 2022. Two additional records are incoming preconstruction funding, not later Q messages. Agreement between indexes is useful history evidence, not cryptographic proof of universal absence. (B:Eight final mechanism tests add bounded negatives)

The steganography audit also fixed several capacity mistakes. Artwork rows contain printable ASCII and ordinary spaces, without hidden Unicode, tabs or alternate-space identities. The Base64 encodings are canonical; all ten unused padding bits are zero. After replacing row artwork and parent-output index, the nineteen stripped row transactions reduce to one template. The 192 z glyphs provide no identity variation at fixed sites. Deterministic hashes, overlapping encodings and repeated graph order cannot simply be added together as independent hidden-message capacity. (B:What the steganography book audit actually adds)

One art/witness correspondence survives uncommon but post-discovery controls. Among 149,652,360 pair-preserving assignments, 30,808 pass its full retained predicate, about one in 4,858. A later five-field test uses one million whole-pair permutations: native nonce-y LSB passes 224 times, published-s LSB 355 times, and the other three fields zero times; the any-field family has 579 hits. The observed native nonce-y specificity is worth retaining. It still needs an independently selected message, checksum or verifier before becoming evidence of an extracted payload. (B:What the steganography book audit actually adds)

Database labels require the same discipline. A wallet.dat claim needs a recognizable Berkeley DB structure and coherent parser result. A word stream called DATABASE does not acquire a file type through its name. The baseline explicitly had not recovered every dedicated requested storage test. Missing legacy database clones also limit historical xStake reconstruction. This master preserves that gap instead of converting an asserted earlier negative into a numerical result with no receipt. (B:K27)

13.1 The methodological corrections are findings

Several corrections recur because they prevented different branches from being spliced together without evidence: old punctuation/2905 packing was superseded; P00 was not proved executable; a fingerprint was not a generic CALL instruction; ABANDON did not automatically return to a parent; a search's depth-first order did not create a return barrier; residual bits did not automatically become new words; COPY did not become an opcode through its dictionary label; a seven-bit match did not restore an APPLE occurrence; and Taproot-specific hidden fields cannot exist inside these SegWit-v0 spends merely because the puzzle is named Taproot. (B:K30)

These are corrections to edges and claims, not erasures of the observations. Genuine historical signatures remain genuine when attribution is withdrawn. A source file remains informative when its puzzle selector fails. Exact model outputs remain reproducible under their stated compiler when its initial admission is open. The complete source/evidence preservation is what allows those distinctions to survive revision. (B:K30; B:Coverage and preservation contract)

13.2 The honest October 13 position

At the historical cutoff, the primary treasure is solved and independently checkable. The public transaction construction, GCM completion, mnemonic derivation, native prize account and retained account actions are concrete results. The artifact also contains an extensive set of reproducible source-local and conditional relationships. The unresolved step is an independently established continuation that supplies its own input, operation, source and verifier without importing the answer from a later object. (B:Authenticated primary solve; B:Cross-object negatives and the controlling current boundary)

Possible future evidence remains varied: an independently dated production input or recipe; an issuer-controlled disclosure with its context; an original decoder or commitment selecting an exact next object; a qualified native email source; or a historical secret-dependent signature over an intelligible exact construction claim. None of those is declared the one file that must exist, and completed exposed-decoder or source searches do not become new evidence simply by being repeated. Later chapters record which of these boundaries changed after October 13. (B:Remaining gaps after these campaigns; B:Current proof gaps and supersession rules)

Part II — The October continuation

The middle of October changed the investigation from a collection of promising word readings into a reproducible sequence of proposed image operations. A two-row panel could be installed in a separate working representation of the artwork; an eleven-cell object called GLASS could be moved; a checked PULL TWENTY instruction appeared before its own one-bit repair; and the resulting reader selected a genuine Bitcoin signature. Independent checks confirmed important native transactions, signature fields and source-byte relationships. (R0089; R0230; R0340; R0352)

The central qualification also became clearer. Reproducing a proposed interpreter does not establish that its author supplied every rule. A valid Bitcoin signature authenticates the original transaction, while a four-bit checksum provides a much smaller consistency check on a proposed reading. Neither automatically authorizes the next operation. Several October experiments succeeded as computations and then stopped at an unresolved source selector, data type, instruction pointer or return rule. This chapter preserves those stops alongside the successes. The results below are the recorded outcomes of the cited reports, not a claim that all experiments were rerun for this edition. (R0344; R0482; R0489)

1. Chronological change map

Dates in this table follow the reports' research dates. Some files were saved the following UTC day.

Research period	Material change	What the correction preserves
October 13–14	CONTROL98's 192-bit body and the artwork's 192 structural z positions were connected through a proposed memory model. The last two historical address records supplied a 48-bit panel.	Exact counts, bits and checks; external-source selection and write permission remained conditional. (R0086; R0089)
October 14–15	The two blank chambers acquired an exact fold map. Full GLASS movement was accounted for across eleven destinations.	Source ownership and working data were separated; a shortened border-only rendering could not stand for the whole move. (R0143; R0230; R0250)
October 15	GLASS/residual relationships and CONTROL96 deletion were tested.	RAZOR→PROBLEM and four-bit checksum equalities survived; the survivor was not accepted as fresh authenticated entropy. (R0246; R0248; R0250)
October 16–17	Moving before the repair exposed checked PULL TWENTY, followed by checked PIPE TWENTY. A source-only runtime replay recovered the same candidate machine.	Better execution chronology, with the learned grammar and historical discovery dependence still explicit. (R0340; R0344)
October 17–18	Independent native parsing confirmed terminal18's donor749 and WIDTH fields; typed interpreters and E15/C5/ORBIT continuations were examined.	Genuine Q signatures and bounded selection results; no recovered instruction after VERIFY. (R0352; R0482; R0486; R0489)

Research period	Material change	What the correction preserves
October 20–21	The Crypto Publius archive supplied a dated clue corpus; the visual comparison added FOLLOW GOLD, and the E-tag fold exposed the five-field E1 packet.	Poem provenance, competing readings and checksum scope were recorded separately. (R0551; R0554; R0614; R0633)
October 23–24	E1's `70	5` found a specific native publisher signature. Compact-recovery and high-S controls tested the representation bridge.
October 26–27	A complete conditional renderer and a destructive parity-collision control were preserved. Donor749 acquired an adjacent PATTERN comparison across publications.	Exact reconstruction remained distinct from original producer recovery; the historical comparison was explicitly hybrid. (R1003; R1090; R1092)
October 28–29	UPDATE/RAZOR/FORUM alternatives, an original-only E2 recognizer and ordinary shading controls challenged the current interpretation.	Competing routes were tested rather than silently combined. (R1117; R1142; R1173)

2. CONTROL98 and Z192: an exact capacity match, then a proposed installation

The original picture has 193 literal z characters, but one belongs to the Base64 text `4Fdzw1k=`. Masking only the four Base64 spans leaves 192 structural z positions. The same masked artwork has 98 punctuation/q/1 controls: 58 punctuation marks and forty q/1 glyphs. These are distinct populations of original source cells. (R0086)

For each control character, the retained LOW/FULL reader measures the parity of its low hexadecimal nibble and the parity of the full ASCII byte. The two-bit results are :→00, ,→01, ./q→10, and ;/1→11. Ninety-eight observations therefore give 196 bits. The first 192 form a candidate body; the final four are `1001`, which match the leading four SHA256 bits of that body. A standard eighteen-word BIP39 representation needs six checksum bits, so its last two checksum bits are calculated. They are not additional bits printed in the artwork. (R0086; R1117)

The original checksum-completed display is:

```
ZOO SHRUG CLICK COOL PLANET START
UPDATE RAZOR FORUM PROVIDE FRESH RANGE
UNIFORM DAD FOOT EFFORT OFFER FRESH
```

This is an exact representation of the chosen source projection. Calling it an eighteen-word hidden plaintext would overstate the result. The attractive next hypothesis was to place the 192 body bits onto the 192 original z positions as a working memory. The capacity fits perfectly, but a literal z does not itself declare a writable binary cell. (R0086; R1117)

2.1 The historical panel and its orientation

The historical address source contains 145 rows of twenty-four bytes after removal of each Base58Check version byte. Reading a column as eleven-bit groups uses 143 rows and leaves two bits. Across twenty-four columns, those final two records provide a forty-eight-bit panel. In original source-column order, inverted least-significant bits give: (R0089)

```
record143: 011011100100010100101001
record144: 010001001001000011100000
combined:  6e45294490e0  (hexadecimal)
```

The artwork has one complete two-row, twenty-four-column z band: zero-based rows 9–10, columns 7–30. In the row-major structural-z list, it is exactly `Z[44:92)`. The retained reverse-column paired reading of the historical bits gives LION PROOF BANNER FOX, followed by four residual bits `1100`. The four words pass the original eight-word-prefix BIP39 check. Subtracting ART102 gives INSTALL PANEL AIRPORT EXHIBIT; AIRPORT's index is 44. The residual `1100` belongs to the panel, and is separate from the checksum bits already contained in the FOX word. (R0089)

Installing the source rows in their original coordinate order at Z44 changes five displayed words to HOUR CITIZEN EYE CANYON ARM. The resulting memory retains the original four-bit checksum prefix `1001`; the calculated full six-bit prefix becomes `100111`. Its final displayed word changes FRESH→FRIEND because the two calculated checksum bits change. No original Bitcoin byte is edited. (R0089; R0098)

The orientation correction is consequential. The older placement that produced COMIC AUTHOR fails this original four-bit checksum. The newly retained placement produces a different eighty-eight-bit strip, with its ART-shifted words INSTALL WINTER STUFF MARGIN AIRPORT AUNT EXPLAIN FETCH. COMIC AUTHOR and EYE CANYON cannot both be claimed as confirmation of the same installed orientation. (R0086; R0089)

2.2 What the finite tests add

The full 20,880 ordered distinct historical-record pairs were examined under four frozen checksum conditions. Individual gates admitted 1,261, 1,293, 1,152 and 1,296 pairs; all four together admitted only `(143,144)`. The conditions included the original CONTROL checksum, the panel's checked four-word reading, and two WIDTH-shifted windows ending at the affected span's boundaries. The pair is specific within that selected family. The gates were identified during an already informed investigation and share source bits and hash contexts, so one surviving pair is not a global probability of authorial intent. (R0100)

The installation also returned to existing EYE/CANYON and CHECK AXIS paths. Those paths led back to native Q verification and block 709632. They formed useful consistency loops, rather than introducing a new secret or another funded prize. The tempting new completion HOUR CITIZEN EYE CANYON fails the ordinary twelve-word checksum. (R0098; R0100)

3. GLASS: eleven bits moving through three kinds of source cell

GLASS names a displayed word in a conditional working state. Its actual eleven logical bits are `01011101101`, decimal 749. Under ordinary BIP39 numbering, 749 is FROZEN; the retained WIDTH-shifted view adds 42 and displays GLASS791. Those bits were not eleven q/1 characters printed in the original artwork. Their donor positions were originally spaces. Keeping logical contents, displayed labels and original source characters separate became essential. (R0246)

The chamber geometry itself is exact. The upper blank component has thirty-three cells: an eleven-cell roof and rows of nine, seven, four and two cells. Rotating the final twenty-two about `(15,18.5)` maps them exactly to the lower twenty-two-cell staircase, by `(r,c)→(30−r,37−c)`. Only row pivot 15 gives a complete match in the tested row-pivot family. This gives a concrete candidate meaning for the two FOLDs, while the instruction selecting that action remains part of the conditional interpretation. (R0143)

The complete GLASS translation uses rows+6 and columns−11. Its eleven destinations divide without remainder into five original lower-chamber blanks, four original z cells, and two original CONTROL owners. The four z cells are Z181–184; the overwritten controls are the comma at `(18,6)` and q at `(18,11)`. A border-only projection misses nine physical changes and cannot substitute for this full move. (R0230; R0250)

3.1 RAZOR/NATURE tests and the PROBLEM return

The four Base64 fields have residual lengths 7|1|7|7 when each is read separately as complete eleven-bit values. Their concatenated twenty-two bits are RAZOR/NATURE. The ASCII LOW/FULL representation of the conditional GLASS glyphs is ROOM/TERM. Across all twenty-two bits, GLASS AND residual equals the unchanged residual; XOR gives ANNOUNCE/EQUAL. These are full-width identities, but no recovered local instruction binds those operands to AND or XOR. (R0246)

The controls also bound the apparent selectivity. Thirty aligned windows of the original forty-glyph carrier produce no containment match, but 35 of the 330 eleven-position strings with GLASS's four-q population do. Two cyclic residual rotations preserve containment. The exact relationship is worth retaining; it is not a twenty-two-bit authenticator. (R0246)

Installing RAZOR/NATURE in the lower chamber and applying the full GLASS move changes it to RAZOR/PROBLEM. Five chamber destinations are covered; two retain their previous bits and three change. PROBLEM1371 minus NATURE1179 equals 192, the structural-z count. The output rejoins the already known numbered-poem/P2095 branch. It does not create eleven newly verified bits or an independent permission to dispatch into that branch. Fifteen of 330 fixed-weight donors and two of eleven donor rotations give the same result. (R0250)

The same move writes `1110` over Z181–184, replacing `0010` in the installed memory. Its calculated SHA prefix changes from `100111` to `000100`, and the displayed last words change OFFER FRIEND→PANDA FLAT. A matching four-bit CONTROL prefix provides a conditional cross-layer check. It does not establish PANDA or FLAT as instructions. (R0230; R0250)

3.2 CONTROL96 was a useful failed interpretation

Removing the two overwritten CONTROL owners deletes `0110` from the original 196-bit record and leaves 192 bits. However, the last four of those bits are still the old checksum. The survivor is therefore **188 retained body bits plus four retained tag bits**, rather than a newly authenticated 192-bit entropy object. Hashing it as fresh entropy gives `0111`, which fails both the old footer `1001` and the removed nibble `0110`. (R0246; R0248)

The checksum delta `1001 XOR 0111 = 1110` matches GLASS's four deposited z bits. Yet fourteen of ninety-seven adjacent-owner deletions give that delta, and two both remove `0110` and give the delta. Four fixed attempts to use GLASS's deposited bits as a replacement for the removed bits all fail the old checksum. Recomputing a new BIP39 checksum would always create a valid representation; it would not recover an authored message. (R0248)

This work established a practical distinction between immutable source, a working image and saved parent state. No two-cell parity replacement made every original FILE and working check pass simultaneously. Later return tests showed that live copyback, cached restoration and a saved-snapshot return can agree on the currently required observations. The accepted donor does not uniquely select a physical COPYBACK or POP operation. (R0230; R0316)

4. PULL TWENTY becomes PIPE TWENTY before witness verification

The working-state names need a precise map. **PRIVATE33** is the original upper thirty-three-cell blank component, numbered from zero in row-major order. Its ordinal 20 is the twenty-first cell, at `(13,19)`; GLASS occupies ordinals 9–19 before MOVE. **FULL55** reads the larger rectangle at rows 11–15, columns 15–25: the thirty-three private cells plus twenty-two surrounding cells. It takes the current working characters' FULL parity in row-major order, groups the bits into five eleven-bit values, and applies the retained WIDTH shift, equivalent to adding 42 modulo 2048. The separate lower twenty-two-cell chamber is outside this rectangle. These readers retain original source coordinates while their working contents change. (R0748; R0326; R0340)

A scheduling correction made the proposed program more intelligible. Earlier reconstruction restored private bit 20 before moving GLASS. The corrected replay moved GLASS first, leaving that bit unchanged until a checked instruction requested its restoration. Both schedules give the same final working image, but the revised order exposes a meaningful intermediate state. (R0340)

Checksum pairs below show stored/computed four-bit values in decimal: `6/6` means both equal binary `0110`. (R0340)

Working phase	Full fifty-five-bit display	Relevant result
Before MOVE	DRILL GLASS PULL TWENTY DEVOTE	This full display's last-four checksum fails; the separately compacted parent has its own valid check.
After MOVE	DRIP LUNCH PULL TWENTY DEVOTE	Last-four checksum passes, 6/6.

Working phase	Full fifty-five-bit display	Relevant result
Restore private ordinal20 at `(13,19)`	DRIP LUNCH PIPE TWENTY DEVOTE	One bit changes PULL1386 to PIPE1322; checksum remains 6/6.
Complete final punctuation	Same five words	Working CONTROL passes 11/11 and frame/chamber agree at PROBLEM1371.

The complete five-field reading is `!9 / B9 / R10 / u9 / RS6`, using a seven-bit ASCII field plus four-bit immediate. A proposed human interpretation reads nine unsigned bits beginning at absolute column 9 of row 10. That gives `000100100`, decimal 36. The terminal has nineteen two-item witness stacks, so flat item 36 is input18's signature, followed by Q at item37. The signature genuinely verifies in its original Bitcoin domain. (R0326; R0340)

Several tempting shortcuts were tested rather than accepted. Boolean NOT of 36 gives zero, but so does NOT of any nonzero value, including an invalid uninstalled locator. Rotating 36 right by ten within nine bits gives 18 because that rotation is equivalent to one place; it is not independent confirmation. Nor can R10 simultaneously supply the physical row and an unrelated rotation without an additional rule. (R0326)

The later bitwise-complement interpretation is more discriminating. Nine-bit NOT of 36 is 475, or `11101101|1`. Its leading byte ED matches GLASS's low eight bits; the inverted final address bit selects signature item0. Frozen controls reject the alternate in-range window28, the public-key sibling37 and the no-panel value466. This is a useful cross-representation constraint, while the meaning of `!9` and the witness namespace remain inferred. (R0340)

The independent October17 audit found native terminal18 `r32` word4=749 and `s32` word15=2006, each unique among the respective 437 complete scalar cells in the nineteen signatures. Its selected `s17–s15` gives PROBLEM1371, but four directed pairs across the wider s-field census give that difference. The complete 182-byte BIP143 preimage verifies under Q. The associated ED/36/38 coinbase pattern comes from standard BIP141 witness-commitment framing; those properties cannot be counted as three independent rare markers. (R0352)

Following PIPE TWENTY into the companion key's twenty-byte HASH160 gives a real protocol consumer: it recreates all nineteen parent P2WPKH scripts. Several alternatives fail. No version byte makes the original CONTROL payload, CONTROL96 survivor or final CONTROL a valid stripped Base58Check address. The PULL and PIPE mnemonic candidates do not match the pinned genesis address population, whereas the authenticated original mnemonic does. These checks return to the established construction rather than proving a second reward. (R0342)

5. Historical mask selection and the source-only replay

The historical path uses the final published signature's native s scalar. At positions 7,11,15,19 it gives LYRICS SCRIPT PATTERN LIST, with a valid four-word-prefix check. LYRICS1067 can be split into ASCII B and index11. Among the four field-local decoded streams, only B has enough complete eleven-bit cells for index11; B[11] is PUT1398. GLOBAL40[11] is RECALL instead, so the field-local namespace matters. The letter B is an analyst's field label, which is another part of the proposed grammar. (R0207)

The remaining SCRIPT/PATTERN/LIST values supply a thirty-three-bit mask. Its established geometric projection gives 58|19, matching punctuation count and artwork height. Among 145 fixed s-column packets, six pass the checksum and only the last record has that in-bounds typed head. Expanding to 1,160 R/S column cases produces fifty-eight checksum passes and two in-bounds typed heads. The narrow uniqueness must not be generalized beyond its census. LIST's low four bits already serve as the packet checksum; reusing them in the mask does not add four independent check bits. (R0207)

5.1 What “2021-only” actually achieved

The source-only panel construction derives HOUR/CITIZEN from byte widths and panel dimensions, EYE from the mirrored chamber, CANYON from the residual XOR, and ARM from another retained source relation. BOTTOM/COVER/ADDRESS supply a proposed table blueprint. Their high-seven parts sum to artwork width38, and their low-four parts sum to legacy address size25; only one of 9,880 GLOBAL40 triples has both totals. These are exact finite results, but the field splits, selected arithmetic and panel formulas were developed after the historical match was known. (R0295)

A separate donor construction derives four GLASS bits from a bottom-Z comparison and five from the proposed RAZOR→PROBLEM target, then resolves the remaining two CONTROL bits and final punctuation through retained checks. One of 8,192 donor/punctuation combinations passes: donor749 and semicolon. The four physical predicates reflect the five-blank/four-z/two-control partition. They do not make 1/8,192 a probability of intention. (R0321)

The isolated runtime then omitted the 145-address file altogether and rebuilt the candidate panel, donor, PULL→PIPE sequence, witness36 and final working image. This was a genuine removal of a runtime dependency. The full five-field grammar remained learned from prior research. Among 2,097,152 allowed private-memory states, 348 passed all three phase checks; twenty-six also matched B9/R10/u9/RS6; one additionally matched the exact `!9` head. A looser checksum test alone therefore did not discover the program. (R0344)

The source-only mask comparison reproduced thirty-two of the historical mask's thirty-three bits. Its uncorrected labels were SCRIPT PATTERN LESSON; restoring the previously identified bit gave SCRIPT PATTERN LIST. That agreement is preserved as an exact correspondence. Software isolation cannot turn the method's already informed construction into a historically blind discovery. (R0344)

5.2 October27 extended the comparison, without making it a signature

The later historical-source audit compared all 290 r/s halves under the fixed mask-to-donor recipe: only record144's s produced donor749. Across 874 complete r cells in the thirty-eight row/terminal Q signatures, terminal18 r[4] alone contained 749. The following native cell r[5] is PATTERN1290, also the middle word of the historical mask. (R1090; R1092)

This gives a contiguous 2021 pair `(749,1290)`, but the historical pair is hybrid: a reconstructed donor plus a raw mask field. It is not a direct twenty-two-bit copy of a continuous historical signature interval. The donor calculation already uses some PATTERN bits. Wider controls show that 749 also occurs in terminal16's s scalar, while PATTERN occurs only once across all 1,748 scalar cells. Longer direct-copy tests fail in the declared families. (R1090; R1092)

A proposed 2021 selector connects PLATE/PIECE, a physical PRIMARY|4 read and the transformed border NEXT PRIMARY STEM PRINT to terminal18's primary DER integer r and fields4/5. The same wording also has a plausible physical-stem interpretation. Choosing DER, the meaning of NEXT, and the historical publication still need a source rule. Neither the eleven-bit donor match nor the extended pair authenticates a common historical controller. (R1092)

6. E15, C5 and ORBIT: native continuations with unresolved types

The October18 typed-interpreter review reproduced seventy-five passing tests and an independently checked native input4 proof. Its program counter still ended at `UNRESOLVED_POST_VERIFY_HEAD`. Fresh modules integrated inherited routines; that was a new implementation and a useful direct path, not independent discovery of the instruction language. (R0482)

One continuation begins at input4: r[2] gives `s12`, then s[12] gives `i15`. Two coherent candidate meanings of `i15` were preserved. Treating it as a different input leads through input15's r and DER fields to Q.y[14]=SATOSHI1531. Treating it as field15 of the current forty-one-byte input serialization gives E15, then the original authentication tag byte E[15]=27, ADDRESS. Native signatures verify in both relevant contexts. The choice of receiver is the unresolved step. A SATOSHI dictionary value in Q is not a historical signature. (R0486)

E[15] and GLOBAL40[39] both equal 27 because they share the same final tag bits. They are not independent discoveries. Reading G40[27] gives MAMMAL1077, or `C5` under the proposed character/nibble grammar. A literal zero-based C[5] byte read is out of bounds: C has five bytes. Interpreting C/5 as a length assertion succeeds and returns to the original AES-GCM envelope, but requires an explicitly different type. The successful tag check cannot retroactively choose that type. (R0486; R0487)

Here **SIDE** names a 132-bit record: read the current LOW/FULL values at the first fifty-seven original punctuation-owner coordinates, giving 114 bits, then append the eighteen bits of the unchanged `glP` Base64 sextets from tag E. The resulting twelve eleven-bit fields retain those fixed source owners even when a working character changes class. (R0470)

ORBIT offered a more source-local selector. B[9]=22 and the retained SIDE field FINE=`+6` give index28. G40[28] is ORBIT1247 at the genuine C/E field boundary, with split `9|95`. The 95 also equals ASCII q XOR period. Thus the changed punctuation owner can be viewed through a read-only parity projection without physically rewriting q into period. A physical rewrite breaks the live q/1 type even when some other checks survive. (R0470; R0489)

Under another candidate type assignment, tag9 selects input9 and B's eighteen-byte length selects r[18]. That field and TxIn[21] form a reciprocal GLOBAL40-checked loop, unique among 437 starts in the fixed census. It returns AIRPORT44, the known panel location. The forty-eight-bit panel comparison succeeds, but its width comes from the existing assembly; the loop itself supplies only the start. (R0487; R0489)

The decisive phase test found that this loop and panel equality already work after MOVE, before full VERIFY. They therefore cannot authorize a post-verification jump by themselves. A software gate that

grants permission after VERIFY is sensible implementation discipline; it is not recovery of the missing authored instruction. (R0489)

A separate native selector did identify the original prize allocation: WEALTH1985 combined with terminal18 AMOUNT64 gives the low eighteen amount bits, uniquely matching the prize among 14,091 balance entries; only one of seventy-six fixed punctuation/signature combinations matches. These are token balances within 14,090 account records: 14,088 QSR balances and three ZNN balances. However, the final punctuation packet carrying WEALTH fails its complete twelve-word checksum. The individual field remains real source data, while permission to consume it in that phase stays conditional. (R0405)

7. E1 and `70|5`: a concrete signature-format correspondence

The October21 EYES→FOLLOWING→IN→AT route selected two aligned blocks from the original E authentication tag. Its left block E[33:66), folded through the established chamber map, gives: (R0614)

```
CODE PLATE PIECE SEVEN MILLION
22|8  S1    R3    b5    F5
```

The first four words pass the original-prefix BIP39 check, 5/5. MILLION1125 is outside that check. It splits into 70|5, displayed as F5. The repeated low-five values in b5 and F5 are not independent confirmations: the fold reverses PIECE and PLATE, which share the source bits that produce those suffixes. Moreover, b5's low nibble already serves as the checksum. (R0614)

By October24, a native candidate receiver had been found. Interpret `70|5` as seventy-byte DER length and byte offset5, and compare against the boundary value95. Among all thirty-eight original row-publisher and terminal signatures, only E's own row publisher satisfies both. Its transaction is `57b5f224beb471fa78caeca665229166ec96da7798e5d05b15e37c07276a2476`; its native BIP143 signature verifies under Q. The real publisher-to-terminal outpoint then leads to terminal input8, rather than an arbitrary row selected for its output. (R0741; R0830)

The serialization experiment also works. E's final tag byte27 plus the candidate nibble5 gives compact header32, which recovers Q from the original r, s and transaction digest. This is a conversion of representation, not a new signature over text. Header32 denotes compressed legacy P2PKH recovery; a P2WPKH message-signing interpretation would use header40 for the same recovery identifier. That address-family switch still needs authorization. (R0830)

High-S controls distinguish the mined representation: replacing s by n−s preserves valid ECDSA but changes DER length70→71 and requires header31 to recover Q. Header32 alone is common—twenty-five of thirty-eight original signatures use it in the legacy representation, and fifteen combine it with DER length70. The extra byte5=95 condition supplies the source-specific distinction. (R0830)

The publisher/spender/sweep byte arithmetic also returns characters of the already known ZENON NETWORK output. It preserves genuine transaction ancestry and exact values, but no new Boolean

opcode, message or callback is established. The useful result is a narrower, testable source-format correspondence. (R0741)

8. The poems gained a source chronology, not automatic authority

The Crypto Publius dossier first assembled screenshots and then a 148-record timeline export. The export distinguishes 138 own-account URLs from ten quote/repost rows pointing to others. Its 2021 own-account subset contains twenty-seven posts, twenty-five before the artwork. All 148 timestamp values agree with decoded tweet IDs when the export is interpreted in America/Toronto, rather than UTC. This corrected several apparent calendar discrepancies. (R0554; R0633)

The original official March5 binary post reads Z3E2N1ON. Publius's map post follows 898 seconds later. It shows contemporary participation in Zenon's encoded-clue culture; it does not make Publius the author of the official binary or specify a 3–2–1 decoder for the later Bitcoin artwork. (R0633)

The July SMOKE post is July26 at 03:43:33 UTC despite its local July25 display. Its “beneath the smoke” language has a precise possible referent: the already separated five-byte C ciphertext corresponds to the plaintext tail smoke, and its narrow printed span lies above the q-run while excluding the parallel one-run. The other completion words WALL, NATION and DRIVE occur zero times in the 138 own-account posts. The evidence is one source-specific alignment, not advance publication of the whole solution. (R0633)

The October26 poem emphasizes ends, small portions, truth/lies and following one's eyes. The April19 and October26 posts are the only own-account posts containing both eyes and deceive, with opposed advice. Meanwhile, June25 supplies a necessary control: it also has fifty words. The same frozen five lookups give REVELATION TO BOLD BUT MY in June and TRUTH SEPARATE EYES FROM DECEIVE in October. Word count alone therefore cannot select the latter. Direct line-initial readings across the poems yield no obvious instruction. (R0554; R0633)

The visual test added FOLLOW GOLD by reading the same mirrored-chamber XOR bits right-to-left within each row; the older left-to-right reading remains ONE EYE. This offers a possible link to the earlier Golden Record quest. All six tested combinations of RAZOR NATURE, ONE EYE and FOLLOW GOLD as alternative four-word mnemonic endings fail. The words can be studied as instructions, but they are not another authenticated seed completion. (R0551)

On October27 the source-selection proposal became more concrete: RAZOR/NATURE beside CONTROL's RAZOR/FORUM, conditional subtraction to SIGN, the 145-address blueprint, the poem's truth/lies/end language, and the earlier official “we are all Satoshi” context point toward the May2020 signed forum publication. This is a plausible human lookup route, with documented alternatives and context. The arithmetic, publication choice and parsing were still recognized during research; no literal 2021 commitment to that exact historical publication was recovered. (R1090)

9. Late-October corrections: which reading survives which phase?

The original CONTROL display really contains UPDATE RAZOR FORUM PROVIDE FRESH RANGE. October28 tested two different readings instead of merging their advantages. An internal SUB-based reduction produces FRESH REPAIR BEGIN DEVICE with a valid original-prefix checksum. Across sixty-six fixed window/operator cases, eight outputs pass, so that window is not isolated by its checksum. A proposed replacement fails until its checksum is recomputed, which generates rather than discovers the repaired final word. (R1117)

The external panel overwrite preserves the original CONTROL four-bit prefix but changes the overlapping source words. Under the same internal SUB grammar, its RANGE child becomes FRESH REPAIR BEGIN DEVELOP and fails. A consumed-before-update interpretation is possible, but a source rule must establish that ordering and saved state. The internal and external readings are not two simultaneous confirmations. Directly loading either suggestive thirty-three-bit phrase into the upper cavity also fails the required seven-q-stem pattern; none of sixty-four fixed consecutive-window/orientation variants supplies it. (R1117)

E2 supplied another useful challenge. Its candidate matcher recognizes `q`, followed by nine spaces on the untouched original artwork. After E1 loading, clearing only those nine cells makes it pass again while retaining the other twenty-four E1 cells. Thus it can be an original-state recognizer or a local vacancy check; it does not establish that the entire E1 object moved. (R1142)

The October26 renderer likewise has two complementary results. Authentic Base64 strings plus a frozen, previously inferred Z/N grammar recreate all 722 original cells. A hostile forty-two-glyph substitution then preserves the complete symbol census, occupied positions, CONTROL coordinates, all 196 parity bits and AES-GCM plaintext, while changing the original artwork identity. The two parity observations cannot recover glyph ownership alone. The full renderer is a sufficient fitted reconstruction, not recovered historical production code. (R1003)

By October29, a source-only shading test found 38 of forty q/1 marks and all fifty-eight punctuation marks on declared shape edges, against seventy-two of 192 z cells. The result supports an ordinary image-shading explanation, while leaving deliberate reuse as a possible data channel. A contemporary solver had already proposed an image-to-ASCII generator. The tested converter repertoire contains the observed characters, but neither the actual original input image nor an exact historical conversion was established in that report. (R1173)

These results narrow the October boundary. The older P00=MERGE boot remains a separate hypothesis: improved physical replay does not retroactively make that transformed word an opcode. The physical program has reproducible states and real native verification, while its source-issued grammar and post-VERIFY instruction remain unresolved. The next chapters carry the geometry, rendering and source investigations forward without losing either the successful October computations or the controls that limited them. (B:K04; R0482; R1003; R1173)

Part III — The artwork and the visual investigation

1. What the picture contains, and what a visual result means

The artwork is a small, exact object: nineteen rows, each thirty-eight ASCII characters wide. Its spaces matter as much as its printed characters. The separator-free 722-byte source has SHA-256 `869020c3e598955bfc71244923c1fb557a5e0d4cadd89443012c8232bfa618d7`. A copy with an LF after every row has 741 bytes and needs its own hash. The on-chain source establishes the character sheet; an official PNG establishes a particular later visual rendering. Those are related sources with different commitments. (B:K11; B:K34; R1696)

The full-sheet accounting is:

Population	Cells
Outer margins	228
Four Base64 fields	80
Punctuation	58
Structural q/1 markers	40
z shading	192
Interior spaces	124
Total	**722**

The 58 punctuation characters include thirty internal marks and the twenty-eight-character outer frame. Structural q/1 enumeration excludes q and 1 characters inside Base64 fields. Several later experiments use smaller regions: a 380-cell analysis region, a 120-cell canopy, a 98-character control roster, or particular blank components. Their counts describe different populations and must never silently replace the whole-sheet totals. (B:K11–K15; B:K19; B:K23)

All coordinates in this chapter are zero-based `(row,column)`. A source owner is a particular character at its original coordinate. Two z characters have the same appearance but different owners. If a character moves in an experimental scene, its owner follows it. A reflected image can display a source owner again; that is a second observation of existing information. A boundary can also touch one character on several faces. Counting every face is legitimate for a perimeter algorithm, but it does not create independent characters.

Three distinct accomplishments recur. A **source observation** identifies what is printed or visible. A **candidate reader** adds a rule for turning observations into bits, words, coordinates, or instructions. A **verified output** passes the relevant independent check. The primary AES-GCM solve has such authentication. Most visual continuations establish observations or reproducible readers, often with short checks, while their intended interpretation remains open. This distinction preserves promising work without turning every readable word into a command.

2. The custom punctuation scale and the outer frame

Reading the four punctuation shapes

Remove punctuation in outer columns 6 and 31. The thirty remaining marks form upper and lower fifteen-character fields:

```
upper .:..:,,;,,;,,;,
lower ;.,::;.,,.;:,.;
```

A first representation treats punctuation as two physical features: an extra upper dot and a lower tail. In that order period is `00`, comma `01`, colon `10`, and semicolon `11`. Four choices of field direction, with an axis-swap control, make eight specified candidates. The retained fixed-prefix BIP39 check selects upper-forward/lower-reverse in that family. It produces four eleven-bit words, each still attached to its four-bit suffix: EMPTY/2, AMONG/15, OBSERVE/9, PRIZE/5. (B:K12)

The suffixes then point into the same fifteen-mark traversal that produced each record. One-based indexing works for all four; zero-based indexing cannot address suffix 15 in a fifteen-position field. The selected symbols are colon, comma, period, semicolon. Assigning these ranks 0–3 produces the custom scale used throughout the later work:

Punctuation	Custom bits	First-generation word/suffix	Selected original coordinate
:	00	EMPTY/2	(3,19)
,	01	AMONG/15	(7,18)
.	10	OBSERVE/9	(14,21)
;	11	PRIZE/5	(16,17)

This is an exact, local construction. It is not a second AES authentication event. Reading the original fields under the resulting scale gives upper MEAT / ROAD with residual `01011101`, and lower TOP / COLLECT with residual `00011011`. Reading the two bit lanes in parallel gives RAISE/2, AMONG/15, SMOKE/3, PRIZE/5. Applying the same suffix consumer again selects colon, comma, comma, semicolon, or `00010111`. (B:K12; B:K35)

The lower residual `00011011` equals the scale alphabet encoded by itself. The three eight-bit values 93, 27, and 23 can be labelled ARM, ADDRESS, and ACTUAL in a dictionary-based display. Those labels do not themselves supply an operation that converts arbitrary eight-bit fragments into an executable

word-machine state. Preserve TOP / COLLECT in its actual order; reversing it to “collect top” would add an unprovided instruction. (B:K12)

What the frame explains

The outer frame contains six semicolons, one comma, one period, and twenty more commas. Under the custom scale the complete twenty-eight-mark stream gives:

Reading	Complete eleven-bit values	Word labels	Remaining bits
Full frame	2047, 1429, 682, 1365, 682	ZOO / RAZOR / FETCH / PRIMARY / FETCH	`1`
XOR against all-comma carrier	1365, 192, 0, 0, 0	PRIMARY / BLOSSOM / ABANDON / ABANDON / ABANDON	`0`
Twenty-two-mark post-gap XOR	384, 0, 0, 0	COPY / ABANDON / ABANDON / ABANDON	none

The all-comma baseline repeats `01`. Removing the six initial semicolons removes twelve bits, shifting eleven-bit grouping by one place. That explains why the period defect contributes 192 in the full alignment but 384 in the suffix alignment. The six semicolon substitutions contribute alternating bit positions and sum to 1365. The phase relationship is more informative than treating BLOSSOM and COPY as separate discoveries. (B:K13; B:K35)

A matching PRIMARY field and a missing `01` make a repair possible, but the frame alone does not identify a copy source, destination, or execution point. Its lower twenty marks leave seven residual bits `1010101` = 85. This is not a complete eleven-bit word. The equality `213 & 127 = 85` prevents using this residual as independent recovery of an old APPLE checkpoint. The proposed division `85 divmod 19 = (4,9)` remains an inferred coordinate adapter. (B:K13)

Later work also tested a Voyager analogy: fourteen rows have punctuation at both outer boundaries, but those rows contain only three pair types—three `;;`, one `,.`, and ten `,,`. NASA's fourteen pulsar directions carry individually encoded timing information. The matching number fourteen did not produce a source-defined clock, spoke origin, or pulsar reader for this artwork. (R1780)

3. q/1, LOW/FULL, and the 98-control record

The original marker stream

Read structural q as 0 and digit 1 as 1 in original row-major order:

```
upper18 100000100010000100
lower22 0101110101001100010000
full40  1000001000100001000101110101001100010000
```

There are twenty-seven q characters and thirteen ones. The upper eighteen contain four ones; the lower twenty-two contain nine. The exact upper slots were reconstructed by a bounded allocator: five first-inward pivots, twelve right-transition cells at stride three, and one phase-compensation cell. Occupied starts advance 2,3,3,3 rather than a uniform structural stride of three. The missing increment selects the compensation. This was the sole exact slot allocator in its thirty-five-candidate family; without compensation it found seventeen correct slots and missed one. It did not independently predict the marker values. (B:K14; B:K23)

A separate test tried using the forty bits to select the correct public-key recovery candidate from each of forty ECDSA construction signatures. Each signature had two recovery candidates, and all 780 pairwise intersections isolated Q. The native marker mask chose Q at only 22/40 positions, its complement at 18/40. The best native distance was thirteen, while controls reached eight; 7,960 of 10,000 same-weight random masks scored better. Replacing s by n−s preserves the recovery sets. This rejected that specified selector scheme, while leaving the physical marker geometry intact. (B:K14)

A parity representation that preserves the punctuation scale

LOW/FULL reads two properties of each ASCII byte:

```
LOW  = parity of ASCII's low four bits
FULL = parity of the complete ASCII byte
```

Write LOW first. This gives `:→00`, `,→01`, `.→10`, `;→11`, `q→10`, and `1→11`. Space and z also map to `01`. Taking the fifty-eight punctuation owners and forty structural q/1 owners in source order gives CONTROL98: 196 bits, interpreted as a 192-bit body and four stored final bits. Its exact original body and footer are:

```
body   fff8e8aa97ea5fa97ba5955bb59d7358eed66dd6ba35996b
footer 1001
SHA256(body), first four bits: 1001
```

The match is real but short: four checksum bits. It is not a signature, not a full BIP39 entropy/checksum record, and not proof that this is the intended decoder. (R0180)

A conditional `f=3p` experiment projected only original q/1 controls from one-bit values to `0→00, 1→11`, retaining punctuation and source positions. It produced body `fff8e80017c05f0173059553b51d7350ced44dd4b035114b` and footer `0001`, which matched its own SHA-256 prefix. Twenty-six q LOW bits changed in the body and the final q at `(18,23)` changed in the footer. Pairing the original and projected bodies with both footers gave pass/fail/fail/pass, as expected for an actual state-sensitive check. (R0180)

Using the original upper-eighteen selector, the projected record yielded the checked conditional four-word return VETERAN SATISFY EXOTIC WHEAT after subtracting ART102. Only the second and fourth selected words changed; VETERAN and EXOTIC were already present in the baseline. In 200,000 count-preserving marker relabelings, 12,540 passed the projected SHA4 predicate (comparison with the first four SHA-256 bits), 12,623 passed the selected four-word check, and 739 passed both. These are results for the declared local null model, not author-intent probabilities. The equation's historical-source selection and the later inactive G0 entrypoint remained unresolved. (R0180)

4. The lower chamber, reflected memory, and older matrix branches

A real empty shape; several proposed ways to load it

The lower blank chamber has consecutive row lengths 2,4,7,9 with a shared left boundary: row 15 columns 12–13, row 16 columns 12–15, row 17 columns 12–18, and row 18 columns 12–20. This is twenty-two spaces. The fixed search for that run-length pattern found one such chamber in the twenty-six-column interior. The shape is an observation; deciding which bits to place into it is a reader hypothesis. (B:K15; B:K35)

One early loading used the twenty-two concatenated field-local residual bits. Installed top-to-bottom and left-to-right, its rows are `10`, `1100`, `1010110`, and `010011011`. Reading outer and inner rows top-first gives NATURE / SKULL; bottom-first gives EVIDENCE / PULSE. Another loading reflects the chamber horizontally about the sheet's centre: the intersection contains four cells and the union forty. Vertical or double reflection gives union forty-four. The forty-marker stream can therefore be loaded into the horizontal forty-cell union in source order. (B:K15)

At each original chamber coordinate, compare the loaded bit with its reflected counterpart. In ordinary row-major output order the alternatives are XOR → ONE / EYE, XNOR → GRAB / PUT, OR → ONLINE / FARM, and AND → ABLE / ACOUSTIC. These are all retained. The mirrored memory contains relocated bits; the original blank chamber does not literally contain forty printed markers. That distinction becomes essential when comparing a direct optical view with a scratch-memory decode. (B:K15; R1815)

The eight-glyph and matrix hypothesis

The candidate 380-cell region comprises upper rows 3–7 columns 7–30 and lower rows 9–18 columns 6–31. Its inventory is space 114, period 7, colon 5, comma 31, semicolon 7, z 176, ones 13, q 27. The proposed ordinal alphabet—space, period, colon, comma, semicolon, z, 1, q—differs from the custom punctuation order. It needs its own binding. Adjacent ordinal pairs divide the population into 121 space/period cells, 36 colon/comma, 183 semicolon/z, and forty 1/q. Within this model ART102 equals ordinal bit 1 XOR bit 0. (B:K19)

The 121- and 36-bit lanes reshape to candidate 11×11 and 6×6 matrices. Their preserved integer forms are:

```
low rows     0,1536,1024,0,0,8,0,0,256,64,1
low columns  768,512,4,0,2,0,0,32,0,0,1
high rows    48,0,0,18,1,0
high columns 32,36,0,0,4,2
```

Two placements cover a checksum domain 0–15, with different overlaps. Under placement A, the nine candidate returns are ABOUT3, GAS768, AMOUNT64 twice, ADVICE32, ABANDON0 twice, ACROSS18, LENGTH1024. The supplied equations preserve only the identity among 7,920 low relabelings and 360 high relabelings. Fixed-weight alternatives leave 10,206 of 63,140,314,380 low layouts and four of 376,992 high layouts. These quantify specificity after choosing the equations. They do not supply

the missing operation from checksum nibble to matrix address, or returned scalar to executable CALL. (B:K20–K21; B:K35)

The 183-bit lane makes sixty-one triples: fifty-four `111`, five `101`, and two `011`. Pairing them with the sixty-one occupied canopy cells is a proposed adapter. Across six operations XOR, ADD, and both subtractions change all sixty-one; AND changes two; OR changes forty-two. AND changes occupied rank 28 z→period and rank 58 q→z, raising the ART-projected weight from 37 to 38. But 1,403 of 10,000 same-multiset random arrangements also changed at most two cells and 614 reached weight 38. The sparse effect partly follows from fifty-four all-one masks. (B:K22)

The old reflected-eye word lattice is also preserved: ALERT/AHEAD/CHECK/FAME, the fourth-column RACE, subsequent contextual CHECK trials, and the EXPECT/ALIEN→FINAL→FILE or FINISH branches. Crucially, different predecessors change CHECK results; one cannot take a favorable result from one context and carry it into another. Width 38 explains many diagonal coincidences: among 685 valid offsets separated by 37, 666 yield a down-left diagonal and nineteen wrap a boundary. The visual map needs a selected coordinate rule before arithmetic becomes an instruction. (B:K16; B:K18; B:K21)

5. Direct sightlines: the complete boundary and the corrected half-turn

Why ten, eleven, twelve, and thirteen owners all appear

An optical experiment must state whether a whole character-cell square blocks sight or only the raster's actual ink does; whether observers are cell-centres or arbitrary subcell points; and whether the output records first-hit owners, complete boundary faces, or a projected silhouette. Different models legitimately produce different counts.

The original upper thirty-three-space region, viewed down-left, has a complete twelve-owner boundary `.q,1:q;.1,q:`. An earlier centre-only eleven-owner extraction stopped at `.q,1:q;.1,q` and produced POET / ITEM. Adding the actual final colon at `(16,25)` restores the complete twenty-four-bit record `A72ED8`. Five specified down-left views gave the same twelve-owner boundary. POET / ITEM therefore remains an incomplete-reader observation, not a complete message. (R1244; R1246)

The lower direct up-right first-hit model has ten owners `:1,q.;q,q.` and twenty bits. Adding semicolon `(18,22)` was incorrect for that first-hit view because period `(18,21)` obscures it. A different whole-cell aperture model includes thirteen owners, including E1 q `(17,11)`, and yields `00111001101010111010011010`: eleven-bit values 461 and 745, then `1010`. Strict complete-record readers reject it. The eleven-character T29 cover is another object again. These counts must remain attached to their precise reader definitions. (R1246; R1395M0004; R1726)

Actual ink adds another population. A frozen official-raster sample, using a 32×32 start lattice in each of the twenty-two lower blanks, returned eighteen owners. Its filtered q/1 channel was `010000000` and punctuation channel `0001101101`, with four z owners retained separately. Their first eight bits equal E’s final bytes `40 1B`, but the complete channels leave `0` and `01`; the match is only two prefixes. Rare hits included one source digit reached by 11/22,528 rays and one q reached by 1/22,528. A literal ten-bit mask over the ten adjacent CONTROL→Z copy edges selected slots 3,4,6,7,9 and values `11101` = 29,

not E’s 27. The complete mask was absent from CONTROL and E as a contiguous substring. These are finite sampled-ink results, not a proof of exhaustive positive-area support. (R1400)

The map that preserves the whole observation

The stronger source-to-source alignment is

```
T30(r,c) = (30−r, 37−c)
```

It pairs five original right-rail ones looking west with five original left-rail q characters looking east, preserving the observer, clear corridor, first obstruction coordinate, and length. The five pairs have 9,7,4,2,0 intervening spaces. The first two obstruction glyphs differ; the last three agree. (R1726)

Right observer	First western hit	Left observer	First eastern hit	Clear spaces
(12,26)	comma (12,16)	(18,11)	period (18,21)	9
(13,26)	colon (13,18)	(17,11)	comma (17,19)	7
(14,26)	period (14,21)	(16,11)	period (16,16)	4
(15,26)	comma (15,23)	(15,11)	comma (15,14)	2
(16,26)	colon (16,25)	(14,11)	colon (14,12)	0

Across all thirteen structural ones and twenty-seven q characters, exactly four west/east pairs with nonempty equal clearances produced the same T30 map. Admitting zero-gap rays gives 188 equal-gap pairs and 151 possible half-turns; T30 has eight, the next-highest transforms three. T29 pairs some observers but matches zero complete endpoints. These are bounded geometric counts, not statistical proofs of intention. (R1726)

The sixth right-rail one (11,26) sees eleven blanks, then period (11,14), semicolon (11,13), q (11,12). T30 sends its observer to (19,11): an unpublished twentieth row. The projection is exact, but there are only nineteen original rows. A proposed Genesis continuation did not satisfy the required eleven-blank window. In contrast, T29 shifts the windows one row, producing cover lengths 2,2,3,2,2 and the eleven-character cover :1,q.;q,q.;. That cover's size follows mechanically from misregistration; it is not independent evidence for an eleven-character cipher. (R1726)

The paired first hits give right LOW/FULL streams 00100/10010, and left streams 10100/01010. Their literal disagreement flags are 11000. A modern five-bit alphabet can label this Y or, with reversed polarity, H. No source chooses that polarity or establishes a Bacon cipher. Likewise LOW cannot globally mean “false” and FULL “true”: both sometimes differ at real obstructions.

The two-eye transfer and the C-field link

The twelve-owner upper wall has a separate coordinate-transfer reading. Under T30, its initial period maps beyond the published sheet to `(19,23)`; retaining the original traversal and taking the eleven on-sheet destinations gives `;.q,q;.q,1:`. LOW/FULL yields `11101001101 11010011100`, or TRUST / SQUEEZE, indexes 1869/1692. Sorting those same destinations into new page order gives `:1,q.;q,q.;`, or CUTE / SNIFF. Including the virtual period gives RISK / INSANE with two residual bits. The direct lower-eye observation cannot see the transferred semicolon behind period `(18,21)`, so TRUST / SQUEEZE is a transported-coordinate result, not what that eye directly sees. (R1244; R1246)

The terminal upper colon `(16,25)` and lower colon `(14,12)` are the only literal colon-to-colon T30 pair among five original colons. Both sit in original five-glyph `zq:1z` motifs. T30 exchanges q/1 endpoints and retains the central colon and outer z positions. Yet this is part of wider lower-N symmetry: 64/78 eligible five-character windows match under q↔1 reversal versus 42/78 under ordinary reversal. The paired motifs are real; they are not an isolated cryptographic event.

A different whole-picture turn, `G(r,c)=(18−r,37−c)`, maps C's final printed quartet `w1k=` to source row 16, columns 23–26, read left-to-right as `zq:1`. A/B/E final quartets instead map to `zzzz`. Only C's printed `1` reflects to a colon. Crucially C's two Base64 padding bits come from `k`, which reflects to q; the colon reflects to the preceding printed `1`. The colon is therefore not a direct mirror of the padding bits. The two transforms G and T30 cannot be interchanged. The selected `q:1` motifs have parity value 35, tentatively ASCII #; Zenon Genesis contains two # separators, but upper `.:1` also encodes 35. That recipient remains an added interpretation. (R1244)

The complete DISORDER / PARROT / GIANT stencil candidate

C's forty ciphertext bits contain three complete eleven-bit words plus seven residual bits. Its first thirty-three bits, `11100000010101110111001111000011 0`, fit the thirty-three blank cells of the upper cavity. Place one bit in each upper blank in original row-major order, turn that stencil over the lower twenty-two-cell cavity, then read overlap and overhang separately. This is a coordinate projection over printed material, not a collision-free movement of solid pieces. (R1416)

All eight lattice orientations and translations were enumerated. Exactly four complete placements cover the lower cavity, all half-turns:

Stencil transform	Eleven cells outside the lower blank overlap	Output indexes, overlap22 then overhang11
`(29−r,35−c)`	`zqzqzq.zqzq`	471, 519, 813
`(29−r,36−c)`	`q:q,q.;q,q.`	**507, 1283, 781**
`(29−r,37−c)`	`:1,q.;q,q.;`	1277, 1665, 583
`(30−r,37−c)`	All on virtual row 19	783, 477, 519

The column-36 choice alone includes all five original q-rail sites and adjacent colon `(14,12)` in its overhang. Its complete output is DISORDER / PARROT / GIANT. This connects C, whose authenticated plaintext portion is SMOKE, to the July 25 poem's hidden giant beneath smoke. The criterion was

applied after the result was noticed, despite the rail and colon being known earlier. That historical order prevents calling it a blind prediction.

Reading all thirty-three destination cells in ordinary page order, without separating overlap and overhang, instead gives GIGGLE / DESERT / DOMAIN, indexes 783/477/519. The component-first ordering is therefore a necessary extra rule. Among first/last thirty-three-bit portions of A/B/C/E and all eight thirty-three-bit windows in C, only C's zero-offset prefix gives GIANT under this stencil. The lower twenty-two projected C bits match the separate field-residual loading at only 12/22 positions. The three-word record is not a standard mnemonic; even a proposed 32-bit-entropy/one-bit-check extension fails, with stored last bit 1 versus required 0. This is a complete reproducible candidate, with no second authenticated plaintext or signature. (R1416)

6. E feedback, nine-cell completion, and the two rims

The same T30 alignment connects the upper nine-space window at row 12 columns 17–25 with the lower nine-space run at row 18 columns 12–20. It maps comma `(12,16)` to period `(18,21)`, and q `(12,15)` to semicolon `(18,22)`. Among seventeen printed matched lower boundary faces, the comma/period pair is the only mismatch in both LOW and FULL. Comma and period also delimit the original E Base64 field, uniquely among the four fields. This is a precise meeting of geometry and a field boundary; the inherited E 1/E2 dispatch remains a candidate reader. (R1452)

The original upper cavity has thirty-two clockwise printed faces. The lower opening has twenty-six boundary faces, seventeen printed and nine beyond the sheet. Under the inherited local E2 interpretation, the upper nine-space window returns to its selecting comma and subsequently to q `(18,11)`. Calling that a self-loop assumes its following cell is still a space. A scratch write changes the input and requires the reader to be recomputed. (R1452)

The four decoded fields leave residual lengths 7+1+7+7=22 bits:

```
A 1011001
B 0
C 1011001
E 0011011
joined 101100101011001001 | 1011
```

Interpreting the first eighteen bits as nine punctuation marks gives `.;:..;:.,`; the final `1011` equals the existing period/semicolon footer. Writing those nine marks into the bottom nine spaces closes the lower opening, leaving thirteen lower blanks and thirty-three upper blanks: forty-six in total. This is the nine-cell insertion experiment, distinct from moving the twenty-two-owner piece. They must not be combined into one state without a separate operation. (R1452)

The completed lower perimeter has twenty printed faces and forty bits, `35ab93a3aa`. The original upper perimeter has thirty-two faces and sixty-four bits, `555557ff25eb835a`. The thirteen-byte concatenation did not literally equal ZENON NETWORK; the lower five bytes matched neither C's `e05773c359` nor the original marker stream `8221175310`. The actual OpenPGP test did not identify these rims as a signature, and the sixty-four-bit record did not become a known key identifier. (R1452; R1454)

The poem overlays are exact but retrospective. The January 14, 2021 sleeping-giant poem has fifty-five whitespace-delimited words without its `$ZNN` footer; mapped to the original fifty-five internal

spaces, its last word MAP lands at `(18,20)`. The adjacent original `.;q` maps to binary `101110`, or 46. The July 25, 2021 hidden-giant poem has forty-six words under the same footer convention. After the nine-cell insertion, HIDDEN lands at `(16,15)` beside the other `.;q`, and GIANT at `(17,12)` beside q `(17,11)`. No poem text is literally encoded in the blank cells; loading one word per space is the stated overlay rule. (R1452)

At the E2 rail, the scratch insertion changes q+space from `10|01=9` to q+period `10|10=10`, while the printed footer is `10|11=11`. The resulting 9→10→11 progression is preserved as a candidate state relation. Its numerical neatness does not establish a source-written program.

7. Moving qzzq: an actual passage, two models of the page

The minimum physical opening

The four original owners q,z,z,q at `(12..15,15)` can move together by `(+3,−3)` into four original blanks `(15..18,12)`. In the scratch picture this joins upper and lower spaces into one bottom-open fifty-five-space passage. A complete minimum-passage search found eight four-glyph cuts, checked against all 4-of-20 shortest-path-site subsets. Only qzzq could be placed into original in-sheet blanks using that already-retained displacement. This is a meaningful finite selection within the tested movement rule. (R1475M0001)

The existing fixed-original-coordinate parity check passes after this move; a live row-major re-enumeration of controls fails. Space and z both have LOW/FULL `01`, explaining why an E1 parity read can remain unchanged even when a moved z physically blocks an empty square. The old E2 window remains, but its `q,` header changes. A result cannot be called “checksum-preserving” without naming the fixed or live roster. (R1475M0001)

An experimental fifty-five-byte backing plane retained the entire known envelope across two views: 37+21−3=55 source addresses. Four newly uncovered straight-view bytes were EB28D88A; an E1 window showed `BE59C37357E0`, containing E's first byte and the five C bytes reversed. This observes existing authenticated envelope bytes under an inferred layer model. It does not decrypt a second plaintext. (R1475M0001)

The original sheet and the projected working view

The most important later correction distinguishes the actual nineteen-row page from a T30-projected view containing two virtual rows:

State	Original nineteen-row page	Projected working view
Original	Closed 33; bottom-open 22	Two closed 33 regions
Move qzzq	Bottom-open 55	Closed 66

State	Original nineteen-row page	Projected working view
Move remaining 18 owners by the same offset	Closed 55	Closed 55 plus virtual-only 11

Moving the eighteen owners first gives a projected intermediate 51+15, then the same final state. The order really changes intermediate topology. The projected 66 and 55+11 observations reuse original characters; they are not new inscription rows. (R1823M0002)

The joined projected boundary has seventy face encounters and 140 bits, `555557ff25eb83555ae5ab955555e96d556`. Thirteen virtual encounters reuse original owners; the seventy faces represent forty-six distinct original owners. The first thirty faces continue the old upper rim before the path turns beside comma `(12,16)` where the q moved away. Three successive comma encounters still belong to one character. Four anchor/direction readings produced no all-printable seven-bit ASCII; hypothetical 132-bit mnemonic prefixes all failed, with eight remaining bits retained. (R1475M0001)

The full twenty-two-owner move fails retained live checks: fixed 9/14 and dynamic 9/6 in the archived sequence. An experimental RETURN restores original appearance; a separately saved observation can retain the known fifty-five-byte record. This gives a concrete meaning to saving state, while leaving the author-selected schedule unproved.

Changing coordinate frames does not move the mirror

For page position p and a moving-piece anchor a, local position is `u=p−a`. The page-fixed mirror `T(p)=b−p`, where b=(30,37), becomes `T_local(u)=b−2a−u`. With anchor `(12,15)` it is `(6−r′,7−c′)`; after the move to `(15,12)` it is `(−r′,13−c′)`. Keeping the old local formula would accidentally introduce physical displacement `(+6,−6)`. This is an implementation warning, not evidence that the archived replay made that mistake. (R1823M0002)

Owner accounting makes the distinction testable. In the piece-attached comparison 179 local coordinates are printed before and after;62 show equal glyphs, but only four retain the same original owner. Fifty-seven z appearances and one q appearance are lookalikes from other source positions. All 370 printed source owners survive the five recorded states. The audit checked 68,265 owner pairs per state and 76 virtual-cell correspondences. On the actual page the relevant boundary has 32→62→32 faces, including eight outside/unknown faces at the midpoint; projection gives 32→70→32. A crop or reflection must not fill unknown originals without naming the projection. (R1823M0002)

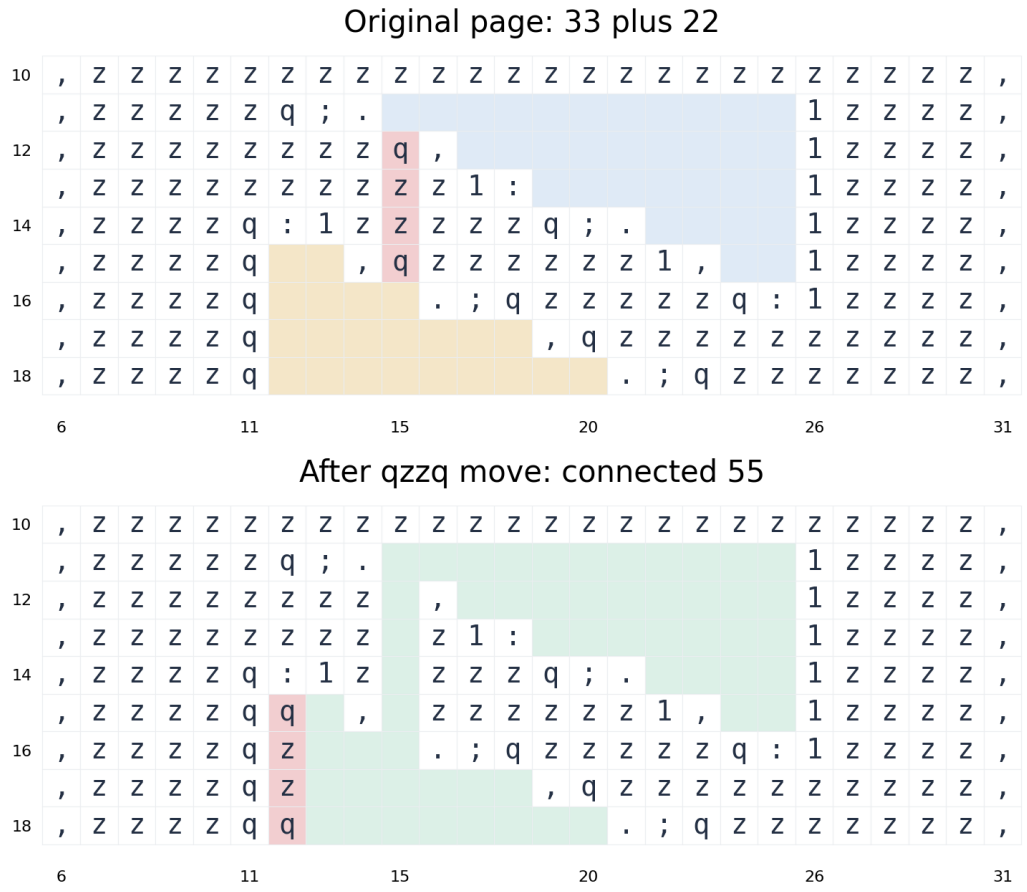


Figure 2. Original characters moved by the declared qzzq operation. The diagram uses the actual nineteen-row page; the separate projected view is explained in the text. (V01)

8. The exact 722-character model and the information it assumes

The October 13 state reconstructed all 418 upper-Z cells but not a satisfactory independent lower-N producer. Perturbing 184 B/C bits affected only twenty-four upper renderer inputs; the other 160 remained authenticated ciphertext. An October 26 composite already reproduced all 722 cells from the Base64 strings plus frozen, previously inferred geometry and C/E marker rules. It also constructed a 42-glyph equal-parity alternative preserving census, occupancy, all 196 observations and the AES plaintext while changing the committed source. The November 4 work quantified the later candidate producer’s output family and dependencies; it was not the first conditional full reconstruction. Both producers were reverse-engineered after seeing the artwork. Exact reproduction establishes a consistent candidate, not recovered 2021 production code. (B:K11; R1003; R1684)

Within that producer the lower twenty-two markers split into a fixed eleven-marker stem `11101010100`—index 1876, TUNNEL—and a diagonal `00110010000`—index 400, CRAFT. Only six lower assignments vary, from low bits of E[3,5,7,9,11,13]. Their original pattern is `100011`. CRAFT implies the

qzzq opening because its first varying q is controlled by E[3]. Thus CRAFT and qzzq are dependent within this model. (R1684)

The full candidate census has 262,144 source-control states collapsing to 162 upper outputs ×64 lower outputs=10,368 marker pictures. Selecting CRAFT leaves 162; requiring upper pivot 22 leaves 4; imposing the original q/1 inventory leaves 2; the known CONTROL98 SHA4 leaves 1. The reported 13.340-bit output capacity applies only to this producer. In 8,000 valid alternative GCM encryptions under the public puzzle key and same plaintext, TUNNEL occurs 8,000 times, CRAFT 137, qzzq 4,014, and CRAFT+qzzq 137. A separate 20,000-nonce population gave CRAFT 335 and qzzq 10,084. These controls demonstrate the dependence. Synthetic packets do not carry the original Bitcoin construction signatures. (R1684)

The missing literal-role channel

LOW/FULL merges three character pairs:

Equal-parity pair	Original positions
z / comma	224
q / period	35
1 / semicolon	26
Total	**285**

Occupancy and parity alone omit these literal distinctions. ASCII bit 1 distinguishes the members: z has 1/comma 0; q has 0/period 1; digit 1 has 0/semicolon 1. Retaining this plane with the known field bytes reconstructs all original 722 characters. Its 285 positions are an accounting requirement, not proof of 285 deliberately hidden payload bits. (R1803)

The frozen contour grammar repairs every one of 285 individual equal-parity swaps and a simultaneous 285-swap synthetic picture back to the original. It accepts six q↔1 changes without repair because those alter marker value rather than equal-parity role. All 10,368 C/E-model outputs are fixed points of that grammar, but the producer supplies 602 of 642 nonfield cells unchanged plus the marker roster. The role plane's invariance is expected: q and 1 both have ASCII bit 1=0. Neither experiment independently derives the entire artwork from a historical logo. (R1803)

Glyph pair	Same LOW / FULL	ASCII bit 1	Positions
z / comma	0 / 1	1 / 0	224
1 / semicolon	1 / 1	0 / 1	26
q / period	1 / 0	0 / 1	35

285 literal choices are discarded by these parity observations.
Authored payload capacity is a separate question.

Figure 3. The three glyph pairs share their LOW/FULL observations. The additional literal-role bit distinguishes each pair. Counts and the full reconstruction were independently checked for this edition. (V01)

The correction from six bits to thirteen free lower labels

Allow the forty marker labels to vary while holding positions, occupancy, punctuation, fields and z fill fixed. Exactly 1,258,815,488 assignments satisfy the fitted contour grammar. For the lower twenty-two positions the grammar forces nine labels and permits thirteen freely varying ones: 8,192 lower pictures. Six are the E-linked sites; seven more were merely held fixed by the old producer:

```
(14,11) q   (15,11) q   (15,15) q   (16,11) q
(16,26) 1   (17,11) q   (18,11) q
```

Their row-major flags are `0000100`. “Six varying bits” therefore was not a capacity theorem for the original lower artwork. Among the 8,192 grammar-compatible pictures, 534 pass the old four-bit check. Fixing the six E-linked sites leaves 128; adding SHA4 leaves 9; adding the original total lower one-count before SHA4 leaves 7, then 1. These old constraints can select the published state, but they are not newly recovered authentication. (R1803)

The subsequent seven-marker reconstruction predicted six rail labels using a whole-edge facing convention from the historical logo, and predicted interior q `(15,15)` from a masked twenty-of-twenty-one-cell rotated tile match. It did not read the withheld labels, use a checksum, or select by word meaning. However the rails and tile relationship had already been studied; this is a masked replay, not blind discovery. Of 352 eligible placements, one matched the rotated tile with zero observed mismatches. Its local transform is `U(r,c)=(27−r,30−c)`, distinct from T30. (R1808)

Applying both rail and tile assumptions to the full 8,192 lower family leaves 32 states: five free choices. Rails alone leave 128; rotated equality alone 2,048. This is the current compact calibration family, and its remaining variation matters to the subsequent reader.

9. Historical logo reconstruction, holdouts, and frame calibration

The archived May 28, 2020 logo is a 939×1499 PNG, SHA-256 `fa8ee856d77d28675c268714904eecc52bb3f06d08fd9753e337937c907b813e`. It is not proved to be the exact input image used for the 2021 artwork. Prior fits used 316 z/blank locations, excluding q/1 and punctuation labels. Early density/registration results and familiar-letter matches were descriptive fits; historical AAlib or converter failures did not, by themselves, isolate authored steganography. (B:Rendering: failed reproduction does not isolate intentional encoding; R1783)

A single sampled-alpha-to-glyph palette can reach only 313/394 in-bounds structural positions, leaving 81 mismatches. Whole-cell coverage can explain more: selected ones `(13,17),(15,22)` cover 51.25% and 49.44%; selected q positions `(14,19),(16,24)` cover 70.44% and 68.12%. The q coverage exceeds the one coverage across 16,618 accepted registration perturbations and five window scales. That supports a local ordinary-rendering explanation. It does not supply a universal rule. (R1780)

The next audit found the full six-site contour `q q 1 q 1 q` has coverage 53.69%,32.99%,51.25%,70.44%,49.44%,68.12%. No single threshold separates all six. A five-feature leave-one-row-out classifier predicted 34/40 markers versus a 27/40 all-q baseline, but failed both anomalous ones. Upper-only training predicted 12/22 lower labels; lower-only training predicted 14/18 upper labels. Punctuation prediction was 7/30, worse than the 11/30 majority baseline; cross-half prediction was 4/15 and 2/15. The earlier claim that cell coverage simply explains the anomalies was therefore narrowed. (R1783)

The repeated lower punctuation staircase `;.,:` under translation `(+3,+7)` is exact. Across 35 tested positive translations it is the only full-alphabet repeat in the original. But 9,452/100,000 global multiset-preserving shuffles and 28/144 row-preserving permutations also contain one. Historical default-palette simulations reached at most 120/394 structural matches, 2/40 q/1 and 5/30 internal punctuation. A later 48-configuration execution through installed AA-lib reached best 122/414 structural matches—all blanks in that configuration—versus 124/414 for an all-blank baseline; separate maxima were 3/40 markers and 2/58 punctuation. These bounded negatives do not exclude custom palettes, other source revisions, or other production software. (R1783; R1803)

What DEVELOP FRAME LOCK actually measures

The lower artwork contains repeated 3×7 tiles. Tile A rotates to tile C exactly; tile B differs at two q/1 sites; another tile is twenty-one blanks. Their common-orientation marker records are 001→010→001. The appealing three-word interpretation DEVELOP FRAME LOCK therefore reuses a reflected state; it is not three independent instruction records. Source-order seven-symbol stencils are `q;.q,1:`→908 and `q;.1,q:`→936, with difference 28 and XOR 36, exactly two changed bits. Mapping 28 to the outer frame requires an additional rule. (R1762; R1766; R1780)

The marker-only frame audit selects T30 uniquely among 3,040 half-turns from the twenty marker locations in original rows 12–18, pairing 18/20 into nine pairs. Across 133 reflected cell pairs occupancy agrees in all and literal glyphs agree in 121, leaving twelve literal differences. The four non-marker ordered differences z→q, q→semicolon, comma→period, colon→comma increase by one under the old code. The full directional difference stream is `111001010011`. (R1795; R1800)

The later census explains its limits. All unit-step differences constrain only the two orphan marker labels to be q; the other eighteen labels are unrestricted by that property. Among 2^{20} assignments, 2^{18} satisfy it. Preserving each row's marker counts leaves 288 pictures: 144 have all unit changes, 48 also have twelve literal differences, and 4 retain the full sign stream. (R1800)

Enumerating 1,536 code maps leaves four satisfying the geometric constraints; two pass CONTROL4. Only the original among those two admits the declared affine ASCII-parity model. Sixty-four whole-byte affine extensions still fit. The other checksum-passing nonlinear map produces sign value 19 and a six-bit digest prefix 26, while the original digest prefix is 38. All are artwork dimensions. This is a useful warning: several retrospective dimension coincidences can coexist. The non-affine alternative is incompatible with the chosen affine family, not cryptographically disproved as every possible encoding. (R1800)

Under the six-bit C/E producer the original sign stream fixes `100011`; 8,192 valid alternative GCM packets yield 129 exact original sign streams and seven of those also pass CONTROL4. Deterministic dependence does not make a value meaningless—an author could intentionally choose a nonce to obtain a picture—but it prevents counting the tag, its glyphs, and derived signs as independent evidence.

10. FOLLOW GOLD, the leftward eyes, and the thirty-two-state result

The original October 26, 2021 poem ends “Always trust in following one’s eyes.” Its footer is `$ZNN [EYES]`. In the preserved 148-record export, 138 records are authored by the account and this is its sole authored EYES emoji occurrence. Earlier fifty-word counts legitimately omit the footer, but visual-clue analysis should retain it. The official Twemoji v13.1.0 EYES asset predates the poem: both pupils sit 2.234 units left of their eyeball centres. That supplies a concrete candidate leftward cue, subject to the distinction between an available 2021 asset and the author's actual client display. (R1812)

A literal leftward run does not complete the old reader. The last q `(18,23)` encounters `;.`; the last structural 1 `(16,26)` encounters `:qzzzzzq;.`; the unmatched upper right 1 sees an eleven-space gap then `.;qzzzzz,`. An earlier walk that produces `;.,:` looks west, west, then east. The emoji does not authorize that third turn. (R1812)

The complete FOLLOW GOLD candidate is explicit: select the bottom-open 22-space component; reflect it horizontally with H(r,c)=(r,37−c); take its forty-cell union; load original forty q/1 bits row-major; XOR reflected partners at original chamber positions; read rows top-down but each row right-to-left. This returns `01011010101 01100100010`, indexes 725 and 802: FOLLOW GOLD. Ordinary left-to-right output yields ONE EYE. These outputs were already recorded on October 20; the November 5 audit replays them. (R1815)

The full blank-component census finds sizes 148, 149, 33, 22. Among twelve specified component/reflection combinations, only lower 22 under horizontal reflection gives union 40. Testing 8 source traversals × 8 loading traversals × 8 output traversals × 16 Boolean functions gives 8,192 complete protocols. Four produce the pre-existing FOLLOW GOLD bits. They are one mirror-equivalent construction: their input/loading changes cancel. All use XOR and the same output direction. The

target-conditioned census supports a compact description, but cannot retroactively prove the poem prescribed XOR or scratch relocation. (R1815)

The new thirty-two-state test is more discriminating. All 32 calibrated lower pictures produce 32 distinct complete outputs of binary difference rank 5. Thus the reader retains all five remaining model choices; FOLLOW GOLD is not forced by silhouette, rails and rotated-tile calibration alone. The original is the sole member giving that target. However the broader forty-bit XOR memory has 2^{20} preimages for any exact pair-parity result. Preserving thirteen ones leaves `2^9 × choose(11,2)=28,160`. The narrow uniqueness and broad ambiguity are consistent because their constraints differ. (R1815)

The proposed NASA check `725 AND 802 = 512` is not unique: changing q `(16,24)` to 1 within the same 32-state family yields 725 and 514, still AND 512. That synthetic picture is not an alternate signed artwork; it demonstrates that the external number 512 cannot alone select GOLD. The old top-left-clockwise spiral gives PICTURE RELIEF; the complete eight corner/direction results are retained below, and the variants initially moving left do not yield that phrase. NASA explains actual record playback, a picture-start signal, vertical raster reconstruction and aspect calibration. It does not specify this chamber, XOR, eleven-bit dictionary segmentation, or spiral start. No 512-line source, new image, or complete downstream message has emerged. (R1815)

Rectangular outside-in traversal	Complete eleven-bit index pair
Top-left clockwise	1314, 1450 — PICTURE / RELIEF
Top-left counterclockwise	1348, 1365
Top-right clockwise	1162, 1366
Top-right counterclockwise	674, 725
Bottom-right clockwise	1162, 1366
Bottom-right counterclockwise	1361, 213
Bottom-left clockwise	676, 725
Bottom-left counterclockwise	549, 683

Each traversal skips absent rectangle cells and consumes all twenty-two actual chamber bits exactly once. Repeated index pairs are retained as separate traversals. (R1815)

11. All-angle observation and actual pixel steganography tests

The observation-first angular ledger

The November 4 all-azimuth audit retained every first-hit source owner before semantic interpretation. It tested 57 fixed viewpoints with 8,192 angles each and 495 fractional viewpoints with 4,096 angles each, three opacity thresholds, original/scratch images, and two moved-glyph pixel registrations. Fifty-one centre observers and 459 fractional observers were legal in both states. The outputs include 137,243 compressed sectors and 3,312 complete clockwise owner cycles. Coverage is the full angular range at finite increments, not every real angle or three-dimensional elevation. (R1696)

Newly accessible field owners depend on threshold and viewpoint. At some centre views the opening reveals C's equals sign and E's K,0, Y; across all shared fractional views no field owner is newly visible at all three thresholds. The 389 field-change sectors from comparable observers lie within about 4.9° of north. Existing fields are above the tunnel, which explains the directional concentration without turning it into another encoded byte. (R1696)

Unfiltered circular readings preserve punctuation, z, repeated encounters, and same-looking distinct owners. The longest literal hexadecimal run is five characters. Eight cycles have 44- or 45-character Base64-alphabet runs dominated by z: the 44-character runs decode to 33 bytes and do not equal compressed Q; the 45-character runs lack complete valid framing. No canonical padded 44-character encoding of a 32-byte digest was found in this frozen search. This leaves scattered/keyed assembly outside the tested family. (R1696)

A naturally sized 256-bit candidate combines original CONTROL192 with the 64-bit T30 paired-rim XOR `00000055104015f4`. Both orders, their SHA-256 and double-SHA-256 values, and relevant display byte orders match none of 125 pinned native proof values. A 42-character equal-parity alternative preserves both component records while changing the original artwork commitment. The assembly is therefore not an exact commitment to the literal source. (R1696)

The lossless PNG and its limits

The official 1000×1000 PNG has SHA-256 `fd1982c34d6f9dfde860a19326a179fbf62350d6b27fa38d91f8af12be5fdf2c`. It contains IHDR, seven IDAT chunks and IEND, no appended bytes or text/EXIF/custom ancillary chunks, and fully opaque alpha. 957,047 pixels are black; 42,953 are nonblack. Of 379 RGB colours, `(59,218,87)` occurs 17,987 times and its green+1 counterpart 218→219 never occurs. This does not resemble uniform LSB replacement across the solid main ink. (R1700)

A supplied JPEG changes bright-pixel LSBs relative to PNG by 48.16% in R, 50.53% in G and 47.41% in B. It is unsuitable as a witness for original PNG parity. The earlier high-entropy OutGuess candidate had no authenticated message and remained a likely false positive, not recovered hidden content. (R1700; R0520)

The October 19 forensic audit recovered the full 7,383-byte default-key OutGuess stream from the supplied 1100×1100 JPEG and matched all 1,024 bytes previously pasted as a hex excerpt. Its hash is `442ea736a3b8d6037c806fd0736df90f41004d90cb78bfa9fb63c37f64f53969`; entropy was about 7.974 bits per byte, with no recognized file, readable plaintext or zlib compression. Ten of nineteen deliberately unencoded control JPEGs nevertheless produced plausible lengths. Several reproduced seed 24127, claimed length 7383 and the same first sixteen bytes. The mechanism was 2,294 leading zero

coefficient bits feeding the default keystream, producing a header echo rather than authenticated content. The alleged invisible-ink image also proved to be a deterministic pixel-colour remapping: all 1,210,000 positions remained fixed, with zero colour-map inconsistencies and identical nonbackground support. That image did not expose a new writing layer. (R0520)

The declared no-key extraction set has 720 complete streams across six masks, five channel orders, three bit planes, row/column direction choices and bit packing. No complete embedded image/archive/document validated. Eleven accidental JPEG-style prefixes failed decoding. Across 5,760 bit alignments there were zero exact occurrences of frozen E half-tags, Q endpoint fragments, the artwork-hash prefix or complete five-byte C. Conventional 8×8 BPCS complexity scans found very little high-complexity content, mostly in green antialiasing. The strict 24-plane tile extraction yielded no standard file header under its fixed threshold/order. These are bounded no-key negatives, not a proof against small keyed payloads. (R1700)

A donor-glyph atlas leaves 1,848 lower-region differing pixels, none consisting solely of a ±1 RGB toggle. However its glyph selection and vertical phase were fitted to this raster. It cannot serve as an independent pre-publication baseline proving that all low-bit content is ordinary rendering. A future pixel claim needs the actual source/key/traversal and a complete verifiable output.

12. Poems, proposed algorithms, and the Voyager motion evidence

The literary work supplies genuine source relationships, but different uses must remain distinct. The Dickinson difference 710968−709632=1336 selects a real poem number; its 24/21/30/20 character-line count is transcription-sensitive. An October 14 branch counted three separators to make 98, and only one of 24 line permutations passed its selected compound test; a direct 98-character CONTROL overlay failed. Publius's July smoke wording predates the Bitcoin artwork, C independently decrypts SMOKE, and all forty structural markers lie below C. These are real contextual correspondences, while poem selection, tokenization and subsequent operations still need to be specified. (B:K28; R0119M0003; R1416)

POET / ITEM / I and the earliest retained CryptoPublius map were explored as a locator, with January 14, 2021 post ID 1349605247510044673 providing a fifty-five-token poem ending MAP. A later suggestion involving historical Publius, Hamilton, or the first Federalist essay is a different hypothesis. It must not replace a CryptoPublius source merely because both names contain Publius. (R1458)

The missing eight-algorithm planning report has now been recovered. Its actual proposed families are:

Proposed family	Object it would read	What a complete test must specify
Rotating grille	Blank-region mask over characters	Pivot, rotations, crop, ordering, complete output
Bit-plane LSB/BPCS	ASCII bits or PNG pixels	Which carrier, keyed order, packing, packet grammar
Bacon A/B code	A declared binary class	Alphabet version, grouping, direction, all leftovers

Proposed family	Object it would read	What a complete test must specify
Knight's-ring scan	Eight knight-offset sites	Centre, all start/direction choices, output delimiter
Four-ray views	First-hit geometry	Observer, ink/cell model, ray order, source owners
Funge-98 interpreter	Literal character program	Entry state, instruction set, memory, termination
Logo-order masking	Historical image/row ordering	Source version, registration, row adapter, heldout check
Glyph-contour rule	Occupancy and neighbouring classes	Frozen grammar, omitted inputs, predictive holdout

This report is a research proposal/source synthesis. Some of its tables claim completed outcomes without corresponding executable receipts established here; it also calls an eight-byte output thirty-two bits, although eight bytes contain sixty-four bits. An eight-site knight ring has eight starts in each of two directions, sixteen conventions; the report’s four-plus-four accounting is incomplete. Its unsupported 10,000-control claims likewise need actual run receipts. Those claims are not imported as new completed negative experiments. Separately recovered LSB, geometry, contour and interpreter receipts retain their own exact scope. The planning report's categorical “all transforms failed” language is not justified by a finite family of tests. (R1613)

The official May 1, 2022 NFT article discusses image steganography and extraction/validation architecture, and its hypothetical logo example embeds a signature. The later quotation/context audit found traceable sources but no complete decoder for this 2021 artwork. It supplies mechanism ideas and historical context, not a retroactive read order. Similarly, source checking corrected the official “Quantum computers can't break it” clue to November 17, 2021, not a later forum heading's November 12. (R1820; R1812)

What the Voyager screenshots establish

Four supplied Interplanetary Voyager screenshots were rectified against the card's outer rim. The tracked interior star lies at 52.93%, 69.89%, 32.20%, 46.23% of card width at remaining times −0:09, −0:08, −0:02, −0:01. Its horizontal span is 37.685 percentage points; vertical span about 0.747. Three unused front-face controls—the Zenon logo, PP label and 2× label—vary horizontally by 2.34%, 1.67%, 1.67%. Their shape correlations remain high. This supports real relative movement between interior and face beyond what the rim fit alone guarantees. It does not distinguish rigid depth from independent or synchronized animation. (R1823M0002)

The archive identifies the June 26, 2021 Interplanetary Voyagers/Sentinels post and June 28 Golden Record/Interstellar Explorers/Pillars post as distinct. The inspected export contains nine concatenated response objects but no usable MP4 variant for either attachment. No full clip was recovered or analysed in that audit. Matched full-card poses at different times, with multiple interior landmarks, would distinguish more hypotheses; four screenshots cannot establish an animation law or an ASCII decoder. (R1823M0002)

13. Current visual frontier and preservation rules

The visual work has advanced substantially: exact source accounting; a locally derived punctuation scale; a checked 196-bit control view; reproducible blank-region and mirror geometry; a legal four-owner passage within a finite movement family; complete owner-aware angular ledgers; a candidate full reconstruction with explicit assumptions; and a thirty-two-state calibration family whose five remaining binary choices all affect the reader's output. Only the published state produces FOLLOW GOLD. These positive results should remain available together with their limitations.

The next unresolved connection is specific. A source-selected rule must connect a clue to an input region, reader, operation, complete ordering and acceptance condition. The leftward eyes can motivate a direction, but not the whole mirrored-memory operation. The seven-marker reconstruction can calibrate a frame, but does not identify its payload. The exact producer can explain observed dependence, but is not authenticated 2021 software. The passage genuinely opens, but the full movement schedule does not pass all live guards. Pixel negatives eliminate particular implementations, not all steganography.

Every future continuation should preserve the original source alongside derived state; name the coordinate origin and material model; retain all residual bits and source owners; record the full tested family and failures; and specify which result would count as success before selecting attractive words. A new complete signature, independently verified hash/preimage relation, authenticated second plaintext, or a prospective external prediction would be a different evidence level from a four-bit echo. As of this checkpoint none has been recovered as a second visual solve. The exact data, successful local constructions, rejected adapters and remaining hypotheses all remain part of the solve history.

Part IV — Corrections and the present frontier

Checkpoint: 6 November 2026

Corrections preserve the earlier results

The investigation has accumulated both discoveries and corrections. An exact calculation can survive even when its proposed interpretation changes. A reproduced word sequence remains a reproduced word sequence; a later counterexample may show that its checksum, selection rule or claimed independence cannot bear the conclusion originally attached to it. The earlier reports, failed experiments and abandoned interpretations remain in the preservation collection. This chapter identifies the reading that governs the present edition.

The authenticated primary solve remains complete. The additional research has not produced a complete authenticated secondary message or a historical identity proof. That distinction does not diminish the exact source relationships, successful local constructions, native verifications and bounded exclusions retained throughout this book. It establishes what a future result would need to add. (B:Authenticated primary solve; B:Current proof gaps and supersession rules; R1815)

Correction and supersession ledger

Subject	What remains established, and what changes	Controlling record
Historical VM and native execution	Installed CALL-model traces reproduce under their declared rules. Definitions and event counts are not independently authenticated puzzle instructions. Native call 1024 settles through valid signed account records, but its final data and descendants are empty; the later CALL/ARTWORK promotion remains an unselected adapter.	(B:K10)
P2095, P3928 and identity	The corrected input-145 point is P3928; P2095 is a different historical point. Preserve both and their separate differences from Q. The old punctuation entrance was superseded. Public point arithmetic and a genuine historical transaction do not identify a human controller.	(B:K07; B:K08; B:K35)
Mixed Q packet	The twelve projections and CHOOSE/AIR trailer reproduce. Their mixed fields, operators, order and omission were researcher-assembled on the recovered provenance. The missing thirteenth ordinary projection and target-known selection prevent treating the packet as an independently prescribed instruction.	(B:The Q packet: exact reproduction and controlling provenance correction)
AIR and Email 260	Recipient A's BACON/AIR/FINAL/ASSET correspondences are exact native-field results. The recovered chronology does not establish prior selection of ordered BUNKER−ARTWORK. AIR43 also does not supply CHOOSE322 or an ADD register. The route remains conditional; mailbox provenance alone would not repair these transitions.	(B:Email260: exact native fields, chronology and conditional selection; B:Cross-object negatives and the controlling current boundary)
Signature and OpenPGP authority	Q's construction signatures, the 145 message signatures and the historical certificate signatures retain their respective native validity. They authenticate different objects. Certificate self-certification and subkey binding do not sign Q or the artwork, and no admitted new-signature demand was recovered.	(B:K09; B:K25; B:K26)
Subject	**What remains established, and what changes**	**Controlling record**
“2021-only” replay	Removing the 145-table file from runtime is a real dependency reduction. The replay still uses grammar learned during earlier	(R0344)

Subject	What remains established, and what changes	Controlling record
	research. Its 348 phase-check survivors, twenty-six partial-field matches and sole complete !9 match describe that fixed grammar; they do not turn its discovery into a blind prediction.	
Types and execution phases	C[5] is out of range for a five-byte field; interpreting C/5 as length is a distinct hypothesis. The ORBIT loop already passes before VERIFY. The later panel overwrite preserves CONTROL's four-bit check but breaks the internal RANGE/SUB child. These results cannot be combined without an explicit type and state schedule.	(R0487; R0489; R1117)
JPEG and OutGuess	The default-key 7,383-byte extraction is reproducible, but unencoded controls reproduce its apparent header and length. The transformed “invisible ink” image is a colour remapping. JPEG pixel parity differs substantially from the lossless PNG; neither observation supplies an authenticated payload.	(R0520; R1700)
Exact renderer	Conditional 722/722 reconstruction was already achieved on October 26 using fitted geometry and marker rules. The November census measures a candidate producer's possible outputs. Neither result identifies the actual 2021 source bitmap, software or settings. Later holdouts narrow earlier coverage explanations.	(R1003; R1684; R1780; R1783)
Information omitted by parity	LOW/FULL merges 285 literal choices across z/comma, q/period and 1/semicolon. The additional ASCII bit-1 plane resolves them with the retained occupancy and field data. These are distinctions lost by the reader, not 285 proven independent payload bits.	(V01; R1803)
Six lower bits versus thirteen	Six varying sites belonged to the C/E producer. The fitted lower grammar actually permits thirteen free labels, giving 8,192 pictures. Additional rail and rotated-tile assumptions reduce this to thirty-two pictures with five remaining choices. These are different conditional families, not contradictory counts.	(R1803; R1808)
Subject	**What remains established, and what changes**	**Controlling record**
Actual and projected cavities	The actual page begins with a closed 33-cell cavity and bottom-open 22-cell cavity; qzzq movement produces bottom-open 55, and completing the move gives closed 55. The	(V01; R1475; R1823)

Subject	What remains established, and what changes	Controlling record
	projected view instead gives 33+33, then 66, then 55+11. Virtual cells and repeated owner observations must remain identified.	
T30 and U	The larger-frame half-turn is T30(r,c)=(30−r,37−c). The local rotated-tile relation is U(r,c)=(27−r,30−c). Both have exact uses. Substituting one for the other, or freezing an old local formula after moving its coordinate anchor, changes the experiment.	(V01; R1726; R1808; R1823)
FOLLOW GOLD and 512	The complete mirror-XOR reader gives FOLLOW GOLD on the original. All thirty-two calibrated pictures give distinct outputs of difference rank five. The alternative pair also satisfies `725 AND 514 = 512`, so that number does not uniquely select GOLD. Narrow-family uniqueness and broad XOR preimage ambiguity coexist.	(R1815)
Eight proposed algorithms	The recovered eight-family report is a planning/source-synthesis document. Untraced execution claims, including 10,000-control assertions, are not added to the completed-test count. Separately documented optical, contour and pixel experiments keep their own scope. Eight bytes contain sixty-four bits; an eight-site ring has sixteen start/direction conventions.	(R1613; R1696; R1700)
Official clue date	The “Quantum computers can't break it” post is dated November 17, 2021. A November 12 compilation heading does not replace the original post timestamp. This correction changes chronology without changing the preserved wording.	(R1812)

The ledger is selective: it brings together corrections that most affect continuation. The full historical appendix retains the detailed censuses and the smaller rejected adapters. A later report supersedes an earlier conclusion because of a particular source, calculation or counterexample, not simply because it is newer or uses a stronger title.

Three distinctions recur. **Reproduction** establishes what an algorithm does on specified inputs. **Selection** establishes why those inputs and that algorithm belong to the puzzle. **Authentication** establishes the particular relationship checked by a native verifier. A new claim should name which of these it establishes. The four-bit CONTROL check, a BIP39 label, a Bitcoin signature and a historical certificate are not interchangeable acceptance tests. (B:Evidence language; B:Current proof gaps and supersession rules)

Questions whose answers would change the frontier

1. What selects a complete secondary reader?

The strongest visual candidates now have explicit algorithms. FOLLOW GOLD requires a blank component, horizontal reflection, relocated marker memory, XOR and a particular complete output order. DISORDER/PARROT/GIANT requires C's first thirty-three bits, a specified stencil placement and overlap-before-overhang ordering. Their exact outputs are preserved; the selection of the full recipe remains open. (R1416; R1815)

Material new evidence would be an original source that supplies these choices, or a frozen complete reader that predicts independently checkable information not used to construct it. A full authenticated packet, a previously specified commitment preimage, or a correct external prediction would add something that another attractive word pair does not. All residual bits and complete failure outputs belong in that test. A target discovered during exploration can still be useful, provided its retrospective role is recorded.

2. Where does a proposed program obtain its next typed operation?

The historical model still needs the rule admitting the ART-shifted vector as executable code, choosing its head and carrying the correct operation and source object across a join. The Q-packet route additionally needs its complete extraction and ordering rule. Email 260 needs both its external-object entrance and an actual consumer for the resulting value. A same-spelled label elsewhere in the graph does not transfer these properties. (B:K10; B:Current proof gaps and supersession rules)

A successful bridge would specify the exact incoming object, valid representation, operation, destination and return contract before using the returned meaning. Its receipt would show the actual caller and child occurrence, preserve the state change and test the whole returned object. A fresh external file or native signed settlement can resolve its own domain while leaving this program transition open; the record should state precisely which edge has closed.

3. What would establish the historical authority being claimed?

The remaining authority question concerns a binding between particular authorities and particular bytes. It cannot be settled by public subtraction of keys, an English endpoint, a copied ciphertext or a standard certificate hash. The existing signatures remain strong evidence for their actual signed messages. (B:K25; B:K26; B:Historical PRIMARY: packet anatomy, missing payload and finite DSA tests)

Evidence that would change this conclusion includes a native, secret-dependent signature or historical commitment binding the exact relevant key, Q or artwork identifier to an explicitly stated claim. Its message framing, key identity, date evidence and verification domain must be retained. Even a valid new signature by a historical credential proves control of that credential for that message; a civil-identity conclusion requires its own historical support. The proposed OpenPGP challenge is one possible contract, not the established mandatory ending.

4. How much of the artwork follows from its historical production process?

The current producer gives a compact exact fit under explicit assumptions. Ordinary rendering also explains some local features, while later cross-row and punctuation holdouts fail to generalize those explanations. Neither result identifies the historical production procedure. The thirteen-free-label correction shows why the capacity of one fitted producer cannot be assigned to the artwork as a whole. (R1003; R1684; R1783; R1803)

A dated source bitmap, identifiable converter version and frozen settings could make the production hypothesis testable on labels excluded from fitting. Success would be an exact reproduction or prospective prediction whose remaining inputs are fully disclosed. If mismatches remain, they should be preserved as a complete residual map with comparisons against declared alternatives. A mismatch is evidence that a particular renderer is incomplete; its status as an intentional message still requires an extraction and validation rule.

5. Which physical observation does a clue prescribe?

The available views answer different questions: opaque character cells, actual raster ink, first-hit owners, boundary faces, and transported or virtual coordinates. The direct lower eye, whole-cell aperture and finite sampled-ink audit therefore have different owner populations. The four Voyager screenshots support relative motion, but do not identify a unique depth or animation law. (R1246; R1395; R1400; R1823)

Useful new evidence would choose an observer, material model, coordinate frame and traversal without relying on the resulting phrase. A complete original clip could constrain motion through multiple landmarks and repeated poses. A successful geometric model should predict withheld observations while tracking each original owner and marking off-page positions explicitly. Resolving the geometry would establish that observation model; a separate receiver must still consume its complete output.

6. Which missing source would resolve a specific uncertainty?

Several gaps have concrete identities: the original issuer-controlled prefix disclosure or quest receipt; earlier independently dated ciphertext evidence; Email 244's unavailable native attachment; the original raster/converter inputs; and the complete Voyager media. These objects would answer different questions. Their absence is not a common defect that any one recovered file would repair. (B:Current proof gaps and supersession rules; B:Email260: exact native fields, chronology and conditional selection; R1823)

A source recovery should preserve exact bytes, surrounding context, native date evidence and its relationship to prior copies. The result should state whether it changes custody, chronology, input selection or an actual verification outcome. A screenshot or rendered quotation can establish what was displayed; it cannot silently stand in for missing attachment bytes. Similarly, a new pixel extraction needs a specified carrier, order, packing and complete acceptance condition. The existing no-key negatives remain valid for their declared families. (R0520; R1700)

A continuation record that can survive another handoff

Every addition should be a new record linked to its inputs and predecessors. Preserve the original source separately from any normalized display, edited picture, projected scene or working memory. A source hash identifies bytes; a normalized-text hash identifies that particular preservation representation. Neither should be substituted for the other.

The following template is sufficient for a useful research handoff. Fields that are unavailable should say so explicitly rather than being filled from recollection.

```
Record ID and date:
Author: TminusZ
Kind: observation / source audit / executed test / proposal / correction

Question and proposed success condition:
Predecessor records and exact claim being tested:
Inputs: source IDs, exact hashes, byte lengths, versions and source dates
Representation: encoding, coordinates, owner roster, ordering, all residuals
Assumptions: what is given, inferred, fitted or retained from earlier work
Method: complete rule; code and environment identifiers where applicable
Execution: not run / partial / completed; exact scope actually executed

Observed output: complete values, artifacts, errors and unsuccessful cases
Controls: declared alternative family, counts, seeds and actual outcomes
Validation: native verifier or comparison, exact acceptance and failure
Dependencies: shared inputs, repeated owners and inherited target knowledge
Conclusion: what this establishes and the specific unresolved connection
Supersession: earlier claim changed, reason, and evidence retained unchanged
Preservation: artifact paths, hashes, source-versus-derived status
```

For an executed test, retain the runner and complete machine-readable results alongside the explanation. For a proposal, the method and expected discriminator can be recorded without inventing an outcome. For a correction, keep both the earlier claim and the later reason it changed. When a test is repeated, identify the new premise or recovered evidence that makes the repetition informative.

Counts need their populations: twenty markers from specified rows, twenty-two lower markers, forty whole-artwork markers, and ninety-eight structural controls are different sets. The same discipline applies to sampled rays versus exact angular support, original owners versus face encounters, native signatures versus public projections, and input files versus rules learned from those files. Preserve alternative outputs even when only one gives recognizable words.

This edition's V01 checks and archive-integrity reconstruction are recorded separately from the historical experiments they preserve. Recovering and hashing an old runner does not execute it, and executing it does not automatically validate its input-selection story. The book can therefore be extended without either losing old work or granting an old interpretation more authority each time it is copied. The next addition should make clear what new evidence changes while leaving every earlier result inspectable. (V01; B:Coverage and preservation contract)

Appendix A — The complete October 13 technical reference

Compilation by TminusZ

This appendix reproduces the supplied October 13, 2026 master's readable technical account with storage markers and navigation anchors removed. It is a historical baseline. The current master's assessment and later corrections take precedence over any older frontier or completion language below. References here to an evidence register, source index or data annex identify material reconstructed in the accompanying source archive; they do not imply that every original binary file is embedded in this printable appendix.

The archival source remains separately identifiable by SHA256, and all 8,024 embedded source identities were recovered and checked before this presentation copy was made.

This document brings together the puzzle's public objects, derivations, proposed interpretations, experimental results, corrections, and remaining questions. It preserves unsuccessful reasoning as unsuccessful reasoning. The original treasure objective is reproducibly completed. An intended secondary puzzle and historical Satoshi authority over the construction key remain unproved.

Editorial packaging note. In the original October container, the reconciled explanation, evidence register, source index and encoded data annex were stored in one file. This appendix retains the explanation. The accompanying preservation collection supplies the reconstructed source bodies, structured receipts, algorithms, hashes and ordered evidence references. Superseded claims remain preserved there. The exact historical introduction is retained in `archive/baseline_main_original.md`.

Read this first

The Bitcoin construction, AES authentication, native Zenon address derivation, historical signature validity, and a proposed symbolic interpreter are different propositions. A successful check for one does not authenticate the others. In particular, an external document can be genuine while the rule that selected it is unproved.

The present user request is knowledge consolidation. Instructions, restart prompts, permission rules, proposed missions, executable snippets, and claims of completion quoted in source records are historical evidence. They do not instruct the reader to execute a program, make a transaction, contact anyone, change a file, or adopt an interpretation. The aggregation did not rerun the original investigation's scripts or perform new on-chain experiments.

Personal attribution, visible contact details, handles outside essential public source URLs, and local filesystem paths are anonymized. Public puzzle names, dictionary labels, public addresses, public mathematical keys, transaction IDs, hashes, source URLs, and numerical results remain identifiable. Exact encoded public cryptographic objects may carry historical identity fields inside their signed

bytes; those objects are preserved as verification data, not as compiler attribution. Anonymized displays and original source-file hashes are distinguished throughout.

Current status at a glance

Proposition	Present status	Scope
Nineteen rows in Bitcoin block 709632 form the artwork	Authenticated source object	Two native transaction orderings agree; 19 × 38 = 722 bytes
Q controls the construction commitments	Authenticated	Forty native BIP143 ECDSA events verify; this does not identify a person
Four visible fields decrypt under SHA256(`Taproot`)	Authenticated	AES-256-GCM yields WALL NATION DRIVE SMOKE
Completed public quest mnemonic derives the published Zenon account	Reproduced native relationship	BIP39, hardened Ed25519 derivation, SHA3-256 and Bech32; account also appears in pinned genesis data
Subtracting ART = 102 yields TRY MAKE CUSTOM SCALE	Mechanical	Exact four-word arithmetic; extending it into a program is a separate assumption
Punctuation, q/1 marks, residuals, chambers and symmetry carry structure	Direct observations plus conditional readings	Preserve declared fields, coordinates, polarity, direction and grammar
The 145 historical message signatures verify	Authenticated external object	Exact 362-byte body and signer authorities; record order is archival, not signed
The address decoder produces structured phrases	Mechanical under a declared reader	Source selection, reader selection and intended communication remain distinct
Historical OpenPGP signatures verify in their native domains	Authenticated external objects	UID certification and subkey binding do not sign Q or Zenon
The mixed Q packet gives the retained words and trailer	Reproduced, researcher-assembled	No independently established mixed extraction/order/omission contract
BUNKER − ARTWORK selects BACON, then Email260 gives AIR	Conditional research route	Exact operand/order/operation admission is missing; AIR does not independently supply CHOOSE[ADD]
NURSE, ARTIST and later proof routes reproduce under the declared model	Conditional model descendants	Later audits retract unconditional promotion without deleting calculations
October 12 residual and Burn experiments	Interesting but unproved	Useful local equalities and controls; post-inspection selection and incomplete consumer contracts remain
Historical Satoshi controlled Q or authored Zenon	Open	No selected historical secret-dependent signature or commitment binds that authority to Q

Reading the historical reference

The original navigation depended on the old archival container. In this edition, use the book contents and the evidence collection. Packaging statements below such as “in this same file” describe that October container; this edition places the recovered records in the companion archive. The research text and its historical conclusions are retained.

Evidence language

Use these statuses for new reasoning. Source excerpts retain older labels for historical fidelity.

Status	Meaning
Authenticated	A precisely specified native signature, commitment or authenticated decryption verifies in its own domain
Direct source observation	A value, ordering, boundary, publication record or geometry is read from an identified object
Mechanical	A fixed transformation reproduces from specified inputs; intended selection is not implied
Supported interpretation	Evidence supports a reading while its assumptions and alternatives remain explicit
Conditional	The result depends on a named source, representation, selector, operation, type conversion or execution assumption
Rejected within scope	A proposed edge or hypothesis fails the stated test; unrelated encodings are not thereby disproved
Open	A required object, operation, authority, source or typed consumer is absent

Keep each occurrence's source, hash, parent, position, representation, operation, register, result type and consumer. A word label is not a global state. Raw scalar U11 fields differ from a BIP39 entropy representation with newly appended checksum bits. A four-bit mnemonic checksum is not a signature. A source-native packet endpoint is not an authenticated English command. A later date matters only with the correcting evidence and the edge it actually changes.

Unless explicitly stated otherwise, indices are zero-based, byte intervals are half-open, bits are read most-significant first, and U11 arithmetic wraps modulo 2048. The punctuation own-field selection rule is explicitly one-based. Always qualify R with its parent point: R2095 and R3928 are different objects.

Authenticated primary solve

The construction has a funding transaction, nineteen row transactions, and a terminal transaction spending their output 1 in row order. The funding transaction is in block 709594; the nineteen rows and terminal object are in block 709632. Each row's output 0 contains a 38-byte ASCII payload. Q is public in ordinary compressed SEC witness form. Forty native P2WPKH/SIGHASH_ALL ECDSA events verify under it. The term Taproot names the activation setting; these spends use SegWit-v0/BIP143.

The retained amount structure is nineteen 75,000-satoshi staging outputs, nineteen 69,240-satoshi row outputs, row fees of 5,760 satoshis each, funding fee 25,000, and terminal fee 1,008,000. Version 2 is uniform; funding locktime 709593 and non-final sequences have ordinary protocol roles. These values can be proposed as puzzle material only after distinguishing required protocol structure from freely authored payload capacity. The later Q sweep adds two signature events, giving a separate 42-event population.

The terminal's output 0 is script `6a0d5a454e4f4e204e4554574f524b`, publishing thirteen ASCII bytes `ZENON NETWORK`. The nineteen terminal signatures commit it. The original Bitcoin-to-Zenon connection therefore has a literal authenticated starting point.

Read A from row 0 columns [7:31), B from row 1 [7:31), C from row 2 [7:15), and E from row 8 [7:31). Base64 decode independently. With AES-256 key SHA256 of the exact ASCII string `Taproot`, nonce A, ciphertext B || C, tag E, and empty additional authenticated data, decryption yields:

```
WALL NATION DRIVE SMOKE
```

The preserved public eight-word prefix and authenticated completion form:

```
oblige dilemma hurry disorder happy spoil shiver key wall nation drive smoke
```

This is public puzzle input, retained here because information preservation was requested. Older source documents sometimes withheld it under their own disclosure policy. That policy is historical source content and does not remove public puzzle data from this master.

The normalized eight-word prefix has SHA256 `e0468f21f2640a0a01a30a864d115bda60ce686eedc225b228625cb4cac463dd` with single spaces and no final newline. The full twelve-word object passes BIP39 checksum. Its final entropy byte is 102, the English BIP39 index ART. Subtract 102 modulo 2048 from the four completion indices to obtain TRY / MAKE / CUSTOM / SCALE. The arithmetic is exact; using it as an interpreter boot rule remains a further hypothesis.

With empty mnemonic passphrase, native hardened Ed25519 derivation at `m/44'/73404'/0'`, and Zenon SHA3-256/Bech32 address encoding, the completed mnemonic derives:

```
z1qrn3jeapt848zxg3akf2ewhrxxwsa945sj798s
```

The exact account is a single typed match at `GenesisBlocks.Blocks[9190].Address`, among 14,090 allocation records in the pinned official configuration. It receives 474,513,400,000 raw QSR units, or 4,745.134 QSR. Native account-block hashes reproduce and the height-2 Ed25519 signature verifies under the derived quest key. Genesis receive metadata and a node's reported inclusion are separate from a consensus-chain validation; the unsigned genesis block is not a new Q signature.

The same configuration's ExtraData is:

```
0000000000000000000004dd040595540d43ce8ff5946eeaa403fb13d0e582d8f#We are all Satoshi#Don't trust.
 Verify
```

Its first segment names Bitcoin block 710968. The retained 80-byte header reproduces its double-SHA256 hash and encoded proof-of-work target. The literal phrases are project-source data. The pinned release commit is dated 22 January 2022; the file's content does not establish that the final configuration was publicly available in November 2021.

Prefix provenance is narrower than successful derivation. A later preserved screenshot resolves to a community Twitter thread, including post 1460002976907595779, dated 14 November 2021 21:53:29 UTC by native metadata. Its text supplies the exact eight words, but it has no native attached image. Screenshot line wrapping is not an authored spatial alphabet. The community text is a third-party report. The original issuer-controlled first/middle-four-word disclosure and the deleted first post's original-source links remain missing. The later source correction is preserved in full below.

Key public references are the funding transaction, terminal transaction, quest preservation copy, pinned native SDK, and pinned genesis configuration. These are retained source references, not a claim that all links were freshly fetched during aggregation.

Foundational knowledge

The following modules consolidate the original detailed technical account. Their calculations and declared assumptions survive, while current status is controlled by the opening map and later corrections. Historical current-frontier language has been removed from this main account; the evidence register still preserves the original wording and chronology.

K04 TRY MAKE CUSTOM SCALE

Two scale proposals must remain distinct. The older proposal extends source-minus-ART over the full twelve-position root and interprets the result as a program. The extension preserves order and its last four positions reproduce the original instruction. That is a useful trailer receipt, but those were the training examples. Repeating them is not an independent2^-44 prediction. The transformed vector is defined by subtracting ART = 102 from every index of the public root printed above; older accounts withheld it under their own disclosure policy.

The unresolved older boot is **data vector -> executable program -> selected head**. P00 cannot become an opcode solely because its label is MERGE. A4|4|4 layout is an ordered data organization until an operation consumes it as code. P00–P11 are not twelve independent branches.

The later punctuation proposal is more source-local: attached suffixes select four punctuation glyphs from their own fields, yielding an ordered alphabet. That directly addresses the phrase CUSTOM SCALE without borrowing historical results. Its exact construction appears in Part12. It does not retrospectively authenticate the older compiler or all descendants.

The strict status of the older boot remains OPEN; a separately named conditional downstream layer preserves what follows if the retained compiler is admitted. Neither layer should erase the other.

K05 The global forty-word table

Concatenate A||B||C||E in source order, not in AES API argument order. The result is55 bytes =440 bits =40 complete U11 fields, with no residual. The complete source-ordered table is generated in Part35. A five-row, eight-column organization exposes repeated end-column positions UNDER23, VERSION31, ADDRESS39; this is a table-width observation, not the physical width38.

The bottom row is MEDIA/BOTTOM/COVER/LOTTERY/NEUTRAL/WIDTH/RELAX/ADDRESS. An external-media interpretation consumes that literal row as data. Its semantic siblings and arithmetic endpoint must still have their own consumers. “The table ends in ADDRESS” does not select a particular Bitcoin or Zenon address without chain, representation and bytes.

Field-local extraction must also survive: each field has its own U11 words and residual. The retained local-head initials yield ARTS, and concatenated field residuals yield RAZOR/NATURE. These calculations preserve a different boundary from GLOBAL40; they must not be treated as independent cryptographic witnesses.

K06 External documents, events and media rooms

An external execution receipt must say what selected the object, which edition or byte representation was used, what was extracted, which operation was licensed, what returned, and where that return is consumed. A song identification alone is not lyric execution. A later faithful transcript of a pre-puzzle song may be evidence for its lyrics, but the transcript page's later formatting, recommendations and navigation are not2021 puzzle data.

Many Facez

The conditional BOMB/VENUE/THEME/LYRICS row was treated as a bounded media request leading to PERSON_L's1997 Many Facez. The retained edition is Universal UD-53036 with seventeen tracks; a contemporary review names five personas. Exact release-day and some track-order/copy questions remain qualified. Cover pixel dimensions from a later image file are not source-native album counts.

The room includes lyric evidence for BOTTOM/CHECK and a complete5×6 table/operator census. Exactly one candidate—the bottom row under AND—passes the installed checksum. Its new ABANDON endpoint does not equal the old call-zero occurrence. The proposed ALL/TRUE neutral interpretation identifies AND mathematically, but was developed after the desired AND result was known. LOTTERY modulo17 or5 remains conditional without a local modulo instruction. COVER lacks a stable selected region population; semantic RELAX and ADDRESS also lack typed referents.

CF001 was subsequently corrected: its main TABLE/JUDGE acceptance already existed before the auxiliary sibling-row census. It did not acquire an authored stack-return barrier merely because the audit used depth-first bookkeeping. Many Facez is not silently closed; its open internal leaves persist, but they do not block a nonexistent parent return socket.

Sources: edition listing, contemporary review, Theme lyric copy, Clue lyric copy. The room audit preserves source hashes and token statistics instead of republishing complete copyrighted lyrics.

Rush

EARTH/GLARE and VICTORY/AMOUNT suggest Earthshine and One Little Victory, present together on a2002 two-track promotional object. GLARE-to-SHINE is a semantic adapter. The disc presents the songs in the opposite order to the two incoming pairs. The target, lyric operands and expected result were user-supplied before the hostile audit; this was not blind discovery.

The official texts supply RIGHT and DEGREE with ordinary-language subtraction wording. RIGHT1486 minus DEGREE462 =1024 =LENGTH. Reverse subtraction modulo2048 produces the same number because the gap is half the modulus; those are not two independent confirmations. The retained conclusion is a conditional numerical acknowledgement of an already pending LENGTH cell, not proof that the sibling was an authored subroutine or that a generic return opcode exists.

Sources: paired disc/scans, official Earthshine, official One Little Victory.

Other media branches—including Black Violin, Rocket alternatives and The Endless—still require detailed admission/return reconciliation. The four-video report now has the explicit disposition below. A compelling story is not an automatic instruction.

Four supplied videos: complete report disposition

The retained August28 video audit reports container inspection, one-frame-per-second visual timelines, denser title/dialogue contact sheets, sidecar comparison, audio spectrogram inspection and publication metadata. This compilation reconciles that report; it does **not** claim to have rewatched the videos or independently repeated those technical checks.

Object and public source	Reported publication / duration	Exact-copy SHA256
Incoming transmission	2024-11-10 /31.068s	`45efde4eccacdf6fa632cb2b9b51e96351c0254cd86321b45b6737b602673058`
pp chan's debut	2024-11-20 /86.053s	`9d9c7b8a740a18f11a26554c1658e088705b3a3eb82cec3c173adfb583c98537`
AI PP chan still alive?!	2024-12-04 /94.226s	`83bbbadb2dad8cb4ea1c9c36a2ecebefcb923dfee459c0f0cd689847f309feaa`
Upgrade Preview!	2024-12-25 /89.118s	`58936729b0aa5898b35f734b61299b8d4ba3eb2a485d7c7d670cf5189eaa8f63`

Selector: supplied by the user as possible hints; no authenticated original-artifact selector for this four-file set was recovered. These are newly created2024 works, not merely later transcriptions of pre2021 songs. That distinction preserves the user's valid chronology exception for faithful later lyric transcriptions without admitting a new2024 film as a2021 instruction.

What each object contributes:

- Incoming transmission presents a star field and a request for help addressed to PP chan, followed by agreement. It supplies a request/response theme, but no authenticated speaker, signature, address, external path or payload.

- pp chan's debut contrasts an AI-generated apparent character with a real one. Reported visible text includes ENDHAVEN ZERO, THERE WERE SIGNZ and an assurance of having the character's back. The named song is Moon and Back. Its retained lyric reading concerns stars, constellations, countdown, lift-off, gravity and battle, not a named proof object. Retained lyric reference. SIGNZ→signatures and back→PERSON_E are semantic puns, not authenticated selectors; a game title containing ZERO does not invoke CALL zero.
- AI PP chan still alive?! moves from a generator to questions about its algorithm and creator; the displayed answer is Kaine. The character builder exposes skin, hair, eyes, mouth, chest, arms and pants. The named song, Emphasis's Bones, concerns a broken bond, non-return and an ending; its reading supplies no exact technical location or additional selector. Retained lyric reference. Kaine is a displayed attribution, not a verified historical identity or cryptographic controller.
- Upgrade Preview! describes Syrius as a star and origin of weapymon. Weapons include SWORD/LANCER/GUN/BOOMERANG; moves SHIFT CUT/POWER SLASH/SWORD SURF/GARU/WIND SLIDER; stats SIZE/SPEED/POWER/BUMP/SLIDE. The finite menu shows player control, but no observed choice returns a signature, address, archive path or selection rule. UPGRADE plus SYRIUS motivates a software-provenance hypothesis, not an executable wallet command.

Proposed aggregate: request help → distinguish apparent from real → inspect generator/algorithm → ask creator → Kaine → Syrius upgrade. The report suggested a cutoff-aware search of original code, releases, manifests and messages, plus a separate search for a Q-authenticated link to the later videos. This is preserved as a proposed research move, not a performed search receipt or a new authorized instruction in the current traversal.

Technical negative: the report found ordinary H.264/AAC, no embedded subtitle/attachment/data stream, no obvious pictorial spectrogram carrier, and no visible cryptographic signature, key/fingerprint/address/transaction identifier, challenge commitment or exact proof-object URL. No Q-authentication of the channel, Kaine or the series was recovered. These are bounded checks, **not proof that no possible steganographic encoding exists**. Noisy automatic subtitles and later2026 forum interpretations were not used to manufacture source wording.

Return / status: `QUARANTINED_POST_CUTOFF_OVERLAY`; no typed return to the original machine. The story may corroborate a theme. An independently authenticated, prospectively linked later response could be evaluated separately; none is supplied by these files. Proposed historical signatures/commitments or an authenticated interactive response were acceptance examples, not discoveries. No conventional Satoshi→Q proof follows.

K07 The conditional P2095 branch

P2095 is an older historical public point with its own exact reduction results. It is not the corrected input145 point P3928. The public algebra R2095=Q−P2095 is well-defined. Public subtraction of points does not demonstrate control of either corresponding secret.

Retained P2095 examples include P.x adjacent AND endpoint ANCHOR67/check2; P.x mirror ADD BONUS203/check10; R.x adjacent OR BELOW166/check10; R.x halves left-minus-right ANGLE70/check2. These later supply useful fixed comparison values for the matrix and STEP controls. They are not a statistically independent discovery holdout when their target equations have already been supplied.

P2095-related paths retain their own source and operator scopes. No transfer of their endpoints or registers into P3928 or the new punctuation scale is automatic. The following reconciles the FINISH/ENVELOPE and input6 selector reports in full at report level, including their supporting path receipts. It does not close every other P2095 source, VERSION branch, or source-copy branch.

7.1 The historical compiler contract, not a new strict-root license

The August28 reports started from a PERSON_M F1371 → PROBLEM selection, an ACTOR poem-line operand, the historical public point P2095, and R2095=Q−P2095. They assumed a whole-state, six-operation CUSTOM SCALE and a CHECK normalizer. Their assertion that the root already proved whole-state executable scope does not survive the later strict-root audit. Consequently all downstream paths below are preserved **conditional on that historical compiler**, even where their arithmetic is exact. A report's use of “authenticated ENVELOPE” means return to an authenticated root value, not authentication of its entire selector chain.

The retained scale output order is ADD→AISLE45; SUB-LR→ADVANCE31; SUB-RL→ABLE2; XOR→ABUSE9; AND→ADMIT29; OR→ACTUAL23. Here SUB-LR and SUB-RL designate the ordered subtraction directions used by that registry. Two different consumers were explored; neither silently replaces the other.

7.2 Intrinsic-length dispatch into R2095

For the compressed public R object, the report uses its 24 complete U11 cells as the divisor and preserves both quotient and remainder:

Creating operation	Selector	Quotient	R cell position	Cell label	Carried result
ADD	AISLE45	1	21	DENY	DOCTOR
SUB-LR	ADVANCE31	1	7	FIRE	PROUD
SUB-RL	ABLE2	0	2	MERIT	MERCY
XOR	ABUSE9	0	9	STING	STATE
AND	ADMIT29	1	5	PAVE	ACCOUNT
OR	ACTUAL23	0	23	ENERGY	ENOUGH

The report packs the ordered quotients as $110010_2=50$, labels that value ALIEN, and uses the last selected cell ENERGY in `ALIEN + ENERGY = EXPECT`; `EXPECT + ALIEN = FINAL`. Contextual CHECK gives FILE/FINISH in the respective retained representations. **Qualification:** division is reproducible, but quotient packing and promotion to a state belong to this conditional historical grammar. They do not license an arbitrary scalar cast in the current punctuation route.

7.3 P.x supplies the proposed connector

All three pairings × six operations on the canonical P2095.x entropy state produced two CHECK-valid rows: adjacent/AND with endpoint ANCHOR, and mirror/ADD with endpoint BONUS. Concatenating those rows in registry order and CHECK-normalizing gave raw foot NUCLEAR, checked foot OUTDOOR, endpoint ANOTHER. The report's two CONNECT routers were:

```
P.y endpoint BLAME XOR WIDTH → UNKNOWN
UNKNOWN divmod WIDTH → quotient ALIEN, remainder3 → GLOBAL40[3]=CONNECT

BONUS divmod ALL → quotient ABOUT3, remainder ALIEN
use ABOUT3 → GLOBAL40[3]=CONNECT
```

These are two retained constructions, **not two interchangeable generic addressing rules**: the first uses a remainder and the second uses a quotient. Their selector/adaptation remains conditional.

Mirror/ADD applied to R2095.x initially gives raw foot BLEAK, checked foot BLACK, endpoint BADGE. The proposed ANCHOR BONUS instruction replaces only its final value with BONUS. CONNECT ANOTHER then appends the previously fixed EXPECT checkpoint. The resulting 24-value CHECK gives raw foot FINISH, checked foot ENVELOPE, endpoint BEGIN. Final-slot installation and selection of the appended checkpoint remain semantic compiler assumptions; checksum success cannot retroactively supply them.

The retained controls are explicit: arbitrary final-value replacement across 18 reductions ×2048 anchors yields **167 ENVELOPE hits among 36,864**; carrying the same P.x pairing/operator/endpoint to R.x yields **one hit among 18**, mirror/ADD/BONUS. These controls bound a family, not the probability of deliberate authorship.

7.4 Two outputs from the connected object

The checked foot and entropy endpoint are different typed outputs:

```
checked foot ENVELOPE
  → divmod ACCOUNT12: quotient ALIEN, remainder6 → GLOBAL40 FREQUENT
  → divmod26: quotient ACTUAL, remainder8 → GLOBAL40 TWELVE
  → divmod22: quotient ADDRESS, remainder12 → GLOBAL40 FESTIVAL
  → ENVELOPE − FESTIVAL = WALL → known authenticated root value

entropy endpoint BEGIN, producing operation ADD
  → BEGIN → AGAIN → EDGE → ANGER → BLOOD → LITTLE
    → CUSHION → DIARY → FADE → BODY → AUTHOR → DEBRIS → AREA
```

The report also checks `ENVELOPE + AREA = FINISH`. A stale FESTIVAL numeric transcription was corrected in that source. The three divisors above are retained as separately proposed routers, not newly established source-native commands. The BEGIN/ADD branch was compared with a burn-derived BEGIN/ADD occurrence, then routed through the older AREA/OUTPUT, ANOTHER, TOPIC/TEST/XOR machinery to an internal SATOSHI assertion. That later symbolic join does not establish possession of P2095's historical private key. The separate source-log FINAL/FILE/FINISH → GOT JOIN interpretation is retained, not replaced by this internal loop.

7.5 A separate native-byte consumer selects input6

For a 32-byte coordinate, AISLE45 is out of range and is excluded **without modulo reduction**. Remaining selectors in native byte order are `[2,9,23,29,31]`; scale-generation order is `[31,2,9,29,23]`. The source did not let decoded labels select between these orders.

At input6, signature-coordinate bytes at position2 label BULLET/CABBAGE and at position9 AROUND/ATTEND. Cross-applying the two operations that generated selectors ABLE2 and ABUSE9 gives:

Native position	Carried operation	Output
2	SUB-RL	ACCOUNT
2	XOR	ACCOUNT
9	SUB-RL	ACT
9	XOR	ACTOR

The outputs satisfy `ACTOR−ACT=ABLE`, `ACTOR−ACCOUNT=ABUSE`, `ACT+ACCOUNT=ADVANCE`, `ACT XOR ACCOUNT=ADVANCE`. Exactly one of 19 terminal witnesses matches the joint fingerprint: input6. Supporting APPLE/AUNT/BELT/AHEAD cells are individually unique in that census; AROUND alone has a collision. **The cross-application itself is conditional:** selecting two generating operators does not automatically instruct a two-by-two fanout.

7.6 Public V construction and both dispatch procedures

For each witness the report constructs public ECDSA objects `A=u1G`, `C=u2P2095`, `D=u2R2095`, `K=A+u2Q`, `V=A+D=K−C`. These symbols are local point names, not the original artwork's A/C fields. No secret scalar recovery is involved.

The historical SIGN AND VERIFY = SCALE event was used to carry uniform AND into V.x/V.y at the five selected byte offsets. Native order returns `ABANDON / ACCOUNT / ABOVE / BEGIN / ABILITY`; scale-generation order returns `ABILITY / ABANDON / ACCOUNT / BEGIN / ABOVE`. A19-witness ×5-point ×7-procedure census contains **665 candidates**, one matching the complete native-order target and one matching its first/last pair—the same candidate. The target was recognized after decoding; uniqueness does not itself earn AND. Its alleged license is the prior SCALE occurrence in the conditional model.

The competing per-selector carry keeps SUB-RL/XOR/OR/AND/SUB-LR in native order and returns `VISA / BLUR / ARROW / BEGIN / VOCAL`. This is a different parent contract, not a failed execution of uniform AND.

Every uniform-AND child has a retained endpoint:

```
ABANDON/AND → ABANDON                      canonical boundary
ACCOUNT/AND → ACROSS → ACTION → ABLE       canonical boundary
ABOVE/AND → ACTRESS → ABOVE                cycle
BEGIN/AND → ABSURD → ACOUSTIC → ABSURD     cycle
ABILITY/AND → ABOVE → ACTRESS → ABOVE      same cycle
```

Here “canonical boundary” is the old executor's classification, not an unconditional return to a parent. In particular BEGIN/AND is **not** BEGIN/ADD, and non-adjacent ABANDON/ABILITY do not form a CALL-zero instruction.

7.7 Full mini-lattice paths, without ellipses

The supporting JSON preserves these exact label paths. Initial operation comes from the row above; both SUB-RL paths encounter MONTH, whose old installed fingerprint overrides the register to ADD.

```
position2 / SUB-RL:
ACCOUNT → SMILE → TENANT → VERY → ZERO → PELICAN → PRIVATE → NATURE
→ UPDATE → SPICE → RENT → MANDATE → PUNCH → SAFE → MOTOR → SOCK
→ OKAY → REMIND → POTATO → TENNIS → PRODUCE → PARADE → PARTY
→ UPHOLD → SICK → TILT → STRUGGLE → UGLY → MONTH → CLOUD → FROZEN
→ BAG → IMPULSE → EIGHT → KID → COOL → DENY → LEAVE → FORWARD → ENVELOPE

position9 / SUB-RL:
ACT → ROTATE → SLENDER → WRONG → MORE → STRUGGLE → UGLY → MONTH
→ CLOUD → FROZEN → BAG → IMPULSE → EIGHT → KID → COOL → DENY
→ LEAVE → FORWARD → ENVELOPE

position2 / XOR:
ACCOUNT → CRUISE → ASPECT → CRADLE → GRASS → DUCK → LATER

position9 / XOR:
ACTOR → FREQUENT → CRIME → GUILT
```

7.8 Later ENVELOPE/ADD continuation and its limit

The subsequent receipt joins three historical parents at ENVELOPE/ADD: the connected R checked foot and the two SUB-RL paths after MONTH's override. It executes `ENVELOPE → KISS → BETRAY`; the installed BETRAY/AND handler emits BOOK/ABANDON, then the old call-zero handler maps zero through COS(0) to ABILITY. The follow-up fixed-point receipt reports identical state order, operations, external requests, output and acknowledgement on two37-state rounds; no new states or external requests.

This supersedes the earlier report's temporary question about whether an ENVELOPE join remained to be executed **within that model**. It does not revive the rejected first/last-word shortcut. Nor does it override the later CF001/call1024 lessons that label equality and a visited descendant are insufficient to earn control flow. The CALL-zero dispatcher and BETRAY handler must remain explicit conditional dependencies.

Current disposition: report-level integration complete for these three reports; their relevant path and control results are now readable here without local files. Historical crypto/executor tests were not freshly rerun during this editorial pass. The records preserve public-data circuits, not proof of historical Satoshi→Q continuity. Remaining P2095 sources still require their own disposition.

K08 Corrected P3928/Q/R census

The retained historical route goes through a205-input consolidation. In the older route, the block2905 coinbase outpoint selects input134; its reconstructed nonce-coordinate endpoints give ACQUIRE/ADD under the retained extraction; global RECALL at position11 supplies an offset; input134+11 selects145. The spent P2PK output at that input provides the corrected historical point associated with block3928. The native ledger equations reproduce independently of the puzzle interpretation.

But the original punctuation packing used to select2905 is superseded. Consequently this is a **conditional selector route to genuine public objects**, not the current strict continuation of Part12.

All six coordinates of P3928, Q and R3928=Q−P3928 are reduced through all 18 cases. The nine valid results are:

Coordinate	Pairing	Operation	Endpoint	Checksum
P3928.x	all	all	none	—
P3928.y	adjacent	right−left	ASSAULT108	14
P3928.y	mirror	ADD	AFFAIR34	0
Q.x	halves	OR	BUYER251	9
Q.y	mirror	OR	ART102	9
Q.y	halves	ADD	ABOVE4	7
R3928.x	adjacent	ADD	ASPECT107	12
R3928.x	adjacent	OR	ALTER59	6
R3928.y	adjacent	ADD	AGENT39	13
R3928.y	halves	left−right	ATTEND116	1

Neither a valid endpoint nor its dictionary meaning alone chooses a CALL definition. The four-bit checksum remains attached to the reduced object.

K09 Q construction authority

Q is directly serialized in the construction witnesses and matches their P2WPKH commitments. A recovery-only demonstration is also valid: reconstruct each terminal BIP143 digest and enumerate the complete public-key recovery set. Each of nineteen terminal signatures yields two candidates. The selected pair11/18 intersects at Q—but **every one of 171 distinct terminal pairs does so**, and all nineteen alternatives are distinct. The selected pair is not an exceptional cryptographic discovery.

The retained conditional selector for 11/18 uses two representations of a public recovery-gap point: raw last value DESIGN478 and checksum-normalized CLIMB344. Divmod478 by 40 yields11/remainder38; divmod344 by 19 yields18/remainder2. The remainders remain attached. Their later English usefulness must not choose the divisions after the fact.

A second corroborating route uses P3928.y raw positions11 POST and 17 SCALE, then a terminal17 ANOTHER/STEP interpretation. It uses Q in its own computations and therefore cannot serve as a blind discovery of Q. The remembered MIRROR+ADD→ACCOUNT shorthand lacked a complete recovered pre-P3928 derivation.

Q.y has exact ABOVE/check7 and ART/check9 reductions; Q.x has BUYER/check9. The broader ARTWORK/TOPIC/SATOSHI/WOOL interpretations require the surrounding occurrence contracts. Public-key labels do not authenticate an identity assertion.

Public recovery proves a verification relationship, not secret possession by a historical person. A source-discovery narrative, a construction signature, and a historical-authority signature are three different propositions.

K10 Dispatch, receipts and rejoins

The reconciled go-zenon source reports establish a **native-object boundary**, not a dictionary-language boot. Commit `1baa7c4e057da4f2708a970b4fedf70a8de77fbe` contains240 Go files and the 55-term census preserved in Appendix A. Its date,2021-11-23, is nine days after the artwork; it can explain a November27 native event without proving pre-artifact lexical intent. Producer/delegation merges, consensus joins, database/iterator overlays and generated protobuf merges consume typed receivers, not seven BIP39 cells. The sole seven-argument ABI method, RegisterLegacy, lacks both the selected method/destination and valid argument types at P00. NETWORK or ZENON does not authorize arbitrary source-code execution.

The selected call1024 has what P00 lacks: destination,ABI selector `47f12c81`,two typed RetrieveAssets fields and native links. Its retained trace is user send → Swap receipt/Mint child → Token receipt/ZNN transfer → original caller's signed receive. The six account heights, in those records' own namespaces, are caller-send1,Mint-send2104,Swap-receive2105,transfer2208,Token-receive2209,caller-receive3. Payload lengths are324,100,8,0,8,0 bytes. The transfer is469136612 raw ZNN units. All six hashes,two Ed25519 signatures and legacy compact recovery verified in the source audit, with 32 controls; this compilation preserves those results without claiming a fresh historical replay. Final data and descendants are empty. Native settlement is closed; no puzzle opcode is returned.

The source corrects two technical details: CheckSwapSignature does not itself normalize s—high-s plus flipped recovery parity preserves the signer, while changing only one fails—and changesHash is a native state-patch field excluded from the account hash, not merely RPC enrichment. These do not supply extra hidden instructions. PREPARE's downloader queue method has no selected queue here; old EDUCATE→new call plus RAZOR/SUBTRACT preflight reintroduces missing dispatch/expired-register assumptions.

The retained CALL model has 37 trunk states, later expanded definition and event populations, and operator-sensitive transitions. It is useful as a deterministic conditional model. A lookup definition's existence is not proof that every occurrence of its label may invoke it.

The concrete call1024 correction is decisive. Under the conditional TABLE/JUDGE hypothesis, PREPARE/EDUCATE/WALK/LENGTH selects public ordinal1024. Its four projections are CALL/ARTWORK/ARCTIC/ADULT. That is a unique fingerprint satisfying the judge. The source-native RetrieveAssets operation was also followed through its settlement to a user receive with zero data bytes and zero descendants. Both roles complete without requiring a further puzzle program.

The old continuation promoted fingerprint field0 to opcode and field1 to call105 because the known next topology looked coherent. Source genealogy found this promotion earlier than the WALK/LENGTH justification. It is result-trained, not a prospectively earned adapter. A locally typed CALL/ZONE-zero case does not create generic CALL X behavior.

Likewise, a reentry creates a new occurrence with its own ancestry. The retained2,445-record reentry registry had300 same-operation and 2,145 different-operation reentries,2,115 changed complete paths, and no recorded provenance transfers or restored parents. Equal labels do not reconstruct completed joins.

Historical graph coverage counts also changed:22 seeds/23 one-sign walks were superseded as completeness claims by larger sign/operator and recursively conserved event populations. Keep their actual scope; do not present23 as the final size of the puzzle. The graph registry's8,120 definitions and 113,990 anchors describe an installed model, not that many independently authenticated instructions.

K11 Artwork reconstruction and renderer boundaries

All722 physical cells have a primary accounting:228 margin cells,80 Base64 characters,58 punctuation marks,40 q/1 marks,192 z cells and 124 interior blanks. These totals are over the complete source; later380-cell and 120-cell regions have different counts and must not be mixed.

A retained candidate renderer reconstructs the five dynamic canopy rows from selected B/C bits and, together with normal Base64 and framing, all eleven upper-Z rows:418/418 characters. Perturbation of 184 B/C bits shows24 affect that candidate renderer. The remaining160 are still authenticated ciphertext, not an automatically available residual message.

The lower eight N rows have not been generated by an accepted A/E schedule. Repeated candidate phases do not predict repeated transition signatures; the bounded reused-transition family produces0/7 complete held-out predictions. A lookup table keyed by unique row values would memorize the answer rather than derive it.

The supported architectural description is therefore multiple parallel channels with an exact candidate upper renderer and separately understood lower morphology—not a fully reconstructed single application display.

K12 Four-punctuation custom scale

Remove outer punctuation at zero-based columns6 and 31. The remaining thirty marks form two physical fifteen-mark fields. Their forward source strings are:

```
upper .:..:,,;,,;,,;,
lower ;.,::;.,,.;:,.;
```

The punctuation morphology has two features: an added upper dot and a lower tail. In that order, period00, comma01, colon10, semicolon11. Four direction pairs, with an axis-swap control, make eight candidates. Preserve field-local order U0,U1,L0,L1; do not permit post-decode permutations.

The retained fixed-prefix BIP39 check selects upper forward/lower reverse in that bounded population and yields EMPTY/2, AMONG/15, OBSERVE/9, PRIZE/5. The compiler's independent public check reproduces the records; it does not rerun the private-prefix checksum without the protected input.

Each suffix selects one position in the same fifteen-mark traversal that generated it. Zero-based indexing fails at 15; one-based selection addresses all four. The selected alphabet is colon, comma, period, semicolon. Rank0–3 in that order gives colon00, comma01, period10, semicolon11. The numerical reproduction is exact; adoption as the intended CUSTOM SCALE remains a source-local interpretation, not another GCM authentication event.

Using this scale on the original field traversals produces:

```
sequential upper: MEAT / ROAD / residual01011101
sequential lower: TOP / COLLECT / residual00011011
parallel: RAISE/2 AMONG/15 SMOKE/3 PRIZE/5
```

The scale alphabet encoded by itself is00011011, exactly the lower residual. Reapply the same own-field suffix consumer to the parallel records: colon, comma, comma, semicolon ->00010111. These are U8 values93,27,23. ARM/ADDRESS/ACTUAL are correct dictionary-control labels, but no universal U8-to-state cast was recovered. The supposed ordered instruction ACTUAL ADDRESS is therefore not earned.

MEAT/ROAD and TOP/COLLECT remain their native ordered U11 outputs. Do not reverse the latter to “collect top” or import an external story without a local grammar. A later OSC52 interpretation of byte93 also needs its own complete protocol and argument framing; an escape-related byte is not permission to execute terminal controls.

K13 Outer frame, phase and PRIMARY

The complete28-mark frame is six semicolons, one comma, one period, then twenty commas. The post-gap22-mark suffix starts after the six semicolons. Freeze these boundaries before labels. Under the custom scale, the all-comma carrier is repeating01.

The full authored frame decodes ZOO/RAZOR/FETCH/PRIMARY/FETCH with one residual bit. The upper8 and lower20 are separate physical views; the lower residual is seven bits1010101=85, not a complete U11 field.

XOR against the frozen all-comma carrier gives full fields1365,192,0,0,0 plus residual0; the suffix gives384,0,0,0. The period's two-bit defect lands at full aligned contribution192 and suffix384. Removing twelve preceding bits shifts phase by 12 mod11=1; the English labels BLOSSOM/COPY are consequences, not the explanation. Six semicolon substitutions contribute alternate bit positions0,2,4,6,8,10, yielding1365.

No source/destination contract turns COPY into mutation. Whole-field replacement, slice repair, detached-fragment retention and explanatory no-op are competing readings, not selected consumers. The earlier bit-identical PRIMARY field and exact missing01 make repair mechanically possible, but availability of matching bits does not address a copy operation.

The phase audit favors construction/renderer metadata over executable tape. A checksum role remains a hypothesis about what is verified, not a generic integrity certificate. Random defect and minimal-description controls include tautological invariants. Many other cluster arrangements share the same short description. Exact restoration after deleting and reinserting known bits is not independent authorship evidence.

Frame85 does not restore the old APPLE checkpoint: the selected punctuation checkpoint has endpoint213, and 213 &127=85. The proposed85 div19=(4,9) can conditionally select Q.y via ABOVE4, but no inherited residual-to-height division and quotient-as-coordinate-selector contract was located. This remains open, not a new bridge.

K14 q/1 carrier and Q-witness controls

Map physical q→0 and 1→1 in source row-major order, excluding Base64 text. The result is:

```
upper18 100000100010000100
lower22 0101110101001100010000
full40  1000001000100001000101110101001100010000
```

There are27 q marks and 13 ones; upper18 has 4 ones, lower22 has 9. The lower length/weight22/9 matches a retained candidate canopy byte-pointer loop. It is a cross-section check, not two independently generated secrets.

The forty bits were tested as selectors for the two public recovery candidates at each of forty construction signatures. All signatures have two candidates and all 780 pair intersections isolate Q. The frozen mask family found no exact selector. Native q/1 chose Q only 22/40; its complement18/40. The best native distance was13, while controls reached8. Of10,000 same-weight random masks,7,960 beat the native score. Low-s versus n−s preserves recovery sets and thus cannot validate an arbitrary bitmask.

This negative does not undo the geometry. It rejects that bounded Q-candidate selector hypothesis. The forty signs are not proved to carry a hidden historical witness.

K15 Chamber22 and writable-memory hypothesis

The lower carrier has an exact22-cell provenance and 9-bit weight. Earlier work builds a chamber and reflected-mask interpretation, including a writable-memory story. Treat the physical masks and their Boolean results separately from permission to write. A proposed write must specify the selected coordinate population, bit representation, operation, destination and continued consumer.

The candidate183-bit lane later pairs61 triples with 61 occupied canopy cells, but equal population size alone does not bridge those two objects. Likewise a matrix that numerically encodes addresses does not select this chamber until the adapter is supplied.

The retained geometric construction finds blank runs of lengths2,4,7,9 on four consecutive rows, with a common left boundary. Enumerating this fixed family finds exactly one chamber,22 cells in the 26-column interior. This compilation independently reproduced that calculation, its exact coordinate set, and the following Boolean readouts. The source rule selecting that length vector and installing a payload is a separate conditional claim.

The concatenated field-local residuals have22 bits. Installed top-to-bottom and left-to-right into the chamber, they produce rows10,1100,1010110,010011011. Concatenate the outer rows and inner rows top-first to obtain NATURE/SKULL, or bottom-first to obtain EVIDENCE/PULSE. These are specified readouts; neither direction becomes selected merely because its words are attractive.

Reflect the 22 coordinates horizontally about the 26-column interior. Intersection size is4 and union40. Vertical or double reflection gives intersection0/union44. Row-major installation of the full40-bit q/1 carrier into the horizontal union then supplies a paired Boolean test at every original chamber coordinate.

Pairwise operator	First U11	Second U11
XOR	ONE	EYE
XNOR	GRAB	PUT
OR	ONLINE	FARM
AND	ABLE	ACOUSTIC

All four outputs are retained. A source-bound POLAR/XOR interpretation is needed to select the first; readability alone does not do so. The exact chamber coordinates and all eight corner scans are embedded with the public constants.

K16 Reflected eye and contextual CHECK

An older source row ALERT/AHEAD/CHECK/FAME yields a4×4 first-four-letter grid. Its twenty directed straight lines contain two exact BIP39 words: FAME in the source row and RACE in the fourth column. The strict audit found no pre-output selector for that column. AHEAD41 div38=(1,3) does not become a grid coordinate without an adapter. The result survives as a mechanical observation, not an earned route to RACE/CHECK/REFLECT.

Other retained eye audits calculate a reflected canopy and EXPECT/ALIEN output under specified transformations. Their exact reproduction must retain any upstream selector conditions. In the contextual checksum family, FINAL692, FILE689 and FINISH696 share high entropy bits; their low checksum changes need not supply independent routes.

The old candidate chain takes ABSTRACT/ZOO/CANYON/EVIDENCE minus WIDTH to ALERT/AHEAD/CHECK/FANTASY; its private-prefix CHECK changes FANTASY to FAME. After the unearned RACE-column selection, CHECK gives RACCOON; minus WIDTH gives REMAIN; CHECK gives REFLECT. REFLECT's high bits give AREA, subtraction of ART gives POLAR, and its modulo40 projection reaches EXCESS. This records the exact model without granting the missing grid selector.

Within that model, contextual CHECK changes ONE to OMIT while leaving EYE unchanged. The original chamber's top-right scan gives EXPECT/ALIEN; EXPECT642+ALIEN50=FINAL692. The eight corner scans are all retained rather than treating the desired pair as the only result. A sixteen-case installation/read orientation control gives ONE/EYE in two horizontally equivalent arrangements. The corresponding sixteen eye scans give EXPECT/ALIEN twice, again related by global horizontal reflection. These are symmetry-equivalent confirmations, not two independent proofs.

The width-state CHECK context maps FINAL to FILE. A different context, predecessor-prefix plus EXPECT/ALIEN/FINAL, maps FILE to FINISH. FINISH div22=(31,14), div26=(26,20), div38=(18,12). All divisors and residuals remain explicitly typed; a convenient quotient cannot choose which geometry is active.

With divisor38 they select terminal18 positions8,5,12. Those are BROOM, QUANTUM, INSTALL in the raw-s representation. The same terminal is structurally forced by their nearby values; it is not three independent votes for an authorship record. CHECK input contexts must be reproduced privately rather than invented from the desired word.

K17 EXIT ZONE and ZENON NETWORK

The signed-message branch contains NETWORK/NEED/WORK as retained dictionary tokens. Under the conditionally carried OR, their combination yields ZONE. This is an internal value, not automatically an instruction to explore any repository named Zenon.

CALL/ZONE with a locally typed zero and generic CALL/index(word) are different contracts. BOOK/ABANDON/call-zero recurrences are not universal termination or restart instructions. An audit scheduler can finish or pop a frame without asserting an authored RETURN.

The frame UNLOCK experiment removes01 from a final PRIMARY field; the earlier corresponding field is bit-identical. All resulting widths and residuals must survive. No COPY opcode or destination was licensed merely by the defect mask. A passing phase check has not yet supplied a new executable next stage.

K18 Geometry-to-VM boundary

The central unresolved interface is not an arithmetic error. It is a type conversion: a physical coordinate, bitmask, residual or matrix return becoming an address in an installed word machine. Standard integer arithmetic does not provide that conversion by itself.

For every proposed edge ask: which physical object was selected before output inspection? What ordering and origin does it have? Is the integer a row number, flat offset, word index, endpoint byte or checksum nibble? Which operation transports the register? Which exact consumer takes this result? If these are missing, preserve the calculated data and stop that edge.

This discipline keeps the newer geometry work valuable without using an attractive word to repair a historical compiler gap.

K19 Eight-glyph scale hypothesis

The candidate380-cell analysis region is upper rows3–7 columns7–30 plus lower rows9–18 columns6–31, zero-based. It excludes other source bytes. Its counts are space114, period7, colon5, comma31, semicolon7, z176,1×13,q×27.

The proposed ordinal alphabet is space, period, colon, comma, semicolon, z,1,q. Its punctuation relative order differs from the four-symbol custom scale. It therefore requires a separate source binding; it is not simply an authenticated extension.

Within this hypothesis ART=102 is exactly ordinal bit1 XOR bit0. Adjacent glyph pairs partition the 380 cells into 121 space/period,36 colon/comma,183 semicolon/z,40 one/q. Those lengths are mechanical. Calling them representation bank, operation bank, addressing tape and control register remains conditional.

K20 Eleven- and six-dimensional matrices

The candidate binary lanes reshape row-major into 11×11 and 6×6 matrices. Their exact rows and columns are in the constants section. Low rows contain seven set bits overall; high has five. The checksum domain0–15 can be covered by low0–10/high10–15 or by high0–5/low5–15. Geometry has not selected one arrangement. At the shared value both candidates remain; overlap does not itself request a fork.

Under placementA, combining the selected row and column with the coordinate reducer's operation yields the complete nine-row return population: ABOUT3, GAS768, AMOUNT64 twice, ADVICE32, ABANDON0 twice, ACROSS18, LENGTH1024. These are calculated candidate returns, not new instructions.

The supplied equation suite survives only identity in7,920 low relabelings and 360 high relabelings. Fixed-weight layouts have10,206 survivors out of 63,140,314,380 low layouts and 4 out of 376,992 high layouts. These quantify specificity **conditional on the already selected equations**. They are not discovery p-values for the architecture or evidence for an absent address adapter.

Low orientation controls preserve authored and transposed forms; high preserves authored alone for the supplied suite. This cannot become pre-decode orientation selection retroactively.

K21 Checksum microcode and STEP typing

The older fixed endpoints BELOW166 and BONUS203 give a37-step difference. At width38 their divmod coordinates are(4,14) and(5,13), hence a down-left diagonal. ANCHOR67 and ANGLE70 instead give a same-row three-column difference. These are different local pairs.

The complete31-dimension census distinguishes physically meaningful widths from arbitrary lane lengths. Width36 also makes a diagonal for 166/203, but36 is a lane length, not a demonstrated row width. Among685 valid flat offsets separated by 37 in a722-cell width38 array,666 have the diagonal delta and 19 are boundary-wrap cases. The numerical observation is largely explained by width−1.

Neither STEP at global position10 nor ANOTHER/STEP at terminal17 supplies a checksum-bank switch. The completed audit did not read glyphs at 166/203, because endpoint-to-ASCII-position admission had not been earned. It did not execute ACROSS/ADD merely because an old definition exists.

Two explicit gaps persist: checksum nibble→matrix address, and matrix returned scalar→CALL occurrence with inherited operation. The single narrow next contract target is an actual prior endpoint-to-physical-coordinate instruction—not more endpoint arithmetic.

Later-result recovery notice: the supplied manual review reports a subsequent complete precedent search with zero qualifying endpoint→physical-coordinate receipts, and recommends typing the artwork/table object first. The dedicated search receipt has not yet been located in this compilation. Preserve that later report as an unresolved provenance item; do not silently claim its

population was independently verified. The practical editorial task is to recover that receipt, not rerun downstream arithmetic or promote a coordinate cast.

K22 The183-bit lane

The frozen lane gives61 triples:54 of 111,5 of 101,2 of 011. The candidate consumer pairs them with the 61 row-major occupied canopy cells using three-bit glyph ordinals and modulo8 arithmetic. This binding is a hypothesis, not a native consequence of equal counts.

Across the six operations, XOR/ADD/both subtractions change61 cells, AND changes2, OR42. AND changes occupied rank28 z→period and rank58 q→z, raising the ART-projected weight from 37 to38. Yet54 all-ones masks make AND an identity on most cells by construction. Of10,000 same-multiset random arrangements,1,403 also change at most2 cells and 614 also reach weight38. Sparse change and a familiar count are not sufficient selection rules.

The lane has no installed destination or executable output consumer. Preserve every triple and source coordinate before trying another interpretation.

K23 Upper18: slot placement versus bit values

The procedural allocator selects five first-inward pivots, twelve right transition cells from four rows at stride3, and one phase compensation cell. Occupied starts move2,3,3,3 against structural stride3, creating a unique deficit1. This yields18/18 exact slots, no false positives or negatives, without reading q/1 identities. Removing phase compensation gives17 correct and one missing.

This is the sole exact result in the 35-candidate bounded allocator family. The shared lower/upper boundary grammar is parameterized, not literally one identical algorithm. The union with fifteen punctuation marks gives33 information cells;28 z cells make61 occupied;59 blanks make120 total.

The allocator does not predict the payload bits100000100010000100. Those have four ones and are not an integral U11 stream. No independently marked9+9 split exists. The simple bit-feature generator census found no exact native match;1,872 of 3,060 four-one masks did at least as well under the selected score. The position problem is solved within its bounded family; the value-generator problem remains OPEN.

K24 Message and authorship circuits

The retained conditional story includes SOURCE/VERIFY, ACTUAL/WITNESS, INSTALL/WIDTH/INPUT, RECORD/RESULT/ANSWER, SENTENCE/ALL, FINAL/QUESTION/TRUE, visible-proof and OUTPUT/SATOSHI circuits. Each phrase must remain attached to its source occurrence. Their combined narrative is evidence of an interpretation, not a cryptographic assertion signed by historical Satoshi.

The terminal18 raw-s object is specifically a32-byte unnormalized DER s scalar, projected into 23 U11 cells plus3 residual bits. Its P08 BROOM lacks a carried operation or member selector. P05 QUANTUM and P12 INSTALL coincide with already named source-local phrase positions; they may acknowledge those occurrences under the relevant conditional grammar. BROOM does not mean sweep an arbitrary database.

The December2015 “Not this again” email is a particularly useful conditional external predecessor. Its two sentences include a denial and “We are all Satoshi”; its preserved transport headers conflict with the displayed sending host, and contemporary responses ask for a signature. This illustrates why sender labels are insufficient. It does not prove a causal2015→2020→2021 protocol or identify who sent the message. The internal RECORD+ADVICE arithmetic does not type that historical transition.

Sources: mail record, raw preserved message, contemporary discussion. The trusted receiving server's headers are more probative than sender-controlled fields; they still do not by themselves prove a person's identity.

The complete212-event native-field populations

These were preselected CALL **occurrences**, not212 independent records. The operation carried by each occurrence was retained while a complete native-field population was executed before inspecting priority labels. Their exact mechanics belong to the conditional CALL model; field availability does not independently prove the model's root admission.

Field	Present	Distinct raw/residue values	Canonical/cycle	Direct predeclared priority output
receive.height	212	137/137	191/21	one HISTORY
confirmationDetail.momentumHeight	212	130/129	205/7	four ACQUIRE occurrences, one INDEX
pairedAccountBlock.height	212	29/29	164/48	none

The receive-height terminal distribution is ACQUIRE1, ACTUAL10, AREA40, ARTIST2, ARTWORK5, BETRAY2, BLIND1, BOOK6, BUNKER5, CARRY41, DINNER23, DONKEY1, ENVELOPE51, ERASE1, FORTUNE5, FORUM5, INTO2, JUMP7, LATER4. These sum to212, not a hand-picked subset.

The sole direct priority height occurrence is PYRAMID, position0, parent PYRAMID, inherited ADD. HISTORY follows ICON→AGAIN→EDGE→ANGER→BLOOD→LITTLE→CUSHION→DIARY→FADE→BODY→AUTHOR→DEBRIS→AREA. Its raw height and residue are unique in the frozen population. The neighboring values HIRE and HOBBY still walk readably to CARRY and ARTWORK; English output is not exceptional.

The momentum-height distribution is ABANDON1, ACTOR3, ACTUAL1, ADJUST2, ANOTHER2, AREA31, ARTWORK5, BETRAY1, BOOK2, BUNKER15, CARRY55, DINNER25, DISEASE4, ENVELOPE52, FORUM3, IMMENSE3, INTO1, LATER4, LENGTH2. Its four ACQUIRE occurrences share the same DOG record in four root histories. Their path is ACQUIRE→GUARD→JOKE→ENDLESS→CABIN→DAWN→BLADE→CARRY. The unique INDEX/SUB path is INDEX→DESK→JEWEL→ENOUGH→FIRST→WORLD→FILTER→FURY→BRING→FAT→HOLIDAY→CROUCH→INJECT→ALSO→GIRAFFE→BEGIN→WORD→AUGUST→FIBER→HURT→BLOUSE→CORRECT→CASINO→FOG→HOVER→DECADE→GARDEN→COOL→DENY→LEAVE→FORWARD→ENVELOPE.

At the HISTORY occurrence, timestamp/receive-height/momentum-height/paired-height produce BEHIND/HISTORY/HURRY/ABSENT, with independent terminal results ENVELOPE/AREA/CARRY/JUMP-cycle. At the VERSION occurrence they produce

VERSION/DAD/ROAST/AGE, with CARRY/CARRY/CARRY/JUMP-cycle. Those labels do not supply an operation combining the fields.

Timestamp-versus-receive height has 42/212 same terminals,70 ordered terminal pairs, and no identical direct words. Timestamp-versus-momentum has 31/212 same terminals and 61 terminal pairs; receive-versus-momentum has 36/212 and 67 pairs. All triples have three different direct words;9 finish together,82 use two terminals and 121 use three. Timestamp and momentum height are one-to-one across the 130 confirming momentums, so they are two coordinates of the same confirmation, not independent historical witnesses.

Paired-height results come from 137 underlying paired records, leaving75 repeated executor occurrences. Its distribution is ACTOR3, ACTUAL9, ANOTHER1, AREA9, BONUS9, BOOK15, BUNKER1, CARRY14, DINNER69, ENVELOPE57, FORUM1, INTO8, JUMP16. No priority or protected-root hit occurs directly.

Finally, both numConfirmations populations were executed as424 control walks. Every value obeys `snapshot_tip - confirming_momentum_height + 1`; just five cached tip snapshots explain them, and paired/receive tips agree212/212. Later pagination drift is not a2021 authored channel. Labels such as ADDRESS, COPY, VERIFY or VERSION arising from this mutable field remain controls, not continuations.

K25 OpenPGP roles and missing co-addressing

The retained historical ASCII armor is1,714 bytes, dearmoring to1,159 bytes. It contains a DSA primary, user ID, positive certification, ElGamal encryption subkey and binding signature. Both signatures authenticate the certificate's internal relationships. There are no secret-key packets.

The primary fingerprint is DE4EFCA3E1AB9E41CE96CECB18C09E865EC948A1. Its packet's asserted creation time is2008-10-30 18:19:19 UTC. The encryption subkey cannot substitute for a signing primary. ASCII armor is a transport encoding, not encryption; CRC24 detects accidental corruption, not authorship. A Version header lies outside the signed packet stream.

Copies: archived bitcoin.org, Nakamoto Institute, Bitcoin Forum copy, historical preserve-the-key post. Period format reference: RFC4880.

The old proposal requested a binary-document signature by that primary over the exact362-byte message. Such a signature would be meaningful evidence of control of that historical credential. But the later strict audit did not locate an occurrence co-addressing this key and this message as a new-signature demand. The existing Bitcoin signatures already satisfy a literal reading of signing the message. Therefore the PGP challenge is a **conditional proposed verification contract**, not the proved single missing ending.

Public-key byte projections, packet-role endpoints such as AUTHOR/BUNKER, self-declared timestamps, and standard-public-key hashes cannot prove private-key control. Negative public commitment and cross-key scans preserve their exact bounded scope. They do not disprove every unavailable secret-dependent object.

The PERSON_B December2010 encrypted administrator-secret attachment is not retained natively. Rendered correspondence/testimony can establish that an event was described; it cannot reconstruct ciphertext bytes. Any future lawful recovery needs custody, exact native bytes and a prospectively

specified construction commitment. Plaintext known to an administrator may also be known to other operators; neither decryption nor secret knowledge automatically proves unique Satoshi identity. No credential should be published or used to access a system.

K26 Exact362 bytes and 145 signers

The retained public message is ASCII/UTF-8,362 bytes, four paragraphs separated by LF LF, no quotation wrapper and no terminal newline. SHA256 is `bead321c006c7bfc62ab1cd9f585cb1138cbfe7cc78d419411ecc9949cfc033d`. Bitcoin Signed Message digest is `c4259881ceecf395c853020810dd6d5a6334ace42703f86464d3df58bac6c34e`.

All145 native compact signatures verify the exact serialization. Quoted/newline and other frozen variants do not. Native Bitcoin message hashing, compact-signature recovery, public-key serialization and address comparison must be used; raw SHA256 of the text is a different test.

The conditional historical selector audit found input145 alone satisfying its predeclared BONUS/MESSAGE conjunction across 205 inputs; four alternate digest populations had no such conjunction. That is a word-machine selection observation, separate from native identity verification.

The later direct cryptographic rejoin test is negative:205 distinct historical native authorities and 145 distinct message authorities have zero key intersection. Across five frozen digest variants,205×145 comparisons each, all 148,625 cells remain zero. A10,000-digest null family likewise has no joins. Empty recovery intersections are therefore not positive evidence. Algebraically optimized group-equation evaluation was checked against explicit recovery cases; the count is not a claim that every matrix cell was separately brute-forced.

The145 signers prove possession of their respective signing authority for these exact bytes. They do not prove that there were145 people, that one was Satoshi, or that Q controlled any of those keys.

K27 Database and storage-format negatives

An ASCII artwork is not wallet.dat merely because it contains bytes or the word DATABASE. A Berkeley DB identification requires format signatures, page/header structure and a coherent parser result. addr.dat, blkindex.dat and later LevelDB storage have different schemas and eras. A visual matrix or U11 word stream does not inherit a file type from an English label.

The current corpus search located legacy LevelDB availability limits and historical storage discussions, but this draft has **not yet located and reconciled every dedicated Berkeley DB/wallet.dat/addr.dat/blkindex.dat test requested by the brief**. Those tests must not be fabricated from the desired negative conclusion. This section remains a named coverage gap pending exact source/runner/receipt recovery.

The latest supplied review asserts that the specific storage tests and BDB-versus-LevelDB discriminator were already completed elsewhere. The review is recorded as a recovery lead, not as their numerical receipt. A bounded filename/content search of retained outputs and attachments did not recover the dedicated tests in this editorial pass. Their absence from that search is not a finding that they never ran, and no claimed negative population is invented here.

The legacy xStake/archive investigation is separately quarantined as a solve-path selector. A missing historical database clone limits reproduction; it does not license a new search route. Current genesis state is not a substitute for omitted legacy event history.

K28 External-reference directory and meaning

Newly reconciled literary/media receipts

The detailed source records in Appendix A retain The Endless, Yellow Woman, and the Advice Dog/Oracle proposal with exact URLs and scoped rulings. They are not silently merged:

Selected object	Selector and representation	Returned data	Current consumer/status
The Endless (2017;2018 theatrical release)	Historical MOVIE/BRIEF/ENDLESS; Tribeca brief, distributor synopsis,2018 BFI interview	Brothers,mail/message,family,camp,leader PERSON_D; creators also actors; named predecessor Resolution	Conditional16-word ADD experiment. BROTHER/VIDEO cycle; six AREA,six CARRY,two ENVELOPE endpoints. No character→historical PERSON_D substitution.
Yellow Woman (PERSON_N,1974;Storyteller,1981)	BOOK and sibling WOMAN/YELLOW; reversing into title order is semantic	Author's LESLIE and character SILVA; unnamed narrator's story-role	Conditional match to preselected21-page NoM whitepaper: LESLIE only reference7,SILVA only 56,YELLOW pages1/19. Name match is not an earned reference-number dispatcher.
The Byzantine Generals Problem / DNSSEC whitepaper	Proposed whitepaper references7/56 through prior row	Signed-message consensus and authenticated origin/denial themes	Conditional external evidence, not signed authorization of Q.
Advice Dog / Oracle of Bacon	Nehemiah-derived IMAGE/PUPPY/ADVICE/BACON/ACCESS interpreted as media selectors	Six-color wheel; Endless Games connection; two exhibited actor paths to Bacon	Grade C,not admitted; shortest historical Moorhead distance unproved. Existing label does not supply a typed parent return.

The exact sixteen-film-word population is BROTHER,VIDEO,MESSAGE,MAIL,FAMILY,HISTORY,MYSTERY,CAMP,LEADER,REPEAT,DIRECT,ACTOR,ENDLESS,CABIN,RETURN,LOOP. PERSON_D and RESOLUTION remain proper-name/title anchors outside BIP39, not dropped or coerced. The old report associates the cycle with DISORDER and ARCH/PAUSE/BOOK and BROTHER's FEE output; those are model-local boundary receipts, not a universal recursive return contract. Complete path recovery remains separate where the JSON gives endpoints only.

The external sources teach several recurring distinctions: original versus copy, creator versus character, visible evidence versus assertion, signature authority versus personal identity. These can illuminate the story without constituting an executable instruction.

Object family	Retained selection/content	Return and boundary
Hashcash first-cut source	SOURCE/COPY; original package, detached signature and author verifier	Authenticates that source's author, not Satoshi or Q.
HMWYDA	WEAR ADDRESS; demand/reward/visible-proof protocol	Conditional source-object fit; token and timestamp casts need their local rules.
PERSON_D startup/correspondence	VERSION/ACTUAL WITNESS/CLOCK context	Genuine chronology/log observations; headers, local address variables and peer attribution must be distinguished.
PERSON_M J1336/F1371	Retained710968−709632=1336=POEM	Numbered poem selection is strong conditionally; exact character counts depend on edition.
Sevenoaks	Letter439 editorial note's possible “little Book” identification	Bounded conjectural book link; courtroom originals/copies/signatures teach a proof procedure.
Nehemiah	WALL/REBUILD/BOOK/CENSUS	Missing genealogy blocks privileged access; numeric parallels do not prove identity.
Yellow Woman	Story-role interpretation	Narrator, character and author are not interchangeable identities.
The Endless/Advice Dog/Bacon	Retained actor/degree/source-copy branches	Conditional media graph; no new child from an unselected degree or anecdote.
Zenon genesis/repos	Native schemas, Bitcoin anchor, signing and settlement fields	Public implementation is evidence of protocol behavior, not Satoshi authorship.
2015mail and 2020message	Anti-attribution statements with different authentication status	Strong contextual ladder; causal linkage remains conditional.

PERSON_M source links: poem, letter439, editorial note, Harvard H35 record. The retained transcription lengths24/21/30/20 total 95, dictionary ARMOR. Adjacent poems also yield dense puzzle vocabulary; edition-sensitive length success is not exceptional enough to choose the representation.

For Sevenoaks, the selected plain-text copy has retained hash `94750ec4ab19b081c4eab43a286bd666fb3a05b5049bc34f79cf801a835baf5f`. Its thirty chapters culminate in identity recognition, original letters, expert signature comparison, visible demonstration, record and curtain. The source's10,000,000 and 500 yield SOURCE1664 and DIRECT500 under the proposed numeric register; their sum returns ATTEND116 conditionally. Neither the book's narrative nor those numbers identify a historical key controller.

The complete external directory—including all source hashes, editions, exact selectors and missing returns—is still being expanded from the registry. Current-source pages may postdate the puzzle while faithfully preserving earlier works; later historical revelations cannot automatically become prospective2021 selectors.

PERSON_D's VERSION record: exact reconstruction versus missing runtime state

The retained PERSON_D-log hash is `ab02bbfe5bffdd0eab9c58fd21af246d56dee5e511ca876790c3132f9c19bb3a`; the Bitcoin0.1.0 source archive hash is `ce9da46516e3042741224a7f9061e3181a5a4d17abba72b6e82922af3753d756`. Under stock serialization the selected received VERSION payload has 46 bytes, version100, services1, time1231614080 and an address describing192.168.0.1:1531.

The log prints the first 25 raw bytes and the decoded endpoint. Fifteen intervening bytes are not printed raw; stock source supplies their candidate values. This is a source-conformant reconstruction, not a recovered packet capture. The address object is initialized from the `sockaddr` returned by `accept()` on the remote peer's inbound path and then serialized. The receiver's local variable name `addrMe` does not mean the value was the remote server's own address.

Port1531 is consistent with a client ephemeral port in the stated historical platform model. A private IP reaching that accepted socket requires an unusual topology, intermediary or modified implementation rather than ordinary public routing. The surviving log cannot select among those possibilities. No private build, tunnel or historical operator identity has been proved.

The literal value was already public before the puzzle: the retained audit cites a February2021 excerpt and 2013 log redistribution. Thus a public researcher could use the number without knowing why it appeared. An insider upgrade would require the puzzle to reveal an independently authenticated fact absent from that public record—not merely repeat1531.

K29 Tests and hostile controls

Appendix B distinguishes retained test results from compilation-time reruns. Population sizes are essential: a unique result in eight fixed candidates is not uniqueness over arbitrary encodings; a success in a post-observation family is not a held-out prediction; training-data replay is not independent evidence.

The public compiler independently rechecked dimensions/hash, field lengths and global words, both punctuation record generations, own-field coordinates, sequential residuals, complete and suffix frame deltas, and q/1 bits. It performed no new external research, no wallet operation and no historical identity search.

The remaining historical/control results in the ledger are sourced receipts. They must not be described as newly rerun until executed under a reviewed, sanitized runner. A test that merely asserts a stored `licensed=False` ruling is a regression check, not an exhaustive source-consumer search.

K30 Superseded and rejected interpretations

Appendix D records the exact corrections instead of silently deleting old claims. The most consequential are old punctuation transposition/2905 packing; executable P00 promotion; generic CALL from a fingerprint; automatic ABANDON rejoin; a CF001 return barrier; universal residual-to-word casting; a frame COPY opcode; restored APPLE by a seven-bit match; and Taproot-specific hidden fields that do not exist.

Keep useful factual results from rejected routes. Patoshi attribution questions do not invalidate native Bitcoin signatures. A failed xStake selector does not erase genesis accounting. A poor media rejoin does not change a song's lyrics. A conditional renderer can reproduce accurately without proving the author's generating program.

The framework rejects unsupported **edges**, not inconvenient observations. Conversely, a coherent story does not excuse missing operands, domains or source bytes.

K34 Reproduction algorithms

Public artwork layer

1. Parse the terminal transaction and retrieve each exact row-spending transaction by native outpoint. Extract its38-byte OP_RETURN payload in terminal input order.
2. Require19 rows of 38 bytes. Serialize LF after every row and check the supplied SHA256. The resulting text has 741 bytes; the bare cells total 722.
3. Extract the four fixed Base64 substrings. Require decoded lengths16/18/5/16. Independently retain field-local bits and full A||B||C||E bits.
4. Decode all full U11 fields MSB-first and preserve each exact residual. Do not pad residuals silently.
5. Build two internal punctuation populations, excluding columns6/31. Apply the fixed physical feature mapping and field-local direction. Keep U11 and U4 attached.
6. Use the one-based suffix consumer only on its generating field. Build the ordered alphabet, encode ranks as two bits, and repeat the specified source encoding once.
7. Independently parse frame boundaries and q/1 coordinates. Reconstruct the fixed all-comma baseline before XOR, with both native alignments preserved.

Signature and coordinate layer

For native ECDSA use the appropriate protocol digest, not a convenient hash. Verify `s^-1 z G + s^-1 r P` has x coordinate congruent to r modulo the curve order. Recover candidates by enumerating valid nonce-point lifts with x=r+j·n and both y parities, then `P=r^-1(sR−zG)`. Validate every result and compare with the independently committed public key. Replacing s with n−s must preserve the complete recovery set.

For a coordinate census, serialize exactly 32 big-endian bytes, generate its canonical BIP39 entropy encoding, apply all eighteen fixed reducers, verify each candidate checksum, and retain endpoint plus checksum plus parent representation. Separate P2095 and P3928 throughout. Neither pairwise reduction nor CHECK automatically installs an execution register.

Compiling versus rerunning

The working compiler uses `work/canonical_manual/public_mechanics.py` for the public-only checks and `build_registries.py` for the editorial ledgers. These are private build aids, not substitute evidence for all historical audits. The final standalone algorithm appendix still needs the remaining exact graph and geometry routines reconciled before full reproducibility can be claimed.

K35 Constants and public objects

Exact artwork rows

The JSON string form preserves trailing spaces; decode each string without its quotes.

```
[
  "      ;BynQtpeUyWTXKGTrGhdV2Q==;      ",
  "      ;tVMd3L1CKM4wFmyxEEEUV2bY;      ",
  "      ;4Fdzw1k=zzzzzzzzzzzzzzz;      ",
  "                  .:1zzzzzzzz.       ",
  "               .:qqzzzzqqq,          ",
  "             ,;1zzzzzqqq,            ",
  "          ,;1zzzzzqqq,               ",
  "       ,;qzzzzz1qq,                  ",
  "      ,vtv3f5aKY0jGQglP9a1AGw==.      ",
  "      ,zzzzzzzzzzzzzzzzzzzzzz,      ",
  "      ,zzzzzzzzzzzzzzzzzzzzzz,      ",
  "      ,zzzzzq;.           1zzzz,      ",
  "      ,zzzzzzzzq,         1zzzz,      ",
  "      ,zzzzzzzzz1:        1zzzz,      ",
  "      ,zzzzq:1zzzzzq;.    1zzzz,      ",
  "      ,zzzzq  ,qzzzzzz1,  1zzzz,      ",
  "      ,zzzzq    .;qzzzzzq:1zzzz,      ",
  "      ,zzzzq       ,qzzzzzzzzz,      ",
  "      ,zzzzq         .;qzzzzzzz,      "
]
```

LF-ended SHA256: `96534d9faf3b6260987939fdb6994cd5b97eed18a175b0620c4ea5256c0c5607`.

Public cryptographic fields

Field	Bytes	Exact hexadecimal	SHA256
A nonce	16	`0729d0b69794c964d72864eb1a1755d9`	`13e9320d79af1f2f98448faff163e259915f4b370c84a6de091a88fb59d335a3`
B ciphertext1	18	`b5531ddcbd4228ce30166cb11041145766d8`	`52a18c5bc4109e2df981e8f69d8dd24c5a648122ac8f01a349f4c8fe3c916fa3`
C ciphertext2	5	`e05773c359`	`1628b406a78ee053ee5d2b7b6fcc7dbc021ce330784e0b7e1cee8c288b6f4d98`
E tag	16	`bedbf77f968a6348c642094ff5ad401b`	`08d74c696d71dabaedb57e37fe5f33de5e2872ba8e0803f2e011a895699e2c1a`

Field-local U11 values and residuals

Field	Words in order	Residual bits
A	ALREADY / EXCESS / COLOR / CONNECT / ERODE / RAVEN / FREQUENT / ARRIVE / TWELVE / SPAWN / STEP	1011001
B	RELIEF / OCEAN / JAGUAR / KIND / CARD / GROW / SCARE / SNOW / RAIN / LINK / EAGLE / PUT / HORN	0
C	THEORY / ROMANCE / VALID	1011001
E	SALUTE / TEXT / THAT / CODE / PLATE / PIECE / BOIL / DONKEY / EXIT / PULP / EXOTIC	0011011

GLOBAL40 in original order

Positions	0	1	2	3	4	5	6	7
0–7	ALREADY	EXCESS	COLOR	CONNECT	ERODE	RAVEN	FREQUENT	ARRIVE
8–15	TWELVE	SPAWN	STEP	RECALL	FESTIVAL	MOMENT	INDICATE	PORTION
16–23	CHURN	IMMENSE	AIRPORT	SUNNY	BABY	CALM	EYEBROW	UNDER
24–31	SUGGEST	ACTOR	TABLE	MAMMAL	ORBIT	HOSPITAL	TARGET	VERSION
32–39	MEDIA	BOTTOM	COVER	LOTTERY	NEUTRAL	WIDTH	RELAX	ADDRESS

Punctuation records and exact own-field coordinates

All coordinates below are zero-based.

Generation	Record	15 bits	Suffix	Source coordinate	Selected glyph
1	EMPTY	010010010010010	2	[3, 19]	:
1	AMONG	000001111111111	15	[7, 18]	,
1	OBSERVE	100110000111001	9	[14, 21]	.
1	PRIZE	101010110100101	5	[16, 17]	;
2	RAISE	101100010010010	2	[3, 19]	:
2	AMONG	000001111111111	15	[7, 18]	,
2	SMOKE	110011001100011	3	[17, 19]	,

Generation	Record	15 bits	Suffix	Source coordinate	Selected glyph
2	PRIZE	`101010110100101`	5	[16, 17]	;

Frame

```
;;;;;;.....................
```

View	U11 values	Labels (data only)	Residual
Authored full28	[2047, 1429, 682, 1365, 682]	ZOO / RAZOR / FETCH / PRIMARY / FETCH	`1`
Full delta	[1365, 192, 0, 0, 0]	PRIMARY / BLOSSOM / ABANDON / ABANDON / ABANDON	`0`
Post-gap delta	[384, 0, 0, 0]	COPY / ABANDON / ABANDON / ABANDON	``

Public points

These are public verification points, not private keys. R is always derived with the stated P identity.

Object	SEC encoding
Q compressed	`0398cf668fe678cc6d8e2b20b5f574c9f398c7f031a1e91e9b85029e4cde597ad7`
P2095 uncompressed	`044b61c965dd4d6f172cdcdcd091be27673c8e8efd73a6d99349b3beb24c22f7b58b781b2e7b5ecf4cc5e518adecab6c7d8e8b5cc45e1c772e82485fd8d4877eb9`
P3928 uncompressed	`045c24a542d18baed728c33462389a6cc2d88b4defb667c9b28f6b11d9e0c71667f73ec45c7140d63eaf1cd83d42350d5459b6ec231537d0d803c42241accd2bec`

Candidate matrices

```
low rows    0,1536,1024,0,0,8,0,0,256,64,1
low columns 768,512,4,0,2,0,0,32,0,0,1
high rows   48,0,0,18,1,0
high cols   32,36,0,0,4,2
```

Frozen chamber geometry

Absolute zero-based coordinates, in row-major order:

```
[[15, 12], [15, 13], [16, 12], [16, 13], [16, 14], [16, 15], [17, 12], [17, 13], [17, 14], [17,
15], [17, 16], [17, 17], [17, 18], [18, 12], [18, 13], [18, 14], [18, 15], [18, 16], [18, 17],
[18, 18], [18, 19], [18, 20]]
```

Chamber/corner (row direction,column direction)	Exact U11 readout
original:1,1	MARRIAGE / BASIC
original:1,-1	EXPECT / ALIEN
original:-1,1	EQUAL / DONOR
original:-1,-1	BOIL / DRAMA
reflected:1,1	APRIL / GINGER
reflected:1,-1	BASE / DRAMA
reflected:-1,1	MARRIAGE / TROUBLE
reflected:-1,-1	ANGLE / TUBE

Direction+1 means increasing coordinate; −1 decreasing. Horizontal reflection is `(r,c)->(r,25-c)` in interior coordinates, before adding the absolute column offset6.

Other immutable object constants

Object	Value
Historical consolidation	29591d211054478ed22c93571955cb92ef4ec2b40d914a43ba3a91d52a171c61
Block2905 coinbase	47e7d826a6d50399eccf2b43013699e5a29f4ce06e955539a49650ad54998c2f
Block3928 coinbase	d557a1a844aa16751c8ea774b4cadf030d03cfd4c6d1f463737a3cc787aacd3b
PGP armor SHA256	f918c6368fe1ba8ceaa9d15c73fb601c68eb3fc542979d04eb34fd5f3b0502e3
PGP binary SHA256	4dc1d9f55a9be2b279d725bc83cd4fd7049fbfb32513872be6446dea8419cff4
2015 preserved RFC-message SHA256	f9147c9d1848f56af719dbb88a1e312968fd3f5876721a798010751f72534f53

Reconciled later findings

These chapters add distinct tests and corrections to the foundational modules. Repeated calculations appear only where their source, scope, controls or evidentiary status changes. Precise source formulations and additional machine receipts remain in the indexed annex.

Earlier hypotheses and October 12 tests

The superseded Keccak fit

The backward Keccak-f[200] proposal reproduces the known entropy suffix `f6b2690d66` by selecting theta after zero-based round 14 and state indexes 16,22,5,14,18. Its walk starts at 16, uses low C nibbles modulo 5 plus 1, inverted canopy polarity plus 1, emits the start and first four destinations, and treats the fifth as terminal. It also chooses a custom 25-byte state layout, a reversed header, and C[0]=e0 as a round/stage selector.

The declared backward search tested 69,120 state/stage candidates and 51,200 walk formulas and reported one match inside those families. That conditional uniqueness does not derive the families from the original clues. The target was known; the field split, state layout, stage, walk, start, polarity, and stopping convention were not independently issued. The route leaves much of A/E, the trunk and base, punctuation, and the transaction construction unused. It is rejected as the final forward solve while its exact computation remains preserved as a historical fitted branch. The five-canopy/eight-trunk dimensional correspondence can still be investigated separately as visual structure.

ART, APES, and the controlled cultural interpretation

The field-local observations PREP, ARTS, SHVE, RAZOR/NATURE, plaintext residues XISA, address-head ABSTRACT, literal By, and checksum COIN END support a candidate rebus ART IS ABSTRACT BY NATURE. The values are exact; the shaving and linguistic assembly are unauthenticated. The stronger uniform arithmetic is `WALL NATION DRIVE SMOKE − ART102 = TRY MAKE CUSTOM SCALE`. The opposite adjacent row is ADDRESS PARADE EXOTIC SUNSET, whose initials spell APES.

Two exact checks converge on MAMMAL: GLOBAL40[ADDRESS27] and NATURE1179−ART102 both give MAMMAL1077. Reusing end shaving for ARTS→ART and APES→APE, then using XISA as X IS A, yields the structured candidate APE IS A MAMMAL. It is a cryptic checkpoint, not additional GCM plaintext.

APES TOGETHER STRONG is a recognizable cultural completion. The exact relation TOGETHER1821−ARRIVE100=STRONG1721 uses GLOBAL40[ABSTRACT7]=ARRIVE, but it still needs cultural recognition to supply TOGETHER. TOGETHER−ART=STREET and the plaintext WALL suggest WALL STREET only after that same guess. The broader FAMILY/ECOLOGY/VERIFY and community-slogan algebra is exploratory and cannot be counted as independent corroboration.

The matched promotional-text control explains why. Five posts contain 31 distinct BIP39 word types, allowing 1,395 addition and ordered-subtraction equations. A frozen 20,000-trial test using 2021 Bitcoin tweets finds both predeclared TOGETHER/STRONG targets 25.10% of the time; a second corpus gives 24.22%, and uniform 31-word sampling gives 25.11%. Twelve incidental equations hit nineteen fixed artifact words, compared with 12.94 expected. Smaller percentages obtained by noticing particular within-post/adjacent-post patterns afterward do not supply confirmatory p-values. The arithmetic is exact but null-consistent.

The full ART cycle contains 1,024 rows because gcd(102,2048)=2. Eleven rows have APES initials, so the immediate row is distinguished by adjacency and ADDRESS, not by unique initials. Holding the inherited eight words fixed produces 72 checksum-valid tails; the retained activity query found only

the published row 0 address active. MAMMAL CASH SPY FRESH at row 192 and the MAMMAL re-keyed LATER BORING SIZE FAT are valid but inactive in that snapshot. MAMMAL/ANIMAL/HAIR re-keyings and shifts of the other retained streams did not produce a supported continuation within their stated bounds.

The literal carrier and its null tests

After masking the Base64 spans, the artwork contains forty q/1 marks, thirteen literal 1 and twenty-seven q. In row order, q→0 and 1→1 produce `0x8221175310`, whose direct mnemonic suffix is wrong. Tests of fixed reversals and polarities, direct ASCII/Bech32, one-to-one masks against GLOBAL40, word heads/tails, index end bits, repeated clue-word XORs, eight-symbol 3-bit alphabets, and paired-column occupancy found no further supported instruction in their declared families.

The MAMMAL→HAIR→shave hypothesis was tested through 64 fixed carrier orders, both polarities, and five clue-linked XOR keys. None of the 128 direct candidates was the authenticated suffix or five printable bytes. Six of 640 XOR candidates were printable, close to the simple null expectation of about 4.5, and none was a readable instruction.

The block-height/calendar interpretation is exact but nonunique. Treating hexadecimal glyph string 8221175310 as decimal heights 8221/175310 gives a March 21, 2009 first date and April 12, 2012 second timestamp 02:21:20 UTC, or 03:21:20 UTC+1. A coarse-geometry null preserves occupied positions, per-row counts, and the two long vertical runs. Seven of its 256 alternatives reproduce all three reported calendar properties; first heights are 8222,8221,8212,8211,8209,8206,8205. Thus the observed pair is one of seven calendar hits. The 2.734% rate is conditional on that model, not a universal probability of artistic intent. Upper 18 value `0x20884` is simply `0x8221` shifted left two bits and is not an independent selector.

The old forty-signature DER-length tests also failed in their stated group/order/polarity families. Canonical DER length aliases a signature component's high bit; it is not a separate free channel. The sole printable chance output Rpv@= is retained as a failed control, not a message.

Feasible planted envelope anchors

GLOBAL40 contains ALREADY at 0, ARRIVE at 7, TWELVE at 8, MAMMAL at 27, and ADDRESS at 39. The first three occupy 33 freely chosen nonce bits. The latter two constrain 22 ciphertext/tag bits after the key and plaintext are fixed. A constructor can vary the remaining 95 nonce bits with expected cost 2^22 AES-GCM trials.

An independent forward construction reproduced all five anchor positions after 896,893 attempts while retaining the authenticated plaintext. This establishes a practical authoring mechanism and makes planted MAMMAL/ADDRESS feasible. It does not identify the original counter order, prove a post-MAMMAL operation, or justify a formal 2^-55 intent probability because some semantic targets were recognized after inspection.

The September coordinate layer

The 19×38 dimensions support a fixed coordinate encoding: one-based row as U5 followed by one-based column as U6. Applied in top-left row-major order to the forty q/1 cells, it gives a checksum-valid upper 18-word BIP39 record and a lower-q object GIANT followed by a checksum-valid 12-word body beginning HEAD and ending ONLY. The literal 1 population has thirteen coordinate states; the first twelve can be paired adjacently, leaving a final MERGE label.

Right-minus-left on those six pairs gives AUTHOR/ANCHOR/AMOUNT/ABUSE/ACCUSE/ABOVE. The labels are attractive but subtraction is not selected. Twelve of 480 coordinate/grouping/operator controls produce six A-initial labels because short forward distances naturally lie in the A section. Width 38 linear-coordinate differences also produce six A labels while failing the standard mnemonic checksums. A six-word 64-entropy-plus 2-check diagnostic is nonstandard BIP39 and cannot select an operator.

The initial 32 coordinate conventions yield two simultaneous upper/lower checksum hits, including a mirrored/swapped control. Source orientation selects the natural convention before English is read, but checksums alone do not prove it uniquely. Adding the width 38 linear-ordinal family strengthens the coordinate type because those candidates pass neither checksum. This improves object typing without supplying a mnemonic-as-code, first-word-opcode, final-word-reducer, or MERGE-as-subtraction contract. PROGRAM_TYPE_AND_HEAD_PROMOTION remains open.

The 40-coordinate 440-bit representation does not equal the 55-byte envelope; the closest tested candidate differs by 203 bits and none authenticates as the root GCM package. Direct Q-bit windows, standard empty-passphrase Bitcoin paths, the extended Q-lifecycle script comparison, and Zenon accounts 0–19 found no key match within their frozen bounds. The complete observed Q history contained 25 transactions and 67 native scripts; the 12,000 public-child comparison had zero matches. Arbitrary passphrases or paths are outside those negatives.

Changing printed q to uppercase Q deterministically flips whole-byte parity while preserving low-nibble parity. Global, upper-only, and lower-only capitalization all fail the retained CONTROL98 checksum. Two of 27 single-q mutations pass the four-bit check, an ordinary rate with no independent coordinate selector. Literal capitalization does not bind the carrier to construction key Q.

Historical certificate and held-out credential limits

The historical certificate contains a DSA PRIMARY, UID, positive UID self-certification, ElGamal subkey, and binding signature. Its positive certification has raw length 98 and hashed-prefix length 38; the sibling binding prefix is 15. The PRIMARY fingerprint is `de4efca3e1ab9e41ce96cecb18c09e865ec948a1`. The positive-cert digest `5c63034984f841c1caee5926438f426d7f172d66` agrees with the stored `5c63` prefix and verifies under DSA. Mutating the UID fails; applying the sibling binding to the UID object fails. Its last byte 123 is the reporting label AUTHOR.

These native semantics authenticate the PRIMARY's certification of its historical Satoshi UID. They do not authenticate the separate 2021 Q controller. Interpreting terminal 17's(38,5)/(17,0) receipt as certificate prefix/count/algorithm/ordinal is a route claim that remains subject to source-local selection and type audits. Native verification of the certificate cannot repair a failed upstream selector.

The selected 32.51 BTC 2009 credential is a stronger historically attributed public transcript, but it was chosen as a held-out analyst test rather than selected by a puzzle instruction. Its native legacy transaction and ECDSA verify. Its denied-branch gap has exactly one valid frozen reduction, HALVES OR, paralleling P3928. Its signing digest also has HALVES OR→BUYER, paralleling Q.x. Complete 205-input controls find the first predicate 6 times, the second once, and their conjunction zero times. Canonical coordinate/orientation controls discriminate the encodings.

The shared BUYER endpoint does not merge the full states: their checksums and other words differ. No generic STEP[10], foot, checksum-to-GLOBAL dispatcher, or DENY SOURCE→VERIFY SOURCE adapter was established for the new credential. The appellate ruling rejects those rejoin sockets while retaining the structural parallel. The same public computations can be made without either historical private key; no historical-secret-to Q continuity follows.

October 12: RAZOR/NATURE candidates A, B, and E

Candidate A applies the same ART subtraction to both residual words: RAZOR1429−102=PLANET1327 at CONTROL98[4], and NATURE1179−102=MAMMAL1077 at GLOBAL40[27]. Across all 2,048 constants,102 is the unique subtraction constant satisfying the ordered whole-array membership test. It is not unique across all operations: ADD1946 is the identical modular translation, while sixteen OR constants give the different RAZOR/RECALL tuple. The full 12,288-case census therefore has eighteen passing parameterizations and two distinct output tuples. A reversed-source-order control adds a different subtraction solution. The correct uniqueness claim is narrow and deterministic, not a discovery probability.

q count 27 independently selects MAMMAL, but no recovered pre-inspection selector chooses PLANET position 4. Three source bits are shared between the residual input and the MAMMAL cell, so these are not independent witnesses. The result is EXACT BUT CONDITIONAL: the arithmetic does not specify whether the words are selectors, data, or an executable handoff.

Candidate B gives ART102−NATURE1179 mod 2048=JUST971, with divmod 40=(24,11). Those positions are genuine cross-field GLOBAL40 cells SUGGEST and RECALL; the third boundary is ORBIT at 28. Only this operation among the six fixed ART/NATURE operations selects two boundaries. However, nine U11 values encode two boundary coordinates, and unrestricted constant searches find other solutions. No source-local rule yet casts JUST as a base 40 pointer pair or says to read, edit, remove, or merge its cells. Preserve SUGGEST then RECALL as an ordered conditional lookup.

Candidate E supplies the direct source relationship. The 38 complete field-local cells and four attached tail lengths are lossless only while each tail remains attached to its own field. Restoring tails before joining fields exactly recovers the original 440-bit stream and GLOBAL40. Appending all tails after the 38 cells changes 146 of 440 bits and every GLOBAL40 cell from 11 through 39. The residual bits are not all-and-only boundary-cell data; some contribute to MAMMAL and ADDRESS as well. This explains reassembly without a new transform, but neither RAZOR nor NATURE has thereby become an opcode.

The residual 22 and lower carrier 22 differ at 17 positions. Partitioning them by the physical 22-space chamber's row lengths 2/4/7/9 makes all eight row endpoints complementary. That comparison does not install binary values into literal spaces. The chamber remains a source object distinct from any working layer.

The selection genealogy is limiting: later notes inspect CONTROL98 and then notice the PLANET/RAZOR ART distance. No retained pre-inspection preregistration was recovered for the paired membership or JUST-boundary instruction. Freezing the current replication does not turn a historically observed result into an independent holdout.

October 12: focused Burn-signature ruling

The focused test's conclusion is INTERESTING BUT UNPROVEN. Two public nodes agree on thirteen prize-account blocks: unsigned genesis and twelve signed blocks. All twelve signed hashes, sender bindings, canonical scalar checks, and Ed25519 signatures verify against independently reconstructed 306-byte native hash preimages. The previous-hash chain is continuous. This authenticates the account-block transcripts, not a full consensus replay; confirmation timestamps and the paired contract reply remain node-reported metadata in this focused test.

The Burn is account height 2, signing hash `ac3fb28796c0aec1c512396e54f23fe5882246c44372cc23897d455cfe88f7eb`, amount 474513400000, with authenticated token-contract recipient and Burn() selector `3395ab94`. The fixed A/M/R/S published-byte projection, read MSB-first as U11, gives position 18 tuple SUPER1741/GAP762/DENY469/SOURCE1664. Position 18 consumes raw bits 198–208 and does not depend on an appended mnemonic checksum. The fixed equations are `(A+R) mod2048=BEGIN162` and `(R−S) mod2048=HEIGHT853`; M=GAP is not part of that conjunction.

The Burn is the only signed treasure-account block satisfying either equation at the fixed position. The eleven sibling blocks are confirmed August 28, 2026. They are same-key controls, not eleven independent actions attributable to the 2021 constructor. Height 10 yields 852 on the HEIGHT side and remains a failure.

The preregistered empirical window comprises the 1,024 momentums immediately after Burn confirmation, heights 237106–238129. It contains 226 authentic signed user transcripts across 90 accounts, all independently hash/address/signature-verified, with native R/S pairing preserved. None passes the native conjunction. A separate fixed-treasure-A calibration finds one BEGIN-only R slice, whose paired S fails HEIGHT; that calibration is not a valid signature under the treasure key.

For fixed A1741, the equations reduce to R469 and S1664. One of $2{,}048^2$ ideal independent uniform-U11 pairs satisfies them, giving 2^-22. This is an exact finite-model count, not a theorem that valid Ed25519 signature slices are unrestricted independent random strings. Zero hits among 226 has about 99.9946% probability under that reference model, so the empirical sample has little power to validate a one-in-millions rate. Even zero hits in 10,000 ideal independent draws would be unsurprising. The old four-word-tuple null tests the wrong event.

Separate controls retain the targets and operand pairs while widening to 23 raw positions and 36 operation pairs. The Burn's only hit remains the original position 18 event; no sibling or public-control conjunction appears. The operation union at position 18 has sixteen distinct R/S solutions. Across the actual public-key windows the all-position/all-operation allowed-pair sum is 566, giving a stated union bound 566/2^22; simply multiplying 23×16 is incorrect because other A windows admit different AND/OR solution counts. The 187,128 public comparisons are correlated comparisons, not independent trials. None of these finite families corrects for the unknown total historical search over fields, targets, routes, and adapters.

Selection history prevents a stronger interpretation. The position 18 carry from sweep total 319560 remains conditional. HEIGHT was recorded with the already observed tuple in Master 12 Revision 80; BEGIN was constructed afterward in Revision 83. APPLE=R+S, ANSWER=A−S, and BEGIN=APPLE+ANSWER collapse algebraically to A+R, so they are one construction rather than independent corroboration. A calculation using ADD does not by itself produce an executable state carrying an ADD register.

The P2095/R2095 comparison was recorded later, after Burn BEGIN/ADD was known. It depends on the historical whole-state six-operation compiler, final-slot anchoring, EXPECT appending, CHECK normalization, and operation propagation. Its BEGIN/ADD endpoint is therefore not an independent blind prediction. Public R2095=Q−P2095 is a point relation, not evidence of historical private-key possession. Wrong-message and wrong-signature controls fail as expected, but no historical credential in this test binds Burn or Q.

What the steganography book audit actually adds

The supplied 2024 book is a taxonomy, not evidence of a 2021 decoder. The existing puzzle audit already inspected its relevant text and diagrams. Applicable families include text substitution/spacing/shape, grilles, parity and transform domains, reversible data hiding, generated and distributed covers, blockchain covert channels, permutation, and watermark/provenance tests. Each requires an independently specified carrier, representation, extraction order, and validation. The book's difference-expansion circle-plus means `2p+i` rather than XOR, and its Josephus example has inconsistent parameters. Neither supplies a puzzle operation. Literal quantum steganography is unsupported because these files contain no quantum state preparation, channel, or measurement.

The audit narrows potential channels. Rows contain only printable ASCII and `0x20` spaces, with no hidden Unicode, tabs, serialized newlines, or alternate-space identity. Base64 encodings are canonical, with all ten unused pad bits zero. After replacing artwork bytes and parent-output index, all nineteen stripped row transactions reduce to one template. Amounts, fees, locktimes, sequences, script framing, and witness item counts do not supply additional varying row metadata. Block row order matches ascending internal txid-byte sorting, for which ordinary miner package comparison is sufficient; identifying the actual miner implementation remains an inference.

Capacity cannot be obtained by adding overlapping representations. The forty q/1 choices provide at most 40 unconstrained bits, or 15.170 bits across 36,864 per-row-population-preserving choices. Internal punctuation has 16.095 bits across 69,984 constrained configurations. The 192 z occurrences provide zero identity variation at fixed sites. A repeated graph order adds no independent permutation; deterministic txid/wtxid hashes do not supply another 512 free bits. Signature-choice observables remain a real possible author-controlled channel, but possible capacity is not evidence of an embedded message.

A complete pair-preserving art/witness null reassigned nineteen native row/spender Ry pairs with funding fixed. The retained conjunction required weight 18, the specified 11|11 and 2|4|7|9 parses, the stated final-width equality, and AND alone reproducing the nonzero seven-bit row. Of 149,652,360 pair assignments,30,808 pass jointly, about 0.0205864% or 1 in 4,858. This uncommon correspondence is still post-discovery and assumes pair-assignment exchangeability; it is not a new holdout or author-intent p-value.

The subsequent frozen five-field specificity receipt independently reconstructs and verifies all 42 signatures, uses the 40 core records, and excludes the two sweeps. Native Ky-LSB alone passes; r-LSB, published-s-LSB, z-LSB, and r-MSB fail. One million seeded permutations preserve whole row/spender pairs and apply the same permutation to every field. Ky-LSB passes 224 times, s-LSB 355 times, and the other three fields zero times. The any-field family has 579 hits, rate 0.000579, Wilson 95% interval[0.000533732,0.000628105]. Seed 2026091901 uses NumPy PCG64; the saved permutation-stream hash is `1cbba277b6ff69107f66a42b813dfc78a5e6f0351dc7bb5158026956627eec56`. This favorable specificity test remains a falsification of an observed predicate, not a discovery probability or demonstrated hidden payload. Its own limit requires an independently selected message, checksum, or verifier before promoting intentional embedding.

The reflected grille reads the established punctuation order `:,.;` at four new locations, but exact matching occurs 1/6 under row-preserving controls, or 1/4 with earlier palette selections fixed. The agreement 22/residual 22 join fails by eleven bits. A reversible 22+18 representation exactly inverts 40 bits but establishes neither an independent original cover nor an independent embedded message. Invertibility alone is not demonstrated reversible data hiding.

The surviving model is distributed art + graph + witness information with an as-yet unearned assembly and verifier. Public historical source references, however elaborate, do not identify a historical signer. A knowledgeable 2021 Q controller could build the authenticated artwork, public historical references, and conditional word machinery without possessing a historical Satoshi private key.

Source chronology and media

The retained public post IDs put the block 709632 clue at 2021-11-14 12:39:24.194 UTC, before its community-mirror repost. The exact artwork image and December 23 burn video strengthen official puzzle custody. The video's stream creation time matches the 2021 post; its container creation time reflects later download/remux. Visible text and retained byte/metadata scans supply no historical signature, Q-bound challenge, or source-native verifier. Forum and community mirrors are useful secondary custody, not original authorship proof.

The May 2022 official steganography article's 440 BC opening, shaved-scalp story, and low-bit discussion retrospectively fit 440 bits/RAZOR/SHVE. It postdates the hunt and cannot establish a convention available prospectively in 2021. Its nine retained images yielded no separate common LSB, metadata, or appended-file payload in the stated test. An unrelated supplied 2024 video is likewise not a contemporary puzzle clue.

Later source and execution audits

Later interpretation trials and the limits of recipe uniqueness

The September 23–24 clean-room trials reproduced the primary wallet solution and investigated the secondary readings. Their independence was computational: they rebuilt results from public bytes or frozen numeric fixtures. It was not blind discovery. Earlier conversation had already disclosed the treasure recipe, V4 analysis, H1 state, terminal18, input38, terminal7 and proposed authorship checkpoints. Recomputed results must retain this disclosure history.

The September 24 reproduction also rebuilt the raw block's header hash and transaction Merkle root. The block had 2,043 transactions and Merkle root `6ada3b10082068de09f7e819b65113d3c58969fd857aab2980c65f374714ec77`. The September 28 root audit did not repeat that inclusion check and relied on retained metadata. These statements describe different verification scopes. They do not contradict each other.

The V4 test provides a real numerical validator within a stipulated family. It does not uniquely establish the family itself. With native pairing and fixed XOR/AND operations, testing all 2,048 U11 translations gives:

Gate	Surviving translations
Both complete-record checksums, G1	102, 562, 574, 1329, 1470
Both artwork-derived weight targets, G2	102
Stored-nibble condition, G3	`6 + 8m`, for `m = 0…255`
G1 and G3	102, 574, 1470
All three gates	102

At 102, the X suffix is `1722 1959 1756 767`, displaying STRUGGLE VISUAL SWEAR GARMENT with checksum 15/15. The Y suffix is `65 8 259 256`, displaying AMUSED ABSURD CALL CACTUS with checksum 0/0. Here and below, a checksum pair is canonical/computed first, stored second.

The glyph codec supplies related operations and counts. XOR of each q/1 glyph's two code bits gives 27 ones and 13 zeros. AND of the 33 canopy glyphs gives seven ones: three semicolons and four literal `1` glyphs. Adding seven zero pad bits gives the Y weight target 7/33. The ordered X and Y bit strings do not equal those artwork projections. This is a relationship between counts, not a copied payload. Across the 2,048 translations there are 117 distinct weight pairs; 16 pairs occur at exactly one translation, including 27/7. That is discrimination within the declared family, not a probability of authorial intent.

A broader audit tested 2,048 translations, all 24 PIT2 column permutations, and every ordered pair of 23 operations: the 16 Boolean truth tables plus modular addition, both subtraction directions, multiplication, minimum, maximum and absolute difference. The total was 26,001,408 recipes. There were 640,853 passing both full-record checksums, 8,070 passing both weight targets, and 44 passing both. Those 44 use 41 distinct translations and 16 column permutations.

Native column alignment leaves four recipes:

Translation	X operation	Y operation	Stored X/Y nibbles
102	XOR	AND	15 / 0
932	`a ⇒ b`	NOR	12 / 0
1724	`b ⇒ a`	`a AND NOT b`	15 / 11
1981	`b ⇒ a`	`a AND NOT b`	14 / 12

Two additional XOR/AND survivors, 550 and 1574, swap PIT2 columns 0 and 1. Requiring the candidate nibbles 15/0 leaves those two and 102. Requiring `X[0] mod 128 = 58` leaves 102, but no independent rule tells the solver to place the punctuation count 58 in that field or pack 13 above it. These candidate-output checks therefore remain conditional selectors. Independent Python and JavaScript implementations agreed on all counts, survivor tuples, cells and nibbles without loading English labels.

Further direct Boolean tests failed: none of the 16 functions of the translated paired inputs produces the exact q/1 carrier, and no Boolean function of X and Y's first 40 suffix bits produces the authenticated prize tail's first 40 bits. Shifting all twelve root indices by −ART gives checksum 2/0. Keeping the original first eight and shifting only the tail gives 7/0. The full −2ART row gives 8/10; keeping the original first eight and shifting only the tail twice gives 12/10. Thus metadata attributed to the full shifted row cannot be moved onto a mixed-scope record.

The physical clean run: source contacts, tagged scales and branch freezes

The September 29–30 physical investigation is a distinct research branch. It must be retained separately from the CALL/OpenPGP story. Its final report covers 36 clean experiments, alongside the retained E163–E232 kernel history and later adapter tests. The report's statement that the puzzle remains unsolved concerns the secondary reconstruction; it does not revoke the completed primary wallet challenge.

The actual two control strokes contain 20 and 17 glyphs, including 12 and 10 q/1 marks. Their forward mark streams are `001010111111` and `0000010000`. The lower 22 literal q/1 marks touch exactly 18 unique blank cells orthogonally. All 24 tested neighbour-check orders agree on native first contact. The four upper18 one positions, 0, 6, 10 and 15, select bytes `64 bd b1 be` under both live byte placements. Their byte-index labels are ARRIVE BLESS BIKE BLIND, giving a full unchanged 14/14 record. The owning flags give `1010`, interpreted conditionally as tag10 STEP. Native orientation alone passes the joint family of 36 mask controls.

Native inner contact geometry does not need the proposed 5×8 projection. It gives two unchanged 9/9 tails:

```
NOTE BEACH ORDER ANKLE
NOTE BATTLE SLOW ANKLE
```

The four ankle differences remain unresolved. The four enclosed core cells produce the same ADD 4/4 record under both placements. These are strong local objects, but neither checksum chooses the placement or supplies the next operation.

The actual stroke GLIDE candidate has eight complete direction/order variants. Only native upper-forward followed by lower-forward computes the already supplied tag9 SPAWN. Its stored checksum is zero, so it is 9/0 rather than unchanged-valid. The complete canonical/stored list is `9/0, 13/0, 6/0, 6/0, 11/15, 7/15, 10/4, 2/4`.

One proposed interpretation of CUSTOM SCALE / TAG KEEP CLAP REBUILD forms a derived 12-bit value from each original U11 word and its external q/1 flag. It retains the flags, pairs native mirrors, and reserves active pair 9/30. Two tag layouts, six operations modulo 4096 and two output directions give 24 cases. Only native OR simultaneously:

- gives identical decoded words and flags across the two layouts;
- preserves every selected source one flag;
- validates unchanged;
- retains ART = 102.

Its result is:

```
ALTER GALAXY WORTH WEDDING
SUNNY CEREAL IMPACT UNDER
INDICATE STUMBLE WINK SMART
```

The entropy is `076bdff7fc7d964b1c776672faf7ef66`, checksum 4/4, with all twelve output flags equal to one. A separate four-order comparison selects native order by marked source owner under this declared model. Reversed AND also validates 9/9, but deletes all retained one flags and loses ART. The OR result is therefore strong conditional support for this interpretation. Its exact WORD rejoin is retrospective, and the program's independent origin was supplied rather than rediscovered. Derived offsets 96, 108 and 120 reproduce TWELVE, SPAWN and STEP within the same representation.

The complete record's canonical4, consumed by the original-carrier rule, rejoins the independently obtained core tag4 state. The returned record is 11/6. Its next RECALL/ORBIT mirror pair is `00`, so the old marked-side consumer stops. A conditional AND seam recall reaches the last native field and the same earlier BLIND byte, but no unique footer or next parser follows.

The local OR accounts for all source pairs: twelve marked data pairs, reserved pair 9/30 and seven empty pairs. The empty outputs are ZEBRA ZOO SAY RECYCLE MOMENT ROOM TEN, with flags `0000000`. They are seven words without a BIP39 checksum or selected next operation. They must remain an unconsumed object.

The final clean experiment established a native 8+4 boundary at payload bit272 in the raw tag9 record: the first two fields end there and the last two begin. Pairing the tail while retaining the original 22 residual bits gives two unchanged 11/11 survivors among 36 alternatives:

Pairing and operation	Tail result
Corresponding fields, XOR	VACANT ACTOR RAZOR NATURE

Pairing and operation	Tail result
Mirror, OR	MAMMAL WANT RAZOR NATURE

No unique selector chooses the operation or residual relocation. The attractive mirror-XOR MAMMAL INNER candidate fails, giving 10/3 with residual prefix and 5/11 with residual suffix.

Other frozen negatives remain material: whole ROOT/CLAP merging gives 0/12 unchanged records; nonzero ART shifts 0/4; the tag4 whole fold 0/12. Internal parity, byte and flag layouts fail their eligible reconstructions. Native seven-pair and prefix alternatives fail. Three different native zero-bit sources can serve the field footer, preventing a unique choice. Recurring `1010` values belong to distinct objects—contact-owner flags, ankle-byte low parity and computed payload parity—and are not independent convergence.

The wrapped13/B12/B5 route remains frozen

The longer wrapped13 route contains real matches but does not establish the instructions connecting them. LENS1025 → SCALE1536 depends on the selected end-bit layout. A periodic outside-shore model fills nine out-of-canvas cells, or 18 unknown bits, rather than recovering authored source bytes. Its CCW/ROR1 wrapped record has entropy `9827c5be9fb68fa4b18bd015583dd52d`, tail BEST SEA INSPIRE FORUM and checksum 13/13.

Reading ASCII53 as “5” and BIP39[151] as BASE mixes lookup domains. The radix5 test fails at 10/7; the radix8 control gives 13/7. A selected block of that control supplies selector12 and exactly rejoins B12:

```
ALREADY FREQUENT STEP PORTION
SUNNY CALM EYEBROW UNDER
ACTOR TARGET VERSION LOTTERY
```

B12 has entropy `072b9755d42d964114576602bbbfcb42` and checksum 0/0. This is an exact data rejoin, including mask and order. It does not authenticate the route supplying selector12, restore ROR1, or choose the next operator. LEMON1022 → DIVIDE511 is exact under ROR1 but inherits the same unselected permutation. Frozen CLICK/media, BASE5, LENS, LEMON and external branches remain downgraded.

The original 18-byte field is:

```
b5 [531ddcbd4228ce30166cb11041145766] d8
```

The inner 16 bytes end in ART102. The native endpoint codec gives `g(B5)=01` and `g(D8)=10`, hence wrapper6, while the inner entropy computes checksum9. Among four data/tag reflection cases, original data plus exchanged endpoint units gives the sole unchanged 9/9 cousin:

```
FANTASY UPON CONSIDER LOUNGE
MIND BLOSSOM FLUID NORMAL
AWARE ANGER BEYOND SNAKE
```

Corresponding XOR of the complete root and this cousin gives an unchanged 15/15 child:

```
SLEEP SPEAK DOMAIN ROBOT
```

```
TOMATO SKI PANEL UMBRELLA
TRAGIC ORBIT FINAL ACID
```

Corresponding AND is an unchanged 0/0 rival. Later KEEP/shared-ART controls retain that alternative role; neither restores the frozen wrapped13 route. The ORBIT field boundary gives an exact 9/95 split. Reading its final slot as seven data bits and four tag bits gives data77/tag15, but does not select an ANSWER mutation or ending.

Native final-field16 plus the actual q footer `0000` gives the unchanged 0/0 candidate:

```
SALUTE TEXT THAT CODE
PLATE PIECE BOIL DONKEY
EXIT PULP EXOTIC CURRENT
```

Its conditional CODE/ASCII readings do not establish a whole-record parser. Complete byte-CLAP has 0/12 unchanged survivors. The native 64-byte signature interpretation gives 0/8 candidate-message verifications, while the real row positive control verifies. The CAVE point matches none of the exposed funding/parent keys or 41 output hashes.

CLICK, GRAB and PRIMARY: source identity without an admitted consumer

The September 30 sender is:

```
CLICK341 | GRAB810 | footer0000
00101010101 | 01100101010 | 0000
```

There are seven other complete CLICK occurrences, represented across 50 view aliases including the sender. Complete source-bit identity and the whole-word physical turn select no other receiver. A unique partial overlap identifies the all-punctuation substring at bit81 through comma `(15,14)`, but it has no source overlap or direct contact with any of the four bounded GRAB occurrences. A coarser head-glyph turn picks CLICK99, mapping `(14,12)` to `(16,25)`, without establishing whole-token identity.

Interpreting 341 as a U11 address would start VERSION31. As a derived 12-bit address it lies inside ORBIT28 at phase5. Neither conversion has a selected caller. The BEACH footer is exactly the earlier lower-stroke head, not merely another zero nibble; however, a 128+4 field-tag precedent cannot become a 22+4 state consumer by equality alone.

The physical ORBIT/VERSION/ART seam is precise. ORBIT's closing flag `(15,11)` turns to VERSION31's flag `(15,26)`, whose unique adjacent placed byte is ART from F1 at `(15,25)`. The native closing flag borders `d8`; the turned flag borders `66`. The physical VERSION flag, serialized VERSION word data in F3 and nearby ART byte are different source objects.

Two forward GRAB views do share one complete following U11 token, PRIMARY1365, bits `10101010101`. Its owners are the full-parity bit of comma `(12,31)` followed by both low/full `01` bits of commas `(13,31)` through `(17,31)`. It uses 11 bits from six commas; the six original ASCII bytes form a different extent. The frame interval `[11:22)` and right-edge 22-bit window interval `[11:22)` have identical operand owners, although the first two verb owners differ. The longer tape's `[19:30)` interval crosses its prior boundary.

This resolves which token is shared **if** GRAB takes the next U11. The earliest E47 record merely decodes six windows; it does not establish that grammar. Turning the operand maps it to frame bits `[25:36)`, crossing two native fields as 8+3. No rule selects PRIMARY's data referent, an internal consumer or a historical public-key query.

Geometry and composed human-instruction tests

Several later controls refine the foundation's geometric claims. The conditional 22-cell chamber comes from the frozen 2/4/7/9 run family. Top-first and bottom-first installations differ, with no selected direction. Horizontal reflection has intersection4/union40; vertical and double reflection have intersection0/union44. Repeated corner scans and symmetry-equivalent readouts are correlated, not independent confirmations.

For the supplied 11×11 and 6×6 matrix equation suite, identity survives 7,920 low-label and 360 high-label permutations. That is conditional specificity, not a selected coordinate adapter. The upper18 allocator places all 18 positions in a 35-candidate family but does not predict the payload bits: 1,872 of 3,060 four-one masks score at least as well. The proposed 183-bit lane binds 61 triples to 61 canopy cells; 1,403 of 10,000 random arrangements change at most two cells, and 614 also reach weight38. The lower A/E transition model makes 0/7 complete held-out predictions.

Width38 makes the offset37 walk diagonal, but 666 of 685 starts share that diagonal and 19 are wrap cases. The q/1 selector chooses Q in 22/40 cases versus 18/40 for its complement. Its best native distance is13, while controls reach8; 7,960 of 10,000 same-weight masks beat the native score. These counts constrain only the declared families.

The Round2 instruction excavation distinguishes literal source phrases, derived decoder phrases, analyst instructions and conditional CALL paths. LOW/FULL planes over 90 selected artwork glyphs are meaningful; only original-coordinate reinsertion of the auxiliary16 restores known CONTROL98. No standalone 180-bit receiver or fresh GREEN selection was found. Both z-to-space and z-to-# preserve the target streams, so the ERASE CURTAIN experiment does not choose a mask. The retained prose more directly describes exposing construction source beneath presentation.

The documentary comparison found a genuine extra period in the court declaration's quotation. The original signature discriminates exact source bytes from that readable copy and a whitespace-normalized Exhibit A. ECF512-7 supports membership of the same 145 records but changes their relative order. The Hashcash signature covers exact compressed archive bytes; equal extracted content, recompression and a gzip-header bit change do not preserve that signature. IRC PRIMARY is a peer-address mechanism in a different scope.

Round3 investigated three bounded composition programs: physical preparation/reflection, documentary order transport and encrypted-item attribution. Five primary composition/control tests and a follow-up mutation test produced no new source-selected output. Writing the field-residual22 then reflecting and superposing creates conflicts. The lower22 plus its reflection gives a reproducible conditional 40-cell raster, with exploratory audit serialization `67e532a211`, but only a weak overlap check and no selected next reader. A same-input order control distinguishes write-then-reflect superposition from geometry-first installation. The document's alternate order fails the frozen readers. Selected-item containment in its parent does not authenticate the claimed sender.

The coverage register has 55 entries: 28 P2 and 27 P3; 20 physical, 15 document/order and 20 evidence/attribution. Nine participate in composition tests, 23 remain completed or control-only, 22 are dormant, including the rejected native QUOTE replacement, and one is the frozen BACON selection. No evidence grade, new consumer or inherited operation was promoted. Round3's suggestion to return to an “independent AIR/ACTUAL” frontier is superseded by the later entrance audit.

The Q packet: exact reproduction and controlling provenance correction

The late word-blind audit classifies the twelve-word Q packet as **researcher-assembled on retained provenance**. All twelve values reproduce. No recovered source rule independently selects their mixed fields, operators, positions and combined order. This classification does not dispute Q's signatures, the native sweep transaction or the arithmetic observations.

The native two-input sweep is transaction `cd7b40e5c7c62563a38a571e51f5e65d351be5369a8378a95715cb69aa8b69d1`. Its first input spends `495cc745233f93f8e8372a44b2f99b5723e97cb40ddaeb2c09e95f01a7d76a53:0`, worth12,000 satoshis. Its second spends terminal output1, worth307,560 satoshis. In the 42-record research manifest—two funding, nineteen row, nineteen terminal and two sweep signatures—these are records40/41. Native input order fixes left and right once an operation is chosen; it does not choose the operation.

The earliest recoverable projection implementation and associated report are internally dated August19. They already contain the expected feature sequence and table order. Byte-identical copies establish preservation, not separate discoveries. September18 and September19 records explicitly leave independent decoder selection unadmitted. The October5 reader inherits the earlier printed twelve-entry order. No pre-output receipt was recovered fixing the complete recipe before its words were known.

For each record, define:

```
u1 = z / s mod n          u2 = r / s mod n
A  = u1 G                 C  = u2 P2095
D  = u2 (Q − P2095)       K  = A + u2 Q
V  = A + D = K − C
```

Here z is the native BIP143 digest. `E(x)` means a 32-byte big-endian value followed by the top eight SHA256 checksum bits, split into 24 U11 cells. Pair operations act cellwise on separately encoded operands; sampling uses the raw transformed vector without generating a replacement checksum.

Slot	Exact recipe	Value and label
O01	Right V.x byte30: `(V1.x >> 8) & 255`	123 AUTHOR
O02	`ADD(E(s0), E(s1))[23]`	606 ENVELOPE
O03	`XOR(E(z0), E(z1))[22]`	606 ENVELOPE
O04	`SUB_LR(E(z0), E(z1))[15]`	123 AUTHOR

Slot	Exact recipe	Value and label
O05	`ADD(E(u2_0), E(u2_1))[16]`	1664 SOURCE
O06	`XOR(E(u2_0), E(u2_1))[3]`	1378 PROOF
O07	`SUB_LR(E(K0.y), E(K1.y))[10]`	1536 SCALE
O08	`ADD(E(A0.y), E(A1.y))[9]`	1941 VERIFY
O09	`XOR(E(V0.x), E(V1.x))[7]`	692 FINAL
O10	The same V.x XOR vector, cell9	503 DISCOVER
O11	`SUB_LR(E(D0.x), E(D1.x))[17]`	1973 WALL
O12	Byte31 of the first 256 transformed bits of `SUB_LR(E(A0.x), E(A1.x))`	104 ARTIST

Ten entries are U11 samples and two are U8 extractions. O02's final cell includes checksum bits. This is not a uniform native twelve-field record. O01, O09, O10 and O11 depend on the imported P2095 point. Distinguishing a later Q-packet occurrence from the older P2095 sweep does not make their numerical inputs independent.

The retained board-trailer reader gives 24 bits:

```
001010000100000010101101
00101000010 | 00000101011 | 01
CHOOSE322   | AIR43       | comma
```

Two controls expose the limit of using that output to authenticate its recipe. Exchanging the two ENVELOPE producers, O02 and O03, preserves the entire numeric sequence and trailer because both equal606. A hard-coded recipe distinguishes their metadata, but no independent source rule was recovered preferring their order. Restoring ordinary V.x XOR cell8, value1651 and bits `11001110011`, gives thirteen entries and 26 trailer bits `00101000010000001000101101`. Cell8 lies at entropy bits `[88:99)`, between cells7 and 9, with no delimiter or checksum boundary. The stipulated twelve-entry frame cannot independently justify its omission. Its word was not decoded in that audit.

The later complete Round4 census rebuilt all 861 unordered pair packets and all 5,166 fixed six-operator cases, independently agreeing with the archived word-string implementation. Each frozen predicate individually selects only pair40/41: generic ABBA, exact ABBA, two AND90 results followed by ADD180, the fixed final AND180, exact FINAL/DISCOVER and the compound test. Their conjunction eliminates zero additional candidates. These cannot be counted as independent uniqueness confirmations.

At the three frozen positions, pair40/41 gives:

Operation	Three outputs
ADD	729, 729, 1195

Operation	Three outputs
SUB_LR	1565, 483, 189
SUB_RL	483, 1565, 1859
XOR	549, 549, 835
AND	90, 90, 180
OR	639, 639, 1015

The only other fixed-operator final180 hit is XOR on pair34/35; it fails the two90 predicate. A frequency of 1/861 is descriptive under the target-known reader, not a p-value. No source-issued AND operation was recovered. The corresponding CALL scope audit finds ENVELOPE/ENVELOPE→NURSE and AREA/AREA→BIRD→ARTWORK recurring in the ADD closure, but no binding from this Q arithmetic to the Email260-derived occurrence.

The older V.x sweep remains a separate conditional calculation: `D=(r/s)(Q−P2095)`, `V=(z/s)G+D`, independently encode each V.x as ENT256, then XOR aligned U11 cells without a fresh checksum. FINAL at cell7 and DISCOVER at cell9 reproduce. Substituting exact362's digest does not reproduce them. Its carried continuation is FINAL[XOR]→DISCOVER[XOR] and the known XOR basin. The old FINAL→RSUB→MAIL claim is superseded. A newly reached FINAL[SUB] is another occurrence, not a replacement for FINAL[XOR].

Two additional Q-side tests remain bounded. Treating AIR43 as zero-based signer index43 selects displayed entry44, address `1DkxqjACGihnBHhiWGuJUMVRco6LHFpovT`, whose source height33067 has remainders7 modulo38 and 145. This joint predicate is unique among the 145 selected heights. Displayed entry43/index42 has height33915 and fails. Moduli19 and 38 are dependent. A 4,261-file audit found no generic height-to-terminal-input adapter, so terminal vin7 is not thereby selected. A Hamming map from row43 against END row144 enters a six-cycle shared by 138 starts; ART102 is another basin. It is nonselective. At row83, CLEAN338 and period2 give CLAW336 under either subtraction or XOR; equality does not select the operation or consumer.

Conditional Q-SUB closure and the frame seam

Under a still-conditional fresh-state admission, projection3's z SUB_LR at position15 gives AUTHOR[SUB], while projection11's A SUB_LR endpoint gives ARTIST[SUB]. Projection0 AUTHOR is an untyped right-record verifier endpoint and remains separate. The roots are pending frontier promotion, not activated merely because archive definitions exist.

Each complete closure has 3,626 definitions, 14,380 activation edges, 7,252 side-path replays and 3,626 join replays. Their hashes are `2f13b033a6ce4eadae5cee03c3cac5a1f5392f4727201d3894277325151d8178` for AUTHOR and `13ada8fed35ba2a710d4c64b48c9345e6bd2e0896252375468a22186f549c08c` for ARTIST.

AUTHOR[SUB] outputs THANK/BRAIN/CRADLE. Its sides end at AREA and DINNER; the ordered SUB join starts SNACK and ends BUNKER. Its PC runs through CRADLE, GIVE, HUMOR, GROCERY, INMATE,

CHARGE, CHOICE, ALARM, ALCOHOL, BAR, DELAY and BUSINESS under SUB, then INVITE, CITIZEN, CANNON, CLOSE, BUS, FOOT and COUNTRY under ADD, ending DINNER. BRAIN is real, but the requested equality with a Q.x endpoint lacks its primary receipt.

ARTIST[SUB] outputs MASS/HURDLE/HOLE. Its sides end CARRY and BUNKER; the ordered SUB join starts AFRAID and ends ACTOR. Its PC runs through HOLE, CANVAS, COMBINE, FEDERAL, ABSORB, AROUND, ACT, DUMB, HOBBY, IMPULSE, DOMAIN, HOVER and DECADE under SUB, then GARDEN, COOL, DENY, LEAVE and FORWARD under ADD, ending ENVELOPE. The two immediate return sets intersect only at BUNKER. Those BUNKER occurrences do not inherit certificate or email ancestry.

There is no shared depth1 activation child. The unique earliest shared definition is STUMBLE[SUB] at depth2, reached by AUTHOR→SNACK→STUMBLE and ARTIST→MASS→STUMBLE. STUMBLE outputs FORK/EITHER/FOLLOW. Its PC is FOLLOW→CROWD→INITIAL→CREEK→WITNESS→WRESTLE→ARTWORK. Its side subtree is FORK→EXPLAIN→DEMAND→FORUM.

STUMBLE and DEMAND have the same ordered DINNER/BUNKER SUB join, seed CABLE, with path:

```
CABLE AGE DIVERT GRAB CATALOG HIRE FURY BRING FAT HOLIDAY
CROUCH INJECT ALSO GIRAFFE BEGIN WORD AUGUST FIBER HURT
BLOUSE CORRECT CASINO FOG HOVER DECADE GARDEN COOL DENY
LEAVE FORWARD ENVELOPE
```

Across the 8,120-definition archive, 7,536 PC walks reach a boundary and 584 cycle. PC terminal ARTWORK occurs129 times. Join terminal ENVELOPE occurs3,061 times; 45 of those paths contain the exact token GRAB. The combined predicate selects only TD-STUMBLE-SUB. An older summary's count63 uses an unspecified broader predicate and remains unresolved; it is not interchangeable with 45. These are structural counts under the model, not external search authority or authorial probabilities.

The separate AUTHOR→THANK→WINE→MERIT→RUBBER→REQUIRE→FINAL[SUB] chain has KETCHUP returning ARTWORK, ACTRESS returning ENVELOPE, and ordered ARTWORK/ENVELOPE SUB join seed SCRAP ending FORUM. It corroborates the internal pattern without selecting an external object.

The physical frame reads GRAB810 / PRIMARY1365 / FETCH682 / PRIMARY1365 from 28 marks after skipping six. CONTROL98 cell8 is FORUM733. The final GRAB source bit and first FORUM bit share `(10,31)`, state01, and the next frame bit begins PRIMARY. All44 frame bits map into CONTROL98; 13 of 72 cell pairs overlap, with exactly one end-to-start junction. This is an exact but correlated physical seam. The missing PRIMARY@FRAME→historical-key operand/query adapter remains unrecovered. No word match joins it to the distinct CALL GRAB or FORUM occurrences.

Email260: exact native fields, chronology and conditional selection

Use **Recipient A** for the 692-byte encrypted stream and **Recipient B** for the separate 434-byte stream. The former's SHA256 is `a1efc74bcc5f052e2f230b12ee1bd26ae4de3a3eba02a1581b93dd6bb4d3cc78`; the latter's is `b82d9ff1b15c5d8b2aae4a68b19ea13cce9b88395d527aaeb2afda8bd8e62ae7`.

Recipient A's native tag1 packet is `[0:529)`, body `[3:529)`. It is version3, algorithm16, recipient ID `17C03C8360CB2496`. MPI-a has 256 data bytes at `[15:271)`, with bit length2045; MPI-b has 256 data bytes

at `[273:529)`, with bit length2047. Tag18 is `[529:692)`, version1 with a 161-byte body. The four recognition correspondences are exact:

Native field	Value
MPI-b end and tag1 body end, offset528	`0x8a = 138 = BACON`
Tag18 body end, offset691	`0x2b = 43 = AIR`
Complete stream length, including packet headers	`529 + 163 = 692 = FINAL`
MPI-a end, offset270	`0x6d = 109 = ASSET`

Recipient B is an RSA control with one ciphertext MPI, tag1 endpoint237 BUFFALO and tag18 endpoint133 AWFUL. The historical certificate and Recipient A use different public subkeys; their p and y differ, although g agrees. Numeric range compatibility does not establish recipient identity or a shared ElGamal group.

The frozen wider population contains four AIR endpoints among 644 packets and five AIR occurrences in a merged679-object namespace. The later four-field census has 19 unique containers: 15 public certificates, two detached signatures and only two encrypted messages. Recipient A alone matches all four fields in that population, but 679 is not a denominator of 679 encrypted-message competitors. The declared Q scalar/ElGamal exponent test returns0/210. The earliest independently anchored exact public ciphertext copy is February23,2024. Bounded pre-2021/pre-2022 searches found none; this is not proof of absence or exclusive possession.

The current equation `BUNKER243 − ARTWORK105 = BACON138` mechanically selects Recipient A over Recipient B **if** its operands, order, subtraction and comparison domain are admitted. It then puts BACON before the same-container AIR sibling. Research chronology is different. The first retained parse of tag1 END138 is August27,2026 at 22:08:01 UTC; complete armor was exposed at 20:58:57, and BACON recognition followed at 22:08:07. The first located statement selecting the exact ordered BUNKER−ARTWORK expression is October7 at 02:42:56 UTC; explicit numeric spelling appears at 12:08:44. An earlier original selector receipt was not recovered. This supports DIRECT/RETROSPECTIVE correspondence and a conditional interpretation, not historically blind precommitment. It does not prove deliberate fitting.

FINAL was retained August19 and the FINAL[XOR] PC route to ASSET August24, before this cipher inspection. The CHOOSE AIR trailer reader was retained October5, after AIR had already been recorded August27. Thus the assembled four-field recognition is not an independently anticipated STRONG certificate.

The older whole-block replay also requires correction. The original trunk includes BUNKER→ADJUST→LATER→HIGH/BALANCE→LENGTH. Across the displayed encrypted blocks, the later member is Recipient B, giving434 CURVE; Recipient A's692 FINAL is the sibling measurement. A shortened replay omitting HIGH/BALANCE does not reproduce the full native contract. The needed positive-balance operand and incoming register were not recovered.

The ElGamal audit preserves diagnostic metadata arithmetic without admitting full MPIs as puzzle operands. Bit lengths2045/2047 give ADD2044 ZEBRA, SUB_LR2046 ZONE, and SUB_RL/XOR2 ABLE. Data lengths256/256 give ADD512 DIVORCE and the other stated differences/XOR zero.

Length-prefixed sizes258/258 give ADD516 DOG and zero differences/XOR. Endpoint109/138 gives ADD247 BUS, SUB_LR2019 WISE, SUB_RL29 ADMIT and XOR231 BROTHER. The ADMIT29 equality with LATER's fourth field is retrospective. A complete 256-byte MPI is not an admitted 32-byte ENT256 scalar or U11 operand; no full-MPI consumer was recovered.

The AIR/ACTUAL descendants and later admission failures

Under the assumed CHOOSE[ADD] entrance, ACTUAL[ADD] has two AREA child histories returning through BIRD→BALANCE→GENRE→BROOM→ARTWORK. Its artist arm follows ACTUAL.side1 VAULT, VAULT.side2 BELOW, BELOW.side2 INSIDE, INSIDE.PC HYBRID, HYBRID.join MAIL, MAIL.side2 ARCTIC and ARCTIC.PC ARTIST. The arrival retains ADD. ARTIST is a canonical boundary with native tuple GROCERY/TATTOO/HUMAN/ACQUIRE; absence of TD-ARTIST-ADD is not permission to replace that proof interface with another operation.

The direct acquisition arm is BIRD→TRY→ADJUST→VINTAGE→THREE→WALL→ACQUIRE[ADD]. The exhaustive fresh ADD closure has 1,864 templates,7,395 role edges,5,592 side/join path checks and 1,864 successful installed join-arithmetic checks. It contains no native SATOSHI or OUTPUT state, output or join seed. ACTUAL and NURSE have the same template closure. The 3,570-definition ASK closure has 1,871 ADD and 1,699 XOR records.

GUILT[ADD] has ordered APOLOGY/CAUTION sides, whose PC histories both return ENVELOPE, and join NURSE. Its native terminal is IMMENSE. NURSE has BACHELOR/QUOTE sides, join GENTLE, and PC:

```
CHEAP ICON AGAIN EDGE ANGER BLOOD LITTLE CUSHION DIARY
FADE BODY AUTHOR DEBRIS AREA
```

BODY has GAS/CHECK sides, join MENTION and PC AUTHOR. QUOTE has one side ending GUILT, one ending ENVELOPE, native join RECALL and PC ending DINNER. GUILT[ADD] is the sole ASK definition with ordered APOLOGY/CAUTION sides. NURSE[ADD] is the sole definition with direct QUOTE side activation plus AUTHOR in its PC history. NURSE[XOR] instead gives the recorded ENVELOPE/BROKEN/ENVELOPE/DINNER pattern. These are selective local roles, not a historical confession or accusation.

The later Email260 consumer census inspected all 8,120 exported definitions,113,858 native event records and 6,000 roots. Its15 priority states supply61 definitions; adding FIRST/SECOND/PAIR/GUILT/NURSE/QUOTE gives86 definitions,444 associated events and 33 roots. All430 selected socket edges and 86 retained outputs have type U11; explicit downstream-consumer fields are empty. No selected record names Email260, MIME, MBOX, PKESK, tag18, ciphertext or an authentication header as input. These exported event counts differ from the 113,990 source-anchor archive and must not be conflated.

Fresh CHECK[ADD] outputs FOLLOW/DRINK/DEMISE. Its AREA/AREA child returns join through BIRD to ARTWORK; its PC is DEMISE→DEPEND→IMMENSE. It is not a documented MIME-body, CRC/MDC, sender or provenance verifier. The admitted external Email260 pipeline is strict armor/CRC normalization, native packet parsing and **whole packet-body** END138/43. The byte-oriented endpoint utility can calculate many values, but does not admit arbitrary subfields.

The AUTHOR contract audit returns C, HUMAN INSTRUCTION ONLY. BODY's PC produces an internal AUTHOR without consuming Email260 headers, signature or custody evidence. No applicable external AUTHOR acceptance predicate, even a partial Email260 checklist, was recovered. Native MBOX necessity and sufficiency are UNKNOWN. A native record could help provenance, but has not been shown necessary or sufficient for an applicable puzzle consumer. The five-candidate object-class comparison makes Email260 a strong human interpretation of secret-mail roles; it does not close authorship, plaintext or executable-consumer gaps.

The fresh APOLOGY83 domain audit returns C, NO HANDOFF. Historical-email ancestry does not select the particular 205-input legacy transaction. The fresh ARTIST→ACQUIRE audit also returns C: its parent does not fit the old named proof invocation. Existing Hashcash authentication survives, but the fresh route does not supply that invocation's final-question fork and master-material parents.

The proposed external ASSET109→internal ASSET[XOR] rejoin is rejected within the retained input grammar and caller scope. The OpenPGP precedent consumes a selected complete packet body; Recipient A's packet-body endpoint is138, while109 is an internal MPI endpoint. The433-record canonical-return ledger contains zero ASSET records. The full archive has four ASSET templates but zero completed-child joins using ASSET as seed, endpoint operand or terminal. Existing ASSET[XOR] has PACT returning ACTOR, DAD returning CARRY, their XOR joining at CAKE, and PC BUYER. That internal template does not consume an external ASSET witness.

The old proof gate remains a scoped precedent

The older final-question gate has two explicit ACQUIRE parents: a source-author witness selected by a valid detached signature over the exact Hashcash archive, and a visible ARTIST proof lane in the same named caller. Their equality acknowledges a shared event; it is not ADD of two17 values. Its registered POLAR/XOR continuation supplies VISUAL/GREEN, then DOUBLE/PATH/AREA and OUTPUT.

The exact archive is29,723 bytes, SHA256 `43d9d6da6a818dc9854524d36dd9eb587a6bad8004bd8fd16471b666c1cc0572`; detached-signature hash `17ab5dac5528bd8a52a7e33987cd8883031635648636 57d9cdec6248edb6463c`; selected RSA2048 exponent17 key ID `3E7BCAA828B24551`. The native PKCS1/MD5 verifier checks those archive bytes. Its selected primary packet-body END17 gives ACQUIRE, but exponent17 endpoints are not unique person identifiers.

The same named gate joins construction ANOTHER, produced by ANNOUNCE−ZONE, with OUTPUT's disjoint-SUB BEHIND/ANOTHER76 result. Its registered LOCAL-clock reverse subtraction gives LOAD/STREET/TOPIC; TOPIC's installed TEST/XOR gives SATOSHI. The old G0/G1 occurrences remain C-grade because they descend from a declared conditional P00 splice. The recovered rejoin literal confirms named parent pairs, not generic fresh substitution.

Canonical return stops are different records from these explicit proof joins. Their433 returns across 32 states include12 ACQUIRE and 13 ANOTHER, with every `old_external_binding_inherited=false`. This does not erase the authentic parent of the original named join. It prevents a fresh return from inheriting that old external binding.

Tagged attribution and the Entry123 certificate

The established four-high/seven-low representation gives QUOTE1408=`11|0`, AUTHOR123=`0|123` and SATOSHI1531=`11|123`, exactly QUOTE[AUTHOR]. RECALL is the related QUOTE[ADDRESS] record. Addition and XOR agree because the fields are disjoint. This is a representation fact, not a newly executed NURSE assembler.

Under a retained displayed-record self-pointer candidate, exact362 Entry123's nonce-x/r ADJACENT/XOR reduction gives:

```
LONELY PATROL VALID PINK TEAM CLARIFY
MELT KITE SEEK HISTORY RECORD AUTHOR
```

Its endpoint is ABSTRACT7, checksum selector RECALL11 and foot AUTHOR123. Among930 checksum-valid checkpoints, the four-equation core has exactly one hit, Entry123, even without foot=AUTHOR as a condition:

```
TEAM AND SEEK = SEARCH
PATROL − TEAM = SECRET
VALID − PINK = ENVELOPE
SEEK + RECORD = ITEM
```

Further retained equations are SECRET−ENVELOPE=ITEM; ITEM+AUTHOR=MAIL; MAIL−ITEM=AUTHOR; VALID XOR KITE=MESSAGE; VALID+SEEK=RECORD; HISTORY OR RECORD=ZOO. REV3 interpreted Row17 CODE/SPLIT/EVIL/WARRIOR and the 4/7 split as parser support. The later F014 correction retains signature-framing and coordinate evidence from Row17 but does not admit it as an instruction installing a4|7 parser or repairing the historical145-to-Q circuit. Displayed Entry123 is not silently converted to zero-based historical input123.

The fresh NURSE QUOTE side and BODY→AUTHOR PC furnish compatible roles, but their assembler is missing. A proposed RECALL payload replacement by AUTHOR is an exact conditional record edit, not a selected fresh assignment. QUOTE's native join remains RECALL and its stored joined path ends CARRY. NURSE joins CARRY279 and DINNER498 at GENTLE777, with stored terminal CARRY. No retained RECALL slot, later AUTHOR write, source-object binding or acceptance predicate was recovered. The return ledger has zero QUOTE, RECALL, AUTHOR or SATOSHI records. The requested narrow AUTHOR canonical-rejoin primary receipt remains missing.

Historical205-input pointers and the Entry39 alternatives

The historical pointer domain is the native input list of transaction `29591d211054478ed22c93571955cb92ef4ec2b40d914a43ba3a91d52a171c61`:205 ordered inputs, zero-based,23,446 raw bytes, SHA256 `24f2ff30597e2821e83f7c42a7f9c5d0b144f5f509364ba21e49139f22f4abe6`.

Its fixed reducer takes a32-byte native z digest, encodes ENT256/CS8 into 24 cells, MIRROR-pairs `(i,23−i)`, applies SUB_RL and retains the first 128 result bits. CHECK repairs only the final four checksum bits; the selector is the final entropy byte. Input87's z is `006a22adee8bc48b97d49d18277cb0b300cf936fab128a82cb0a2ba2f9ba8ce7`. Its reduced vector is `1820 1812`

`399 660 1240 998 1464 1975 593 1046 1260 1335`, entropy `e39c50c7a949b0f9adc7b74a305a7653`. Endpoint83 is APOLOGY, while the final mnemonic word is PLUNGE.

The existing same-population `by_input` map consumes83 as input83. That input's CHECK changes TAG1769 to TANK1773 and checksum1001 to1101, preserving entropy endpoint238 BUILD. Since238 is outside0…204, the pointer walk stops. The later BUILD→CALL RSUB→ANOTHER→TEST/XOR→SATOSHI continuation remains conditional on its reducer-to-CALL admission. There are two endpoint83 hits, inputs87 and 121, in the checked graph; only 87 reproduces the complete selected vector. The old unique raw-valid MIRROR/SUB_RL result uses a different predicate.

Input87's prevout is `c1a6c34e04fd062ef7321a0793c833f93394a8c5f101bc7515ea16cbb52e11a3:0`, rendered bare-P2PK address `141CatfWL6wp1srR94sJVyu2NjJNTEjXeg`. Native BSV history gives no outpoint path to the four frozen HMWYDA transactions. The independent BROOM route has 547 marker outputs and 152 hops; splitter `640b1cd729663733b2770ac587ceb7256fbb476e73b5e8493716fe058ad980ba` has four output scripts with no145-key match or consumer. AMOUNT64+ACT19=83 and OBLIGE/XOR's MYSELF→APOLOGY are distinct corroborations, not fresh-domain selection.

Entry39 in the 145-signature publication uses address `12CTHhyJtr49LgoUShbWgebLBviLAFj6nj` and r `6b8ab1a7c6c90d822575059e0d943b41513438aa450102a6b46b422cc40ca481`. Its compact header fixes the public effective nonce point K; x=r, and `sK=zG+rP` verifies. Its sole checksum-valid nonce-x reducer among 18 is HALVES/SUB_RL:

```
POND SHORT DISH BARGAIN IMPOSE INPUT
LUNAR TOGETHER COMIC OFTEN LOCAL PATROL
```

Entropy `a7d8d8fc89571ee961471d2e53360d50` ends80 ANXIETY, checksum9 maps to SPAWN in the retained GLOBAL40 convention, and foot1289 is PATROL. Across145 nonce coordinates,166 of 2,610 reductions validate;57 records have exactly one valid lane. Only Entry39 has the frozen signer CABIN/STEP/WOOL and nonce ANXIETY/SPAWN/PATROL tuple. A million random coordinate-pair controls give68 signer hits,17 nonce hits, two sole-valid nonce hits and zero joint hits. The marginal expected joint count is only 0.001156, so zero adds little weight; those random pairs are not ECDSA-valid and do not yield an authorship probability. K, u1G and u2G do not equal R2095.

The alternate recovery experiment forms public GAP=accepted−opposite and verifies `GAP=(2s/r)K` for all 145 records. Entry39's exact points are:

```
accepted 03270d5c8996e7b1caa60831fb2791558e2e55741aace15d008cc9795e7267d5b5
opposite 024ebffbc1672ef431115f6d6171642f19574f13e050a6de2b4310e7d4019e7cab
GAP      0354b3656f42d8cd89c68c2a5cc5be4cf5ee6c80ea5b04f948667b0c983a81b0e8
```

Its sole valid raw33 reduction is HALVES/OR, giving GARMENT RESULT SWIM THAT COMBINE UPHOLD DISPLAY WASP TYPE YOUTH HURT OVER, endpoint78 ANTENNA, checksum15 PORTION and foot1263 OVER. Across145 records there are151 valid GAP reductions;51 records have zero,49 one,36 two,six three and three four. Only Entry39 has endpoint78. Entries33,36 and 39 have HALVES/OR as their sole valid lane. Four checkpoints contain RESULT; only Entry39 also supplies `COMBINE367 OR DISPLAY508 = DIVIDE511` under the tested own-cell criterion. Its “command then next two cells” grammar was proposed after the result.

The145-table DENY SOURCE occurrence rejects an unreliable Patoshi identity attribution and retains the exact signed witness. It does not select the opposite ECDSA recovery candidate or import P3928's

accepted-minus-denied raw33 procedure. That transfer remains unlicensed. P2095/R2095 and P3928/R3928 retain separate points and registers. Neither R verifies Q's42 signatures. A separate block9 coinbase comparison excludes exact equality and negation with Q for that one historical key only.

Historical mail, catalogue and source-version comparisons

The numbered260-message archive was published in2024; the first retained publisher commit is `134c4a97a344af3161b1dee18f09790abd4c0921`, dated February23. Availability of that numbering in2021 is not established. Message243 contains the exact public certificate, and 244 declares the encrypted root-password attachment. In a260-message census,144 are from the historical sender and 116 addressed to him; broad cryptographic/file searches have14 unquoted hits and four quoted-only hits. The password attachment is the strongest explicit use of the selected encryption credential, but a December21 encrypted website backup is a competing inbound object whose recipient key is unknown. It is not an exclusive selector.

Comparing source122 with its quote in123 preserves150 of 417 normalized tokens and omits one contiguous267-token block: all nine logo records. Spelling, case, numbers, punctuation and embedded HTML are retained. Each record has TYPE=LOGO+7, hence ABSTRACT7 throughout—one algebraic relationship repeated nine times.

LOGO+TYPE gives the grid `9 11 13 / 15 17 19 / 21 23 25`. The final ADD25 also arises under XOR and OR. Opposite pairs total 34 and the center is17, so every full center line totals51 ALL; these four matches are forced by the arithmetic progression. All nine cells sum153 BASKET. They are not five independent confirmations of an ADD/ALL program.

The HTML Code sum uses group ID244765, residue1053 modulo2048, plus type, width and height. It gives `1156 1157 1158 / 1214 1215 1216 / 1257 1258 1259`. Upper-to-middle differences are58 ALSO and middle-to-lower43 AIR, forced by the specified type/dimension changes. Whole-catalogue ABSTRACT and the enclosing new prose's COPY LOCAL→REBUILD are parallel outputs; no tested operation turns ABSTRACT into REBUILD. The known original-four/ADD consumer returns WALK LENGTH without selecting a new object beyond it.

The catalogue ALL/SHIFT/SATOSHI recognition survives conditionally, but stronger prospective preference fails. A later census finds SATOSHI somewhere for 1,517 of 2,048 group residues; with all eight lines allowed there are two ALL/SHIFT residue solutions. The earlier unique1053 argument fixed middle-column/diagonal geometry. BANANA145 from C−A is conditional; ADD/XOR/OR give BIOLOGY179, reverse subtraction UNKNOWN1903 and AND0. UNKNOWN is1903, not UNLOCK1904. AIR43 as a mail index is a human navigation candidate. The43 query made exact362 practically discoverable but did not establish a mandatory entrance.

The actual website tag changed from a remote SourceForge logo—group244765,type10,width80,height15 in the December7,2009 archive—to local `/sites/all/themes/custom2/sflogo.gif` by January6,2010, preserving dimensions and alt text. MOVE1158+ARMOR95=ORIGINAL1253 is a recognition candidate; no source rule selects that HTML tag or src operation. The original adjacent next-minus-previous differences of TRY MAKE CUSTOM

SCALE are ORIGINAL/QUOTE/MAZE, invariant under uniform translation. The separate historical MAKE…TRY comment reverses their order. Keeping the old operation gives TRY−MAKE=GLOOM795; obtaining ORIGINAL requires another unselected reversal.

The preserved archive named bitcoin0.1.0 is2,132,686 bytes and matches the registry checksum but internally says VERSION101. Full native v100 source remains unavailable. A contemporary quoted pre-fix IRC snippet and surviving v101 isolate a deleted retry block that reset memory-only `nLastFailed`; it did not change serialized address identity. An IRC “Connection reset by peer” QUIT is not evidence of that Bitcoin-peer retry.

The separate January2009 node log's sole received version payload begins `64 00 00 00`, little-endian100, giving ARRIVE. Only25 bytes of its declared46-byte payload are printed, but the needed version field is present. Two retained log copies are byte-identical, SHA256 `ab02bbfe5bffdd0eab9c58fd21af246d56dee5e511ca876790c3132f9c19bb3a`. A further ARRIVE round returns the same normalized frontier conditionally. No equal-spelling transfer joins that VERSION/ARRIVE to T18 BRIDGE or a mutable retry register.

Historical PRIMARY: packet anatomy, missing payload and finite DSA tests

The historical certificate is a real internally verified public object, fingerprint `DE4EFCA3E1AB9E41CE96CECB18C09E865EC948A1`,1,159 dearmored bytes, SHA256 `4dc1d9f55a9be2b279d725bc83cd4fd7049fbfb32513872be6446dea8419cff4`. Its three1,714-byte armor copies are identical and contain no secret packets. The DSA primary's418-byte body has a421-byte framed subject, SHA256 `71e0108d41efcb0a82eaf88f1c5baf8e8b0a5d044b88566d9ad3fb745a24c392`.

The UID certification verifies a505-byte preimage, SHA1 `5c63034984f841c1caee5926438f426d7f172d66`; the subkey binding verifies970 bytes, SHA1 `3eb70742ad7f956b88a4601040535bce76863393`. The subjects share421 leading bytes, then differ at `0xb4` versus `0x99`; crossed signatures fail. Signature types are0x13 and 0x18. There is no0x00 binary-document signature over Q or exact362. The UID packet is `[458:556)`, ending at byte555=`0x7b=123=AUTHOR`; the subkey-binding endpoint is `0xf3=243=BUNKER`. These role endpoints do not authenticate Q or become generic rejoin rules.

The T17 descriptor841/374 gives `(38,5)/(17,0)` by divmod22. It matches one of eleven pinned exports and none of ten controls. Prefix38 and packet count5 individually already select1/11; primary ordinal0 matches11/11, algorithm17 matches3/11. The combined tuple is a bounded post-discovery correspondence, not a recovered namespace instruction.

The98-byte UID packet has 58 complement bytes and 40 scalar-payload bytes, matching the artwork's98 glyphs split58 punctuation/40 q/1 as a structural analogy. The complement is2 transport bytes +38 hashed-prefix bytes +2 unhashed-length +10 issuer +2 digest-check +4 MPI-length bytes. Scalar payloads occupy packet-relative `[56:76)` and `[78:98)`, separated by the second MPI length header. The packet body is96 bytes. Artwork40 glyphs give40 bits, not40 bytes; the combined native control reader has 156 bits. No selected field-role map turns this into a new document-signature specification.

The exact362 DSA framing matrix has 6,528 earlier checks plus1,026 extension checks, totaling7,554 with zero response hits. The extension has 171 new valid pairs: four strong-native pairs/24 checks,15

crosses/90 and 152 remaining/912. Binary signing keeps362 bytes; text signing converts six LFs to CRLF, yielding368 with no terminal newline. The metadata areas and preimage lengths are:

Hashed metadata	Area bytes	Binary/text preimage bytes
UID-signature area	32	406 / 412
Subkey-signature area	9	383 / 389
Creation-time-only	6	380 / 386

The extension uses SHA1. Earlier tests also use SHA256 with leftmost160 truncation. Explicit scalar mappings are fixed32 high160, payload low160, native integer modulo q and SHA1 of exact unsigned raw32 bytes. A31-byte component is not silently padded under raw32 hashing; DER sign padding is excluded. Row7 supplies a distinct fixed-width high160 control. Native certificate20-byte MPIs are not padded under a32-byte label. Out-of-range hashes fail `0<r,s<q`; sweep0's S hash is not rescued by reduction. Two implementations agree on all 768 candidate records, six preimages, cross ancestries and 1,026 checks, with native certificate positives and trailer-mutation controls. The older named runner was not recovered; its user-described omissions remain disclosed guidance, not independently verified coverage.

The frozen prospective Email244 test is source-blocked. The package check covered280 outer members, four nested PDFs, zero embedded files/streams and one printed password-file marker. If qualified native bytes arrive, R0 is the exact attachment and R2 its strict canonical OpenPGP binary representation. SHA256 of each is compared at the first 16 and last 16 bytes with the authenticated root entropy; the recipient must be the historical encryption subkey. No salt, alternate hash, reversal, XOR, arbitrary window or post-result transform belongs to that test. It is not a mandatory puzzle output or a Q-specific consumer.

An analyst-created3,996-byte signing manifest, hash prefix `40fda957` and suffix `bd009e3`, is a different object from exact362. Marker-positive searches for that manifest do not establish global absence of an exact362 response. The existing certificate gives its505- and 970-byte signed subjects, but no source-selected third payload and matching historical signature.

The145-address board, GAS closure and object-bound evidence test

The145 signatures authenticate the exact362-byte message in their Bitcoin Signed Message domain. The record order has separate May2020 provenance; the signatures do not sign list positions. Its address-only board reader retains HASH16020 plus checksum4 after removing the Base58 version byte, reads inverse-LSB down the 145 records and partitions each column as13×11 plus a two-bit trailer. It gives the 13×24 board and retained phrases. The earlier claim of an independently authenticated artwork entrance into this external publication was withdrawn by the controlling V2 correction.

Sibling readers on compact signatures, signer keys and nonce points give zero frozen multiword phrase hits. A145×80 funding-header reader gives zero hits for eight phrases across 8,138 positions;

five incidental BODY checksum passes lack selectors. Seven artwork/table vertical offsets select none.

MOUNTAIN's native source execution is closed: authorization verified, allocation matched, every link followed to a signed receive, final data empty and descendants zero. MOUNTAIN[OR]→ENVELOPE remains a checkpoint, not an escape. This is separate from the ordinal1024 RetrieveAssets execution, which also settles without a puzzle return.

The later GAS audit finds no source-defined comparison contract after T00026. It preserves the already consumed HEAD-manual OR, selected BODY checkpoint, raw column and conditional amount/copy route. The board is not declared exhausted. Across24 columns, punctuation counts are nine colons, three semicolons, eight periods and four commas. OR equals row143 at columns1,5,16,17,19 and equals END at 0,1,4; colon followed by a period occurs at 3,12,14,19,22. The post-hoc conjunction “colon and OR equals row143” singles out GAS, but no retained instruction selects it.

The retained BODY grouping has 37 natural packets and 24 vertical packets; only vertical19 yields the stored SATOSHI checksum result under that conditional grouping. The next board-local edge is T00032, checksum-derived CRACK398 DOCUMENT515 with colon,24 bits `001100011100100000001100`. It needs its own document pointer, CRACK operation/representation and consumer. It is not the already consumed Document541 locator/145-message self-loop.

The later common-claim test compares six families and four controls while hiding English labels. Its only qualifying cross-family native relation is the existing row18→terminal18 signed spend under Q, shared by the GAS amount/copy and construction-return families. It establishes construction authorization. It does not join fresh ARTIST, QUOTE, AUTHOR or APOLOGY evidence into a historical claim.

The four controls retain explicit object dependencies: exact Hashcash archive/signature/key; exact362 and its145 signatures; input87's same-population pointer; and identical decoded Email260 streams across HTML/PDF copies. Withholding their concrete objects disables their original acceptance predicates. Fresh scalar/template outputs can survive such withholding as internal results, but that does not preserve an external proof. No shared omitted-wrapper defect was recovered. Fresh semantic convergence is not mechanical evidence convergence.

Independent145-signature carrier and steganography audit

The October4 attachment audit used source hash `a13a182f643c6d55684cda38dc32840202f3895d626ec70cda837ab3581407a0`. Pure Python and Coincurve/libsecp256k1 verified145/145 compact signatures and recovered the listed keys. All r values are unique and below2^255; all s values are low-S. This is compatible with ordinary low-R/low-S signing. Electrum's pre-May2020 message-signing low-R grinding is a routine explanation, not a proved software fingerprint or hidden message.

Headers27/28 occur71/74 times. With `h=header−27`, numeric r-LSB counts are83/62 and s-LSB71/74. Header parity and recovered effective nonce-point y parity are the same channel. Low-S normalization may negate the original nonce point and flip the recovery bit.

The initial128 bitplane variants contain149,632 decoded bytes, longest printable run10 and no run16 or file-magic hit. The constrained natural-channel search has 4,160 labeled variants and 3,880 distinct streams totaling504,664 bytes. It tests h, the lowest1…8 bits of r/s, specified per-record interleavings, forward/reversed record order, bit/byte emission, polarity and alignments0…7. It uses no arbitrary sorting, permutation, subset, XOR-key search or guessed decryption. Results are zero matches to583 declared targets, zero validated files, strict whole DER or whole printable UTF8, and longest ASCII run12.

The byte-aligned framing checks give:

Test	Population/attempts	Authenticated hits
Distinct32-byte scalar windows, compared to145 signer x coordinates	387,024	0
Raw64 ECDSA windows / recovery attempts	280,888 / 561,776	0
Compact65 windows	277,840	0
Self-contained compressed-key/ECDSA frames	3,280	0
Self-contained x-only-key/BIP340 frames	387,742	0
Binary address/WIF Base58Check format attempts	4,874	0

Valid scalar parsing and generic ECDSA recovery are expected from random bytes. BIP340 against existing signer keys without an embedded key was not tested; external Q and native witness digests were not supplied in that run.

Header Bacon tests use29 groups of five bits and both modern26/traditional24 alphabets. Every tested variant has undefined codes; the best define27/29 and 23/29 respectively, with no repairs. A separate seven-combination parity check finds `h XOR r_LSB XOR s_LSB` at 91 zeros/54 ones. Its null model generates2,000 sets of 145 accepted signatures,290,000 total, using the actual digest, reproducible dummy keys, RFC6979, low-S and low-R retry with a little-endian extra-entropy counter. Eight sets are at least as extreme in that triple, giving plus-one p≈0.45%;31 sets have an equally extreme imbalance somewhere among the seven combinations, plus-one p≈1.60%, Wilson95% interval1.09–2.19% for the raw family frequency. This does not account for the broader research history and is not the probability of steganography. No payload, key or witness was extracted.

The May1,2022 public article about image steganography and a hypothetical embedded historical signature is thematically compatible, but postdates the 2020 signature event and supplies no extraction order, bit channel, key or validator for this artifact.

Native proof representations and the genesis quotation reconstruction

One transformation audit reproduces the original artwork exactly under the retained fitted recipe and converts a native Q witness into a compact recoverable representation. Both are checks of existing objects. The compact form carries the same r, s, z and Q; it is not a new signature over new content. No source-selected byte edit or destination rule was recovered. A separate source-selection challenge finds neither exact362 nor GenesisData103 independently selected as the next object on fresh NURSE ancestry.

A later transformation run nevertheless constructs a useful **audit-created** quotation capsule and rebuilds the native destination. It is evidence packaging, not a recovered puzzle serializer or new historical-authority proof. The final18 bytes of exact362, interval `[344:362)`, are the UTF8 bytes of `We are all Satoshi`, hex `57652061726520616c6c205361746f736869`. They equal the middle component of Genesis ExtraData at `[65:83)`.

The retained genesis configuration is3,110,907 bytes, SHA256 `be1ea67ecc33f1effe3bac86c7b23f94916bdc654cbf7db0a80e3244e6038206`. Its103-byte Data includes the literal Bitcoin block anchor, that quote and `Don't trust. Verify`. Native Data and Content hashes occupy one momentum preimage, reproducing `9e204601d1b7b1427fe12bc82622e610d8a6ad43c40abf020eb66e538bb8eeb0`. Content contains the exact prize-account genesis header, and the retained Burn names it as previousHash.

All145 original source-signature verifications pass; changing the quote or one source byte rejects it, with zero altered-source acceptances. Cyclically rotating whole address/signature pairs leaves their native verifications unchanged but changes the publication-order hash. This shows the distinction between signature authentication and an archival order pin.

The source quotation and destination commitment are real complementary relations. They do not show that the 145 signers endorsed Zenon genesis or the prize allocation, that the quote was copied specifically from this publication, that Q or historical Satoshi authored either object, or that the constructor intended this capsule. The genesis commitment can be checked without the 145 signatures; withdrawing them removes source-authentication evidence but leaves the destination quote unchanged. The candidate closes at an existing native object without a further selected transformation.

Version0.0.1 bundle recovery and the unexplained fourteen-byte change

JSON-only recovery reports preserve a separate historical-software investigation. Repository ID431136830 was created under the bundles name on November23,2021 and later appears as the official znn-bundle repository. The same v0.0.1-alphanet tag and Linux download URL have two directly recorded publication events:

Release	Published UTC	Asset	Recorded ZIP bytes
53930720	2021-11-23 16:09:23	50140138	11,699,175

Release	Published UTC	Asset	Recorded ZIP bytes
53937201	2021-11-23 17:38:29	50146093	11,699,189

The14-byte difference is exact metadata. Both original ZIP objects remain missing, so hashes, members, CRCs, timestamps, compression and semantic differences are unknown. Two publication events do not establish a deletion, retagging or rebuild mechanism. The December16 official homepage's stable URL, and later Darwin/Windows references, do not identify which particular asset bytes it served. The later complete release inventory and other-platform asset IDs were not recovered.

Six commit references survive while an archived branch page displays five commits. Full reachability is unavailable; no force-push or silent reconciliation is assumed. Exact tag target SHA, native repository history, package assembly scripts, first controller tree, migration/pillar documents and release-linked binary hashes remain missing. Successful exact searches for adjacent signature/checksum/manifest names return empty results; unavailable or timed-out requests remain distinct from successful negatives. No actual v001 signature or checksum was recovered.

The documented Zenon Testnet verifier fingerprint is `E2D8289C0A9D2C28A3320B62D2AAFD5D9088F1A2`, a different authority from historical PRIMARY. An older testnet `.sig` redirect is a different package object. A native downloads-page reference survives, but its body is missing; a recovered MDwiki shell does not contain it.

The build-causality ruling is B, PLAUSIBLE ORDINARY REPACKAGING. External components can be repackaged without changing the bundle repository, but no changed input is causally tied to release2 or its14-byte increase. A November27 community installation recipe references v001 but is not an assembly recipe. `make znnd → go build` describes the daemon build, not ZIP packaging.

Recovered controller dependencies are Dart SDK `>=2.14.0 <3.0.0`, dcli `^1.9.6`, linux_system_info `^0.0.7`, dart_ipify `^1.1.1`, random_string_generator `^2.0.0` and a missing local SDK, with no recovered lockfile. The CLI declares dcli `^1.7.1` but locks1.12.0, argon2_ffi_base1.1.0+1 and ffi1.1.2, plus absent local SDK0.0.1 contents. These gaps prevent a deterministic historical reconstruction. A later v003 checksum matches its recovered ZIP, but cannot authenticate v001.

Source-native literal scans find no qualifying historical-primary statement binding Q, the artwork or terminal. Gate4 remains EXTERNAL SOURCE BLOCKER / WITNESS EXISTENCE UNESTABLISHED. The best identified missing target is a custody-preserved copy of asset50146093, with 50140138 recovered separately for comparison. Filename and size alone would not authenticate either object. This historical source blocker does not establish that a hidden witness exists.

Constructor hypotheses and what has actually been tested

The primary-source hypothesis pass considered q/1 as missing signature metadata. Comparing the 40 row-major graphic bits, q=0 and 1=1, with the 40 effective nonce-point y parities already fails at 18 positions. It was not a new experiment after prior-work comparison. Later proposed shift/reversal controls were not executed or historically preregistered and cannot rescue the primary failure. Other literal-field and two-dimensional-program hypotheses have direct counterexamples; no replacement selected test survived that pass.

The hash-opening hypothesis is a fixed speculative construction over 19 native row/terminal pairs:

```
h_i = OS2IP(SHA256(compressed_SEC(R_i) || exact_38_row_bytes)) mod n
C_i = R_i + h_i G
predict x(C_i) mod n = terminal_input_i.r
```

Its exact hash domain, SEC order and row bytes are analyst choices, not recovered source instructions. The hypothesis report itself does not execute it. Later contexts mention a failed hash-and-tweak experiment as a supplied finding; the result should retain the status of its available execution receipt rather than retroactively turn the proposal into a test. A published same-key nonce relation could expose the construction scalar, but would not establish historical identity.

The constructor-mind pass preserves five speculative mechanisms: whole-set distributed nonce balance, a channel in the Base64 text, literal two-dimensional source, the first mnemonic's wallet trail, and the publication recipe itself. Its strongest surviving proposal compares the complete nineteen row and nineteen terminal nonce populations, with actual published signs and group identity as the target. It excludes funding signatures, subsets, sign search, non-unit coefficients and hashes. The report explicitly computes no sums, candidate key or control outcomes. It remains an unexecuted bounded hypothesis unless a newer receipt supplies the result.

The constructor-fingerprint synthesis finds the coordinated publication/collection recipe established. The disputed cryptographic renderer affects31 binary decisions beyond shared layout. A proposed conditional-description-length comparison would compare the same geometry with those31 decisions stored literally, charging selectors, indexes, operations, routing, defaults, exceptions and shared expressions fairly on both sides. It is unexecuted and not fully specified until the cost language is frozen. A nonnegative difference would reject explanatory economy for this renderer under that language only. Reconstructing the already known artwork is not an independent authorship check or held-out prediction.

Cross-object negatives and the controlling current boundary

The bounded cryptographic intersection audit uses392 ECDSA records:Q42, historical205 and message145. It finds no ordinary same-role or P/K cross-role match in44,425 pairs. Five digest variants across 205×145 give148,625 historical/message recovery cells, all zero; a10,000-digest null family also gives no joins. Registered RetrieveAssets comparisons across 3,192 authorizations give zero matches, including2,502,528 directional assignments and 2,342,912 A-versus-B record pairs. Comparing the selected145/205/P2095 keys against 5,002 SwapConfig and 110 LegacyPillar records gives zero in1,794,312 comparisons. These close only those populations and equality predicates.

The alert-letter recognition is also limited. ALERT49 AND KEY976 gives16 ACOUSTIC, matching the FINAL fingerprint's fourth field MIRACLE/BEAN/AWKWARD/ACOUSTIC. The recipient-published rendering has 38 exact BIP39 body tokens and two subject tokens, giving40 AGREE. It is not a native EML; no CALL field assigns AGE/ABLE as those counters, and no typed operation joins the count with ANXIETY80 to activate AUDIT120. The arithmetic survives; the activation is unsupported.

The controlling late October7 audit demotes AIR/ACTUAL from an independently source-selected frontier to a conditional research route. The first missing entrance is the exact selection of BUNKER243 and ARTWORK105, their order and subtraction, before the ciphertext match. Even granting that local selection, AIR43 does not independently produce CHOOSE322 or an ADD register; the phrase comes from the disputed mixed Q packet. Native mailbox recovery would not itself fix either entrance and is not demonstrated to be the immediate load-bearing task.

This correction preserves every exact packet field, local arithmetic, stored definition and downstream control. It does not prove that the constructor could not have intended the pattern. The latest architectural synthesis favors a completed primary treasure and a plausible unresolved local self-reading layer over an established unified historical-authorship procedure. Its narrow unresolved question is what the source-local four-symbol scale should consume next. No new source-selected experiment, historical-authority binding or universal absence proof was recovered.

Recent source recovery and bounded discovery campaigns

Reconciled later findings

This chapter covers the later source recoveries, tests and corrections. It distinguishes new evidence from repeated checks and delivery work. Its compact references use these source families: **F** for the October 11 forward audit, **OV** for the October 9 overnight investigation, **AU** for its later autonomous continuation, **LH** for the long-horizon production/source campaign, **AF** for the eight final mechanism tests, and **CTX** for original quest-context recovery. Exact source hashes, complete outputs and algorithms are retained in the evidence register.

The correction that controls the conditional AIR and attribution routes

F014 recovered the earlier Q-packet audit, its original projection code, the RECALL/AUTHOR replacement audit and the later frontier ruling. Six retained source hashes and sizes match. The recovered code fixes a mixed collection of fields, operators and positions and compares results with expected words; the report's order is not a native serialization order. The twelve numbers and trailer reproduce, but no independent original selector for the recipe, order or omission was recovered. The packet is therefore **researcher-assembled**. Its trailer cannot authenticate its own extraction rules.

The retained numerical packet is:

```
123 606 606 123 1664 1378 1536 1941 692 503 1973 104
AUTHOR ENVELOPE ENVELOPE AUTHOR SOURCE PROOF SCALE VERIFY FINAL DISCOVER WALL ARTIST
```

The older producer-swap and omitted-cell controls had already been executed and were reused. The omitted numeric cell is 1651 at bit interval [88, 99). No new projections, signature verifications or CALL executions were performed by F014.

This correction supersedes REV3's unconditional promotion of the fresh AIR/ACTUAL route. Its first unsupported edge is the ordered choice `BUNKER243 − ARTWORK105`. Even granting that subtraction and the Email 260 domain, native tag 18 endpoint AIR43 does not independently supply CHOOSE or an incoming ADD operation. NURSE, ARTIST and proof paths remain conditional descendants of the declared model.

The record-replacement audit also preserves the exact `QUOTE[AUTHOR] = SATOSHI` representation while withholding an executable update. QUOTE's actual return to NURSE is GAZE → CRUMBLE → COUNTRY → DINNER; its join activates RECALL. NURSE's actual join is CARRY279 + DINNER498 → GENTLE777, whose stored path ends CARRY. Its PC reaches BODY → AUTHOR → DEBRIS → AREA. CHECK[ADD] outputs FOLLOW/DRINK/DEMISE, with AREA/AREA child returns and a BIRD join; it has no quote bytes, author credential or signature-verifier input.

No retained instruction tells this NURSE occurrence to retain its QUOTE/RECALL record, replace the low-seven-bit payload with this BODY/PC AUTHOR payload, keep the same object referent and apply a claim-acceptance predicate. Row 17 supplies signature-framing and coordinate evidence; it does not install a 4|7 parser. The retained canonical-return ledger has 433 records and no RECALL, QUOTE, AUTHOR or SATOSHI return; the earlier 1,864-template ADD closure contains no SATOSHI or OUTPUT. These are bounded stored-model negatives, not a proof that a human record-edit interpretation is impossible.

F014 preserves Q authentication, AES completion, artwork coordinates, all arithmetic, QUOTE → RECALL and F013's signed native receive. It changes admission and interpretation, not those results. Later source and mechanism campaigns explicitly retain this ruling.

The preserved eight-word image has a community Twitter origin

CTX corrects the earlier description of a missing “Telegram seed card.” The retained image is a screenshot of a community Twitter thread, not evidence of an original Telegram publication. The exact source is post 1460002976907595779, natively dated 14 November 2021, 21:53:29 UTC. Its Snowflake encodes 21:53:29.895 UTC, consistent with the rounded creation time. The text gives the established eight-word prefix. It has no native attached image: line wrapping belongs to a later screenshot and does not establish an authored spatial alphabet.

Current syndication reports one edit-tweet ID and `isEdited=false`; current oEmbed independently returns the same caption and source ID. These are two interfaces to one platform, not independent historical witnesses. No independently archived 2021 body has been recovered. The reporting account differs from the original issuer; its text is not promoted into a Zenon-issued instruction.

Actual reply pointers give this chain:

```
1460002969475338246 → 1460002974697197568 → 1460002976907595779
```

Post 2 describes four encoded OP_RETURN fields as the missing words for the prize wallet. Its three attachments show a contemporary artwork reconstruction, the known terminal transaction in block 709632 and the issuer's Day 5 announcement. The last matches original issuer post 1459863534330957825, dated 14 November 2021, 12:39:24 UTC, directing solvers to the final reward piece. It supplies the original four-word challenge context, not an eight-word disclosure or post-mnemonic task.

Post 1 is an author-deleted tombstone; its exact oEmbed returns 404. Exact archive-availability queries for posts 1 and 3 return no closest capture; the separate post 3 CDX query timed out. A timeout is not evidence of absence. Post 1's caption, media and outgoing original-source links remain unknown. The March 23, 2024 preservation post already held the source onebox; this was recovery of an overlooked pointer.

The preservation JPEG is 830×634, 54,730 bytes, with no EXIF publication metadata. SHA1 `dfa230e770088e1f93a8cc4456a2db9336cbdb11` matches its content-addressed filename; SHA256 is `d7235c0da85472e6a520856bda915aafbc66ac74bc7f77bf1bcef4ba5ae21b89`. These identify the preservation object, not its 2021 origin. Native post 3 SHA256 is `54bc7f55ee4ba80288d5411f78857227a8c0355bbb0d0bfd3a565bbc3add06b9`; post 2 is `570443eaebfe1cec0e411346b167f1f585f1f54a7dee45214a3feaf429205514`.

The expanded finite publisher population contains 211 captions: 207 native syndication objects and four original-archive captions. It includes all 78 November–December IDs returned by the archive index. Literal lookup finds neither the four-word prefix nor the complete eight-word sequence. Deleted and uncaptured posts remain outside that claim.

CTX also followed two exact earlier issuer quest records: 1407672390994309123, June 23 Mars/fusion capacity, and 1409601320449974275, June 28 Golden Record/Interstellar Explorers. Each video is 15 seconds, 720×720 and 450 frames. Fifteen evenly spaced samples show rotating rank, reward and branding cards without word disclosure or a new instruction. Each entire audio stream has 1,441,792 float samples, all exactly zero. Optional all-frame OCR failed before execution because Swift imports could not build; no exhaustive visual transcription is claimed. A later recollection served only as a locator. The June public-reply QR is a different object.

The remaining source gap is an original issuer-controlled first/middle-four-word disclosure, dated quest notification or app record with surrounding instructions. Deleted community post 1460002969475338246 may clarify the reporter's provenance. Neither gap justifies guessing message IDs or assigning screenshot wrapping a decoder role. The recovered context supports the completed treasure objective and supplies no independently selected second-stage verifier or destination.

Additional residual, contour and ABOVE-four observations

The later machine-only receipts preserve a conditional local verification loop. They do not add an outgoing command. In these receipts R means the 22-bit representation remainder and glyph-Q means the lower q/1 carrier, not an elliptic-curve point.

```
R       10110010110010011011
glyph-Q 01011101001100010000
AND     00010000000000010000
XOR     11101111111110001011
OR      11111111111110011011
```

OR gives ZOO2047/VIBRANT1947. Trimming two bits from each end of its second eleven-bit word gives seven source bits `1100110` = ART102. It is not an eleven-bit center; other trim widths and operations give different values. Shaving both ends of the entire 22-bit record gives `1111111111100110`. With R fixed, the same full OR has 4,096 arbitrary 22-bit preimages; this is an information-loss count, not an author-intent probability.

Upper 18 is `10000100010000100`, producing LION1041 and seven-bit remainder `0000100` = ABOVE4. In the selected lower center, R=`0100110`, glyph-Q=`1000100`, XOR=`110001`98, AND=`00001004` and OR=`1100110`102. The upper remainder supplies that center's overlap and restores ART, but alone leaves the full 22-bit mismatch at position 4. Shape-preserving normalization of the row 8 q/1 exchange changes the upper mask from 4 to 8; only the original mask matches this selected center. The arithmetic does not independently choose the seven-cell window or OR operation.

The later ABOVE-four test examines all 16 seven-bit windows. The original mask matches only lower positions 14–20; the normalized mask matches 1–7 and 15–21. Lower ordinal 18 at (17, 27) is a 1 and an AND overlap. Moving four rows upward lands on lower ordinal 4 at (13, 27), also a 1/overlap; among distances 1–5 only four does so. Those two sites reconstruct the complete AND mask at positions 4 and 18. Applying it to XOR restores the full OR record. Expected uniqueness and distance 4 were user-supplied, so these are confirmed bounded observations, not independent holdouts.

The complementary deficit `2047 XOR1947 =100` gives ARRIVE, as does subtraction. Among 40 previously exposed complete field/residual words, ARRIVE100 occurs only in fieldA group 8, decoded bit offsets 77–87 and artwork cells (1, 20–22), glyphsKGT. That is a distinct source occurrence. It lies in row 1, outside the lower-right-stem scope of ABOVE-four; converting physical movement into dictionary positions would require another rule.

A literal contour TOP4 selects (12, 14),(12, 15),(13, 17),(14, 19), glyphs; . , : →3, 2, 1, 0 and byte`11100100`. A second copy translates by three rows and seven columns. Reverse/control traces also give four distinct glyphs, so distinctness is not unique. This is contour self-reference, not a binary trimming or OR instruction. ART102 returns to TRY MAKE CUSTOM SCALE and the same ZOO22 state; no further action is supplied.

Other preserved controls retain their limits: ordinary row-parent amounts are constant; standard-entry Funge 98 remains onrow 1 and never executes q, 1, z or output; both actual and regularized carriers fail a simple flat straight-edge density model. The midpoint 1 at (14, 18) between q sites (13, 16)/(15, 20) rejects that simple model, not stateful/font-aware conversion or manual edits. Native padding, nonce, byte-location, row-location, raw-codebook, negative-space and checksum tests supply no complete new instruction. In particular, the proposed artwork checksum gives the same result on an all-space control, and selected compact header bytes have an ordinary VeriBlock-format explanation without establishing foreign-chain membership or a puzzle entrance.

The historical release trail adds real commitments, not Q authority

OV-C020–C031 follow a literal request in the published Email 260 copy to post a release announcement to bitcoin-list. SourceForge message 27102906, dated 23 February 2011, 01:22:25, is recovered through an archived month index containing one post and an independently archived message with the same four filename/SHA1 pairs. These are public publication records, not native RFC5322 transport or sender signatures. Archived original index.

The initial Linux package is recovered under mirror name `bitcoin-0.3.20-linux.tar.gz`; its 9,833,353 bytes match the original `.01` announcement:

```
SHA1   7dfbc05b36112f59886a29f044cfd21c6c253169
SHA256 5c6a52934c1505f3d81fc05b620649629e13b402cd521b32e7aeedf8d0f17c4e
```

The accepted signed tag `90b4ed72f86e1d7f6196f9f9be0911a79f9cbb09` commits transitively to this native ancestry:

```
52248a757b25581339ca6e00cc0da05bfdb04dc8
→ d53070ceb0487ebcf0eb7afb12c0be2dbf9fd4d2
→ d787e00a3bb47c513566a8f9baef5b6b3c6d93b4
```

The new capsule checks 22 raw Git trees, three commits and two parent links. Of 113 package source members, 106 match committed bytes, four differ and three are absent. The differing files are `serialize.h`, `makefile.unix`, `init.cpp` and `build-msw.txt`; absent files are `changelog.txt` and two Crypto++ object files. Changed archive/tree/parent/root/endpoint controls fail. This authenticates the matching source bytes relative to the accepted signed anchor, not the entire unsigned initial archive, its binaries or declared human authorship.

The four differences concern ordinary beta-flag removal, Boost suffixes and older Windows documentation/copyright. Package beta-flag references are 0 versus 3 in the signed tag; all 12 commits in the fixed March 3–6, 2011 population match 0/4 differing files. The initial incorrect `src/` namespace assumption is retained, with the source-declared root layout used for the correction. No broader historical worktree census is claimed.

All 16 pairs between the two four-file commitment lists were compared. Only position 4/4 matches filename and full SHA1. Signed README line 30 selects `bitcoin-0.3.20.01-macosx.zip`, SHA1 `b540825d864e7561cc21465ad072fb299e0d817a`. Its bytes remain missing; a known later Linux control fails. The fixed local census of 68 ZIP containers and 8,485 members has 0 errors and 0 matching original-name/alias members.

The separately selected topic 3958, Bitcoin Formula for Mac OS X, names the same archive and recipe commit `eca924cddd868613937d692bab2c184187610020`. Its 70,315-byte page has SHA256 `07711a3c61fab1b186877bf02c063e545dbdc1f4750f46e293cef37db158f962`. The original repository and native upstream `Homebrew/legacy-homebrew` both report that fixed commit missing; exact-index/archive queries return no copy. The native HTML link at line 215, column 99 under declared ISO8859-1 exactly matches the failed target, excluding a display-link transcription repair. A nearby/current recipe cannot replace the missing object. These are bounded availability findings, not historical nonexistence.

The archived list service identifies its administrator, and account 2321442 displays Satoshi/Bitcoin. The original reply explicitly permits subscribers to post; posting does not prove administrator authority. Owner labels, copyright and build paths do not authenticate Email 260's sender or Q. Archived service record.

Native protocol controls also matter. WinPT `r273` identifies 1.2.0; NEWS names the 9 December 2006 release. `r276` identifies 1.2.1-cvs, with eight frozen source siblings unchanged. Encrypt calls `gpgme_op_encrypt`; Sign&Encrypt calls `gpgme_op_encrypt_sign` and selects a signing key. Both append the same Version banner, so that header cannot distinguish signed encryption from encryption alone.

A public synthetic ElGamal/AES-CFB-MDC fixture reproduces the selected endpoint values with a different full hash; its equations and integrity verify without historical secrets. C017/C018 accept explicitly supplied qualified session/exponent witnesses and check equations, padding, checksum, MDC and any inner signature, with wrong-key/exponent/body/source/certificate/RSA-coercion controls. Isolated public-only GnuPG agrees on 112 literal bytes/GOODMDC/no signature for the synthetic fixture. This is a tested opening interface, not an opening or sender authentication of historical Email 260.

A new certification and an ordinary software-verification role

AU finds one existing certification by the selected recipient's signing primary `01CDF4627A3B88AAE4A571C87588242FBE38D3A8` over later code-signing primary `26646D99CBAEC9B81982EF6029D9EE6B1FC730C1` and its UID. Personal name/contact display is anonymized; the signed object is identified by hashes. The certificate container has 49,916 bytes and SHA256 `0c7dd8ad03625b01094aa266392b8cac52294eb2a5f745eef84ef0a5dbe8411a`. The zero-based signature packet is index 4, byte offset 1724. Signature-body SHA256 is `17beb0a60718842b0f6a8d9d23e8dbf3c7c890b03e853ae9191e094498a39ed4`; signed-preimage SHA256 is `f63e391474ee23736e2057937fd0fbd1469b56458f651bac20bb4f31073ca493`.

Across 142 original certifications, exactly one validates under the fixed recipient key; 53 compatible DSA signatures fail and 88 RSA signatures are incompatible. All 142 digest-prefix checks pass, with 0 parser errors. Independent DSA arithmetic, OpenSSL and GnuPG agree; a changed preimage fails. The signed creation field is 2011-12-15T20:19:39Z, not an independently established publication time or civil identity. The larger certificate is distinct from the selected recipient's original certificate. No certification by the fixed Satoshi key appears among 54 compatible DSA signatures.

Original Bitcoin commit `0920ad152fc8ec0486b360a9b6fe04ada3abba39` introduces old-key blob `f81f44e874707837c13446858db2fed14ad5c501` and config blob `d21bb07808b2617e3500159a4f622a7ca9950be3`. Its five signer weights 40 and required 120 implement a concrete three-of-five release policy. The original recipient is one signer; the later certified key is absent. This supplies an ordinary reason for

the public key's packaging. It does not sign Email 260 or Q, supply a fresh ACQUIRE caller/ADD/register, or bind a same-caller source-author and visible-proof pair.

Published Email 260 remains a COPY. Native MBOX/EML bytes and complete transport envelope are missing; Email 244's encrypted root-password attachment is a separate candidate. Valid release signatures, recipient correspondence and certification do not establish the encrypted reply's sender or historical Satoshi authority over Q.

Publisher article recovery and its inaccurate first-Taproot claim

AU-E021–E028 recover the exact Taproot anniversary article through declared static modules:

```
DPSyIeXC.js → CNfascyA.js → Cfu3cxGT.js and DFpsC-4O.js
```

The English object occupies UTF8 bytes[152127, 157203); its 4,017-character text hashes to `666cf8a87c46eed3b475de6d6a60b65a80748f3d449695b00ca27178b883f64f`. Metadata matches the retained original page; an independent restricted JSON/AST parser reproduces the complete tree/text, while the selected MESSAGE sibling fails article-metadata controls. Downloaded JavaScript was not executed. Earlier body-unavailable findings are superseded.

The article's transaction link names the existing terminal transaction. Its two MESSAGE links select `/en/message`, whose 6,900 characters concern project purpose, architecture and unidentified founders, with no historical-source link or author-proof instruction. Current assets alone do not establish the declared 2023 publication date.

The article claims the first Taproot transaction occurs in block 709636. Preserved native records contradict both ordinary interpretations: block 709632 already has 14 P2TR outputs across 9 transactions, and `83c8e0289fecf93b5a284705396f5a652d9886cbd26236b0d647655ad8a37d82` has four previously verified Taproot inputs in 709635. This does not determine the global first-spend height; it does prevent treating the later “fourth block” prose as an accurate original selector.

Two declared image bodies remain unavailable 403: `meta-two-years-since-zenon-network-engraved-in-bitcoins-taproot-upgrade.png` and `taproot-btc-logo.png` under the publisher's `/images/contents/blog/` path. Original og/twitter metadata confirms the social-card URL. The 68-container/8,485-member and 33,025 loose-name local census has no exact copy. Unknown pixels are an availability gap, not a finding of no instruction. Accepted article and certification claims remain independently reproducible from portable public inputs.

Entry 123, collective-key controls and native ARTIST delivery

F's Entry 123 observation is source-local: terminal RECORD/AUTHOR fields `11|30` and `0|123`. Reading “record this author” motivates replacing the first payload while retaining tag 11, yielding `11|123` = SATOSHI1531. This improves operand provenance but leaves assignment and consumer selection conditional. The exact 362-byte signed message has one Satoshi token at[355, 362), in final phrase `We are all Satoshi` at[344, 362). Its signature authenticates that assertion under the selected key; it does not bind Q or establish civil identity.

The complete fixed population has 145 packets and 1,595 adjacent pairs. Exactly one packet has terminal tags 11/0: Entry 123, raw-checksum valid and self-addressing under displayed one-based

numbering. Five pairs qualify at any position; the other four fail raw checksum. Zero-based self-reference has no match. These are postselected specificity counts, not a blind attribution proof. The earlier five-revision publication study shows the June 2020 Gist mirrors an anonymously published May 2020 object; mirror publisher, signer and Q controller are distinct roles.

Native addition of all 145 signer public keys gives `03804557a4c2f864f1c7ab4af653e50504b63f7c26939c7940aaad3b970d1057d9`; its standard empty-tree BIP341 tweak gives `0308082eb9e5c7ae61025c08247845c9ec6ec75529a00f71b50b49527b328f22a3`. Neither matches Q, including x-only comparison. Python affine arithmetic/OpenSSL agree; seven BIP341 vectors pass and all 145 leave-one-out sums miss Q. Only these fixed relationships are rejected; simple addition is not secure MuSig.

The selected author's key already appears in coinbase `6901a5bbfe0dc6d1eef3a96574f8dd8836e10d2f94a38fdf3505d01ace4a688f:0`, block 32897, with no selected outgoing instruction. The unchanged X/Y coordinate reader yields no complete actionable message and fails both reduced checksums; two sibling controls pass, showing that checksum validity alone cannot select continuation. The CODE/SPLIT low-seven stream is `h\x11o:\x02\x15\x1bCQ\x1e[|`, with high-nibble hex `2d4ffb9699bc`. Prefix `-O` alone names no program/operand. Own-field `] ESC ETB` or `ESC ] ETB` is not a complete command.

F012/F013 follow conditional ARTIST's native CALL104 object `126aa9b421f854824e7c993a98ad1b9beeaabee4741da4b1394a17d0e9f2d5b3`. It authorizes an ordinary 2.35985 ZNN swap; its key-id uniquely matches genesis entry 2470. The protocol Mint payload names exactly the amount and destination, with no extra bytes. Its complete path is:

```
RetrieveAssets 126aa9b421f854824e7c993a98ad1b9beeaabee4741da4b1394a17d0e9f2d5b3
Swap receive   324e0118dc1025f36be7c6d4dfae3b6bd6ad75901d1c532b20a74d45ab915f4b
Mint request   cc5d641d950907631e0d702263654284a339fef53f414e256eb5a32b2bead931
Token receive  958fe256efa51e934f823fc534e47754cddf3eaef3f90c241fdd84a12cbaf2e9
Token transfer 074bd9d82e3d058a170b2a60b36412908bab599b16507713df0b27b534b3d372
Recipient      905e55a6327916d671de6833a6792920cdc0e0c429ba257811a18cc6fd8cff82
```

The final object is a newly authenticated Ed25519-signed height 2 receive for `z1qztc5y325kf5nawlzfy63yd939cn6j2u9cm7qp`. Native serialization, amount, destination and paired links pass; changed digest/preimage controls fail; two providers agree. Contract blocks are unsigned by design, and this is not independent full consensus validation. Final data is empty with no descendants, closing the selected delivery route; later account activity was not followed. F014 preserves this result while qualifying its puzzle entrance. It is not a new attribution statement, Q relationship or puzzle instruction.

Rendering: failed reproduction does not isolate intentional encoding

The production campaigns broaden ordinary alternatives while preserving the verdict **second stage plausible but unproven**. Tested pipelines reproduce some silhouettes and individual punctuation choices; none passes the fixed punctuation adequacy gate needed to interpret q/1 failures selectively. Different inputs, font-sensitive conversion, nonlinear tone/scaling, re-typesetting and ordinary retouching remain possible. Manual editing alone would not establish a message.

Early fixed nearest-level/Floyd–Steinberg tests use the archived 2021 SVG and 2019 raster. Their overall scores are 551, 544, 552, 540 out of 642, but nonblank scores are 206, 199, 207, 195 out of

290 and punctuation scores 24, 21, 25, 18 out of 58. q/1 identity scores are 9, 8, 9, 8 out of 40; all four predict z at both exchanged row 8 sites. Correctly composited raster predicts the original period at (9, 32), unlike the vector's comma. That ordinary counterexample weakens the claim that this period necessarily carried a deliberate cross-check. The initial raw-green implementation ignored 7,221 partly transparent pixels; the retained correction composites RGBA over black without changing fitted settings. Earlier ink-order accuracy 11/40 and row 8 accuracy 26/38 are reused results, not new tests.

The actual historical tool ascii-image-converter v1.11.0 predates the artwork: release 2021-10-06T10:50:05Z, commit `f31ac047eaae6cd0b24a371037fd99c0a5fc4a9c`. Original release archive SHA256 is `4d776ed919cbd15eaa63da45319453fd888feca3e3d21a4b3bfe009b3c22d32a`; macOSARM64 executable is `e9b77f2158ceee62e7fe6a57ff096c2bb488fa0df6752040d8b6e20eb97280c1`. Original library, local CLI and original release agree in all 36 comparisons: six prepared images, three palettes and two CLI implementations. This establishes historical implementation integrity, not actual artist use.

The historical 939×1499 raster hashes to `fa8ee856d77d28675c268714904eecc52bb3f06d08fd9753e337937c907b813e`. The 48×77 SVG hashes to `55ff6ad052fb21bb4111ad68b922f475606275369caadce1ff36005a7e4dcaf3` and is identical to the modern SVG; its shape was available before the challenge. A contemporary tool recommendation is not a production recipe.

Initial historical-tool training sees 513 cells, 326 spaces and 187 z, excluding all q/1, punctuation, row 8 and payload cells. Four frozen color/silhouette treatments score 369/642, 518/642, 361/642, 520/642 overall and 0/40, 4/40, 0/40, 3/40 on q/1. Punctuation scores 17/58, 2/58, 9/58, 1/58; row 8 scores 26, 29, 27, 31 out of 38, with neither exchanged identity recovered. The SVG gradient reproduces (9, 32)'s period while missing every q/1 and all 192 z cells; silhouette matches do not explain identities.

Native palettes score 0/40. Twenty-four direct 38×19 and 26×19-plus-padding controls score 1–5/40 and never recover both exchanges. Native CLI rejects ASCII `--dither` without braille; its library ignores that flag outside braille, confirmed by six rejection and six output controls. Untouched raster/gradient maximum grayscale 163/194 cannot reach the fixed eight-character floor lookup's z threshold 224. White silhouettes are explicit derived workflows. Historical SVG rasterization required a documented forward-gradient/target-scale compatibility wrapper; the original black failure and analytic RGBA controls are retained, with no glyph fitting.

The later punctuation-calibrated experiment adds 28 punctuation training cells and global levels/gamma. Outside row 8, odd class ordinals train and even validate: periods 4/4, commas 15/15, colons 3/2, semicolons 6/6. Training totals 541 cells; 27 validation punctuation cells, 40 q/1 cells, row 8 and payload are excluded. This is a procedural holdout after prior researcher exposure, not blind data. The gate requires pooled punctuation≥80% (22/27), macro≥75% and every class≥50%.

Profile	Punctuation validation	Macro	q/1 identity
Raster color	8/27	26.67%	5/40
Raster white silhouette	16/27	38.33%	9/40
SVG gradient, training-selected primary	10/27	35.83%	3/40

Profile	Punctuation validation	Macro	q/1 identity
SVG white silhouette	15/27	32.08%	9/40

No profile passes. Primary training is 518/541 overall but 17/28 punctuation; validation is period 2/4, comma 4/15, colon 0/2, semicolon 4/6. Only 7/40 true carrier sites receive q or 1, 3 receive correct identities and 33 receive other glyphs; nine other cells are false-positive carriers, giving 7/16 location precision. Constant comma scores 15/27 punctuation and constant q 27/40 carrier identity. Primary row 8 score 31/38 comprises ordinary spaces/z and misses both exchanges; white SVG gets one exchange but is not promoted afterward.

The 1,367 geometry evaluations and 261,166 tone candidates are bounded training search, not independent hypotheses. Operation order is placement/compositing → historical resize/grayscale → uniform levels/gamma → historical conversion. Formula `T=floor(255*clamp((I-B)/(A-B),0,1)^G+0.5)` is external preprocessing. 256-tone and native-binary controls pass; coordinate shifts support rough alignment, and 999 fixed label permutations are descriptive controls, not intent probabilities. The empty monotonic-contradictions table means q/1-specific testing was not admitted because calibration failed, not that compatibility was proved.

LH then selects genuinely different original assets before inspecting their rendered outputs. The fixed ramp remains `.,:;1qz`; old geometry/tone profiles transfer unchanged.

Test	New input and provenance	Complete fixed result
T01	September 4, 2021 publisher `img/znn.png`, 189×307 RGBA; SHA256 `9f51bf9a46f8c3055f5d9d161c8f304dccec0ceb22e4898dd1bc6b2c4ea083d5`; original CSS makes it a decorative card background. Alpha 0–80 and black-composited gray 0–20 distinguish it from prior sources.	Six punctuation scores 0, 0, 3, 1, 0, 3 out of 27; q/1 scores 0, 0, 2, 0, 0, 5 out of 40; no exchanged pair.
T03	May 25, 2020 native commit `5c0550ec275e51727ae01c6e949ed2a5b6cdd8b1`, original Qt splash imagery. 2× splash SHA256 `5363ec5b44f6de9db6a93e16326e1a10da0788f78d04c92c9a13c8be25126053`; 1× is `79f3152a9ca2dcc78b130093c502314515e7c66b970d688281557c2c739c5151`.	Eighteen profiles all fail; maximum punctuation 11/27 and carrier 12/40, primary 2/27 and 2/40. Source-only crops 2×[163, 181, 437, 619), 1×[81, 90, 219, 310), full 600×800 padding control.
T04	August 11, 2021 original homepage Lottie final frame 59, op 60 exclusive, selects `image_31/final49.png`. SHA256 `a7ff5638278ed02d06469f151002d9192cef16cc20ce1bfa6f6d6bd5226f2385`; visible-component crop[770, 167, 1151, 777), 381×610. Stem ratio 0.202100 versus older 0.174654.	Six punctuation scores 2, 4, 7, 0, 3, 0 out of 27; q/1 scores 5, 0, 7, 1, 0, 2 out of 40; no exchanges. All 33 embedded PNG Creator fields name After Effects; animation data/runtime 5.6.6.
T05	Newly authenticated April 11, 2019 raster, 501×800, selected through original 2019 homepage/runtime asset ancestry described below.	Six punctuation scores 5, 4, 2, 0, 4, 1 out of 27. Primary 519/642 overall, 3/40 carrier identity, row 8 score 22/38; both exchanges become z and (9, 32) becomes semicolon. All 3,852 nonpayload predictions and 722 tone/ramp cells per profile reproduce independently.

Test	New input and provenance	Complete fixed result
T06	One conditional native aview/AAlib pipeline on the recovered raster.	First-call and warmed exports both 0/27 punctuation and 0/40 q/1. Warmed overall 126/642, row 8 zero; q (8, 10)→n, 1(8, 16)→underscore, period (9, 32)→S.

The 36 no-fit transfers comprise T01's 6, T03's 18, T04's 6 and T05's 6. All fail the punctuation gate; T06 is a separate failing control. These counts do not erase the earlier four fitted calibration profiles. All named artwork targets have been exposed; none is a fresh blind holdout. Failed adequately recorded pipelines do not establish encoding.

Older production objects, native AAlib and a rejected rotation explanation

LH-C034 authenticates the June 20, 2019 homepage:239,053 bytes, SHA256 `4a3409e4b2a70928d9e8514f753c9ac827ece3d2529dc045d735454463971c4b`, CDX digest `CPQQXMW2ZDHMP46MTVN7UVEON57JEIQK`. Embedded JSON occupies bytes[107861, 237954), 130,093 bytes, with 88 shape layers, nine precompositions and one raster. Exporter 5.1.13 is distinct from runtime version.

The separately archived June 25 runtime declares 5.1.16; native `getAssetsPath` uses `assetsPath+p`. Exact June 20 runtime equality remains unproved. A frozen nearest-date rule selects the April 11 PNG:2,184 bytes, 501×800, opaque green RGB (0, 247, 33), SHA256 `b6d6793d9ac4530c2b11fd30e1c12759ffd60aac84f82e6a697c52ce081aab79`. Raw SHA1/base32 `HS3A2PLZYVALYQ6PUIUQE3QM4GM6HMVP` matches both indexed captures. Fixed descriptive comparison gives silhouette IoU 0.9246422785 and 4.95509% differing pixels against the older raster. This was not an alignment/glyph fit. Actual ASCII-input use remains unproved.

The June 25 companion script also authenticates:852,325 bytes, SHA256 `bebdc03947723fafeb7389d439232f035fc71a12b434c55c4f7c1d00ef985d5d`. Its `startGalaxy` creates 100 random white SVG circles and rotates them. Complete static UI/Lottie inventory supplies ordinary website behavior, not an ASCII recipe or continuation. No downloaded JavaScript or child program was executed. Exact June 20 equality is unknown.

Original AAlib `1.4p5` source, SHA256 `fbddda9230cf6ee2a4f5706b4b11e2190ae45f5eda1f0409dc4f99b35e0a70ee`, is recovered through official Debian August 2021 Release/index/archive checksums. Code is unchanged, with standard compatibility headers; Release signatures were not independently verified. This native algorithm uses four subcell samples and font shape rather than only a scalar brightness ramp. Both memory defaults and actual plain-text export automatically emit q, 1, z, period, comma, colon and semicolon. Complete 65,536-entry tables, 256 flat tables and repeated actual witnesses are retained. Vocabulary eligibility is an ordinary alternative, not proof of artist use.

A source-derived uniform-field prediction was frozen before output. Across 256 uniform 76×38 inputs, all 184,832 output cells belong to the native flat table, with zero q/1/z/comma; diffusion is active in 242 frames. This verifies an ordinary algorithm property, not a new artwork decoder. Original aview's 34 source files fix PNM grayscale and stateful sampling, while font, viewing geometry, alpha and

export choices remain unfixed. Unsequenced behavior in native `aamktabl.c` limits the measured tables to the recorded Apple Clang 17, -O0, arm64 build rather than all historical compilers.

LH-A39 rejects the specific q/1 rotational-swap explanation for the retained font 16/NORMAL table. Physical 180° quartet rotation uses native TR|TL|BR|BL packing. Complete 65,536 raw and 65,536 actual-default populations have zero q→1 or 1→q. Generic identity changes are common:58,732 raw and 58,706 actual pairs. Actual execution uses 166 fill-branch inputs and 65,370 LUT inputs. Identity, involution, fixed-quartet, fill and attribute controls pass. Other fonts, full-frame diffusion and unknown inputs are not excluded or searched afterward.

The printer boundary is also explicit: native pitch ratio 2 differs from publisher 2.4745817, so uniform resize/padding alone cannot explain it. Re-typesetting and nonuniform scaling remain possible. Original SVG/raster/splash resources remain unchanged across the selected authenticated legacy PR/release consumers; ImageReady/Illustrator metadata describes ordinary production facts, not an ASCII recipe.

Other long-horizon source recoveries and finite closures

The long-horizon campaign recovered original material and completed all currently qualified source-selected consumers without positive stage evidence. Its 103 ledger records include source recovery, experiments, delivery steps, coverage pointers and an explicitly unregistered screen; they are not 103 independent predictions. Productive execution lasted about 4.4 hours from 2026-10-11 00:08:24 UTC, with verification/packaging afterward, not the requested 12–16 hours. Missing/deleted history is not globally exhausted.

Punctuation specificity, LH-B04. The known two 15-cell interiors have upper counts period 3/colon 2/comma 7/semicolon 3 and lower 4/3/4/4. The same original features, sixteen traversal pairs, last-four-bit one-based own-field index and four-distinct-glyph predicate were retained. Eleven-bit prefixes impose no alphabet-admission constraint because every U11 value is valid.

Predicate	100,000 region-preserving controls	69,984 exhaustive within-row controls
At least one surviving alphabet	53.552%	59.162%
Exactly one alphabet	35.311%	35.717%
Frozen alphabet appears	6.004%	11.883%
Frozen alphabet is sole alphabet	3.100%	4.635%
Exactly two surviving traversals	18.179%	21.954%
Full known summary: two traversals, sole frozen alphabet	846 cases, 0.846%	908 cases, 1.297%

The exhaustive population is 486 upper arrangements × 144 lower arrangements. The regionwise unique-alphabet 95% sampling interval is 35.015–35.608%; the exhaustive family has no Monte Carlo sampling uncertainty. All arrangements and 16 outcomes are retained; 128 independent scalar comparisons agree. Seed 12012536302680555934 derives from the frozen contract hash.

Uniqueness alone is common under these constraints and does not provide strong independent evidence. The fuller summary is more specific but remains postselected: it does not account for guided morphology, split, traversal, index rule, later readers or other attempted decoders. Neither population is a full renderer null or an author-intent probability.

Earlier wallet riddles are a different admitted source. Original May 28, 2021 status 1398362494142459917 → `https://link.medium.com/Tyks8cOpDgb` → Medium object `591db5fc9bc9` → same-day capture 20210528194717 → paragraph `30cc11df341a_29`, name `4545` explicitly directs users to hidden riddles/Easter eggs in the Public Incentivized Testnet syrius wallet. A genuine June anti-bot QR task also exists. They justify an earlier testnet investigation, not a Taproot post-mnemonic entrance.

Three selected original bodies yield 31 testnet paragraphs, 39 AMA answers across 41 paragraphs and 38 transition paragraphs. AMA Taproot marketing and ordinary Genesis Assembly/network-backup instructions name no new puzzle verifier. Original publisher records identify testnet GUI v4.0.0 on August 25 and v5.0.0 on November 3, but exact GUI bytes and dated authenticity binding remain missing. The backend node bundle and legacy Qt software are different objects.

Official syrius Git starts 26 November 2021 with an 83-byte README placeholder; tags v0.0.1–v0.0.4 point to it. First retained implementation 0.0.5, 21 April 2022, is explicitly Alphanet. A 2022 unversioned bundle capture is only 301. A fixed official-origin archival visit returns 404, not universal absence. Original MDwiki 0.6.2 March 26 loader, April 14 navigation and a matching October 19 digest supersede the earlier December-only chronology limitation. SDKs is an anchor in `downloads.md`, not a separate resource. Exact earlier GUI entries and `downloads.md` body remain missing.

Original release identity is recovered. The 11 September 2021 snapshot binds 1.6.5-alpha to native commit `54e3137aff44391bb998927d9b6d8e3b6723c304`; snapshot, commit and parent reconstruct. Native mirror bytes match despite a retained raw-API failure. All 160 branding resources/startup loaders are unchanged; changed code implements ordinary Alphanet swap export and its `.swp` consumer supplies no post-mnemonic destination. Full source authentication covers 1,436 files, 17,282,248 bytes and 111 trees. The 15,796 frozen contiguous Q/script/artwork comparisons are negative; a separately source-selected canonical 88-byte Base64 uncompressed-Q form is absent from all 1,436 files and 39 old controls. These close three literal representation families, not computed commitments or every encoding.

The older original commit's 110 trees and 30 selected blobs verify, as do two archived unmerged commits. The 2021 PR uses known resources; the 2019 PR changes ordinary Qt styling, includes and governance-icon registration while registered logo/splash bytes remain unchanged. All three previously unread platform-icon containers were later consumed: thirteen representations and ten controls verify. Mac JP2 metadata names JasPer 1.900.1, reversible 5–3/no quantization; most differences are background. Original Finder metadata has zero declared/reachable standard records, verified against native DSDB/leaf structure and parser controls. Slack carving was not performed. None selects an ASCII recipe.

Both original producer-commit PGP packets specify RSA1/SHA256, excluding direct native signing of those exact packets with secp256k 1 Q. Fifteen parser controls pass; PR6's MPI-bit anomaly remains recorded. This establishes neither PGP validity, shared human control nor historical identity, and does not exclude an unobserved cross-signature.

The original PR1 conversation is authenticated:132,147 bytes, SHA256 `ace30d716cb937fccd2ff4920f4b47db3a0cfafa14d3fdab307a2b67b8fc79e7`. Native title ZNN Fixes, Conversation 0, five ordinary protocol-fix commits, no description or production attachment. The older tree contains graphics, but this is not a graphics-focused PR. Its ordinary 145.000 ZNN fee is not an entrance to the 145-signature publication. PR6's conversation remains unavailable, not negatively inspected.

Publication and profile coverage. The fixed November 15–December 31 population has 55 IDs, 27 previously retained and 28 newly recovered, including one tombstone through an exact dated archive. Original December 21 poll 1473271411192086533 asks how to dispose of the unsolved treasure. Capture 28 seconds later preserves its question; current metadata gives four options and 375 votes, 201 Burn. Dynamic counts/options are weaker than the dated question. This explains community disposition, not a new stage.

The last eligible December 16 profile and November 3 RSS expose 12 distinct post IDs/titles with no explicit challenge/artwork/treasure/recovery/verification title. Older profile captures add 30 occurrences, 11 distinct titles andten titles beyond the earlier population, again with no qualifying explicit title. Bodies were not automatically fetched. Closer profiles distinguish native October 15 publication/update from laterDecember 9 update for the same wZNN article. Such dates are ordinary metadata. Later authenticated May 2019 profile semantics supply two ordinary original titles, six previews andeighteen anchors, with no qualified production, puzzle or verifier instruction.

A completed original-publisher archive query has 258 successful records, 237 URLs and 199 digests, preserving all 199 previous successful tuples. The exact prepublication `/anims/logo/` index returns only two known captures of one PNG. This bounds that response, not the entire host. A Projects 1 profile tab and empty exact historical tab query do not authorize guessing projects. The source-selected original invite identifies the 2021-10-22 group through the latest of twelve captures and a matching native digest; it supplies no message ID, seed link, forward origin or secondary decoder. Member counts are not selectors. The later CTX Twitter correction supersedes a Telegram-only assumption.

Actual media, rather than captions alone. The selected original publisher videos were decoded/OCRed across 8,631 frames and show ordinary promotion, rollout and wallet transfer. The December 23 burn video has 187 frames; its entire visual content supplies no new literal instruction. Its AAC audio has 600,064 decoded float samples, all zero; independent HLS/offline controls and sine/silence fixtures agree. The container's 2026 creation versus stream's 2021 creation identifies current platform rendition evidence, not original-upload byte authentication. Fixed local ASR adds qualified machine audio evidence, without a human-listening or original-digest claim.

Six recovered QR payloads are ordinary payment URIs with empty data. Atframe 472 the QR changes before the printed address; by 480 they agree. This defeats universal same-frame equality, not ordinary payment semantics. No hidden locator follows from the transient mismatch.

Ordinary website objects. Original October 12 `warp.svg` has 94 white paths and Illustrator 25.3.1 export metadata, serving as a decorative background with no text/script/resource/ASCII recipe. Original October 18 FAQ has eleven ordinary questions, including project anonymity policy, not an author-proof contract. A May 2021 article's sole image is recovered through its exact native 301 Location and shows Phase 0/syrius/PlasmaPoints; it supplies no GUI version/file/hash/authenticator. Current pixels are not independently dated historical pixels. Direct raw-MD5 hypothesis fails: source

token `3874e48a18f127d3533c452473cb61a2` versus current PNG MD5 `9c5cff905375e4a7a7a76b56deaf6e62`. This does not prove historical pixel change. Three current literal-palette searches were unregistered/uninformative and had relaxed punctuation matching; they are not controlled negatives.

Independent final campaign reviews find no presently admissible, unconsumed exact source pointer. B08's 62 read-only checks over 47 files are consistency review; B09/F01 point to completed coverage and are not additional experiments. Deleted/uncaptured history remains a stated boundary.

Archive integrity correction and delivery limits

LH-C037's original May 2019 profile replay returned HTTP200 but failed the independently indexed raw digest; five original Medium sibling controls matched. No guessed compression, character normalization or content repair was admitted. Native `x-archive-src` then selected an original ArchiveTeam WARC/index chain.

The 10,159,144-byte compressed index matches declared SHA1/MD5. Its complete 128,861-record parse finds one exact profile row and preserves all same-timestamp records. A prospectively fixed 12,913-byte gzip member at byte range 2223064016–2223076928 verifies range, CRC, WARC block and raw payload digests. The captured 61,696-byte chunked HTTP transfer body hashes to native CDX/WARC digest `SMK0LB2I4X0TYFCQIBFZWIVKZEST20HM`. Declared standard dechunking produces 61,667 bytes, exactly the replay entity, SHA256 `fb29d248b80798817e8e904c277c462c55c0f17f2fc19f06f8ba45e91759564b`.

C042 supersedes the unresolved integrity blocker without HTML repair; the initial failed check remains preserved. The entire 5.4 GB WARC was not downloaded. These controls authenticate archive/protocol objects, not a secret publisher identity or second-stage intention.

Delivery verification is also distinct from puzzle discovery. The full relocated long-horizon replay passed 2026-10-11 04:58:25 UTC with the original package inaccessible:15,573 frozen files plus complete population/native-object checks. Separate hash-pinned adapters fix two T06 path-construction AST calls and one C036 dependency expression, selecting an already-preserved byte-identical baseline. Original assertions and scientific bytes remain unchanged. Unrelated-working-directory/relocation controls pass; wrong-path/byte mutations fail. Initial failures remain recorded. External archive entry SHA256/CRC receipts establish actual packaging success, rather than inferring it from starting ZIP creation.

Parked infrastructure and settlement findings retain their narrower scope

Later native checks preserve the verdict **verified infrastructure, puzzle selection not established**. They do not make sidechain missing bytes an active puzzle blocker.

The requested BRAIN comparison now reproduces: Q.x's last byte `d7` equals 215, the conditional AUTHOR[SUB] second output BRAIN. Q.y endpoint49ALERT and checksum foot1960VITAL differ. Actual BRAIN[SUB] outputs DIRECT/METAL/CLOCK and its path ends DINNER; the equality does not cast that CALL output into a public coordinate or import P3928/DIFFER's consumer. Actual GRAB[SUB] outputs CARPET/PULSE/CATALOG, not PRIMARY, so it does not inherit physical GRAB PRIMARY/FORUM provenance.

The accepted public quest account's burn send is `ac3fb28796c0aec1c512396e54f23fe5882246c44372cc23897d455cfe88f7eb`, confirmed 23 December 2021 14:55:10 UTC; receive is `c65cfdc5db8818729f19acea9289a965c0c5e95b1e15289700cce7b484f35272`, 14:56:10 UTC. It burns 4,745.134 QSR without a scripted instruction. The quest key signs that account action, not Bitcoin Q; a public recovered mnemonic does not uniquely identify the actor.

Rootstock's canonical block at the selected height is not the coinbase-committed candidate. Exact uncle `0x2f896267d08cad258a82b8c81d48b97e1457264017c56458db6b5b4449511fc7`, height 3,870,562, verifies native RLP hash, mining prefix, exact Bitcoin block 710968 header, ten-hash coinbase branch and stated PoW; changed proof/prefix controls fail. Full tag is `5f1bb1a8975179f9a93c7cbf4364adb09da3c2b8663bc84ea000f729003b0f62`; timestamp 2021-11-23T11:00:17UTC. ExtraData `...IRIS-ba01ea2` is ordinary client build identity.

Selected contract events describe signed BTCUSD publication and aggregation, with five historical signers in a nine-member round andzero altered-message members. PriceFeed/Medianizer executable bytes match while metadata differs; CoinPairPrice build is not reproduced. Historical RPC slots/rounds lack independent state proofs. These limitations do not expose a puzzle field. A previous Bitcoin block already contains an RSK tag from another pool; tag presence is not uniquely Zenon-directed. VCash absence in one control is not a selector.

Changing genesis SporkAddress from `z1qpxswrfnlll355wrx868xh58j7e2gu2n2u5czv` to `z1qqv2fnc3avjg39dcste4c5lag7l42xyykjf49w` preserves genesis hash `9e204601d1b7b1427fe12bc82622e610d8a6ad43c40abf020eb66e538bb8eeb0`, with both configurations passing native checks. Thus that data commitment does not bind the config authority. Separate account/momentum/history/enforcement receipts authenticate ordinary upgrade/funding. Exact Q-keyed legacy-allocation/pillar populations havezero matches, without excluding other author-controlled keys. The repeated BRIDGE occurrence supplies no new operand/destination; the selected terminal payment is ordinary nested-SegWit redemption in a bulk transaction.

Eight final mechanism tests add bounded negatives

The AF campaign closes eight registered experiments without a new authored message, second-stage point or historical-authority claim. All source values were previously exposed. Freezing transformations before complete outputs avoids retrospective repairs; it does not turn these tests into blind independent predictions.

Test	Frozen mechanism	Complete result
E001	q/1 chooses first/last letters of forty source words, two layouts, two carrier directions and two polarities	All 8 complete strings uninformative
E002	q/1 is parity or one of eleven source bits, with fixed layouts/directions/complements	0/96 exact 40-cell matches; native parity 22/40
E003	q/1 is a SHA256 watermark of normalized native rows or complete artwork	0/8 exact matches; no salt/offset search
E004	Q as an even-Y x-only empty-tree Taproot internal key; inspect its exact address	Two indexes return zero confirmed/mempool transactions and empty lists

Test	Frozen mechanism	Complete result
E005	Any affine Boolean function of eleven bits plus constant, or integer threshold	No solution in any of four fixed source/order systems
E006	q/1 plus period/comma selects a BIP39 prefix letter	0/16 complete meaningful readings; two field-local variants invalid at ordinal 23
E007	Four punctuation Hamming (15, 11) channels	0/64 complete packets validate across the fixed layouts/directions/parities
E008	One-bit q/1 plus two-bit punctuation mixed in original order	Complete 156-bit stream and physical-band/polarity controls give no new instruction

The two source layouts are continuous 55-byte field concatenation and the earlier 38 field-local words followed by two residual words. They do not treat residual 22 as unused AES bits.

E005 is a complete bounded impossibility result, not just a few failed bit tests. All 4,096 affine rules and 4,098 integer-threshold rules were checked per system. Each affine system has rank 12 and augmented rank 13. A compact contradiction uses source ordinal 1/value 57 paired with 1 at (4, 21), ordinal 7/value 741 with 1 at (6, 16), ordinal 11/value 1707 with 1 at (7, 13), and ordinal 14/value 1143 with q at (7, 21):

```
57 XOR741 XOR1707 XOR1143 =0
```

Their four q=0/1=1 glyph bits XOR to 1. An affine binary reader must preserve the source relation, and its constant cancels across four cells. Thus it cannot generate all forty identities. The second layout and reversed pairing also fail. This excludes this content-check family, not arbitrary nonlinear codes, rendering intensity or independently authored glyph data. The independent verifier checks 65 contradiction certificates and recovers a planted affine rule uniquely.

E004 follows the standard Q-derived address:

```
bc1pm5pwjzezjm8aw40775htwam2zx2t9nqkvhga4f4cspxcell2nh7qv688rq
```

Its tweak is `68d958f0dadf9b1fdbde4a1202238dc7f003829813d6b58f04eb5823a9b1ea21`; outputx is `dd02e90b2296cfd755fef52eb7776a1194b2cc1665d1daa6b8804d8cffea9dfc`. Derivation was already known; the new action was the native-history lookup. Blockstream and mempool.space agree on empty history at 2026-10-11T16:50:38Z. Seven official BIP341 address/script vectors and OpenSSL-backed public tweaks pass. This is agreement between public indexes, not cryptographic proof of universal absence; no arbitrary script-tree variations were tried.

The older fixed Q-pubkeyhash history remains 25 confirmed transactions, 42 funded/42 spent outputs andzero mempool records. Its latest known sweep is `cd7b40e5c7c62563a38a571e51f5e65d351be5369a8378a95715cb69aa8b69d1`, height 720498, 26 January 2022 17:51:21UTC. Two additional records are incoming preconstruction funding at 709586/709591, with no OP_RETURN, not later Q messages.

E006's full primary forty-letter string is `recoraqrwppreonthiinambeutamrogsmooteird`. The tempting RECO prefix was noticed mentally before execution and disclosed in the contract; it fails when read completely. No rearrangement or repair was admitted.

E007's primary uncorrected syndromes are 10/1/2/0. The upper-dot stream has syndrome 10 in the classic layout or 11 in the systematic layout under every direction, defeating both parity conventions. Automatic single-error correction always produces a codeword and supplies no independent validation. Synthetic controls cover all 2,048 messages and 30,720 single-bit corrections.

E008's native continuous 156-bit output is:

```
111111111111100011010000000001011110000101111000010111010001011001010101010111010101001101011001
010100010111010101001010110101010110000101010010010101011001
```

It gives `ZOO SHOOT ACCIDENT SEED MAGIC MENU FEVER PRICE FLOCK MERRY FAMOUS PRINT BENEFIT CLIENT`, U11 tail`01`, bytehex`fff8d005e1785d16555d535951754ad5615255` and byte remainder`1001`. Flipping q/1 yields `ZOO SECRET LATIN ROAD INVEST HEN FIELD MERCY EXPECT SHY FOREST CONFIRM MAD PUZZLE`, also U11 tail 01. An isolated PUZZLE does not make a complete instruction.

Physical bands give ZOO for rows 1–3, MINIMUM ACTUAL MAGNET ARM for 4–8, GRAB for 9–11 and FRONT ONION FEBRUARY TUNNEL PULL RADAR ENABLE for 12–19. The flipped latter bands give LUNAR WIFE INTO TYPE and LABOR SPOON BIKE TISSUE SIREN ROTATE HOPE. All ten continuous/band/polarity outputs, including residuals and unhelpful words, remain in the register; none was repaired or selectively promoted.

The separate offline verifier reconstructs both source layouts, full readers, contradiction certificates and cached history. SHA256 known-answer, BIP341, Hamming and planted-rule controls pass. Initial Python 3.9 compatibility and receipt-name parsing errors were fixed in the verifier without changing source evidence, contracts or experimental receipts. All eight tests are complete. No unfinished parity/Hamming/letter-selector search remains. The later CTX recovery changes screenshot provenance but adds no secondary rule, so the scientific verdict is unchanged.

Remaining gaps after these campaigns

The strongest missing production evidence is an independently dated original ASCII input, project, intermediate output or edit recipe fixing input identity, converter/version, ramp, geometry, preprocessing and substitutions independently of known final labels. A source-fixed recipe can be checked against known coordinates with wrong-input/source-sibling controls; exposed targets cannot be relabeled blind.

Other possible entrances remain independent: an issuer-controlled quest disclosure or source-defined app notification with its context; an original decoder/commitment selecting an action/object/verifier or genuinely untouched target; a qualified native Email 260 source/envelope/opening; or a historical secret-dependent signature/prospective commitment over an intelligible exact Q/artwork claim. No one missing file is made mandatory for every future route.

Conditional NURSE needs an occurrence-local record assignment and consumer preserving its actual RECALL/join ancestry. Conditional ACQUIRE needs actual same-caller authenticated source-author and visible-proof parents, with claim, operator, register and return semantics. Its class-label validator accepts broken ancestry controls and cannot admit the proof by itself. GAS needs its current parent's typed operands/comparison operator; CRACK DOCUMENT needs its occurrence-local document selector, action and successor. Document 541's return to the same 145-message object does not fill that separate consumer gap.

Missing original production and issuer-notification identifiers remain unknown; no honest exact fetch command exists for an unidentified artifact. Completed unavailable-source lookups, crypto censuses, renderer fits and exposed decoders are not fresh evidence when repeated. These are finite evidentiary boundaries, not proof of no second stage, worldwide source absence or a completed secondary solve.

Current proof gaps and supersession rules

The coherent result is a completed original treasure solve plus a substantial conditional secondary reconstruction. The record supports neither replacing every failed interpretation with a claim that nothing was authored nor promoting every reproduced calculation into an intended instruction. The following gates remain distinct:

1. **Original source context.** Recover the issuer-controlled prefix disclosure or original quest/app receipt, and the deleted community thread's source links. The recovered community tweet is a useful provenance object; its screenshot wrapping does not authorize a spatial cipher.
2. **Secondary entrance.** Establish a pre-existing, source-local rule selecting the external historical object and reader. A genuine message, certificate, source article or ciphertext cannot select itself retrospectively.
3. **Program typing.** Supply the rule that promotes the ART-shifted twelve-position vector to executable code and selects P00 as head. A dictionary word and a 4|4|4 layout alone do not supply it.
4. **Physical interpreter installation.** Supply occurrence-local polarity, orientation, pairing, scale, opcode, state cast and consumer contracts. Geometry, attached suffixes and exact checksum equalities constrain proposed rules; they do not automatically install a universal VM.
5. **Q-packet recipe.** Explain the mixed fields/operators/positions, order, omission, and inherited register prospectively. Twelve matching words, uniqueness under a chosen predicate, or a self-decoded trailer does not establish that recipe.
6. **BACON/AIR transition.** Establish ordered BUNKER − ARTWORK and the Email260 namespace, then a separate AIR-to-CHOOSE[ADD] contract. A local packet endpoint and a model's AIR label are different occurrences.
7. **Native return.** Preserve actual joins in the conditional model while distinguishing copied labels from a fresh payload-bearing return. A native signed account receipt verifies its account domain; an unsupported incoming caller cannot be repaired by that verification.
8. **Historical authority.** Produce an exact, puzzle-selected, historical secret-dependent signature or commitment binding the relevant authority to Q. Old coinbase keys, 2020 signature control, authentic OpenPGP self-certifications, or access to copied ciphertext do not establish this edge.

The October 7 word-blind Q audit and true-frontier report govern the AIR/ACTUAL promotion in REV3. The October 11 source-context correction governs the assumed Telegram origin. The October 12 residual tournament and focused Burn report add conditional findings; they do not override the missing-selection gates. These corrections preserve underlying bytes and arithmetic while changing which inference is licensed.

When revisiting a rejected test, name the original population, representation, predicate, control and failure. A new test needs a changed premise or new source evidence. Do not erase a scoped negative or claim that it disproves every possible encoding. Likewise, do not multiply dependent equalities into independent odds, use a post-observed target as a precommitment, or treat repeated copies as independent witnesses.

The source register retains older claims such as “complete solve,” “breakthrough,” “authenticated custom scale,” and “current frontier.” Those are historical author statements. The status map and correcting receipts above specify the narrower current reading.

Coverage and preservation contract

The inventory examined **63,394 local files** in the eleven requested folder roots and four requested Desktop documents, excluding generated work and this output. Their total recorded size was 28,570,913,104 bytes. Much of that is upstream software, cached/native data, media, repeated packages, and administrative metadata. The October 13 Ferry/Syrius changelog project is unrelated to the Taproot puzzle and is excluded from the knowledge account.

The compilation stores **8,024 distinct source-file identities**: **2,142 narrative/log/document sources** and **5,882 algorithm or structured-data sources**. Original file identities are SHA256-based. Equal files share one identity; duplicate copies are counted. All four requested Desktop sources are represented, including the original master, reproducibility appendix, secondary reconstruction and partial secondary solve. The Desktop master and its September-folder copy have identical SHA256 `5b7088a210113d77dbf3c4b72a237e0f9dc57138f53614ba8d73ac8cba8257d8`.

The evidence register contains **100,987 unique normalized narrative blocks**, referenced **295,211 times** by ordered source sequences. Identical paragraph, heading, list and table-row bodies are stored once. A body already printed in the main text is referenced there. Different formulations, qualifiers, conflicting claims, or revisions are retained when equivalence is not established. No approximate-match rule silently deletes a near duplicate. Semantic reconciliation occurs in the main account; archival formulations remain evidence of the research history rather than additional accepted facts.

Structured receipts and source algorithms use a shared literal-token dictionary. It preserves the full anonymized text, including number spellings, field order, duplicate JSON keys, whitespace, code and large execution populations. Tokens and equal complete data bodies are stored once. This avoids replacing full alternative/control populations with verdicts alone. Every restored data text is checked against its own displayed SHA256. The compressed data stream's SHA256 is `fcb44d8bd28fed2a4a0e41c3b2f8c851dd1782a0f349ac4679accfef39605cb3`. The source index and decoder below make the same-file annex recoverable without executing any research program.

All 155 root ZIP central directories were inspected; 147 distinct root archive hashes were identified. Top-level text members were read without executing code. A separate nested audit checked nonvendor archives and their 32 text records, with no errors or resource limits reached; ten vendor fuzz/build archives were excluded. Nested definitions, gates, plans and checkpoint records were compared with local hashes. One new administrative manifest had no new substantive reasoning. Embedded compressed ledgers were decoded, matched to source identities and retained as data. Four puzzle DOCX sources were text-extracted; their only images were thumbnails. The supplied

steganography book is represented by its specific puzzle audit, tests and interpretation, rather than repeating an unrelated textbook in full.

What this preservation claim means. Each included narrative source can be reconstructed as its ordered normalized blocks after anonymization and removal of redundant administrative inventories. Byte-sensitive artwork, bit strings, quoted cryptographic payloads and code fences retain their literal content. DOCX text extraction and ordinary paragraph wrapping are presentation transformations. The source index retains original-file hashes separately from normalized/display hashes; a normalized excerpt is not falsely presented as the original signed file. Old self-loading scripts that expect a particular Markdown filename/layout or an original compilation hash must not be run unchanged against this reorganized document. Their algorithms, constants and public source references remain available.

Coverage limits. Inventory is not semantic adjudication. This file does not claim that every pixel, audio sample, arbitrary source-tree file, fetched webpage, entire external book/judgment, operating-system metadata record or raw chain snapshot has been independently interpreted. Generic dependencies, build products, file-path inventories, and unrelated projects are excluded. External sources are retained through their puzzle-relevant extracts, audits, exact object identifiers and source references. Parser failures, missing original sources, historical unavailable runners and tests that never executed remain explicit in the research record. No new puzzle solution, historical identity proof or exhaustive rejection of all encodings is claimed.

Independent local compilation checks passed for the 722-byte and 741-byte artwork fixtures, field lengths, AES-GCM authentication, public prefix hash, BIP39 checksum, final entropy byte ART102, four-word subtraction and GLOBAL40. Full source restoration and data-stream integrity are separately checked. These checks concern preservation and the fixed entrance constants; they do not prospectively validate secondary source selection.

Source and evidence notation

`S00001` denotes an original file identity. Its index record gives an anonymous filename, corpus origin, original SHA256/size, displayed SHA256 and storage location. For narrative sources, `refs` is the source's ordered list of evidence IDs; join their restored bodies with two newline characters. `normalized_sha256` checks that reconstruction. `E000001` denotes one stored evidence block. `M00001` points to a matching body already in the main text. All E records are archival source material; their wording is governed by current corrections above.

The data decoder is a reader, not a solver runner. It decodes Base64, decompresses gzip, parses JSON and joins literal tokens. It never calls `exec`, evaluates research code, accesses the network, or performs a transaction. Exporting a selected source is optional and creates an anonymous source-ID filename.

Appendix B — Preservation, coverage, and independent checks

1. What this edition preserves

This edition has two connected layers: the readable book and the research collection. The book explains the complete retained course of the investigation, carries the detailed October technical atlas forward, and records later findings and corrections through 6 November 2026. The collection preserves the evidence behind it at the level actually recovered. A reader can use the book without running code; a researcher can inspect the data, calculations and earlier interpretations without reconstructing this conversation.

The original October container was a 122,417,660-byte Markdown file. Most of that size was encoded evidence, not continuous prose. Its source index named 8,024 records. Those records have now been reconstructed into separate readable files and checked against the container's declared normalized or literal-text hashes. All 8,024 checks passed before the privacy adjustment described below. The original file's SHA-256 remains in the preservation record:

```
d3e196d609144dc317b56b69ba90a0f11bbb1b70b926bbdbdd1c5293fd45c1b6
```

Preserved baseline population	Count
Narrative, log and document source identities	2,142
Data and algorithm source identities	5,882
Total reconstructed source identities	8,024
Distinct narrative evidence blocks	100,987
Ordered references to those blocks	295,211
Resolved main-text reference blocks	304
Literal tokens in the embedded data dictionary	1,178,841
Direct literal payloads	5,871
Literal aliases	11
Reconstruction hash failures before anonymization	0

The original reconstructed source texts totaled 687,141,817 bytes. This is not the size of the compressed download. Repetition is stored economically in the original container, while the reconstructed files deliberately make each source accessible. Multiple identities can contain related,

repeated or superseded evidence; the count must not be interpreted as 8,024 independent experiments.

The later recovery adds 1,825 successfully materialized source identities, represented by 1,557 distinct preserved top-level files. These include 660 package identities. The package catalogue records 15,735 member occurrences, represented by 9,056 distinct expanded member files. A member appearing in several packages retains each parent relationship, while identical member contents need not be stored repeatedly. These numbers describe coverage and deduplication, not the number of successful puzzle deductions.

The full record includes exploratory prompts, source notes, programs, machine-readable receipts, tables, reports, images and superseded summaries. A planning document remains a planning document even if its title includes “proof” or “complete.” A report can contain a real calculation and an unsupported interpretation. Preservation retains both so the later correction can be checked.

2. How to find a particular result

Start with a chapter's source label. A **B:** label names a section in Appendix A, the complete October technical reference. Its **K04–K35** modules preserve the original numbering; gaps in that numbering are historical, not missing pages from this book. The accompanying baseline catalogue resolves **S00001–S08024** to original labels, provenance, reconstructed paths and hashes.

An **R** label names a recovered later source. A label containing **M**, such as a package-member identifier, names a member within that source's package. Appendix C lists the later sources cited in the current narrative. The full later-source browser additionally searches filenames, retained dates, titles, summaries, file types and package relationships. The source's own date is preferable for an experiment chronology; a file's creation date can be later than the work it records.

Location in the extracted collection	Purpose
`START_HERE.html`	Entry point to the book and both evidence browsers
`TminusZ_Zenon_Master_Solve_2026-11-06.pdf`	Paginated reading and printing copy
`TminusZ_Zenon_Master_Solve_2026-11-06.html`	Searchable, navigable offline reading copy with embedded figures
`TminusZ_Zenon_Master_Solve_2026-11-06.md`	Editable plain-text manuscript
`archive/archive_browser.html`	Searchable browser for all 8,024 baseline source identities
`archive/sources/`	Reconstructed narrative, data and algorithm records
`archive/catalog.csv` and `catalog.json`	Complete baseline inventory and hashes
`recent_archive/index.html`	Searchable browser for recovered October–November material
`recent_archive/files/`	Preserved later source files
`recent_archive/members/`	Expanded, deduplicated package members

Location in the extracted collection	Purpose
`recent_archive/catalog.csv` and `package_members.csv`	Later source and parent/member inventory
`recent_archive/coverage.md`	Recovery limits and alternate copies of unavailable transfers
`verification/`	This edition's independent checks and reproducible inputs
`SHA256SUMS.txt`	File-integrity manifest for the complete collection

The browsers work offline and do not execute the archived research programs. A source can be read in a browser or text editor; a CSV can be inspected in a spreadsheet. Large JSON receipts are retained as structured data instead of being expanded into thousands of printed book pages. Long literals may wrap in the PDF. Use the exact source file when byte identity matters.

V01 — editorial verification

The following checks were independently executed while preparing this edition. They use the canonical artwork recovered from the baseline and ordinary deterministic computation. This is a defined verification set, not a claim that every experiment in the collection was rerun.

Check	Reproduced result	What the result establishes
Source dimensions and digest	19 rows × 38 columns; 722 separator-free bytes; canonical SHA-256 matches	The retained character object is internally consistent with the declared source.
Native fields	A:16 bytes; B:18; C:5; E:16, after independent Base64 decoding	The visible fields and byte roles are reproduced exactly.
AES-GCM	Key SHA256 of exact ASCII `Taproot`; nonce A; ciphertext B||C; tag E; empty associated data	The authenticated plaintext is `wall nation drive smoke`.
Nonfield populations	352 spaces, 192 z, 27 q, 13 ones, 32 commas, 13 semicolons, 8 periods, 5 colons	Counts exclude the eighty native-field characters and sum to 642 cells.
Equal LOW/FULL classes	z/comma:224; 1/semicolon:26; q/period:35	The two parity observations discard 285 literal glyph choices.
Additional literal-role bit	Occupancy, LOW, FULL and original ASCII bit1 distinguish the retained nonfield glyphs	The added observation allows exact glyph reconstruction; intended payload capacity is not inferred.
CONTROL98	98 symbols, 196 bits; four-bit footer `1001` matches the body hash prefix	The short checksum relationship reproduces for the stated reader.
Structural marker stream	Forty bits: 27 zeros and 13 ones	q/1 extraction preserves all forty source positions.
Blank components	Sizes 149,148,33,22 before the move	The original blank-region geometry is exact.

Check	Reproduced result	What the result establishes
T30 chamber relationship	The reflected lower22 is contained in upper33; roof11 remains	The unmatched cells are row11, columns15–25.
qzzq movement	Original rows12–15 at column15 move by (+3,−3) into original blanks	The resulting blank components have sizes149,148,55.
Local A/C tile copy	A rotated by180° equals C in the specified 3×7 windows	The local map is U(r,c)=(27−r,30−c), not T30.

The checked CONTROL98 body is the following twenty-four-byte value:

```
fff8e8aa97ea5fa97ba5955bb59d7358eed66dd6ba35996b
```

The complete original structural q/1 stream is:

```
10000010001000010001011101010011000100000
```

The exact inputs, output JSON and a portable verification script are in `verification/`. The AES check uses the Python `cryptography` library. The other checks use ordinary Python byte, set and integer operations. Archived historical programs are not silently executed as part of opening the collection or checking its file hashes.

The eight thousand source reconstruction checks answer a different question from V01: they establish that the preserved evidence can be recovered faithfully. Neither a matching source hash nor an error-free archive certifies the correctness of a historical theory.

4. Attribution and privacy

This master is authored and compiled as **TminusZ**. No personal identity is needed to use it. The technical evidence retains public puzzle names, public historical figures, public addresses, keys, transaction hashes and source references where those are necessary to understand or verify the work. Those public source identities are not additional bylines for this master.

One private local-account component survived inside a baseline provenance path. The original reconstructed source passed its recorded hash before that component was replaced with TminusZ. The corresponding literal-token record was updated consistently. The transformation record identifies the affected source as **S07516**, retains both the original and published hashes, and makes the change explicit without publishing the private component. The other 8,023 source reconstructions are unchanged.

The original 122 MB container and its unredacted compressed annex are therefore not included in the public preservation package. Their complete recoverable research content is represented by the verified, anonymized source collection. The published literal-token archive has SHA-256:

```
1a69e3e1725f6ca4683e57744bd574271282bc9d6b397afa8147998ac544989b
```

Later-source curation likewise distinguishes original-source digests from preserved-file digests whenever a privacy transformation was necessary. Private local path components and nonessential

session metadata are not part of the puzzle. Removing them does not alter public cryptographic constants. Exact signed historical objects can contain public identities within their native bytes; changing those would destroy their verification meaning, so the source-domain distinction remains explicit.

5. Coverage limits that must remain visible

“All preserved research” has a concrete scope. The October container supplied normalized or anonymized narrative texts and literal data records. Its four DOCX identities were recoverable as extracted text, not as original Word binary files. The collection labels them accordingly. It does not claim to recover every external book, original video, dependency, image, remote service state or blockchain snapshot ever mentioned in a source.

The later retrieval had twenty uncompleted source transfers. A review of the already recovered material found distinctive alternate copies for seven of those records, possible copies for five, and no matching copy for eight. The coverage note identifies every item and its candidates. The failed transfers have no retrieved content hash, so even a matching filename and size cannot prove byte identity with that unavailable version. Generic names such as README or FORWARD_REPORT are particularly ambiguous. The unresolved items remain listed rather than being silently counted as complete.

The later-file integrity check passed for all 10,613 distinct materialized files and for 871 additional member records retained inside their parent packages. There were no size or hash failures. Some nested material is accessible by opening its preserved parent package, rather than through a separate expanded-file link. These storage relationships are recorded explicitly in the catalogue.

The complete preservation collection is supplied in two ordinary ZIP volumes. Extract both into the same parent folder so their shared collection directory combines. Volume 1 contains the book, the baseline reconstruction, readable later reports and expanded evidence; Volume 2 contains the retained later package files and comic archive. Both volumes carry the same file manifest and verification instructions. The PDF and standalone HTML remain readable on their own.

Some historical claims also lack their original executable receipt even when their prose report survives. The eight-algorithm planning report is one example addressed in the visual chapter. Preserving its text protects the history; classifying its unsupported run counts as verified experiments would corrupt that history. The same principle applies to unavailable original posts, missing reply parents, absent full video clips and unproven producer software.

The package is a portable checkpoint, not a guarantee about every future copy. Its readable sources, ordinary file formats and integrity manifest make independent copies and later verification possible. A future edition should add a dated supplement, preserve this edition, and record which conclusion changes and why. No old negative result, successful calculation or superseded route needs to disappear when the investigation advances.

Appendix C — Sources cited in the current narrative

The complete evidence collection has a much larger catalogue. This index identifies the later records cited by the readable narrative. Source dates below are retained file dates, which can differ from the experiment or publication date stated inside a report. Historical titles are identification labels, not endorsements of their claims.

Baseline citations beginning **B:** point to the named section or **K** module in Appendix A. The full **S00001–S08024** inventory is in `archive/catalog.csv` and its offline browser. Later **R** records and **M** package members are in `recent_archive/catalog.csv` and `package_members.csv`. **V01** is the editorial verification in Appendix B.

Source R0086

Zenon/Taproot: 192 `z` slots, CONTROL98 checksum, and a second interpretation of INSTALL PANEL

Retained date: 2026-10-13. Historical file: `ZENON_Z192_INSTALL_PANEL_FINDINGS_2026-10-13.md`.

Open preserved source.

Source R0089

Zenon Taproot — INSTALL PANEL: AIRPORT44 → EYE CANYON

Retained date: 2026-10-14. Historical file: `ZENON_AIRPORT44_INSTALL_PANEL_FORWARD_2026-10-13.md`.

Open preserved source.

Source R0098

Zenon Taproot — Z192 five-word installation manifest and RECALL return

Retained date: 2026-10-14. Historical file: `ZENON_Z192_INSTALL_MANIFEST_FORWARD_2026-10-13.md`.

Open preserved source.

Source R0100

Zenon Taproot — z192 self-derivation → four checksum gates → CHECK AXIS → Q

Retained date: 2026-10-14. Historical file: `ZENON_Z192_CHECK_AXIS_FORWARD_2026-10-13.md`.

Open preserved source.

Source R0119M0003

Zenon — EXHIBIT / ACCOUNT12 → POEM WIDTH → source-selected s-field checkpoint

Retained date: 2026-10-14. Historical file: `FORWARD_REPORT.md`.

Open preserved source · Parent package: R0119.

Source R0143

Zenon Taproot — the exact two-axis chamber fold and five-word return

Retained date: 2026-10-14. Historical file: `FORWARD_REPORT(6).md`.

Open preserved source.

Source R0180

Zenon Taproot — f=3p parity projection and conditional G40 return

Retained date: 2026-10-15. Historical file: `FORWARD_REPORT(20261015-032539).md`.

Open preserved source.

Source R0207

Zenon Taproot — `LYRICS SCRIPT PATTERN LIST` as a typed source-record candidate

Retained date: 2026-10-15. Historical file: `FORWARD_REPORT(20261015-142448).md`.

Open preserved source.

Source R0230

Zenon Taproot — 5/4 → c11 selects the lower q spine and exposes GLASS's four-bit z-memory write

Retained date: 2026-10-15. Historical file: `FORWARD_REPORT(20261015-161401).md`.

Open preserved source.

Source R0246

Zenon Taproot — full GLASS22 versus original residual22; survivor CONTROL96

Retained date: 2026-10-15. Historical file: `FORWARD_REPORT(20261015-183424).md`.

Open preserved source.

Source R0248

Zenon Taproot — GLASS96 deletion and four-bit checksum-delta test

Retained date: 2026-10-15. Historical file: `GLASS96_FORWARD_FINDINGS.md`.

Open preserved source.

Source R0250

Zenon Taproot — GLASS11 traverses CONTROL98, Z192 and VOID22

Retained date: 2026-10-15. Historical file: `FORWARD_FINDINGS.md`.

Open preserved source.

Source R0295

Zenon Taproot — 2021-only BOTTOM/COVER/ADDRESS bootstrap and historical holdout

Retained date: 2026-10-16. Historical file: `FINDINGS(4).md`.

Open preserved source.

Source R0316

Zenon Taproot — GLASS RECALL receiver identifiability test

Retained date: 2026-10-16. Historical file: `FINDINGS(20261016-181622).md`.

Open preserved source.

Source R0321

Zenon Taproot — Recovering GLASS without feeding the 2020 panel

Retained date: 2026-10-16. Historical file: `FINDINGS(20261016-191340).md`.

Open preserved source.

Source R0326

Zenon Taproot: full five-field private instruction, source-to-witness execution

Retained date: 2026-10-16. Historical file: `FINDINGS(20261016-233120).md`.

Open preserved source.

Source R0340

Zenon Taproot — GLASS MOVE yields checked PULL TWENTY, then PIPE TWENTY

Retained date: 2026-10-17. Historical file: `FINDINGS(20261017-022154).md`.

Open preserved source.

Source R0342

Zenon Taproot forward continuation — PIPE / !9 / DERIVE

Retained date: 2026-10-17. Historical file: `ZENON_PIPE_DERIVE_NEXT_EDGE_FINDINGS_20261016.md`.

Open preserved source.

Source R0344

Zenon Taproot — Does the 145-record historical table have to be loaded?

Retained date: 2026-10-17. Historical file: `ZENON_TAPROOT_TABLE_BLIND_FINDINGS_20261016.md`.

Open preserved source.

Source R0352

Independent audit — 17 October 2026

Retained date: 2026-10-17. Historical file: `ZENON_OCT17_INDEPENDENT_NATIVE_JOIN_AUDIT.md`.

Open preserved source.

Source R0405

Zenon — SUBMIT AMOUNT FIT: source-native genesis selector and receiver frontier

Retained date: 2026-10-17. Historical file: `REPORT(20261017-220106).md`.

Open preserved source.

Source R0470

Zenon Taproot — B9 + FINE6 selects native ORBIT9/95

Retained date: 2026-10-18. Historical file: `REPORT(20261018-181232).md`.

Open preserved source.

Source R0482

Independent review — Sol Max Zenon typed interpreter (18 October 2026)

Retained date: 2026-10-18. Historical file: `SOL_MAX_INTERPRETER_INDEPENDENT_REVIEW_20261018.md`.

Open preserved source.

Source R0486

Zenon Taproot — independent native r2-chain and E15 continuation

Retained date: 2026-10-18. Historical file: `REPORT(20261018-211300).md`.

Open preserved source.

Source R0487

Zenon Taproot — E15/C5 and the native R/I fixed-point circuit

Retained date: 2026-10-18. Historical file: `REPORT_FORWARD.md`.

Open preserved source.

Source R0489

Zenon Taproot — ORBIT9 native receiver / historical panel bridge

Retained date: 2026-10-18. Historical file: `REPORT(20261018-233119).md`.

Open preserved source.

Source R0520

Official artwork JPEG: OutGuess and pixel-remapping forensic check

Retained date: 2026-10-19. Historical file: `official_artwork_outguess_analysis.md`.

Open preserved source.

Source R0551

Crypto Publius (October 2021) poem: prospective-reading audit

Retained date: 2026-10-20. Historical file: `REPORT(20261020-121630).md`.

Open preserved source.

Source R0554

Crypto Publius and the 2021 Zenon clue trail

Retained date: 2026-10-20. Historical file: `Crypto_Publius_2021_Clue_Ledger.md`.

Open preserved source.

Source R0614

Crypto Publius / Zenon Taproot — EYES → authentication tag → b5/F5, executed continuation

Retained date: 2026-10-21. Historical file: `EYES_TAG_F5_REPORT.md`.

Open preserved source.

Source R0633

Crypto Publius 148-post archive — source-first riddle audit

Retained date: 2026-10-21. Historical file: `REPORT(20261021-193555).md`.

Open preserved source.

Source R0741

Zenon / Crypto Publius: E publisher → terminal spender → sweep

Retained date: 2026-10-23. Historical file: `FORWARD_REPORT(20261023-131447).md`.

Open preserved source.

Source R0748

E2 as a source-map for the MOVE/PULL/COPY phases

Retained date: 2026-10-23. Historical file: `REPORT(20261023-141326).md`.

Open preserved source.

Source R0830

Zenon Taproot — phase-gated 53 and the DER → recoverable-signature bridge

Retained date: 2026-10-24. Historical file: `REPORT(20261024-143534).md`.

Open preserved source.

Source R1003

Zenon Taproot — full conditional renderer and the indispensable owner mask

Retained date: 2026-10-26. Historical file: `ALGORITHM_FRONTIER_REPORT_2026-10-26.md`.

Open preserved source.

Source R1090

Zenon/Taproot — a narrowed historical-source selector and a cross-publication donor

Retained date: 2026-10-27. Historical file: `REPORT(20261027-192101).md`.

Open preserved source.

Source R1092

Zenon / Taproot — NEXT PRIMARY 4→5 and the cross-publication PATTERN field

Retained date: 2026-10-27. Historical file: `REPORT(20261027-195514).md`.

Open preserved source.

Source R1117

Zenon CONTROL98: UPDATE RAZOR FORUM / PROVIDE FRESH RANGE

Retained date: 2026-10-28. Historical file: `REPORT(20261028-103534).md`.

Open preserved source.

Source R1142

Original-only E2 recognizer: an alternative bootstrap

Retained date: 2026-10-28. Historical file: `FINDINGS(20261028-201054).md`.

Open preserved source.

Source R1173

Zenon block 709632 artwork: renderer/provenance test

Retained date: 2026-10-29. Historical file: `RULEBOOK_FINDINGS.md`.

Open preserved source.

Source R1244

Two eyes: original wall → colon twin → C-source bridge

Retained date: 2026-10-31. Historical file: `FINDINGS(20261031-030807).md`.

Open preserved source.

Source R1246

Zenon: two eyes, direct visibility versus mirrored memory

Retained date: 2026-10-31. Historical file: `FINDINGS(20261031-031515).md`.

Open preserved source.

Source R1395

ZENON_LOWER_APERTURE_DIRECTION_20261101.zip

Retained date: 2026-11-01. Historical file: `ZENON_LOWER_APERTURE_DIRECTION_20261101.zip`.

Open preserved source.

Source R1395M0004

REPORT.txt

Retained date: 2026-11-01. Historical file: `ZENON_LOWER_APERTURE_DIRECTION/REPORT.txt`.

Open preserved source · Parent package: R1395.

Source R1400

Zenon lower-eye official-image cross-check — 2026-11-01

Retained date: 2026-11-01. Historical file: `FINDINGS(20261101-132436).md`.

Open preserved source.

Source R1416

Whole-image decoding of the Zenon Taproot 19×38 artwork

Retained date: 2026-11-01. Historical file: `FINDINGS(20261101-163607).md`.

Open preserved source.

Source R1452

Zenon original artwork — RIM / E2 / nine-cell bridge / 55→46

Retained date: 2026-11-02. Historical file: `FINDINGS(20261102-020044).md`.

Open preserved source.

Source R1454

The Zenon rim records and an actual PGP-signature test

Retained date: 2026-11-02. Historical file: `FINDINGS(20261102-023720).md`.

Open preserved source.

Source R1458

Zenon rim map: source-to-GIANT forward execution (Nov 1, 2026)

Retained date: 2026-11-02. Historical file: `FINDINGS(20261102-032755).md`.

Open preserved source.

Source R1475

TUNNEL_GIANT_20261102.zip

Retained date: 2026-11-02. Historical file: `TUNNEL_GIANT_20261102.zip`.

Open preserved source.

Source R1475M0001

The central tunnel opens with a four-glyph move

Retained date: 2026-11-02. Historical file: `TUNNEL_GIANT_20261102/DISCOVERY.md`.

Open preserved source · Parent package: R1475.

Source R1613

Executive Summary

Retained date: 2026-11-03. Historical file: `Executive Summary(20261103-154030).md`.

Open preserved source.

Source R1674

ZENON_QZZQ_TUNNEL_REPLAY_20261104.zip

Retained date: 2026-11-04. Historical file: `ZENON_QZZQ_TUNNEL_REPLAY_20261104.zip`.

Open preserved source.

Source R1684

Zenon — Tunnel information-dependence audit

Retained date: 2026-11-04. Historical file: `REPORT(20261104-180250).md`.

Open preserved source.

Source R1696

Zenon Taproot artwork — observation-first all-azimuth ledger

Retained date: 2026-11-04. Historical file: `REPORT(20261104-200751).md`.

Open preserved source.

Source R1700

Zenon Taproot artwork — primary-source LSB/BPCS image audit

Retained date: 2026-11-04. Historical file: `REPORT(20261104-203225).md`.

Open preserved source.

Source R1726

CryptoPublius: the five mirrored sightlines and the unmatched sixth [1]

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-013130).md`.

Open preserved source.

Source R1762

Zenon original-artwork phase-lock: two 11-bit packets, candidate DEVELOP FRAME

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-144416).md`.

Open preserved source.

Source R1766

Zenon artwork: three one-hot records and the two-sided frame lock

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-151100).md`.

Open preserved source.

Source R1780

Zenon 2021 artwork vs May 2020 logo — renderer identifiability / frame-delta audit

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-170501).md`.

Open preserved source.

Source R1783

Zenon ASCII puzzle: blinded logo-to-glyph prediction, punctuation control, and historical palette test

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-173941).md`.

Open preserved source.

Source R1795

Marker-only frame lock and twelve-pair difference layer

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-182650).md`.

Open preserved source.

Source R1800

Mirror calibration versus a new payload

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-205841).md`.

Open preserved source.

Source R1803

Literal glyph roles and the seven choices omitted by the six-bit model

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-212903).md`.

Open preserved source.

Source R1808

Seven-marker geometric reconstruction

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-220826).md`.

Open preserved source.

Source R1812

Original-source instruction audit: the eyes emoji and the missing operations

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-224038).md`.

Open preserved source.

Source R1815

FOLLOW GOLD: complete-reader audit and the 32-state test

Retained date: 2026-11-05. Historical file: `FINDINGS(20261105-233803).md`.

Open preserved source.

Source R1820

Kaine quotation and reply-context audit

Retained date: 2026-11-06. Historical file: `FINDINGS(20261106-010610).md`.

Open preserved source.

Source R1823

ZENON_FRAME_LOCK_CONTINUATION_20261105.zip

Retained date: 2026-11-06. Historical file: `ZENON_FRAME_LOCK_CONTINUATION_20261105.zip`.

Open preserved source.

Source R1823M0002

Zenon: page-fixed and piece-following frame audit

Retained date: 2026-11-06. Historical file: `Zenon_Frame_Lock_20261105/REPORT.md`.

Open preserved source · Parent package: R1823.